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Editors

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The Next Frontier of Cybersecurity: Zero Trust for Enterprise IoT Ecosystems



Belal Ali  and Omar Amjad Dib 

Abstract The rapid proliferation of Internet of Things (IoT) devices across enterprise environments has significantly expanded the digital attack surface, exposing organisations to new classes of security threats. Traditional perimeter-based security models are increasingly ineffective in this context, as IoT devices often lack standardised controls, operate in distributed ecosystems, and frequently interact with untrusted networks and services. This paper presents a structured Zero Trust Security (ZTS) framework explicitly tailored for enterprise IoT ecosystems. Grounded in the principles of continuous verification and least-privileged access, the proposed five-step methodology addresses the unique security challenges posed by IoT deployments. Key components include protect surface definition, transaction flow mapping, Zero Trust network architecture, granular policy enforcement, and continuous monitoring. The paper also identifies practical implementation challenges, such as device discovery, authentication, and segmentation, and offers guidance on overcoming them. By aligning ZTS principles with IoT-specific requirements, this work contributes a scalable and adaptable approach to building resilient, context-aware security architectures for future enterprise infrastructures.

Keywords Zero trust security · IoT security · Zero trust framework

1 Introduction

The exponential growth of the Internet of Things (IoT) has fundamentally transformed enterprise infrastructures, enabling novel applications across sectors such as manufacturing, healthcare, and finance. According to the Gartner Machina IoT

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database [1], the number of connected enterprise devices is projected to exceed 18 billion by 2030, outnumbering enterprise users by a factor of four. While this proliferation unlocks operational efficiencies and new business opportunities, it also introduces significant security concerns due to the expanding attack surface.

Research by Palo Alto Networks Unit 42 [2] highlights the severity of this challenge: 30% of enterprise assets in 2020 were IoT devices, 57% of which were highly vulnerable to cyber threats, 98% transmitted unencrypted traffic, and 83% operated on unsupported systems. These figures underscore the limitations of traditional perimeter-based security models, which are increasingly inadequate for managing diverse and distributed IoT environments [3].

IoT devices, often resource-constrained and lacking standardized security controls, frequently interact with cloud services, third-party networks, and remote access points. This dynamic ecosystem dissolves the notion of a fixed network perimeter, necessitating a fundamental shift in security strategy. Zero Trust Security (ZTS), which operates on the principle of “*never trust, always verify*”, offers a robust alternative by enforcing continuous authentication, contextual access control, and real-time monitoring [4].

While Zero Trust adoption is maturing in traditional IT environments, its application in IoT remains limited due to the unique constraints of these devices and the lack of structured implementation models. **This paper addresses this research gap by proposing a five-step methodology specifically designed to operationalize Zero Trust principles within enterprise IoT ecosystems.** The methodology focuses on defining the protect surface, mapping transaction flows, architecting Zero Trust networks, enforcing granular policies, and maintaining continuous monitoring.

The remainder of this paper is organised as follows: Sect. 2 reviews the foundational concepts of ZTS. Section 3 outlines the key challenges of adopting Zero Trust in IoT contexts. Section 4 presents the proposed five-step framework. Section 5 discusses future directions and concludes the study.

2 Zero Trust Security

Zero Trust Security (ZTS) was introduced by Forrester Research [5] to overcome the inherent weaknesses of perimeter-based security models. These models rely on the outdated assumption that entities within a network are inherently trust-worthy. In contrast, ZTS is grounded in the principle that no device, user, or application should be automatically trusted, irrespective of its location [4, 6].

ZTS is a multilayered cybersecurity approach that incorporates continuous verification, strict access control, and dynamic policy enforcement [7, 8]. This paradigm shift helps mitigate risks from lateral movement, advanced persistent threats, and device compromise by ensuring that access decisions are always context-aware and identity-driven.

Traditional security models attempt to secure the entire attack surface, which is increasingly complex with the proliferation of IoT devices [9]. ZTS redefines

this model by focusing on a smaller, always-knowable *protect surface*—comprising critical data, applications, assets, and services (DAAS) [10]. This focused approach facilitates the implementation of micro-perimeters, allowing more precise policy enforcement.

In IoT environments, ZTS requires organizations to enforce least-privileged access [11], monitor every transaction in real time [12], and validate the health and identity of devices continuously [13]. This approach reduces the exposure to threats such as ransomware, unauthorised access, and policy violations.

2.1 *Trusted Elements: The Pillars*

Implementing ZTS in IoT ecosystems involves securing the core domains of: Identity, Device/Workload, Access, and Transaction [3].

- **Identity:** Authenticate and authorise every device, user, and service before granting network access.
- **Device/Workload:** Continuously monitor device integrity, posture, and workload execution.
- **Access:** Apply least-privilege principles to limit interaction to only necessary resources.
- **Transaction:** Ensure that all communications are encrypted, inspected, and policy-compliant.

2.2 *Secure Processes: The Enablers*

Alongside device-level security, operational security processes are essential [14].

- **Visibility and Analytics:** Leverage telemetry, logging, and analytics for anomaly detection.
- **Segmentation:** Use network and micro-segmentation to contain threats.
- **Continuous Monitoring:** Employ ML-driven tools for real-time behavioural analysis [15].
- **Policy Enforcement:** Regularly update and enforce device-specific policies.
- **Asset Management:** Maintain an up-to-date inventory of authorised IoT devices.

3 Challenges in Adopting Zero Trust for IoT

Despite its advantages, implementing ZTS in IoT ecosystems faces several technical and operational challenges [16, 17].

3.1 Discovery and Identification

IoT devices often lack unique identifiers and have limited processing power, making them difficult to detect using conventional endpoint solutions. Many rely on proprietary protocols or prepopulated signatures, which restrict scalability.

3.2 Authentication and Segmentation

Most IoT devices do not support standard authentication mechanisms like 802.1X, necessitating manual onboarding and policy creation. This hinders automated segmentation and increases the risk of misconfiguration.

3.3 Continuous Risk Assessment

Standard vulnerability scanning tools often miss IoT endpoints, and scanning sensitive devices may disrupt operations. This results in unassessed risks and blind spots in the security posture.

3.4 Lack of Intelligent Policy Engines

Existing solutions frequently lack the intelligence to automate Zero Trust policy recommendations. Manual configurations are prone to errors, and many tools are limited to alerting without enforcement capabilities.

4 Overcoming Zero Trust Challenges for IoT Devices

To operationalise ZTS in IoT environments, we propose a five-step methodology that addresses the specific limitations of IoT devices [6]. This approach is illustrated in Fig. 1.

4.1 Define the Protect Surface

Identify and prioritise critical DAAS components to establish micro-perimeters around high-value assets.

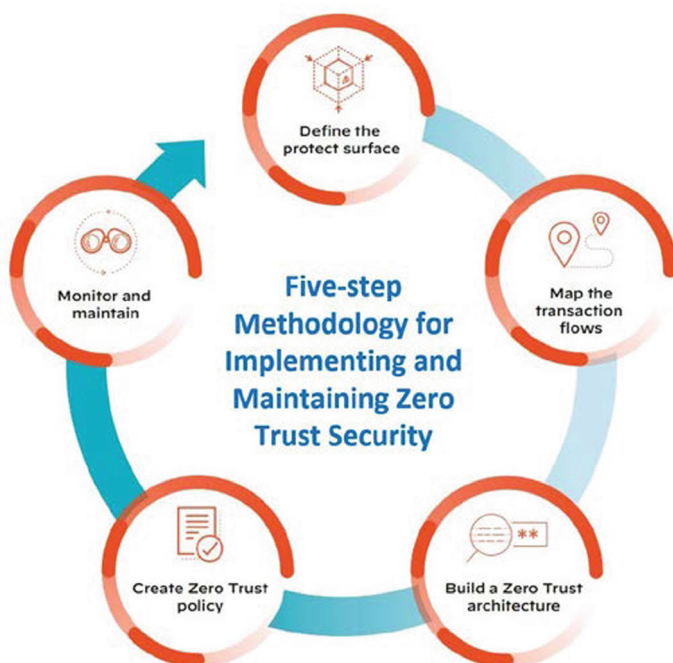


Fig. 1 The five-step methodology

4.2 Map the Transaction Flows

Document device communications and dependencies to inform context-aware access policies.

4.3 Architect a Zero Trust Network

Design a segmented network around the protect surface using gateways and Layer 7 controls to limit unauthorised access.

4.4 Create Granular Security Policies

Develop policies based on contextual parameters (who, what, when, where, why, and how) to limit exposure.

4.5 Monitor and Maintain

Continuously inspect device activity, enforce telemetry-based adjustments, and adapt policies to evolving threats.

5 Conclusion and Future Directions

The complexity of enterprise IoT ecosystems demands a reimagined approach to security. This paper proposed a targeted, five-step Zero Trust framework tailored to the unique characteristics of IoT environments. By focusing on protect surface definition, transaction flow mapping, policy enforcement, and continuous monitoring, organisations can build a resilient security posture.

As IoT deployments continue to scale, the use of automation and machine learning will be essential for scalable Zero Trust implementations. Future work should explore the integration of technical artefacts, such as flowcharts, pseudocode, or decision trees, to provide actionable guidance for real-world deployments. Ongoing research should aim to develop adaptive and context-aware ZTS models that evolve in response to the dynamic threat landscape.

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Next-Generation IoT Connectivity: FSO-RF Integration with High-Altitude Platforms for Reliable Data Transmission



Ramy Samy, Shuang Li, Fang Xu, and Basem M. ElHalawany

Abstract The increasing demand for high-speed and reliable Internet of Things (IoT) connectivity presents serious challenges to existing terrestrial and satellite-based networks, such as congestion, high costs, and scalability limitations. High-altitude platforms (HAPs) are a viable alternative, providing wide-area coverage and low-cost deployment. However, they are constrained by limited radio frequency (RF) bandwidth. We propose a hybrid free-space optical (FSO)-RF communication system for HAP-enabled IoT connectivity to address this limitation. FSO can cover a dense area of IoT communications at high speed through FSO links to access points (APs). At the same time, RF can provide wide area coverage to IoT devices outside the range of APs. We analyze the system's performance by deriving the ergodic capacity for both APs and remote IoT devices. Through Monte Carlo simulations, we validate the theoretical model, while numerical results demonstrate the system's effectiveness in enhancing IoT connectivity.

Keywords IoT connectivity · FSO-RF · High-altitude platforms

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1 Introduction

The Internet of Things (IoT) has revolutionized device communication, enabling a vast network of interconnected devices [1]. IoT applications in healthcare, transportation, agriculture, and smart cities are rapidly expanding, increasing the demand for high-speed and reliable data transmission. As of 2024, approximately 18 billion IoT devices are connected worldwide, with projections exceeding 32 billion by 2030 [2]. This rapid growth places significant strain on existing communication infrastructures, highlighting the urgent need for reliable, high-speed connectivity. As the IoT ecosystem continues to expand, developing innovative solutions to address bandwidth limitations and spectral congestion while improving cost efficiency is crucial for sustaining seamless communication.

High-altitude platforms (HAPs) are emerging as a game changer for IoT connectivity, offering a cost-effective and flexible alternative to satellites [3]. Positioned in the stratosphere, HAPs provide wide-area coverage with lower latency and significantly reduced deployment costs. Unlike satellites, which require expensive ground stations, HAPs can establish direct connections with IoT devices, making them particularly beneficial for remote and underserved areas. Compared to traditional terrestrial networks, a single HAP can replace up to 450 cellular towers, significantly reducing infrastructure costs and power consumption [3]. However, most existing HAP systems rely on radio frequency (RF) transmissions, which are constrained by bandwidth limitations and spectral congestion [4, 5]. As the number of connected devices continues to grow, RF-based systems struggle to meet the increasing demand for high-capacity data transmission, especially in densely populated areas and IoT hotspots.

Free-space optical (FSO) communication offers a viable solution to the growing demand for seamless connectivity, offering high-speed data transmission with license-free bandwidth [6]. By integrating FSO with RF in a parallel configuration, HAP-based systems create a dual-mode communication: FSO delivers high-capacity data transmission in dense hotspot areas, while RF ensures reliable, wide-area coverage. This hybrid approach enhances overall system efficiency and scalability, making it well-suited to meet the expanding demands of IoT and beyond. In urban hotspots, IoT devices typically connect to access points (APs) via Wi-Fi or 5G networks, enabling high-speed, localized connectivity. These APs aggregate data from densely deployed IoT devices and efficiently transmit it to the HAP via FSO links, ensuring smooth data processing and uninterrupted connectivity. However, in remote or rural regions where APs are unavailable, IoT devices establish direct connections to the HAP through RF transmission. The RF link provides reliable long-range connectivity and is less susceptible to environmental disruptions compared to FSO, making it ideal for wide-area coverage.

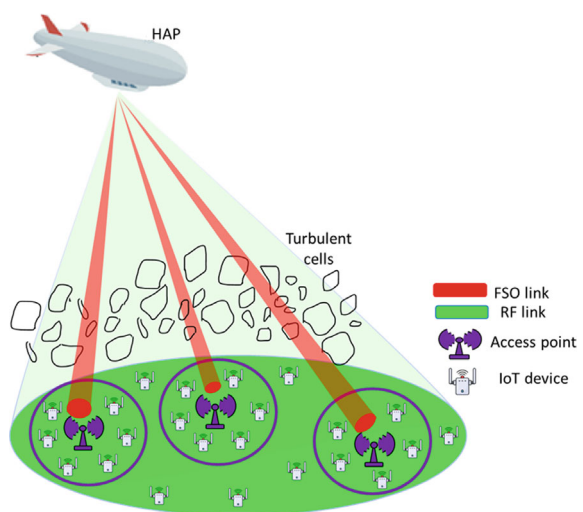
This paper investigates the potential of HAP-based systems utilizing parallel FSO-RF communication and highlights their advantages. Unified analytical expressions are derived for the ergodic capacities of APs and remote devices (RDs) located outside AP coverage. The analysis incorporates intensity modulation with direct

detection (IM/DD) and heterodyne detection (HD) schemes, employing the Gamma-Gamma fading model for FSO transmission and the Rician fading model for RF transmission. Monte Carlo simulations are performed to validate the accuracy of the theoretical findings. The rest of this paper is structured as follows: Sect. 2 introduces the system and channel models. Section 3 analyzes the ergodic capacity. In Sect. 4, we present numerical examples to illustrate key findings. Finally, Sect. 5 concludes the paper.

2 System and Channel Models

The proposed system architecture comprises a single HAP and multiple APs, as illustrated in Fig. 1. The HAP is equipped with dual-mode transmission capabilities, utilizing both FSO and RF resources. FSO links are employed to deliver high-speed connectivity to APs in high-density IoT environments. IoT devices within AP coverage benefit from reliable, high-speed data transmission. Meanwhile, the RF link provides ubiquitous coverage, enabling seamless connectivity for remote IoT devices located outside the coverage of APs. We prioritize data transmission to APs, as they serve a large number of IoT devices. When an AP fails, the HAP dynamically switches to RF transmission to restore connectivity for one AP at a time using the RF payload at the APs. The system operates using a hard switching scheme to simplify receiver design. The analytical model is based on the following assumptions: (1) The HAP serves N APs, utilizing N FSO links and a single RF link for extended coverage. (2) When an FSO link to an AP fails, the HAP switches to RF transmission, which is shared among APs with equal priority. (3) The system operates under

Fig. 1 HAP with FSO-RF transmission for IoT reliable connectivity



a non-saturated traffic condition. (4) FSO transmissions follow the same statistical fading model. (5) FSO links experience independent atmospheric turbulence effects. (6) The HAP accurately receives channel state information for both RF and FSO transmissions through feedback from APs. (7) There is no error-control mechanism implemented between the APs, remote IoT devices, and the HAP.

2.1 FSO Channel Model

In this study, we consider Gamma-Gamma fading as the primary impairment affecting FSO transmissions, incorporating the effects of pointing errors and atmospheric conditions [6]. The unified probability density function (PDF) of the instantaneous SNR, for both HD and IM/DD, is given by [7, Eq. (2)]

$$f_{\gamma_b}(x) = \frac{\xi^2 x^{-1}}{b \Gamma(\alpha) \Gamma(\beta)} G_{1,3}^{3,0} \left(\alpha \beta k \left(\frac{x}{\mu_b} \right)^{\frac{1}{b}} \middle| \begin{matrix} \xi^2 + 1 \\ \xi^2, \alpha, \beta \end{matrix} \right), \quad (1)$$

where $k = \xi^2/(\xi^2 + 1)$, ξ represents the pointing error coefficient [8], $\Gamma(\cdot)$ is the Gamma function [9, Eq. (8.310)], $G(\cdot)$ represents Meijer's G-function [9, Eq. (9.301)], and b is a detection-dependent parameter ($b = 1$ for HD, $b = 2$ for IM/DD). The parameters α and β characterize large-scale and small-scale fading effects caused by atmospheric turbulence [10, Eqs. (54, 55)]. μ_b represents the average electrical SNR, which is associated with the average SNR, $\bar{\gamma}_b$, as $\mu_1 = \bar{\gamma}_1$, $\mu_2 = \frac{\alpha \beta \xi^2 (\xi^2 + 2)}{(\alpha + 1)(\beta + 1)(\xi^2 + 1)^2} \bar{\gamma}_2$, and $\bar{\gamma}_b = \frac{(\eta P_f G_{tx} G_{rx} k I_l A_0)^b}{FL^b \sigma_n^2}$, [7, 11, Eq. (10)], where η denotes the optical-to-electrical efficiency. P_f represents the FSO transmit power. G_{tx} and G_{rx} are the transmit and receive telescope gains, respectively. FL refers to the free-space loss over the FSO channel. σ_n^2 is the variance of additive white Gaussian noise (AWGN). A_0 denotes the fraction of transmitted power collected at the receiver aperture. I_l accounts for atmospheric attenuation due to varying weather conditions and is modeled using Beer-Lambert's law [8].

2.2 RF Channel Model

In HAP-based communication with remote IoT devices and APs, the RF signal primarily follows a line-of-sight (LoS) path with minimal scattering and reflection from the surrounding environment [3, 5]. Due to this dominant LoS component and relatively weak multipath effects, the small-scale fading coefficient of the RF link is well characterized by the Rician distribution. The presence of a non-zero mean in this model represents the influence of a strong LoS component. The PDF of the instantaneous SNR is given by [12, Eq. (2.16)]

$$f_{\gamma_r}(\gamma) = \frac{K+1}{\bar{\gamma}_r} \exp\left(-(K+1)\frac{\gamma}{\bar{\gamma}_r} - K\right) I_0\left(2\sqrt{K(K+1)}\frac{\gamma}{\bar{\gamma}_r}\right), \quad (2)$$

where $I_0(\cdot)$ represents the zeroth-order modified Bessel function of the first kind and K is the Rician factor. The average SNR $\bar{\gamma}_r$ is expressed as $\bar{\gamma}_r = \frac{P_r G_{tx,r} G_{rx,r}}{FL_r L_o \sigma_{n,r}^2}$, [13, Eq. (5.34)], where P_r is the RF transmit power. $G_{tx,r}$ and $G_{rx,r}$ denote the transmit and receive antenna gains, respectively. FL_r represents the free-space path loss over the RF link. $\sigma_{n,r}^2$ corresponds to the variance of the thermal noise at the RF receiver. L_o accounts for additional losses due to atmospheric absorption, polarization mismatch, rain attenuation, and misalignment errors. With the application of [9, Eq. (8.447.1)] and $\mathcal{F} = \frac{K+1}{\bar{\gamma}_r}$, (2) can be rewritten as

$$f_{\gamma_r}(\gamma) = \mathcal{F} \exp(-K) \exp(-\mathcal{F}\gamma) \sum_{m=0}^{\infty} \frac{(K\mathcal{F})^m}{(m!)^2} \gamma^m, \quad (3)$$

3 Ergodic Capacity Analysis

We consider data transmission in the downlink scenario. It is important to note that the communication link between an AP and IoT devices within its coverage is beyond the scope of this paper. The analysis focuses on the ergodic capacity of APs and remote IoT devices outside AP coverage. As all APs are identical, the capacity performance can be analyzed by focusing on a single AP, referred to as the tagged AP. The remaining $N - 1$ APs are referred to as other APs. The ergodic capacity of the tagged AP, $\bar{C}_{AP}$, is calculated as [14, Eq. (8)]

$$\bar{C}_{AP} = (1 - P_o) \bar{C}_f + P_o P_r \bar{C}_r, \quad (4)$$

where $\bar{C}_f$ and $\bar{C}_r$ represent the ergodic capacities of the FSO and RF links, respectively. P_o denotes the outage probability of the FSO link, while P_r indicates the probability that the HAP utilizes the RF link to transmit data to the tagged AP. Based on the system's operational mode, the HAP switches to the RF link to maintain AP connectivity when the corresponding FSO link experiences poor quality, i.e., when the instantaneous SNR falls below the predefined threshold γ_{th} . Consequently, P_o is defined as $P_o = P[\gamma_b < \gamma_{th}] = F_{\gamma_b}(\gamma_{th})$. By applying (1) and [15, Eq. (07.34.21.0084.01)], P_o can be expressed as

$$P_o = \mathcal{A} G_{b+1, 3b+1}^{3b, 1} \left(\mathcal{D} \frac{\gamma_{th}}{\mu_b} \left| \begin{matrix} 1 & \mathcal{B}_1 \\ \mathcal{B}_2 & 0 \end{matrix} \right. \right), \quad (5)$$

where $\mathcal{A} = \frac{b^{\alpha+\beta-2} \xi^2}{(2\pi)^{b-1} \Gamma(\alpha) \Gamma(\beta)}$, $\mathcal{D} = \left(\frac{\alpha\beta k}{b^2}\right)^b$, $\mathcal{B}_1 = \{\frac{\xi^2+1}{b}, \dots, \frac{\xi^2+b}{b}\}$ comprises of b terms, and $\mathcal{B}_2 = \{\frac{\xi^2}{b}, \dots, \frac{\xi^2+b-1}{b}, \frac{\alpha}{b}, \dots, \frac{\alpha+b-1}{b}, \frac{\beta}{b}, \dots, \frac{\beta+b-1}{b}\}$ comprises of $3b$ terms. The ergodic capacity of the FSO link to the tagged AP, $\bar{C}_f$, can be calculated as

$$\begin{aligned} \bar{C}_f &= \int_{\gamma_{th}}^{\infty} BW_f \log_2(1 + \epsilon \gamma) f_{\gamma_b}(\gamma) d\gamma \\ &= \underbrace{\frac{BW_f}{\ln(2)} \int_0^{\infty} \ln(1 + \epsilon \gamma) f_{\gamma_b}(\gamma) d\gamma}_{I_{C1}} - \underbrace{\frac{BW_f}{\ln(2)} \int_0^{\gamma_{th}} \ln(1 + \epsilon \gamma) f_{\gamma_b}(\gamma) d\gamma}_{I_{C2}}, \end{aligned} \quad (6)$$

where BW_f represents the bandwidth of the FSO link, and ϵ is a constant, defined as $\epsilon = 1$ for the HD technique and $\epsilon = e/(2\pi)$ for IM/DD [16]. To compute the integral I_{C1} , we express $\ln(1 + \epsilon \gamma)$ in terms of a Meijer G-function using [15, Eq. (07.34.03.0456.01)]. By applying algebraic transformations and utilizing (1) along with [15, Eq. (07.34.21.0013.01)], the analytical expression for I_{C1} is

$$I_{C1} = \frac{BW_f \mathcal{A}}{\ln(2)} G_{b+2, 3b+2}^{3b+2, 1} \left(\frac{\mathcal{D}}{\epsilon \mu_b} \middle| \begin{matrix} 0, 1, \mathcal{B}_1 \\ \mathcal{B}_2, 0, 0 \end{matrix} \right). \quad (7)$$

To compute the integral I_{C2} , $\ln(1 + \epsilon \gamma)$ is expanded using its Taylor series representation [14, Eq. (32)]. Utilizing (1) and [15, Eq. (07.34.21.0084.01)], the analytical expression for I_{C2} can be written as

$$\begin{aligned} I_{C2} &= BW_f \log_2(1 + \epsilon a) P_o + \frac{BW_f \mathcal{A}}{\ln 2} \sum_{u=1}^{\infty} \frac{(-1)^{u+1} \epsilon^u}{u(1 + \epsilon a)^u} \gamma_{th}^v \\ &\quad \times \sum_{v=0}^u \binom{u}{v} (-a)^{u-v} G_{b+1, 3b+1}^{3b, 1} \left(\mathcal{D} \frac{\gamma_{th}}{\mu_b} \middle| \begin{matrix} 1-v, \mathcal{B}_1 \\ \mathcal{B}_2, -v \end{matrix} \right). \end{aligned} \quad (8)$$

The infinite summation in (8) converges provided that the SNR meets the ratio test condition in [14, Eq. (34)]. Given that γ_{th} is generally small, an appropriate value of a can be chosen to ensure convergence.

For the given tagged AP, let P_n represent the probability that n out of the remaining $N - 1$ primary FSO links experience poor channel conditions. Consequently, the HAP will utilize the RF link to transmit data to these n APs at a given time. Using the binomial distribution, P_n can be expressed as

$$P_n = \binom{N-1}{n} P_o^n (1 - P_o)^{N-n-1}. \quad (9)$$

Let P_r denote the probability that the HAP utilizes the RF link to transmit data to the tagged AP. Based on assumptions #2, P_r is expressed as

$$\begin{aligned} P_r &= \left[P(\text{tagged AP}) + P(\text{tagged AP} + \text{one AP})/2 + \dots + P(\text{all APs})/N \right] \\ &= \left[P_0 + \frac{1}{2}P_1 + \frac{1}{3}P_2 + \dots + \frac{1}{N}P_{N-1} \right] = \sum_{n=0}^{N-1} \frac{1}{n+1} P_n. \end{aligned} \quad (10)$$

The capacity of the RF link can be calculated as $\bar{C}_r = \frac{BW_r}{\ln(2)} \int_0^\infty \ln(1 + \gamma) f_{\gamma_r}(\gamma) d\gamma$, where BW_r represents the bandwidth of the RF link. First, $\ln(1 + \gamma)$ is expressed in terms of a Meijer G-function using [15, Eq. (07.34.03.0456.01)]. The analytical expression for $\bar{C}_r$ is then derived through algebraic manipulation, incorporating (3), [15, Eq. (07.34.21.0088.01)], and [9, Eq. (9.31.2)] as

$$\bar{C}_r = \frac{BW_{\text{RF}}}{\ln(2)} \exp(-K) \sum_{m=0}^{\infty} \frac{K^m}{(m!)^2} G_{2,3}^{3,1} \left(\frac{K+1}{\bar{\gamma}_r} \left| \begin{matrix} 0, 1 \\ 1+m, 0, 0 \end{matrix} \right. \right). \quad (11)$$

Now, we compute the ergodic capacity for remote IoT devices located outside AP coverage, denoted as $\bar{C}_{RD}$. Remote devices can utilize RF resources only when they are not allocated to any APs. Therefore, $\bar{C}_{RD}$ is expressed as

$$\bar{C}_{RD} = (1 - P_o)^N \bar{C}_r. \quad (12)$$

4 Numerical Results

This section presents numerical results to illustrate the performance of the proposed HAP-based solution. Due to its simplicity and cost efficiency, all FSO links are assumed to operate using an IM/DD detection scheme. The system parameters are provided in Table 1. To facilitate parallel transmission, the HAP distributes total power between FSO and RF links, with power allocation based on channel bandwidth.

Figure 2a illustrates the ergodic capacity of APs and remote IoT devices located outside AP coverage as a function of the total HAP transmit power. For comparison, the performance of an RF-only HAP-based solution is also included. In the RF-only approach, the entire transmit power is allocated to the RF link. As a result, the RF-only system achieves higher spectral efficiency at lower transmit power levels. However, the parallel FSO and RF transmissions provide greater overall capacity.

In Fig. 2a, we consider a scenario with 5 APs and a binary phase-shift keying modulation scheme. A threshold of 10.5 dB is set to achieve a BER of 10^{-6} , ensuring a specific quality-of-service [10, Eq. (50)]. At a transmit power of 25 dBW, the RF-only system achieves an ergodic capacity of 8 Gbps for all IoT devices, corresponding to a spectral efficiency of 8 bps/Hz. In contrast, the proposed system attains

Table 1 Simulation parameters [5, 10]

Parameter	Value	Parameter	Value
AP aperture height	1 m	RF carrier frequency	2 GHz
HAP height	20 km	Bandwidth of RF	1 GHz
HAP zenith angle	15°	Rician factor	6
Transmit and receive telescope gains	50 dB	RF transmitting and receiving antenna gains	20 dB
Wind speed	41 m/s	RF receiver noise PSD	−114 dBW/MHz
Optical wavelength	1550 nm	RF receiver noise figure	5 dB
Optical bandwidth assigned to AP	2.5 GHz	Telescope aperture radius	10
Optical beam radius at transmitter	2 cm	Optical to electrical efficiency	0.7
Pointing error	5 cm	Optical noise power	250 μW

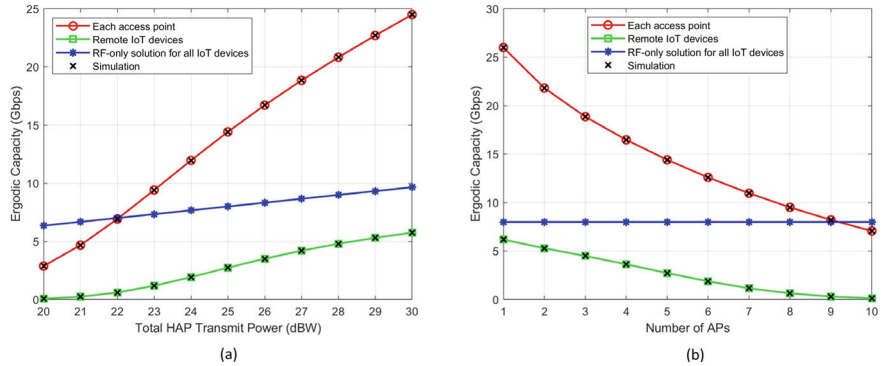


Fig. 2 Ergodic capacity performance: **a** using various transmit power of HAP, **b** with different number of APs

an ergodic capacity of 14.5 Gbps per AP and 2.5 Gbps for remote IoT devices, yielding a total capacity of 75 Gbps, demonstrating the effectiveness of the proposed system in enhancing network performance. This corresponds to spectral efficiencies of 5.8 bps/Hz for each AP and 2.5 bps/Hz for remote IoT devices.

Figure 2b shows the impact of increasing the number of APs on the ergodic capacity. We assume a transmit power of 25 dBW and a threshold of 10.5 dB. As the number of APs increases, the capacity per AP decreases since the total available power is distributed among more APs, reducing the power allocated to each. For example, with one AP, the ergodic capacity per AP is approximately 25 Gbps, but this drops to 7 Gbps at 10 APs. Similarly, the capacity for remote IoT devices also decreases from 6 Gbps at one AP to 0.2 Gbps at 10 APs. This reduction occurs

because the RF link, which supports remote IoT connectivity, becomes increasingly shared among multiple APs, limiting its availability for direct IoT-to-HAP communication. Compared with the RF-only approach, the proposed hybrid FSO-RF solution demonstrates superior performance in total system capacity. Even with 10 APs, the proposed system achieves a total sum capacity of approximately 70.2 Gbps, significantly surpassing the RF-only approach. This highlights the scalability and efficiency of the proposed system.

Figure 3 illustrates the impact of varying pointing error coefficients in FSO links ($\xi = 5, 2$, and 1) on ergodic capacity under different HAP transmit power levels. As expected, pointing errors significantly affect system performance, particularly for APs that rely heavily on FSO transmissions. Lower pointing error values ($\xi = 1$, represented by the dotted lines) lead to noticeable degradation in ergodic capacity. For example, at 25 dBW, the capacity per AP drops from 14.5 Gbps ($\xi = 5$, represented by the solid lines) to approximately 9.5 Gbps ($\xi = 1$), emphasizing the impact of FSO misalignment.

Similarly, for remote IoT devices, the capacity decreases from 2.5 to 1 Gbps under the same conditions. This degradation occurs because severe pointing errors weaken the received optical signal, increasing the likelihood of transmission outages. These results highlight the importance of mitigating pointing errors in FSO-based HAP systems to ensure stable and high-speed data transmission, especially in applications that demand high data rates.

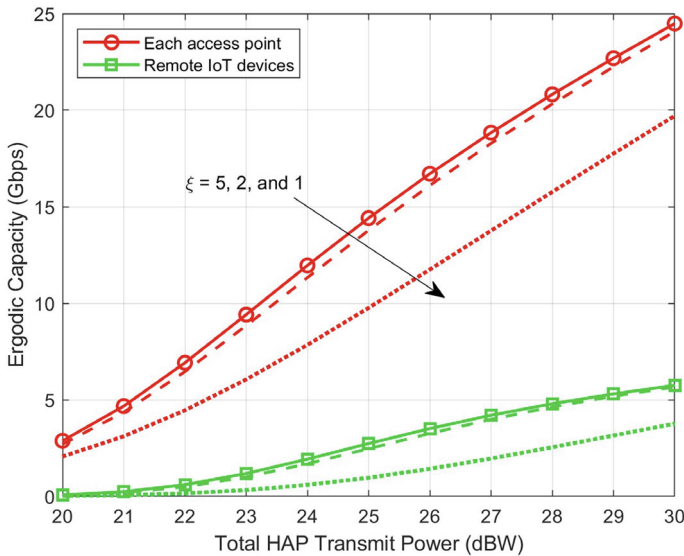


Fig. 3 Ergodic capacity with different FSO link pointing error coefficients

5 Conclusion

In this paper, we propose a hybrid FSO-RF communication architecture for HAP-assisted IoT networks, combining high-speed FSO links with wide-coverage RF connectivity. In particular, we prioritize dense IoT environments through FSO links to APs, while the RF link serves remote IoT devices. We conducted a comprehensive ergodic capacity analysis for APs and remote IoT devices under realistic channel conditions. We showed significant improvements in network capacity through numerical results, even in adverse channel conditions. The proposed HAP-based network can effectively address IoT connectivity challenges in both remote and urban communities. Future research could explore advanced error-control techniques, dynamic resource allocation, and machine learning-based optimizations to further improve system performance.

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Multimodal Pain Recognition: Integrating Facial Expressions and Biomedical Signals with Deep Learning



Arhina Ghosh, Neha Tyagi, Nitin Rakesh, Balamurugan Balusamy, and Akhil Gupta

Abstract Pain expression is characterized by high complexity and multidimensionality. It often uses subjective self-reporting as precise pain assessment remains an essential problem. In this research work, a new methodology for recognition of pain is represented based on a hybrid model that combines convolutional neural networks (CNNs) and particle swarm optimization (PSO) technique along with bidirectional long short-term memory (BiLSTM) networks. This study used video and physiological signals as input data. Experimental results significantly improved the binary pain classification accuracy over several existing models. This study is giving results of 86.3% accuracy and proving the hybrid approach to be very effective. The CNN handles extraction of spatial hierarchies from video data, while the PSO-optimized BiLSTM appropriately models the temporal evolution of physiological responses. Through multimodal integration and advanced optimization approaches, a pathway is paved toward a more reliable and efficient pain recognition model which will eventually help in improving patient care.

Keywords Pain recognition · Multimodal · Facial expressions · Physiological signals · Deep learning

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1 Introduction

Among the cardinal aspects of healthcare, pain assessment truly grips itself to the relevant factors influencing diagnosis, treatment, and subsequent management of the patient. Most of the traditional assessment methods for pain, including various self-reporting scales, are usually subjective and lacks consistency, and often influenced by bias, which altogether makes it difficult to accurately assess the pain experienced by an individual. A few really promising results were achieved in the pain recognition task, utilizing deep-learning approaches with convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers.

This area of research has witnessed such important studies over the years on automatic pain detection. PainAttnNet [1] was proposed as a transformer encoder for multiscale deep learning in pain classification based on physiological signals. Pain recognition was explored through thermal image frame analysis employing vision transformer and video vision transformer, emphasizing the advantages of ROI extraction [2]. A nonlinear dynamic model used galvanic skin response (GSR) for contactless auto-monitoring of the initiation and termination times of pain detection [3].

To address this challenges, this research work follows as:

- A novel model is proposed for pain detection that will integrate video and physiological signals in a most efficient manner.
- The focus of this study will be towards addressing the problem of integrating heterogeneous data using a PSO technique to optimize the selection and fusion of features from video data and physiological data.
- The proposed research study will exploit a Convolutional Neural Network
 - Bidirectional Long Short Term Memory (CNN-BiLSTM) network to take advantage of spatial as well as temporal dependencies found in multimodal data to further improve pain recognition.
- To examine the effectiveness of the proposed framework in terms of reliability, performance will be measured using relevant metrics such as accuracy, precision, recall and F1 score.
- To substantiate the claims that the new approach presented here offers a competitive advantage and helps in augmenting the field, the performance will be compared against some existing studies.

The remaining sections of this paper will present the following information: related works are discussed in Section II; proposed methodology is covered in Section III; experimental results are covered in Section IV; Section V concludes the document with possible future research directions.

2 Related Work

Automated pain detection has made significant advances over the years due to deep learning, multimodal data integration, and advanced physiological signal analysis. In 2025, the researchers explored spatio-temporal deep learning methods for classifying pain using facial expressions, confirming the efficacy of the hybrid spatial-temporal graph convolutional network (STGCN)—LSTM and ConvNeXt-LSTM models [4]. Another study adopted a new AI method called adaptive classification for recognizing children's clinical pain [5]. These previous studies imply that the integration of spatial features and temporal features can be a weapon in more accurate pain recognition.

In 2024, personalized and adaptive neural networks were proposed to deal with inter-subject variability of pain behaviors [6]. Another study introduced a neural ensemble framework that was fused using images from faces and electrodermal activity (EDA) signals [7]. Likewise, PainMeter classified pain levels using physiological signals such as electrocardiogram (ECG), GSR, and electromyography (EMG) [8]. These studies highlight the significance of multimodal data fusion for pain assessment.

Explainable AI (XAI) techniques using deep learning methods have also progressed. In 2024, a deep learning approach to pain sentiment recognition from facial expressions was developed [9]. A logistic regression-based model was proposed for pain intensity detection [10]. An augmentation technique on electrodermal activity (EDA) signals was used to enhance the accuracy of pain recognition models built using support vector machines [11]. These strides clearly illustrate the synergetic potentiality of incorporating classical machine learning approaches with modern regularization techniques for robust pain recognition.

In 2024, synthetic thermal videos were integrated into a vision-multi layer perceptron framework and showcased an improved performance in pain recognition [12]. Another research study investigated the deep transferability of learned features of heat pain to electrical pain [13]. Transformer-based frameworks incorporating facial videos and heart rate signals were also proposed [14]. These works accentuate the flexibility of transformer architectures when dealing with multimodal data for pain detection. All the previous studies clearly shows how these pain recognition methods have evolved from traditional machine learning methods to the deep-learning multimodal framework.

3 Proposed Methodology

This section represents details regarding the method which is proposed for predicting pain. Figure 1 represents the architecture of proposed methodology visually.

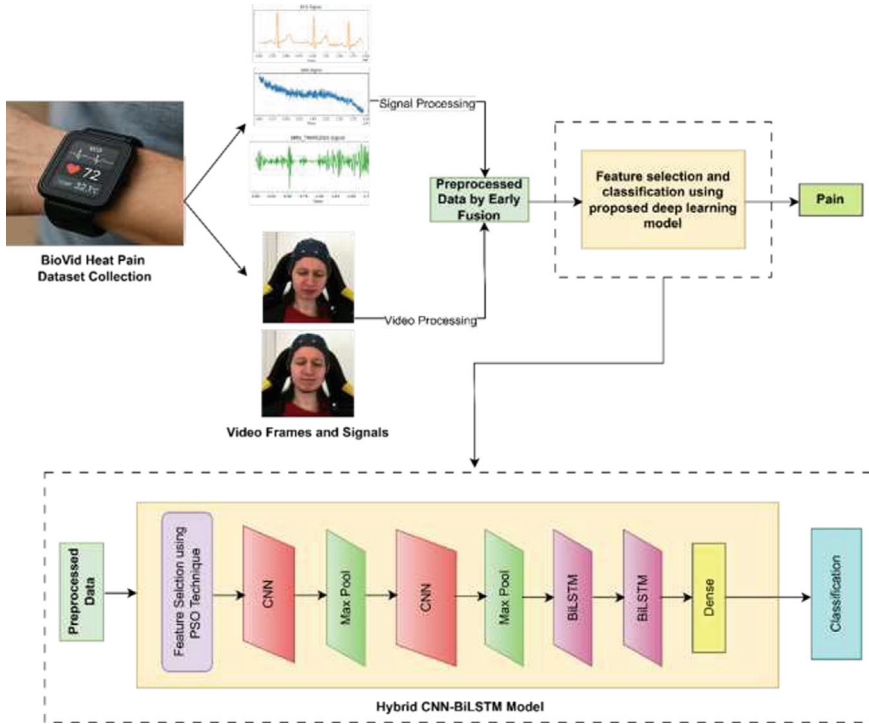


Fig. 1 End to end framework

3.1 Dataset

The dataset used in this research for pain recognition, namely the BioVid Heat Pain Dataset [15, 16], part A, comprises several physiological recordings and video data which were recorded synchronously from healthy participants undergoing an experimental paradigm with carefully controlled heat stimuli. It includes physiological signals in the form of electroencephalogram (EEG), ECG, EMG, and GSR, which offers the physiological description of the pain experience. Besides these physiological measures, including video recordings to assess facial expressions gives tremendous insight into human behavioral responses to pain. Participants were subjected to a validation procedure consisting of varying intensities of thermal stimulation so that pain responses could be assessed over a series of perceived intensities.

3.2 Data Pre-processing and Feature Extraction

This section describes the preprocessing and feature extraction techniques for videos and signals data. The data preprocessing is performed following the methods outlined in a study [17]. Preprocessing for the video frames involved the low-pass filtering of data for smoothing purposes and the calculation of temporal derivatives in order to accommodate the variability of the facial features in time. For the physiological signal data (ECG, GSR, EMG), a butterworth bandpass filter was employed to remove the noise and grasping the true power of the signals in minimizing presence of trends in the signal. At this point, after the feature extraction, an early fusion method was used to merge the resulting features from the processed video and biosignal measurement modalities. This is combining the feature vector of one modality with the feature vector of another so that there is combined 228 feature representing the multimodal pain response in a single place.

3.3 Feature Selection

Feature selection is an indispensable process in improving model performance. In this case, PSO was used for feature selection. PSO is a meta-heuristic optimization algorithm, inspired by the social behavior of bird flocking. It explores the search space by iteratively improving a population of candidate solutions, called particles. A configuration of the particles, that is a candidate solution in the case of PSO, is effectively defined as a position of the particle in a binary vector of equal length to the number of features in the model. A binary '1' indicates that the corresponding feature is selected. A binary '0' would denote that it has been excluded. This was assisting for using the optimal subset of all features that would give best accuracy for the model in the test set.

3.4 Validation and Hyperparameter Optimization

The modeling work was subjected to robust validation via a stratified 80/20 split of the dataset into training and testing sets. The PSO parameter settings were inertia weight ($w = 0.7$), a cognitive coefficient ($c1 = 1.5$), a social coefficient ($c2 = 1.5$), and maximum velocity ($v_{max} = 4.0$) over 10 particles and 20 iterations. The PSO identified 103 features among 228 features.

3.5 *Model Architecture and Training*

The base of this method is a hybrid CNN-BiLSTM model that exploits both a short-range and long-range dependency of a given dataset. The building blocks of CNN-BiLSTM network architecture used were convolutional layers (64, 128 filters, kernel size 3), max pooling (size 2), BiLSTM (64 and 32 units), a dense layer (64 units, ReLU, dropout 0.5), and sigmoid as the output layer. The Adam optimizer was employed during training, coupled with a batch size of 32. The final model was trained for 20 epochs. The binary cross-entropy loss function was used since it fits well in the realm of binary classification.

3.6 *Evaluation of Performance Metrics*

The performance of the CNN-BiLSTM under consideration in this paper was evaluated with reference to the held-out test set and its capacity for generalization testing. This performance is evaluated in terms of accuracy as the proportion of correctly classified instances out of the total number of instances, Area Under Curve (AUC) which measures the ability to distinguish between the two classes using its value as an area under the Receiver Operating Characteristic (ROC) curve, confusion matrix consists of counts of true positives, false negatives, and true negatives, as well as false positives associated with the model's prediction, classification report based upon overall precision, recall, F1-score. Precision indicates the number of accurate predicted positive cases to the total number of positive cases. Recall reveals how many defined actual positive cases were also well predicted. The F1-score is the harmonic mean of precision and recall.

4 *Result and Discussion*

4.1 *Performance Metrics*

The evaluation performance of the pain recognition model concerned a confusion matrix and a number of standard measures. Normalized confusion matrix is presented in Fig. 2, which summarized the results of the classification.

This matrix reflect the model's effectiveness in distinguishing between the pain and no-pain condition. Table 1 shows all the performance matrix for the developed model in this study. F1-score balances the precision and recall of the proposed model confirming the outstanding performance.

The comparison of existing methods of pain recognition is depicted in Table 2. The table shows the different approaches of the previous studies with respect to the models used, physiological modalities and mentioned accuracies. Various machine

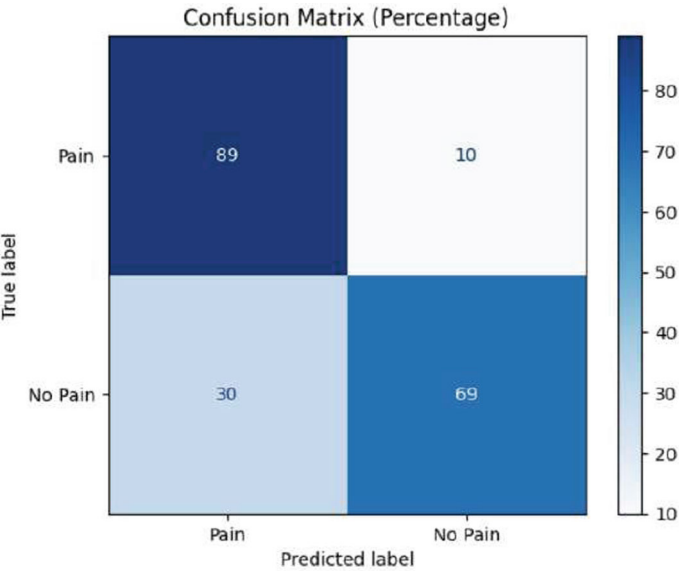


Fig. 2 Confusion matrix for proposed model

Table 1 Performance measure

Proposed model	Accuracy	Precision	Recall	F1 score
	0.86300	0.8968	0.9364	0.9162

learning and deep learning approaches have been tried out in pain recognition, going from very traditional ones such as Random Forest and Logistic Regression, to very advance architectures such as RNNs, CNNs, and Transformer-based models. Also physiologically different modalities are utilized which comprise ECG, GSR, EMG, skin potential, and facial expression video data. The apparent accuracy from these methods ranges from a low of 80.3% to a high of 84.8%. This implemented method is a PSO optimized CNN-BiLSTM architecture using ECG, GSR, EMG, and video data, which has an accuracy of 86.3%, topping all the approaches above.

4.2 ROC-AUC Curve

The area under the receiver operating characteristics curve is usually an indication of the capability of differentiating between the two classes by the model. The attained AUC scores were found to be 0.71 for class 0 and 0.94 for class 1 (where, class 0 represents no pain and class 1 presents pain) from which it could actually be inferred that class 1 have a higher prediction power. The difference in the AUC score may

Table 2 Comparison of pain recognition methods

References	Model used	Modality used	Accuracy
[5]	Wavelet Transform, Machine learning, Random forest	Skin potential	80.3
[6]	RNN	ECG, GSR	84.58
[7]	Ensemble neural networks (fine-tuned, 3-stream hybrid deep neural networks, CNN)	EDA, video	84.8
[9]	CNN, AlexNet	Video	82.41
[10]	Machine learning logistic regression	EMG, EDA, ECG	83.20
[14]	Transformer	ECG, video	82.74
Proposed method	PSO technique, CNN-BiLSTM	ECG, GSR, EMG, video	86.3

suggest class imbalance in the dataset. Figure 3 depicts the ROC-AUC curve for binary classification of this model.

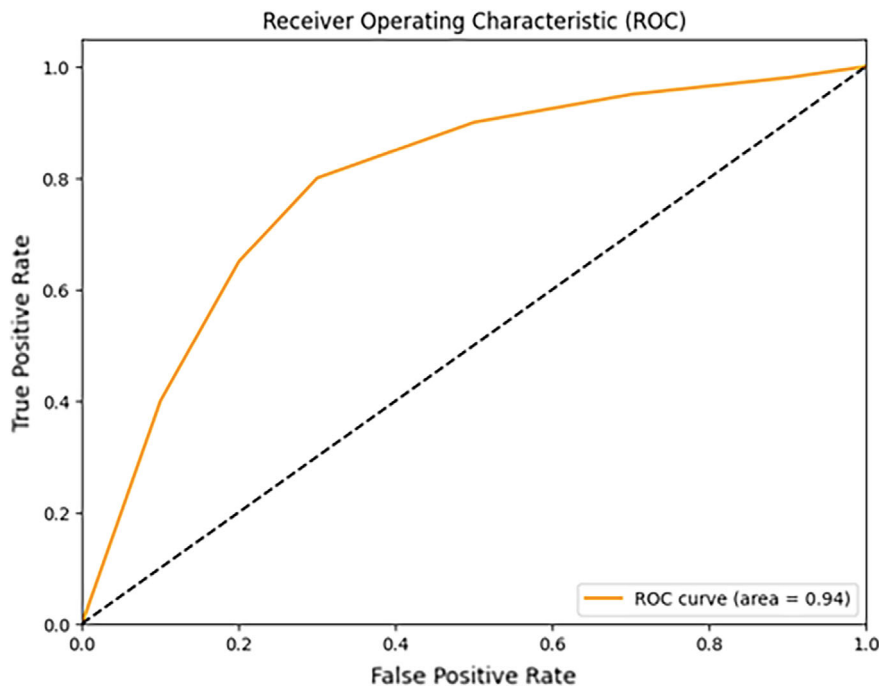


Fig. 3 ROC-AUC curve for proposed model

5 Conclusion and Future Scope

In this research, a new multimodal pain recognition model based on a PSO- optimized CNN-BiLSTM architecture is proposed. The resulting 86.3% accuracy from fusion of video and physiological signals makes this method promising beyond a number of existing approaches. The role of PSO was mainly in the optimization stage of feature selection and fusion so that the model could better learn and discriminate between any related information about pain from the multimodal data. The spatial and temporal dependencies within input data were captured by the CNN-BiLSTM network, boosting the performance of the system. These results also show that the proposed framework can be an effective solution for automated pain assessment. Future works deal with augmenting the dataset with additional and varied samples to generalize the model even more. Various other optimization algorithms and deep learning architectures may improve system performance. More advanced fusion techniques with attention mechanisms may help improve performance as well. Future work will also entail addressing the problem of imbalanced classes in the dataset to improve the sensitivity of the model towards minority classes.

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Detection of Diabetic Macular Edema Using Deep Learning: A Comparative Study



Ismaeel Al Ridhawi, Ali Abbas, and Hasan Fayyad-Kazan

Abstract Diabetic Macular Edema (DME) is one of the major causes of blindness in adults, primarily due to fluid accumulation in the macula derived from Diabetic Retinopathy (DR). In this paper, a Deep Learning (DL)-based diagnostic framework is developed to detect and identify various levels of DR severity, thereby facilitating early identification and management of DME. Our methodology begins with a comparison of several convolutional neural network (CNN) architectures for baseline modeling, including VGG 13, VGG 16, Resnet 18, Resnet 101, ResNeXt 50, EfficientNet B0, EfficientNet B3, EfficientNet B4, and ShuffleNet. Among these, the VGG-13 model was found to be the best performing model with an accuracy of 84.6% in identifying the severity of DR. Based on these outcomes, we introduce a Mixture of Experts (MoE) model that combines the capabilities of several task-specific models to increase the overall classification quality. This ensemble-based method improves the diagnostic accuracy to 93.7%, representing a 9.1% improvement over the baseline. These results underscore the potential of MoE in providing reliable DME diagnosis.

Keywords Deep learning · Mixture of experts · Diabetic macular edema · Convolutional neural network · Internet of things

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1 Introduction

The advancements in computing technology, primarily Artificial Intelligence (AI), has led to a significant shift in healthcare, mainly in biomedical sciences and engineering. Both complex medical devices, such as magnetic resonance imaging (MRI), computed tomography (CT), and pacemakers, in addition to end-user devices such as mobile and Internet of Things (IoT) devices, are benefiting from these technological breakthroughs. AI-enabled solutions provide real-time data analysis, prediction, and automation that would enhance the diagnostic and decision-making processes, as well as streamline treatment workflows, resulting in better patient care. One particular area of application that is tremendously benefiting from AI is medical imaging and diagnostics. The fast developing field of Deep Learning (DL) has exhibited a high-level of effectiveness in various tasks including image classification, segmentation, and anomaly detection. Ophthalmology, is one medical area that is seeing rapid developments in DL-based disease diagnosis solutions relying on medical imaging.

Diabetic Macular Edema (DME) is a condition in Diabetic Retinopathy (DR) that causes tiny bulges, called microaneurisms, in the blood vessels to occur, causing fluids to leak into the retina, which leads to retina swelling [1]. This serious eye disease can cause severe vision problems and ultimately lead to vision loss. These conditions require early detection and precise classification to enable early intervention and avoid potential vision loss. Despite ongoing progress in medical imaging and practitioner awareness, DME disease diagnosis can go undetected or may be categorized improperly. With the rise in the availability of high-quality annotated medical image datasets, particularly for DME conditions, and advances in AI diagnostic tools, trust among ophthalmologists is on the rise. AI-based screening tools can reduce the burden on clinical resources, enable remote patient monitoring with the aid of IoT devices, and thus, expand access to care in under-served regions.

In this paper, DL methodologies are employed for the automated diagnosis of DME based on the analysis of retinal fundus images. In general, DL approaches are based on convolutional neural network (CNN) to learn relevant features from medical images. These features are then passed through fully connected layers for performing the classification tasks. Such systems when integrated into clinical automation can provide numerous advantages, which may include shorter time to diagnosis, timely reporting, secure medical communication, and better patient monitoring. The purpose of this work is to design a robust algorithm that can directly analyze retinal images and localize regions of interest that have abnormalities, and detect degenerative macular diseases. Using the DL techniques, the proposed approach is expected to improve the diagnosis time, cost, and accuracy, and hence reduce the risk of vision loss. The main contribution of our work is the development of a VGG-13 based Mixture of Experts (MoE) model used for the classification of the severity of DR from retinal fundus images. The model is integrated into an E-Hospital platform, which allows both healthcare practitioners and patients to view and handle the diagnosis. Results are stored into patients' electronic health records (EHR), which can be accessed using the E-Hospital's identity management system (IMS).

The rest of this article is organized as follows: Sect. 2 highlights some state-of-the-art work conducted in the area of DME diagnosis using DL. Section 3 provides a deep discussion on the proposed methodology and its integration into the E-Hospital platform. Experimental evaluations and results are looked at in Sect. 4. Finally, in Sect. 5 concluding remarks are drawn from this work.

2 Related Work

The application of DL has greatly improved the detection and classification of DME from various imaging tools, particularly Optic Coherence Tomography (OCT) and fundus photography. The authors in [2] proposed an ensemble DL framework to jointly identify DR and DME from a single image of the retina. Their model was trained on a large dataset of 8739 retinal fundus images. The model exhibited a high diagnostic capacity with an area under the receiver operating characteristic curve (AUC) of 0.992 for referable DR and 0.994 for referable DME. These results highlight the possibility of enhancing the diagnostic accuracy by employing ensemble architectures and thus improve the screening and management of retinal diseases at a clinical scale. In [3], the authors developed a DL model that effectively uses fundus photographs to infer OCT-based severity grading, thus offering a non contact method of predicting the structural retinal changes that are usually only apparent on OCT. This approach enables ophthalmologists to predict the severity of DME without having to rely solely on OCT imaging, which may not always be easily obtainable.

The authors in [4] designed a transfer learning model based on a pre-trained DenseNet-121 architecture to predict the severity levels of DME from retinal fundus images. Using the IDRiD dataset, their model attained a classification accuracy of 86.4%. This approach shows that transfer learning can be adopted for domain-specific medical imaging tasks to reduce the training complexity needed for high-quality DME severity evaluation. The work in [5] used OCT-derived radiomics in conjunction with ML techniques to determine patient outcomes to anti-vascular endothelial growth factor (anti-VEGF) intravitreal therapy in DME. The model they developed, based on a backpropagation neural network, used radiomics features extracted from OCT images and had excellent predictive performance, with sensitivity of 96.2% and specificity of 92.6%. Their research highlights the potential of integrating radiomic features with advanced neural networks to make personalized treatment predictions and improve patient outcomes with tailored therapeutic approaches.

The authors in [6] proposed a hybrid DL model that combines CNN and recurrent neural networks (RNN) to analyze OCT volumetric scans for DME screening. Under real world conditions, their hybrid model achieved very high diagnostic accuracy with an AUROC of 0.94. The authors have also noted that the integration of CNN for spatial feature extraction and RNN for capturing temporal or sequential relationships within OCT volumes significantly enhanced the robustness and predictive accuracy of their framework. These recent studies collectively show significant technical improvements in DL methodologies, ranging from complex ensemble and

hybrid models to very sophisticated radiomics-based predictive systems, all of which contribute uniquely to improving DME detection. Integrating these techniques into clinical practice will improve early diagnosis, personalize treatment, and increase the accessibility of ophthalmological care.

3 Methodology

3.1 Dataset

The APTOS-2019 dataset was used as part of this study, which constituted of 3662 image samples collected from patients in rural India [7]. The samples are divided into five categories, namely, no Diabetic Retinopathy (DR), mild DR, moderate DR, severe DR, and proliferative DR. Figure 1 highlights the different levels of DR severity as can be seen through fundus images. In the no DR case, it is noticeable that there are no hard exudates or pre-retinal hemorrhage. With mild DR, it can be seen that there are a few hard exudates on the center-right. In the moderate DR case, it is noticeable that there is a significant increase in the number of hard exudates. The severe DR case shows that in addition to the hard exudates, we start seeing sights of pre-retinal hemorrhage. Finally, in the proliferative DR case, very large pre-retinal hemorrhage can be seen from the fundus image, indicating the person is likely to be blind.

One key issue of this dataset is that the raw data is divided into 3662 training images and 1927 testing images, where labeling is only performed on the training data. The data set is also imbalanced, where no DR images constitutes 49.3% of the data, followed by moderate DR with 27.27%, mild DR with 10.10%, proliferate DR has 8.05%, and finally, severe DR with 5.26% of the data. This imbalance will lead to training imperfections. Given that the raw testing data was not labeled, it therefore could not be used for testing nor validation. We have divided the raw training data into 80% for training, 10% for testing, and 10% for validation. The size of images in the sample dataset varies, as can be seen from Fig. 2, showcasing the different heights and widths of the image dataset. The raw data was augmented to increase its size.

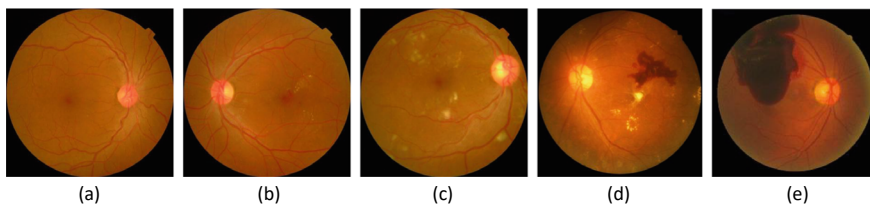


Fig. 1 Levels of DR severity as seen through fundus images, **a** no DR, **b** mild DR, **c** moderate DR, **d** severe DR, and **e** proliferative DR

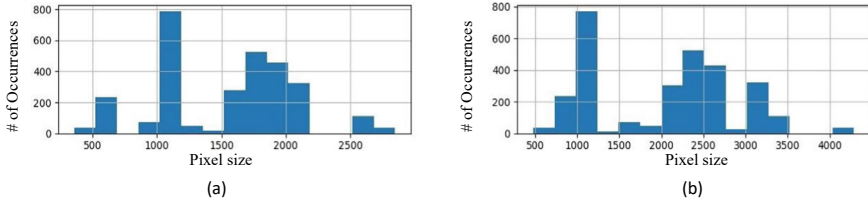


Fig. 2 Distribution of dataset image, **a** width and **b** height

3.2 Data Pre-processing

Due to the reduction in image size and to balance the labels in the dataset, the split training data was augmented to create more samples for each label. The size of each label was increased so that all the labels had the same quantity of data. The *torchvision* package was used for the transformations [8]. The images were first resized to 360×360 pixels to ensure constant height and width across all images. The additional images were flipped either horizontally or vertically with a probability ratio of 0.5. The additional images were allowed to have a color jitter with brightness and hue set to 5%. The images were normalized based on standard practice to $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$ for the red, green, and blue channels of the ImageNet-1K, respectively [9].

3.3 Baseline Modeling and Mixture of Experts

Five pre-trained model architectures were used for the performance evaluation in order to select an appropriate model backbone, namely, VGG, ResNet, ResNeXt, EfficientNet, and ShuffleNet. A total of nine different models were built by applying transfer learning on the five pre-trained models. The MoE model was designed such that the selected pre-trained models perform specific tasks rather than simply performing generalized tasks of predicting all labels. A hierarchical approach was followed to build three expert models. The first expert is based on a binary classification model and is used to diagnose the absence of DR (i.e., classifies no DR). The models only go to the second expert if it predicts there is DR. The second expert diagnoses the presence of moderate DR, and is too based on a binary classification model. The model goes to the third expert only if the second expert predicts it is not moderate DR. The third expert then classifies between mild DR, severe DR, and proliferative DR.

3.4 Deployment

The MDE classification model was deployed into an E-Hospital platform [10]. The E-Hospital has its own identity management system that identifies the role of the user, whether a patient or doctor. Once the login is successful, the user accesses the E-Hospital front-end so that they can view their diagnosis. Doctors on the contrary, in addition to viewing diagnosis results, can also submit request for diagnosis, which is then handled by the diagnosis system. The diagnosis system communicates with the back-end to fetch patient records from the EHR, which was implemented using MongoDB. Figure 3 provides an overview of the E-Hospital platform, and is out of the scope of this article.

4 Performance Evaluation and Results

4.1 Results

The DL models were evaluated based on the ROC, AUC and confusion matrix. Figure 4 depicts the ROC and AUC results for all models. For the confusion matrix, it is important to note that the data is not balanced for each label. The values 0 to 4 represent the labels no DR, mild DR, moderate DR, sever DR, and proliferate DR, respectively. Figure 5 depicts the confusion matrix results for all models. Table 1 provides a high-level summary of the model performance metrics. Results are shown in terms of the overall model accuracy, the weighted average precision, the weighted average recall, and the weighted average F1 score of the model. From the results, it is clear that the VGG-13 model performed best under all metrics, then followed by VGG-16, then Resnet-101, and then ResNeXt 50. It was also noticeable that all of the EfficientNet models performed poorly. Similarly, the ShuffleNet model also performed poorly.

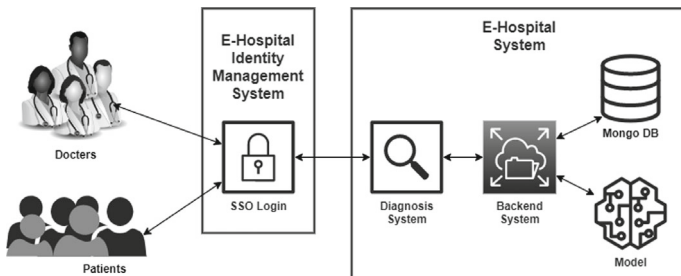


Fig. 3 Deployment architecture of the MDE classification model into the E-Hospital framework

Table 1 Model performance metrics results for all the transfer learning models

Model	Accuracy	Precision	Recall	F1-score
3 MoE VGG-13	0.937	0.935	0.93	0.92
VGG 13	0.846	0.82	0.82	0.81
VGG 16	0.83	0.78	0.79	0.78
Resnet 18	0.788	0.74	0.76	0.72
Resnet 101	0.827	0.78	0.79	0.78
ResNeXt 50	0.805	0.74	0.76	0.73
EfficientNet B0	0.786	0.69	0.74	0.71
EfficientNet B3	0.706	0.55	0.70	0.61
EfficientNet B4	0.736	0.54	0.69	0.60
ShuffleNet	0.731	0.58	0.72	0.64

4.2 Performance Analysis

Baseline Models Results using the VGG-13 model show that the accuracy is high only for no DR and mild DR labels. Otherwise, the performance is not ideal. In comparison to the other models, VGG-13 did perform the best in baseline modeling, where the AUC score for three classes was above 0.9. Similar to VGG-13, VGG-16 performed well and was extremely similar for all classes except for mild DR. Resnet-18 did perform poorly when compared to ResNet-101. We notice that Resnet-18 never predicted the severe DR class correctly and misclassified moderate DR significantly. This poor performance can be seen in the ROC and AUC scores, where the AUC score was more than 0.9 for only two classes. Resnet-101 had the same limitations as Resnet-18, where it misclassified moderate DR most of the time. For the severe DR class, it was only able to make two accurate predictions. Compared to Resnet models, the ResNeXt model was able to perform better in predicting the severe DR class. But at the same time, it performed poorly for the mild DR class.

Among the family of EfficientNet baseline models, EfficientNet-B0 performed the best. But, with that said, its performance was the worst among all models. From the results, we can see that it was not able to classify any sample from the severe DR class correctly. It only classified three proliferate DR images correctly. As a result, this family of models is not ideal for classification in the case of DME. EfficientNet-B3 is the worst-performing baseline model in the family of EfficientNet models. From the results, we see that it was only able to predict two classes, which had the highest amount of samples per class. It is to be noted that even though it never predicted the other three classes, it still outputs five prediction values. The prediction values are so small that the model never selects it in an argmax. Similar to EfficientNet-B3, EfficientNet-B4 is also not a good model for diagnosis. But, it is still slightly better compared to EfficientNet-B3. ShuffleNet is the worst-performing baseline model. It too only outputs two classes similar to EfficientNet.

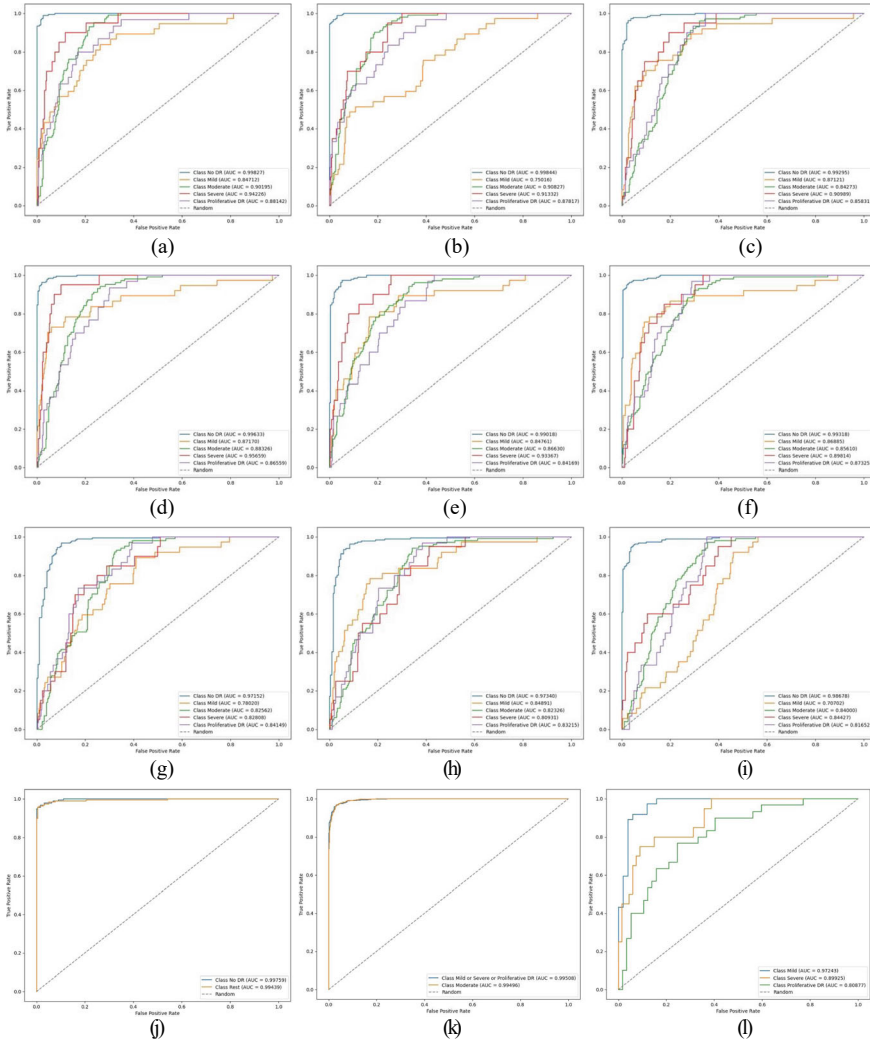


Fig. 4 ROC for each label prediction using the following models, **a** VGG 13, **b** VGG 16, **c** Resnet 18, **d** Resnet 101, **e** ResNeXt 50, **f** EfficientNet B0, **g** EfficientNet B3, **h** EfficientNet B4, **i** ShuffleNet, **j** Expert 1—no DR, **k** Expert 2—moderate DR, **l** Expert 1—the rest

MoE Results for MoE show that for expert 1 significant enhancement is achieved for VGG-13, where the AUC and ROC results are at 0.9975. In the case of expert 2, we see an improvement in the AUC from 0.901 to 0.994, which is quite significant. In the case of expert 3, we see that the mild DR classification has an AUC of 0.972 for MoE, while for VGG-13 it is at 0.847, which is a clear improvement. In the case of severe and proliferate DR, the AUC drops slightly for MoE. Overall, the new accuracy of the model is 0.937 which is much higher when compared to VGG-13 at 0.937.

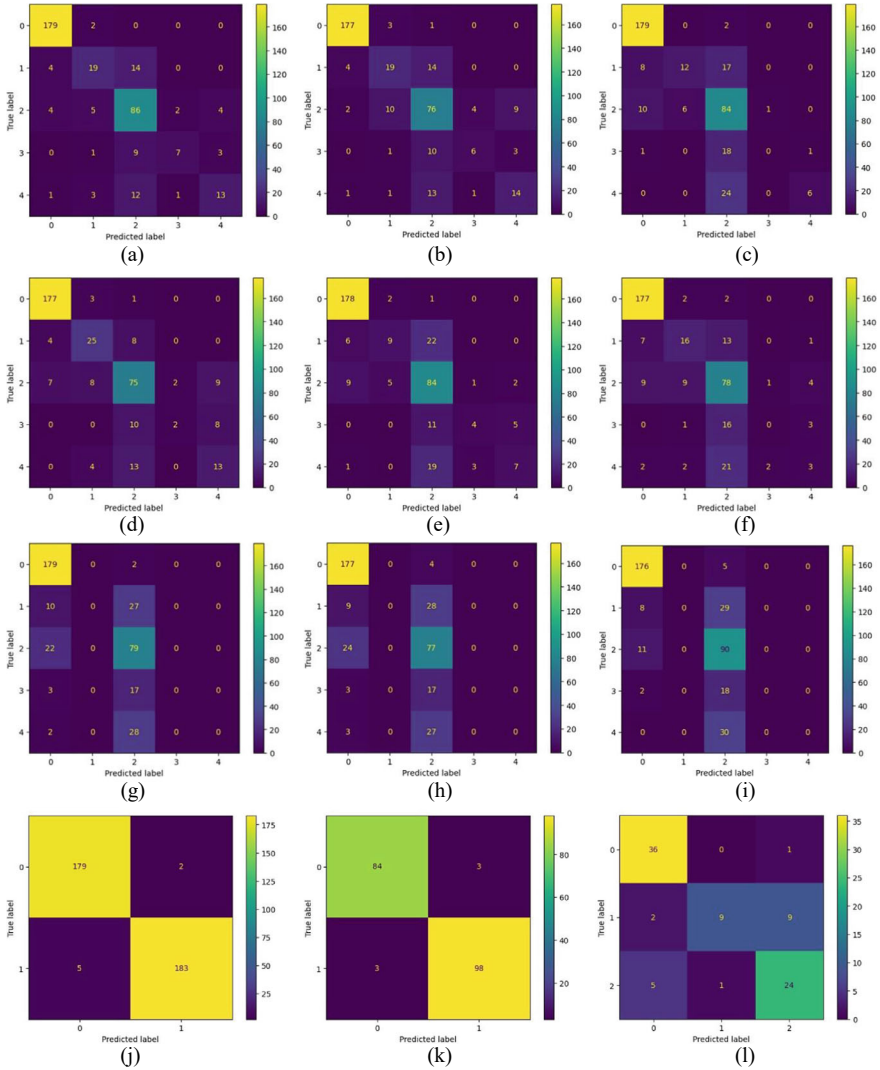


Fig. 5 Confusion matrix for the validation set prediction of the five DME labels using the following models, **a** VGG 13, **b** VGG 16, **c** Resnet 18, **d** Resnet 101, **e** ResNeXt 50, **f** EfficientNet B0, **g** EfficientNet B3, **h** EfficientNet B4, **i** ShuffleNet, **j** Expert 1—no DR, **k** Expert 2—moderate DR, **l** Expert 1—the rest

5 Conclusion

In this paper, we have demonstrated the effectiveness of deep learning for the diagnosis of DME through accurate classification according to DR severity using retinal fundus images. Our proposed approach leverages a VGG-13-based MoE model, which achieved a notable improvement in diagnostic accuracy, thus enhancing performance by 9.1% over the baseline VGG-13 model. This clearly indicates that combining multiple specialized models in an ensemble architecture can significantly enhance the reliability and accuracy of automated retinal disease detection. The developed model was integrated into an E-Hospital platform, enabling diagnosis access to both patients and health providers.

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Enabling Low-Cost Access to Weather Satellite Imaging: A Ground Station Solution for IoT Applications



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Abstract Polar satellites play an important role in gathering information on various weather phenomena. However, real-time access to satellite imagery remains constrained by operational costs and the technical complexity of receiving ground stations (GS). This paper presents the design and implementation of a low-cost, portable, and easily deployable GS to receive signals from weather satellites operating in the VHF band. The GS integrates standard commercial hardware with open-source software for receiving and processing satellite images. We perform detailed link budget calculations using the System Tool Kit (STK) software to ensure reliable data transmission during communication sessions. Experimental validation is demonstrated through the successful acquisition of NOAA-19 weather images. The results highlight a cost-effective and practical solution that improves access to real-time weather information for a wider community and Internet of Things applications.

Keywords Ground station · Weather satellite image · Link budget analysis · NOAA-19 reception · IoT applications

1 Introduction

In recent years, satellite systems have been considered the dominant way to gather weather information due to their unique global coverage [1, 2]. Such information is crucial for planning and mitigating the impact of severe weather events. With recent advancements in technology, satellites can capture high-resolution images for

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daily weather forecasting. Meteorologists obtain near-real-time data crucial for tracking tropical storms, wildfires, hurricanes, and climate change with unprecedented accuracy [3, 4].

At the same time, the rapid expansion of Internet of Things (IoT) devices in sectors such as agriculture, urban infrastructure, and environmental monitoring has reshaped the way real-time data is captured and utilized [5]. Devices such as smart irrigation controllers, soil sensors, traffic systems, and air quality monitors rely on timely, localized data to enable autonomous decision-making. Among the most vital data sources for these systems is accurate weather information, which influences operations ranging from irrigation to disaster response. The integration of satellite-based weather imaging with ground-level IoT deployments is thus increasingly valuable, especially in regions with limited terrestrial weather coverage. In such contexts, low-cost satellite reception systems can serve as a critical data backbone for scalable, weather-aware IoT applications.

Typically, weather satellites operate in two main orbits: geostationary (GEO) and polar. GEO satellites, like GOES-16, are positioned approximately 36,000 km above the equator at a fixed longitude, allowing continuous monitoring of a specific region [6]. On the other hand, polar satellites, such as NOAA-19, orbit much lower, about 800 km above the Earth's surface in low-earth orbits (LEO), passing over the poles to provide global coverage [7]. The lower altitude of polar satellites enables higher spatial resolution, making them ideal for detailed atmospheric data and climate tracking. Meanwhile, GEO satellites excel in real-time severe weather monitoring, such as storm tracking.

The Polar Operational Environmental Satellites (POES) series, operated by the National Oceanic and Atmospheric Administration (NOAA), serves as a primary source of accurate environmental data for research and weather monitoring [8]. These satellites follow a sun-synchronous orbit, ensuring they pass over the same regions of Earth at the same local solar time each day. This consistency is crucial for comparing images over time, allowing for the detection of environmental changes. NOAA satellites are equipped with advanced very high-resolution radiometers (AVHRR), which capture images across multiple spectral bands. These images are then transmitted to ground stations (GS) via automatic picture transmission (APT) signals, providing valuable atmospheric data, including cloud coverage and surface temperatures.

Despite the significance of weather information, access to real-time satellite imagery remains limited due to the high costs and complexity of satellite GS. In this work, we address such challenges by developing a low-cost GS designed to receive and process APT signals from the NOAA satellite series, making weather information accessible to a wider community. To ensure affordability and accessibility, the GS is built using commercial off-the-shelf components. It also utilizes free and open-source software for image reception and decoding. By integrating these components, the proposed solution offers a cost-effective and practical approach, enabling individuals and researchers to access real-time satellite imagery without the need for expensive infrastructure.



Fig. 1 System configuration of the portable ground station, including a HackRF SDR, Yagi-Uda antenna, and portable computer running signal processing and decoding software

Importantly, the capabilities of the proposed system go beyond conventional meteorological use and extend into the rapidly evolving domain of IoT. Integrating satellite-derived weather data with IoT platforms opens new possibilities for intelligent, data-driven decision-making. In precision agriculture, for example, IoT-enabled systems can leverage satellite-derived imagery to automate irrigation, schedule fertilization, and detect pest outbreaks based on vegetation indices and temperature maps. This integration enables more efficient resource usage and improved crop yields, especially in remote or underserved areas. In the context of smart cities, weather-aware IoT systems can help manage traffic flow, mitigate flood risks through smart drainage networks, regulate energy consumption in buildings, and coordinate emergency services during extreme weather events. Real-time satellite-based rainfall estimation, for instance, can enhance the responsiveness of urban infrastructure by feeding accurate data into control systems. Such use cases illustrate the value of a reliable, low-cost GS capable of supporting weather-aware IoT applications.

The cost-effectiveness and scalability of the proposed software-defined radio (SDR)-based ground station make it a transformative tool for both precision agriculture and smart cities, particularly in resource-constrained regions. By leveraging open-source software, the solution can be easily customized and integrated with existing IoT platforms, further reducing costs and increasing accessibility. The main contributions of this paper can be summarized as follows:

- Integration of a low-cost SDR-based ground station using a HackRF receiver, a high-gain Yagi-Uda antenna, and an open-source software stack for image decoding.
- Use of System Tool Kit (STK) to conduct dynamic link budget simulation across multiple orbital passes, validating system feasibility and ensuring signal margin optimization.
- A fully reproducible deployment methodology with detailed hardware and software configuration steps.

The rest of the paper is structured as follows: Sect. 2 presents the system configuration, while Sect. 3 discusses the link budget analysis. Section 4 describes the data acquisition methodology, and Sect. 5 concludes the paper.

2 System Configuration

We consider a satellite communication (SatCom) system where there is one GS and a satellite. The GS can communicate with the satellite directly through radio frequency (RF) transmission. In this work, we build a case study on NOAA-19 that operates at 137.1 MHz.

The GS comprises a Yagi-Uda antenna in the VHF band, a HackRF SDR receiver, a portable computer, and a set of cables, as illustrated in Fig. 1. Note that we utilize the HackRF receiver due to its wide operating frequency range (1 MHz to 6 GHz) and lower receiving sensitivity of -130 dBm, making it well-suited for NOAA APT signal reception. We also utilize a 5-element Yagi-Uda antenna operating in the 135–139 MHz frequency range. It provides a maximum gain of 9.5 dBi at 137 MHz, effectively directing the RF signal to the SDR receiver.

The GS also integrates open-source software, including SDR-sharp for signal processing, WXtoImg for NOAA image decoding, and Orbitron for satellite tracking. These tools, combined with the hardware setup, enable efficient NOAA satellite data acquisition and analysis. Additionally, the GS configuration is highly flexible and can be easily adapted to support other satellite missions. The portable computer runs the software to control the receiver, process received signals, and generate the final images.

3 Link Budget Analysis

A link budget is crucial for computing the efficiency and reliability of a SatCom system, as it evaluates the received signal power [9]. It accounts for key parameters such as transmit power (P_{tx}), transmit antenna gain (G_{tx}), receive antenna gain (G_{rx}), and free-space loss (FSL). Additional losses are also considered (L_o), including receiver feeder loss, antenna misalignment, atmospheric absorption, and polarization mismatch [10]. By analyzing these factors using the power balance equation, the link budget helps assess link feasibility, optimize performance, and ensure reliable data transmission under various operating conditions. The detailed calculation is presented below [11]

$$P_{rx} = 10 \log_{10} \left(\frac{P_{tx} G_{tx} G_{rx}}{FSL L_o} \right) + 30, \quad (1)$$

where P_{rx} is the received power level in dBm. Note that FSL is the primary factor responsible for signal attenuation. As the RF signal propagates, its strength decreases proportionally to the square of the traveled distance. Moreover, particles in the atmosphere absorb portions of the transmitted signal. Such effects become more severe at higher frequencies, resulting in higher attenuation. The FSL is expressed as [12]

$$FSL = \left(\frac{4\pi Rf}{C} \right)^2, \quad (2)$$

where $C = 3 \times 10^8$ m/s denotes the speed of light, f is the carrier frequency in Hz, and R is the slant range in meters.

A SatCom system is typically designed with a 6 dB link margin to ensure reliable transmission under varying environmental conditions. This margin compensates for external factors such as rain and other unexpected signal degradation. By incorporating this margin, the system can maintain stable performance despite fluctuations in propagation conditions. The link margin can be determined as

$$\text{Margin} = P_{rx} - \text{Receiver sensitivity} \quad (3)$$

We use STK software to perform link budget calculations for NOAA-19 and its corresponding GS located at 30.08° North, 31.29° East. The STK simulation covers a 30-day period, from December 1 to December 30, 2024, as shown in Fig. 2. The primary output of STK, P_{rx} , is used to compute the link margin. Table 1 presents the pseudocode for link budget calculations in a typical operating scenario. The process begins by prompting the user to input key parameters, including satellite orbit (apogee = 817.3 km, perigee = 803.9 km, and inclination = 98.8°), transmit power of

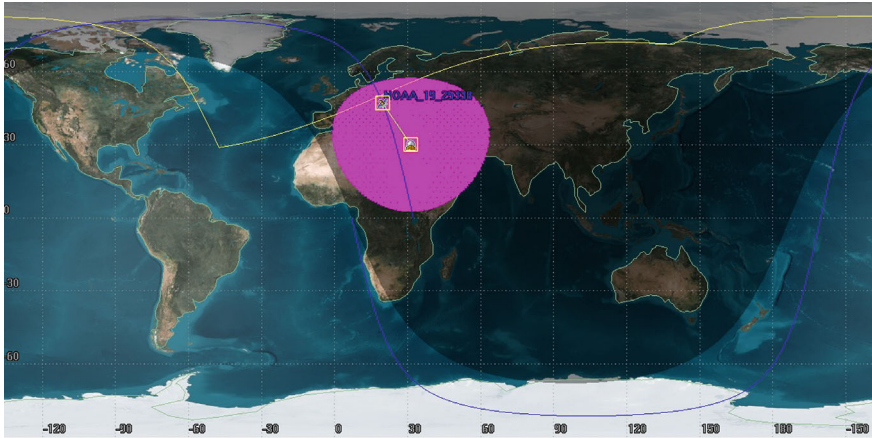


Fig. 2 Visualization of NOAA-19 satellite pass schedule, showing slant range and visibility windows from the ground station

Table 1 Step-by-step pseudocode used for link budget calculation in STK, including orbital and hardware parameters

```

1: Start
2: Input satellite orbit parameters (apogee = 817.3 km,
   perigee = 803.9 km, inclination = 98.8°)
3: Input  $P_{tx} = 5$  watts
4: Input  $G_{tx} = 0$  dBi
5: Input  $f = 137.1$  MHz
6: Input minimum  $P_{rx} = -130$  dBm as a constraint
7: Input GS location (30.08° North, 31.29° East)
8: Input  $G_{rx} = 9.5$  dBi
9: For  $i=1$  : number of sessions
10:  For  $t=1$  : 1 minute : session time
11:    Compute link budget parameters ( $R(i, t)$ ,  $FSL(i, t)$ )
12:    Return  $P_{rx}(i, t)$ 
13:  End
14: End
15: End

```

5 W, an operating frequency of 137.1 MHz, transmit antenna gain of 0 dBi, receive antenna gain of 9.5 dBi, and a receiver sensitivity constraint of -130 dBm. STK then computes the slant range and the corresponding FSL throughout the communication sessions with the GS. Table 2 shows a sample of the link budget calculations between NOAA-19 and the corresponding GS. The results indicate that the receiving GS maintains a positive link margin of at least 20 dB throughout all communication sessions.

4 Data Acquisition Methodology

According to the link budget calculations, the GS can reliably receive and process APT signals when direct line-of-sight communication with the NOAA-19 satellite is established, based on its orbital position. The Orbitron software uses NOAA-19's two-line elements (TLE) and the GS's geographical coordinates to accurately predict the satellite's position [13, 14]. It calculates the satellite's pass times, azimuth, and elevation angles, providing critical information for aligning the Yagi-Uda antenna. This alignment maximizes signal strength during satellite passes and ensures real-time tracking of the satellite's position relative to the GS. Additionally, the satellite pass is cross-verified using the N2YO website, which offers precise pass predictions and real-time tracking capabilities.

Table 2 Sample output of the link budget analysis across different NOAA-19 passes, showing consistent reception margins

Access	Time	P_{tx} (dBW)	G_{tx} (dBi)	Range (km)	FSL (dB)	G_{rx} (dBi)	P_{rx} (dBm)
1	12/1/2024 15:36	7	-1.27	2024.48	141.31	9.49	-105.59
2	12/1/2024 17:15	7	-0.47	1039.42	135.52	9.50	-99
3	12/1/2024 18:57	7	-1.44	3265.29	145.47	9.50	-109.91
4	12/2/2024 3:56	7	-1.41	2544.40	143.30	9.49	-107.70
5	12/2/2024 5:36	7	-0.12	849.13	133.77	9.50	-96.89
6	12/2/2024 7:14	7	-1.42	2667.25	143.71	9.50	-108.13
7	12/2/2024 15:10	7	-1.40	2614.48	143.54	9.50	-107.94
8	12/2/2024 16:49	7	-0.10	852.37	133.80	9.50	-96.90
9	12/2/2024 18:29	7	-1.41	2660.39	143.69	9.5	-108.09
10	12/3/2024 3:30	7	-1.45	3173.71	145.22	9.49	-109.67

The APT signal from NOAA satellites uses amplitude modulation (AM) with a frequency modulation (FM) subcarrier for encoding image data. Specifically, the image data is amplitude-modulated onto a 2400 Hz subcarrier, which is then frequency-modulated onto the 137.1 MHz RF carrier. This FM transmission with an embedded AM signal enhances signal robustness, making it more resistant to atmospheric interference. The HackRF receiver, controlled via SDR-sharp software, continuously scans RF frequencies and successfully detects a strong signal from the NOAA-19 satellite. Figure 3 presents the spectrum and waterfall display of the NOAA-19 satellite signal, captured using SDR-sharp software, demonstrating a strong and stable reception. The captured FM signal is demodulated using SDR-sharp's built-in capabilities, converting the received FM carrier into an audio signal.

The audio signal (.wav file) representing the APT data is then routed to WXtoImg software via a virtual audio cable for further processing. The WXtoImg software processes the received audio signal, demodulating and decoding it into APT weather satellite images. The software automatically detects the start and end of the satellite transmission, ensuring that the entire pass is successfully decoded. This real-time processing enables the visualization of weather data during signal reception, providing continuous monitoring and analysis of atmospheric conditions. Figure 4 shows the received image from the NOAA-19 satellite.

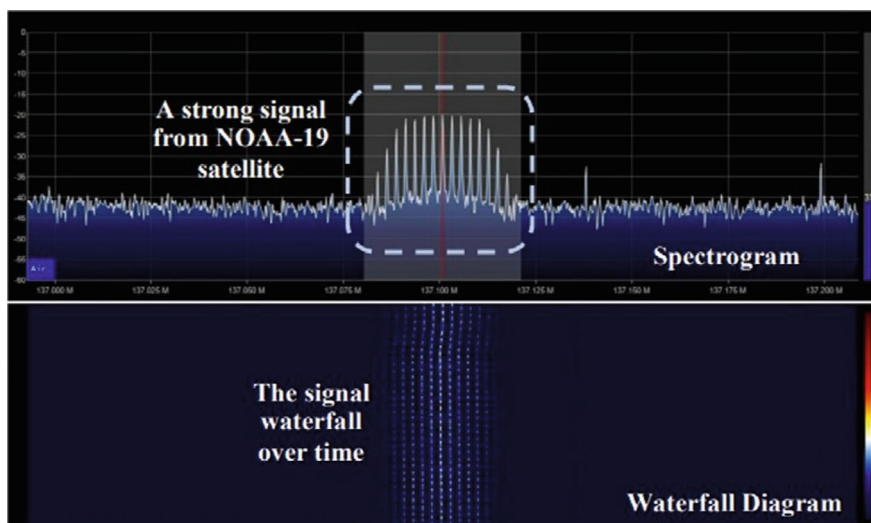


Fig. 3 Waterfall display of the received NOAA-19 signal in SDR-Sharp, showing stable signal tracking

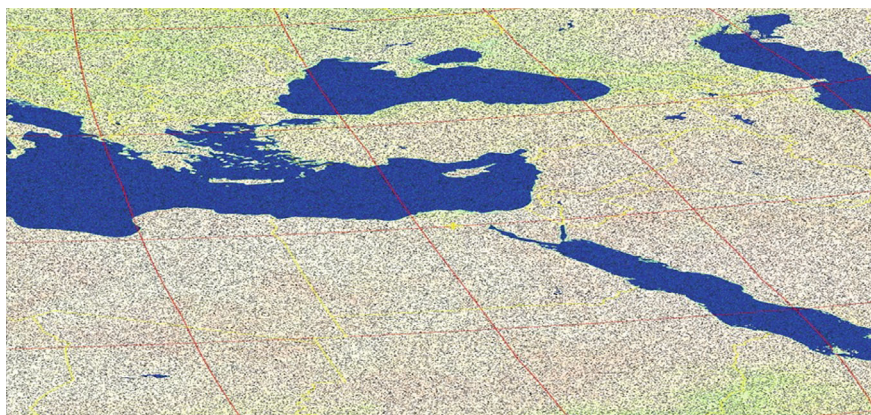


Fig. 4 Decoded APT image from NOAA-19 using WXtoImg

5 Conclusion

This paper presents the design, implementation, and successful operation of a low-cost ground station for real-time reception and display of weather satellite images. A link budget analysis based on a typical operating scenario confirms the reception's reliability. The findings demonstrate that reliable satellite communication can

be achieved with the right combination of affordable, commercially available hardware and open-source software. The results are promising, emphasizing the system's effectiveness in capturing and processing weather satellite images.

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Drone-Aided Agriculture 5.0: A Survey on Machine-Learning and IoT Paradigms for Smart Farming



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Abstract Agriculture 5.0 leverages emerging technologies such as drones, Internet-of-Things (IoT), machine-learning (ML), and digital twins to enhance precision farming and improve crop health management. Drone-based monitoring combined with advanced ML algorithms and IoT sensors has become a crucial tool for real-time agricultural surveillance. This survey explores the integration of Unmanned Aerial vehicles (UAVs) in smart farming. We review various drone-mounted sensor technologies and payloads for plant health assessment and environmental monitoring. Furthermore, we analyze state-of-the-art ML techniques, including supervised, unsupervised, reinforcement, and federated learning, for plant classifications, disease detection, yield estimation, anomaly identification, and privacy preservation.

Keywords Agriculture 5.0 · UAV · Drones · Smart farming · Machine learning · IoT

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1 Introduction

The agricultural sector is undergoing a significant transformation with the advent of Agriculture 5.0, a new paradigm that leverages advanced technologies such as artificial intelligence (AI), the Internet of Things (IoT), Robotic platforms, drones, digital twins, and big data analytics to enhance productivity, sustainability, and efficiency. Traditional farming methods are increasingly being supplemented or even replaced by smart farming solutions that enable real-time monitoring, precision resource management, and data-driven decision-making. Among these innovations, drone-based monitoring and machine learning (ML) techniques have emerged as key enablers of modern agricultural practices [1, 2].

Agriculture 5.0 represents the fifth agricultural revolution, from Agriculture 1.0 (Pre-Industrial Revolution—Traditional Farming), Agriculture 2.0 (Mechanization and Industrial Agriculture), Agriculture 3.0 (Green Revolution and Chemical Inputs), Agriculture 4.0 (Smart Farming and Digitalization), and finally Agriculture 5.0 (Sustainable and AI-Driven Farming). This new phase of agricultural development focuses on automation, connectivity, and intelligence to improve crop yields, resource efficiency, and sustainability.

Necessary Components for Smart Agriculture (Agriculture 5.0): Agriculture 5.0 integrates technologies such as (1) IoT and sensor networks for real-time environmental and soil monitoring, (2) Big data analytics and AI for predictive modeling and decision-making, (3) Robotics and automation for precision farming and labor reduction, (4) Drones and remote sensing for high-resolution field surveillance and data collection, (5) Communication protocols for integrating system components. These advances help reduce waste in both water and fertilizers, enhance disease detection, and optimize farm operations, leading to an efficient and environmentally friendly agriculture [1].

Drones, also known as unmanned aerial vehicles (UAVs), have become widely used across various industries, including military, logistics, disaster response, and environmental monitoring. In recent years, their role in agriculture has expanded significantly due to their ability to provide high-resolution imaging, real-time surveillance, and automated field analysis. In agriculture, drones are employed for multiple tasks, including (1) Crop health monitoring using multi-spectral and hyperspectral imaging, (2) Precision irrigation through thermal imaging to detect water stress, (3) Pest and disease detection using AI-driven image analysis, (4) Yield estimation based on plant growth patterns and density measurements, (5) Automated spraying of fertilizers and pesticides, reducing manual labor and resource wastage. By enabling frequent, cost-effective field analysis, drones enhance decision-making and help farmers react swiftly to emerging threats such as disease outbreaks and nutrient deficiencies [3].

Machine learning (ML), a subset of AI, enables computers to learn patterns from data and make predictions without being explicitly programmed. ML has revolutionized various fields, including healthcare, finance, autonomous vehicles, communication, and cybersecurity, offering insights that would be difficult to extract using

traditional analytical methods [4, 5]. In the context of agriculture, ML plays a crucial role in processing large amounts of data collected from drones, IoT sensors, and satellite imaging. Key applications include crop disease detection, weed identification, yield prediction, soil, and weather analysis. By combining ML with drone-based monitoring, modern agriculture can become more efficient, resilient, and scalable, ensuring higher productivity [6].

This paper explores the role of drone-based monitoring in Agriculture 5.0. We examine various UAV types and their applications in smart farming in Sect. 2. Section 3 explores the key technologies integrated into agricultural drones such as sensor, Navigation technologies, and drones payloads. Machine learning algorithms and applications used to enhance crop health assessment and resource optimization are summarized in Sect. 4. Finally, the paper is concluded by highlighting some interesting future research directions in the conclusion Section.

2 Drone Technologies for Smart Farming

Drone technology empowers farmers with faster and more precise methods for managing their fields compared to conventional equipment. These aerial platforms offer a unique vantage point, enabling a comprehensive overview of agricultural landscapes and facilitating data-driven decision-making. The evolution of drone technology in agriculture has progressed rapidly from basic aerial platforms to highly sophisticated systems. Contemporary agricultural UAVs are now equipped with a suite of state-of-the-art technologies, including high-resolution multispectral cameras, thermal sensors, precise GPS navigation systems, and advanced AI-powered mapping tools. This technological integration allows for the collection of granular data and the execution of complex tasks with remarkable accuracy. Drones are no longer mere tools for capturing aerial imagery; they represent a fundamental shift towards precision agriculture, offering solutions that reduce human labor while simultaneously enhancing productivity and promoting sustainable practices [7].

This section will delve into the key aspects of these drone technologies, exploring the various types of UAVs utilized in agriculture, the crucial technologies integrated within them, and the diverse applications they enable.

2.1 *Types of Unmanned Aerial Vehicles (UAVs) in Agriculture*

The diverse needs of modern agriculture have led to the utilization of various types of UAVs, each offering unique advantages tailored to specific tasks and farm sizes. The most promising types include fixed-wing drones, multi-rotor, and hybrid drones [7] as shown in Fig. 1.

Fig. 1 A sample of drone types for smart agriculture



- **Fixed-Wing Drones:** characterized by their longer flight times and greater endurance compared to other types, making them particularly well-suited for surveying vast farmlands and collecting broad data over large territories. However, a primary limitation of fixed-wing drones is their requirement for a runway or an open space for takeoff and landing. They are also typically less maneuverable than multi-rotor drones, which can restrict their use in tasks requiring close-range inspection or navigation around obstacles. Consequently, the primary applications of fixed-wing drones in agriculture lie in large-scale surveying, comprehensive mapping of extensive farmlands, and broad data collection across significant geographical areas. Their ability to quickly cover large distances makes them invaluable for tasks such as initial field assessments, monitoring crop health across expansive fields, and generating detailed aerial maps for overall farm management.
- **Multi-Rotor Drones:** characterized by their vertical takeoff and landing (VTOL) capabilities and exceptional maneuverability. These drones excel at precision tasks, offering the ability to hover for detailed inspections and execute rapid changes in direction. They offer easier handling, potentially less training required compared to fixed-wing, eliminate the need for runways, and providing greater flexibility in deployment across various terrains. Despite these advantages, multi-rotor drones generally have shorter flight times compared to fixed-wing, and their lower speed makes them more suitable for smaller to medium-sized fields. The applications of multi-rotor drones in agriculture are diverse, including close inspection of crops for health issues, precise applications such as targeted spraying of pesticides or fertilizers, detailed field mapping for localized analysis, and efficient data acquisition in smaller to medium-sized agricultural fields.

- **Fixed-Wing Hybrid VTOL** This type merges the benefits of both fixed-wing and rotor-based drones. This design has rotors attached to the fixed wings, allowing it to hover and take off and land vertically.

The selection of the appropriate drone type is crucial for optimizing its utility in farming practices, depending on the specific needs related to farm size, terrain, and the nature of the agricultural task at hand.

2.2 *Drone Applications in Agriculture*

The integration of diverse drone technologies has revolutionized various aspects of smart farming, offering solutions that enhance efficiency, precision, and sustainability across agricultural operations [7].

Enhanced Crop Monitoring and Health Assessment Drone technology has significantly enhanced the ability of farmers to monitor and assess the health of their crops. By exploiting several sensors, subtle changes in plant reflectance patterns, often invisible to the naked eye, can be detected, enabling the early detection of diseases, pest infestations, nutrient deficiencies, and crop growth stages. This data-driven approach enables farmers to make timely interventions and optimize their farming strategies for better crop yields and resource management.

Soil and Field Analysis: Drones equipped with specialized sensors and advanced data analysis techniques can map various soil properties, including texture, organic matter content, moisture content, and nutrient distribution, providing a comprehensive understanding of field variability. Topographical mapping and analysis of water accumulation patterns visible in aerial imagery help in identifying areas with drainage issues or erosion, allowing for targeted remediation measures.

Efficient Irrigation Management: Drones play a crucial role in enhancing the efficiency of irrigation management in agriculture. By detecting temperature anomalies or variations in vegetation health using thermal and multispectral imagery, drones can identify leaks and inefficiencies in irrigation systems, enabling timely maintenance and repair. They can also monitor soil moisture levels across large areas quickly and efficiently using moisture sensors or by analyzing thermal and spectral data.

Accurate Yield Prediction and Estimation: Analyzing crop health, density, and growth patterns captured in drone imagery, often using AI and ML algorithms, allows for more accurate yield prediction and estimation compared to traditional methods. This capability helps farmers in planning for harvesting, storage, logistics, and market strategies, leading to better overall farm management and potentially reducing losses due to over- or under-production.

3 Key Technologies Integrated into Agricultural Drones

The effectiveness of drones in smart farming is largely attributed to the sophisticated technologies integrated within them. These include a variety of sensors, precise navigation systems, versatile payload capabilities, and advanced data processing and analytics tools.

3.1 Sensors

A diverse array of sensors can be integrated into drones to capture critical data about crops, soil, and the environment.

High-Resolution RGB Cameras: can capture images in the visible spectrum (Red, Green, Blue). These cameras provide visual assessments of crop health, monitor growth stages, facilitate basic field mapping, and are instrumental in generating geometrically corrected aerial images (also known as orthomosaics). While RGB imagery offers valuable insights, its lower spectral resolution limits its ability to measure vegetation-related indices that require data from the non-visible part of the spectrum. Therefore, RGB sensors are often used in conjunction with other types of sensors for a more comprehensive analysis [8, 9].

Multispectral Sensors: capture data across multiple narrow bands of the electromagnetic spectrum, including visible and near-infrared (NIR) wavelengths. This capability allows for the calculation of crucial vegetation indices such as the Normalized Difference Vegetation Index (NDVI) and the Chlorophyll Content Index (CCCI), which are indicative of crop health, stress levels, nutrient deficiencies, and overall biomass. By analyzing the spectral absorption and reflection patterns of vegetation in different bands, multispectral imaging enables the early detection of diseased crops, often before symptoms are visible in the red channel of the RGB spectrum. The widespread use of multispectral sensors underscores their importance in providing quantitative data on plant health [10].

Thermal Sensors: detect infrared radiation emitted by objects, allowing for the measurement of their temperature. These sensors are invaluable for identifying plant stress due to water deficiency, detecting leaks or blockages in irrigation systems, monitoring livestock body temperature for early signs of health issues, and potentially identifying pest infestations or disease outbreaks through temperature anomalies in crops [11].

Light Detection and Ranging (LiDAR) Sensors: utilize laser pulses to measure the distance to the ground and the upper layer of plant life (branches and leaves), enabling the creation of highly accurate 3D maps of the terrain. This technology allows for precise measurement of height and density of trees, and detailed analysis of soil topography, which is crucial for optimizing planting and irrigation strategies.

3.2 Global Navigation Satellite System (GNSS) and Global Positioning System (GPS)

Precise positioning is fundamental to the operation and effectiveness of agricultural drones, and this is achieved through the integration of GNSS and GPS technology. These systems provide accurate location information to the drone, enabling autonomous flight along pre-programmed routes, which is essential for systematic data collection and precise application tasks. Furthermore, the location data is crucial for accurate geo-referencing of all collected data, including images and sensor readings, allowing for seamless integration with Geographic Information Systems (GIS) for subsequent spatial analysis and mapping [12].

3.3 Payload Systems

Agricultural drones are equipped with various payload systems tailored to specific tasks, primarily including spraying systems, spreading systems, and picking systems.

Spraying Systems: are designed for the precise application of liquid pesticides, herbicides, and fertilizers. These systems typically consist of a tank to hold the liquid, a pump to generate pressure, hoses to transport the liquid, filters to prevent nozzle clogging, and nozzles to atomize and distribute the spray. Advanced systems may also incorporate flow meters for real-time adjustments to the application rate and variable rate capabilities, allowing the drone to adjust the spray rate based on sensor data and prescription maps generated from field analysis. The use of drones for spraying offers numerous advantages, including enhanced precision, reduced chemical usage, cost-effectiveness compared to traditional methods, and improved safety for operators by minimizing exposure to chemicals [13].

Spreading Systems: are utilized for the dispersal of granular materials such as seeds and granular fertilizers. These systems generally include a hopper to contain the material and a mechanism for controlled release, often integrated with GPS technology to ensure accurate and even distribution across the targeted areas. The development of drone planting systems, which can shoot pods containing seeds and nutrients directly into the soil, represents an innovative approach to automating the planting process [14].

Automated Fruit Picking and Harvesting Systems consist of a robotic arm and an end-effector, commonly referred to as a gripper. The design of this gripper is of paramount importance, as it must be capable of gently yet securely grasping and detaching the fruit without causing any physical damage such as bruising or punctures. Advanced vision systems powered by artificial intelligence (AI) have to be integrated with the picking system, which is essential for accurately detecting ripe fruits among leaves, assessing their size, maturity level, and overall quality, and subsequently guiding the robotic arm to perform the picking action [15].

4 Machine Learning for Agricultural Data Analysis

Complementary to the advancements in drone technology is the advancement in the field of AI, specifically ML and deep learning (DL), which have been proven to be indispensable in processing the vast datasets generated by aerial platforms. The data acquired through the drone's sensors encompasses detailed visual representations of fields (high-resolution images), spectral signatures that reveal physiological conditions of plants, temperature readings that can highlight anomalies, and accurate spatial models of the agricultural terrain. The huge volume and inherent complexity of the data generated by agricultural drones present a significant challenge for traditional analysis methods and necessitates the application of automated processing and sophisticated analytical techniques to extract meaningful insights. ML algorithms are particularly well-suited to the task of efficiently sifting through large quantities of data to identify underlying patterns and correlations [16].

In the following, we shed the light on a variety of ML algorithms employed in processing drone data for smart farming, tailored to specific agricultural tasks.

4.1 Supervised Learning Algorithms

Supervised learning exploits labeled agricultural datasets collected from drones, IoT sensors, and satellite imagery for several applications such as [16, 17]:

Crop Disease Detection and Classification: Algorithms such as Random Forest, Support Vector Machines (SVM), and Convolutional Neural Networks (CNNs) are trained on labeled images of healthy and diseased plants to automatically identify infections with high accuracy.

Tree Species Classification and Counting: classification models such as CNN enable the automatic classification of tree species, aiding in biodiversity assessment, forest management, geo-locating healthy and diseased plants.

Weed Detection: where supervised learning models such as YOLO (You Only Look Once) and Faster R-CNN segment and classify weeds in real time, allowing precision herbicide application and reducing chemical usage.

Yield Prediction models such as XGBoost and Linear Regression are widely used for yield prediction by analyzing historical crop data, weather patterns, and soil conditions, enabling farmers to optimize harvest planning and resource allocation.

Irrigation and Fertilizer Recommendations: Supervised learning algorithms can process data from drone-mounted thermal and hyperspectral sensors to predict irrigation needs and fertilizer recommendations, improving water and fertilizer use efficiency.

Pest Infestation Forecasting: Time-series supervised models like Long Short-Term Memory (LSTM) networks can be used to predict outbreaks based on past drone surveillance data, allowing preemptive pest control measures.

4.2 *Unsupervised Learning Algorithms*

Including various Unsupervised learning clustering techniques play a crucial role in identifying inherent patterns and anomalies within the drone-collected data. K-means clustering and DBSCAN are widely used to analyze drone-captured multispectral and hyperspectral images without requiring labeled training data for many tasks, such as [17]:

Crop Segmentation: Unsupervised learning algorithms automatically segment crops, soil, and vegetation. For instance, K-means can group pixels based on spectral signatures to identify healthy crops, water-stressed areas, or disease patches, enabling targeted interventions. Principal Component Analysis (PCA) further enhances efficiency by reducing data dimensionality, helping isolate key features for crop health assessment.

Weed and Anomaly Detection: Unsupervised models excel in detecting weeds and anomalies in fields by analyzing spatial patterns in drone imagery. Algorithms like Gaussian Mixture Models (GMMs) and isolation forests identify outlier pixels that deviate from normal crop textures, flagging potential weed infestations or pest damage. This approach reduces herbicide use by enabling precision spraying only where needed.

4.3 *Reinforcement Learning (RL)*

RL has emerged as a powerful paradigm for optimizing autonomous drones and robotic systems to adapt to real-world uncertainties by continuously refining their actions based on environmental feedback and reward signals. In precision agriculture, RL algorithms are being deployed for autonomous drone navigation in complex farm terrains, adaptive pesticide spraying that minimizes chemical usage while maximizing pest control, and optimal path planning for crop monitoring missions that reduce energy consumption. Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) have been successfully applied to coordinate drone swarms for large-scale field surveillance, where the system learns to prioritize high-risk crop areas based on vegetation indices and historical data [18–20].

4.4 *Federated learning (FL)*

Federated learning (FL) is a decentralized machine learning paradigm that enables multiple stakeholders to collaboratively train a shared model without exchanging raw data, making it ideal for privacy-sensitive applications in smart farming. Unlike traditional centralized approaches, FL allows farms, drones, and IoT devices to train models locally on their own datasets—such as crop imagery and soil sensor

readings—and only share model updates (e.g., gradients or weights) with a central server for aggregation. This preserves data confidentiality while leveraging collective insights to improve agricultural AI systems. In drone-assisted farming, FL facilitates cross-farm collaboration for tasks like disease detection or yield prediction without requiring farmers to disclose field-specific imagery. By addressing key challenges like communication efficiency and heterogeneous data distributions, FL is emerging as a scalable solution for secure and collaborative precision agriculture [21, 22].

5 Conclusion

The integration of drone technology with Machine Learning, Deep Learning, Reinforcement learning, and Federated Learning has unpaved the way for a new era of precision agriculture, aligning with the vision of Agriculture 5.0. This survey explored the transformative role of drones in smart farming, covering their diverse types, payloads, and applications. The synergy between drones and advanced ML/DL techniques, such as CNNs for disease classification, LSTMs for yield forecasting, and RL for autonomous drone swarms, has significantly enhanced agricultural operations' efficiency, accuracy, and scalability. Furthermore, FL emerged as a ground-breaking approach to address data privacy and collaboration challenges, enabling farms to collectively improve models without sharing sensitive data. Despite these advancements, challenges remain, including the high cost of drone deployment, regulatory constraints, and the need for robust edge computing to process real-time data in resource-limited environments. Future research should focus on lightweight AI models for edge devices, energy-efficient drone designs, and scalable FL frameworks.

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Adaptive Hyper-Heuristics for Smart Logistics Optimization



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and Jinane Sayah

Abstract The complexity of logistics combinatorial optimization problems including vehicle routing, warehouse scheduling, and dynamic delivery has increased because of rising demand and evolving constraints. Metaheuristics show effectiveness but need problem-specific tuning and demonstrate limited general applicability. This research presents a learning-based hyper-heuristic framework which operates at high abstraction levels to select or generate low-level heuristics through dynamic decision-making based on problem features and real-time performance feedback. The proposed system uses reinforcement learning to select heuristics while pursuing adaptability, scalability and domain independence. Additionally, the framework demonstrates its effectiveness through benchmark dataset experiments, which show better solution quality and improved computational efficiency and robustness compared to traditional metaheuristics. Moreover, the framework shows its capability to perform automated decision-making while minimizing human involvement and demonstrating effective adaptation to changing logistics environments. Finally,

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this research presents an adaptable intelligent optimization system which enhances operational efficiency and resilience in smart supply chain systems.

Keywords Hyper-heuristics · Combinatorial optimization · Smart logistics · Heuristic selection · Supply chain optimization

1 Introduction

The core of many real-world decision-making scenarios in logistics depends on combinatorial optimization problems because they require efficient routing, scheduling and resource allocation [1]. The implementation of smart logistics systems using Industry 4.0 technology and the Internet of Things (IoT) has led to substantial growth in problem complexity, which exceeds traditional optimization methods' capabilities [2]. The optimization challenges are addressed through the use of metaheuristics, which include Genetic Algorithms (GA), Ant Colony Optimization (ACO) and Simulated Annealing (SA) [3]. Unfortunately, these methods need problem-specific tuning and lack broad applicability to different problem instances. Hyper-heuristics function at a higher level of abstraction because they select or generate low-level heuristics dynamically while avoiding domain-specific customization [4]. This research presents a learning-based hyper-heuristic system which addresses multiple logistics-related combinatorial problems through vehicle routing, warehouse scheduling, and dynamic delivery planning. Real-time feedback integration with reinforcement learning strategies enables the framework to select heuristics dynamically, which results in improved adaptability and scalability and enhanced decision quality in dynamic logistics environments.

2 Related Work

The complexity of combinatorial optimization challenges in smart logistics systems continues to rise while metaheuristics and hyper-heuristic methods emerge as successful solution approaches. Metaheuristics such as Genetic Algorithms (GA), Simulated Annealing (SA), Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) demonstrate effectiveness in solving the Vehicle Routing Problem (VRP), Job Shop Scheduling Problem (JSSP), and warehouse operations [3].

Hyper-heuristics emerged as a solution to overcome these limitations through higher-level abstraction techniques, which enable automatic low-level heuristic selection or generation during the search process [4]. These approaches focus on developing problem-independent solutions which need minimal human intervention, provide enhanced flexibility and improved generalizability. Research has

recently investigated the application of reinforcement learning (RL) and deep learning as components of intelligent hyper-heuristics specifically designed for logistics optimization.

Hyper-heuristics consist of two main categories, which include heuristic selection and heuristic generation, according to [5, 6]. The first approach chooses heuristics from an available set according to their performance, while the second approach generates heuristics dynamically through machine learning methods. The field of logistics has experienced explosive growth in hyper-heuristics applications [7] through several prominent frameworks. The applications include Deep Reinforcement Learning Hyper-Heuristics (DRLH), Q-learning with Clustering, Nested Markov Chain Hyper-Heuristics (NMHH) and Large Language Models (LLMs).

The current generation of hyper-heuristics demonstrates strong potential, but they primarily work with benchmark problems while neglecting real-world logistics operations under time constraints. Our research closes this knowledge gap through a practical reinforcement learning-based hyper-heuristic system that ensures scalability and adaptability.

3 Proposed Hyper-Heuristic Framework

The proposed hyper-heuristic framework is a general-purpose solution for combinatorial logistics problems that is robust, adaptive and designed to solve problems such as vehicle routing, warehouse scheduling and order fulfillment. It has two hierarchical levels: the high-level strategy (heuristic selection and acceptance mechanism) and the low-level heuristics (solution perturbation operators). This framework aims to separate problem-specific details from search control, and thus enable greater scalability and reusability across domains such as vehicle routing, warehouse scheduling and order fulfillment.

3.1 Low-Level Heuristics

The proposed framework implements a range of low-level heuristics (LLHs), which function independently of specific problem domains to provide broad applicability across multiple logistics optimization challenges. The organized heuristics consist of two functional categories that include constructive operators, which create initial solutions through greedy insertion methods and perturbative operators, which enhance existing solutions. The perturbative operators consist of five essential techniques which include (1) Insertion for repositioning tasks or nodes to enhance solution feasibility (2) Swap for exchanging pairs of elements to discover new configurations (3) 2-opt as a specialized routing heuristic that reverses subsequences to minimize travel distance (4) Mutation for controlled randomization to escape local optima and (5) Greedy Repair for systematic solution component re-optimization. The heuristics

need only basic cost evaluation and feasibility checking functions to operate since they maintain minimal interaction with the problem domain, thus ensuring both computational efficiency and easy integration into different problem contexts. The framework design enables flexibility while delivering strong solution improvement abilities throughout various combinatorial optimization problems in logistics.

3.2 Heuristic Selection Method

The high-level controller employs a reinforcement learning-based selection strategy. Specifically, a multi-armed bandit (MAB) approach is used where each low-level heuristic is treated as an “arm,” and selection is based on its historical performance (e.g., improvement rate, time efficiency). The selection probability P_i of heuristic h_i is updated using the Upper Confidence Bound (UCB) algorithm shown in Eq. 1:

$$P_i = \overline{R_i} + C \sqrt{\frac{\ln N}{n_i}} \quad (1)$$

where $\overline{R_i}$ is the average reward, n_i is the number of times h_i has been applied, N is the total number of heuristic applications, and C is the exploration parameter.

3.3 Acceptance Criteria

To balance exploration and exploitation, the framework utilizes an adaptive acceptance criterion:

- **Improvement-based Acceptance:** Always accept a new solution if it improves the objective function.
- **Simulated Annealing-based Acceptance:** If the new solution is worse, it is accepted with a probability $p = e^{-\Delta/T}$, where Δ is the deterioration in solution quality and T is the current temperature, gradually reduced over time.
- **Stagnation-based Diversification:** If no improvement is observed over a pre-defined number of iterations, a large mutation is introduced to diversify the search.

3.4 Upper Confidence Bound Heuristic Selection Strategy

The framework uses the **Upper Confidence Bound (UCB)** algorithm to dynamically select low-level heuristics (LLHs). The framework treats each heuristic as an

“arm” in a multi-armed bandit problem, which balances exploration (trying less-used heuristics) and exploitation (reusing high-performing ones).

Each heuristic $h_i \in H$, where H is the pool of low-level heuristics, is treated as an arm in the bandit model. The UCB score for each heuristic is calculated using Eq. 2:

$$UCB_i = \overline{R}_i + C \sqrt{\frac{\ln N}{n_i}} \quad (2)$$

where:

$\overline{R}_i$ is the average reward obtained from applying heuristic h_i ,

n_i is the number of times h_i has been selected,

N is the total number of heuristic applications across all $h \in H$,

C is the exploration parameter that controls the trade-off between exploration and exploitation.

The reward $\overline{R}_i$ can be defined based on:

- The relative improvement in objective function value.
- The quality of the solution compared to the global or local best.
- The computational efficiency (e.g., improvement per time unit).

At each iteration, the heuristic with the highest UCB score is selected bas on Formula 3:

$$h^* = \arg \max_{h_i \in H} (UCB_i) \quad (3)$$

3.5 Overall Workflow

The overall workflow is outlined as follows:

1. Generate an initial solution using a greedy or random constructive heuristic.
2. While termination criteria are not met:
 - (a) Select a low-level heuristic based on the UCB heuristic selection strategy.
 - (b) Apply the selected heuristic to the current solution to generate a candidate solution.
 - (c) Evaluate the candidate solution.
 - (d) Accept or reject the candidate solution based on the acceptance criterion.
 - (e) Update heuristic performance metrics.
3. Return the best solution found.

This flexible, intelligent framework allows for generalization across various logistics scenarios, offering a balance between solution quality and computational efficiency.

4 Experimental Setup

To evaluate the effectiveness and generalizability of the proposed hyper-heuristic framework, a series of computational experiments were conducted on well-established combinatorial optimization problems in logistics, including variants of the Vehicle Routing Problem (VRP) and Job Shop Scheduling Problem (JSSP).

4.1 Benchmark Datasets

The experiments utilized publicly available and widely used benchmark datasets:

- **Solomon’s VRPTW Instances** [8]: These datasets include customer locations with time windows, service durations, and vehicle capacities.
- **Gehring & Homberger Extended VRPTW Instances** [9]: Large-scale versions of Solomon’s problems, featuring up to 1000 customers.
- **OR-Library JSSP Instances** [10]: Standard scheduling datasets, including LA, FT, and ABZ problems, often used to test scheduling heuristics.

Each dataset represents a different class of real-world logistics problems, ensuring diversity in problem complexity and scale.

4.2 Implementation Details

The hyper-heuristic framework was implemented in Python 3.11 using NumPy and SciPy for numerical operations. A modular architecture was adopted to enable plug-and-play of low-level heuristics and heuristic selection strategies. Each experiment was repeated 30 times with different random seeds to ensure statistical significance. The performance of the proposed framework was compared to classical meta-heuristics such as Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Simulated Annealing (SA).

4.3 Parameter Settings

- **Max Iterations:** 10,000
- **Population Size (for GA and ACO):** 50
- **Exploration Parameter (UCB):** $C = 2$
- **Initial Temperature (SA):** 100, with exponential decay
- **Stagnation Limit:** 500 iterations

Parameter values were selected based on preliminary experiments and commonly accepted settings in the literature [4].

4.4 Evaluation Metrics

The following metrics were used to evaluate solution quality and computational efficiency:

- **Total Cost / Objective Value:** Total travel distance (VRP), makespan (JSSP), or combined cost.
- **Average Gap to Optimal / Best Known Solution:** Expressed as a percentage.
- **Computational Time:** Execution time in seconds.
- **Stability:** Standard deviation of solution quality over multiple runs.

Statistical tests (e.g., paired t-test and Wilcoxon signed-rank test) were used to compare results and validate improvements.

4.5 Algorithmic Complexity and Scalability

The proposed hyper-heuristic framework exhibits a modular and efficient architecture designed to support scalability across large-scale logistics optimization problems. The overall time complexity per iteration can be decomposed into three principal components:

- **Low-level heuristic application:** Each heuristic (e.g., 2-opt, swap) has a complexity that depends on the problem structure. For instance, 2-opt has a complexity of $O(n^2)$ in routing problems, where n is the number of nodes or tasks.
- **Heuristic selection via UCB:** The selection of the best-performing heuristic using the Upper Confidence Bound (UCB) strategy involves computing a score for each of the k heuristics. This results in a selection time complexity of $O(k)$ per iteration.
- **Solution evaluation:** Evaluating the candidate solution (e.g., computing the cost, feasibility) typically requires $O(n)$ to $O(n^2)$ operations, depending on whether the problem is routing, scheduling, or packing.

Combining these components, the time complexity per iteration becomes as shown in Eq. 4:

$$O(f(n) + k + e(n)) \quad (4)$$

where $f(n)$ is the time to apply a heuristic, k is the number of available low-level heuristics, and $e(n)$ is the cost of evaluating a solution.

Assuming the algorithm runs for I iterations, the total worst-case runtime is shown in Eq. 5:

$$O(I \cdot (f(n) + k + e(n))) \quad (5)$$

Table 1 Solution quality: hyper-heuristic versus genetic algorithm

Instance	GA (Avg)	HH (Avg)	Improvement (%)
C101 (VRPTW)	845.7	798.2	5.61
R201 (VRPTW)	1215.3	1105.6	9.02
LA16 (JSSP)	998.3	942.8	5.56
ABZ8 (JSSP)	820.5	773.9	5.67

Bold represents results of our method

5 Results and Discussion

The experimental findings of the proposed hyper-heuristic framework appear in this section against a classical Genetic Algorithm (GA), which functions as the performance evaluation baseline. The evaluation takes place across standard benchmark instances for vehicle routing and job shop scheduling. The same computational environment and termination criteria were applied to both methods.

5.1 Genetic Algorithm Implementation

The hyper-heuristic framework demonstrated superior performance than the baseline Genetic Algorithm (GA) in all benchmark instances (VRPTW and JSSP). The main findings included:

- 5–9% better solutions (e.g., 798.2 vs. 845.7 for C101 VRPTW)
- Higher stability (lower standard deviation in repeated runs)
- Generalized performance across problem types (routing/scheduling)

The framework’s adaptive heuristic selection needed 10–15% longer runtime to outperform GA’s fixed operators but demonstrated better robustness especially in dynamic scenarios. (Data from Table 1, Sect. 5.2)

5.2 Solution Quality Comparison

Table 1 shows the average objective values over 30 independent runs. The hyper-heuristic outperformed GA in all test cases, achieving lower total cost or makespan with higher consistency.

5.3 Runtime and Stability

While the hyper-heuristic required slightly longer average runtime per instance (by 10–15%), it exhibited more stable performance with lower standard deviation across repeated runs. The adaptive learning and dynamic acceptance strategy contributed to fewer premature convergences compared to GA.

5.4 Discussion

The results demonstrate that the hyper-heuristic framework delivers substantial improvements above a well-configured GA system. The framework demonstrates better generalization capabilities through its autonomous heuristic selection and adaptation mechanism, which uses real-time feedback. The hyper-heuristic provides a scalable automated solution for real-world logistics environments because it adapts to dynamic problem characteristics better than GA with its domain-specific tuning.

6 Sensitivity and Robustness Analysis

This section investigates how the exploration parameter C affects the hyper-heuristic framework together with its performance across various problem instances. The exploration parameter C determines the equilibrium between exploration and exploitation in the UCB strategy. The graph in Fig. 1 how different C values affect the average objective value during 30 runs for the C101 instance.

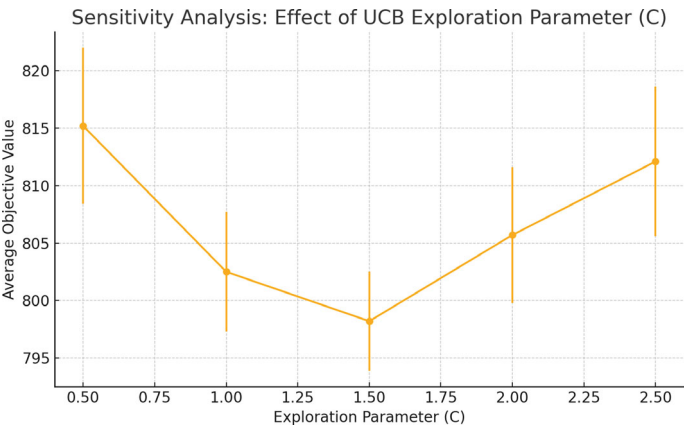


Fig. 1 Sensitivity analysis: effect of UCB exploration parameter C

The results indicate that the best performance is achieved when $C = 1.5$, with an average objective value of 798.2 and the lowest standard deviation of 4.3. Smaller values of C result in insufficient exploration, while larger values lead to excessive exploration, degrading performance.

6.1 Robustness Across Instances

Figure 2 shows the standard deviation of the objective value obtained by the hyper-heuristic across four benchmark instances. Low variance across repeated runs indicates that the algorithm produces consistent results regardless of problem scale or type.

The results confirm that the proposed framework maintains stable performance across both routing and scheduling problems. The average standard deviation ranged from 4.3 (C101) to 6.1 (LA16), highlighting its robustness in dynamic and stochastic environments.

7 Conclusion and Future Work

The research introduced an innovative hyper-heuristic framework which solves complex combinatorial optimization problems in smart logistics systems. The proposed method gained high generality and adaptability toward different problem types including Vehicle Routing and Job Shop Scheduling through its abstraction of domain-specific heuristics. The framework used a learning-based heuristic selection

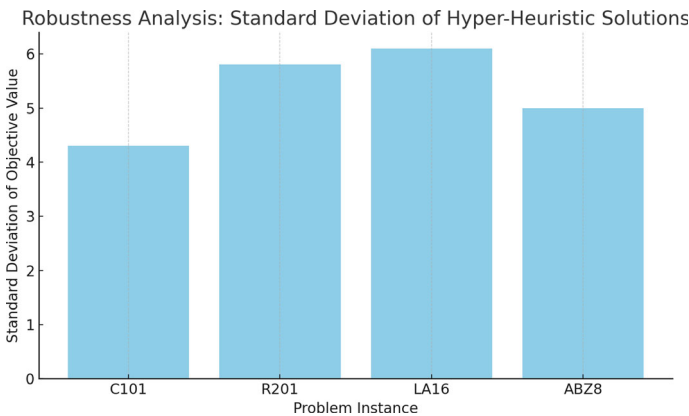


Fig. 2 Robustness analysis: standard deviation across benchmark instances

strategy based on Upper Confidence Bound (UCB) to achieve effective exploration and exploitation balance.

The experimental findings showed that the hyper-heuristic produced better solutions than a well-tuned Genetic Algorithm while demonstrating stable and robust performance. The exploration parameter sensitivity analysis confirmed its effect on search performance while robustness evaluation showed the framework maintained consistent results across various benchmark instances.

The upcoming research will concentrate on developing the framework to manage real-time and stochastic logistics problems. Our future work includes developing multi-objective optimization functions and implementing deep reinforcement learning to generate adaptive heuristics in dynamic systems with uncertainty.

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Metaverse-Driven Immersive Therapy: Reducing Air Travel Anxiety for Individuals with Autism



Mariam Thaher, Shurouq Khan, Fatima Alqattan, Tala Alhammouri,
and Ismaeel Al Ridhawi

Abstract Virtual reality (VR) functions as a substantial transformative tool which delivers therapeutic interventions particularly in exposure therapy. This paper investigates the implementation of metaverse-based virtual reality to create flight simulations for children with autism spectrum disorder (ASD) and social phobia. The proposed solution provides a personalized controlled space through which users can experience airplane boarding, in-flight interactions, and disembarkation with the support of VR tools. The system aims to expose users to these situations within a controlled space to minimize their stress levels before they face actual air travel. The system would adapt the environment to meet the specific needs of the user, thus offering an individualized therapeutic approach. This adaptation is performed with the aid of visual and audio feedback by incorporating verbal and a non-verbal user behavior determination models. The models would identify the user's attitude through speech analysis, facial recognition, physiological reactivity identification, and gesture recognition. The system is a work-in-progress and is expected to offer a personalized and real-time adaptive approach once complete, making it more effective than other methods in the literature.

Keywords Metaverse · Virtual reality · Immersive therapy · Autism spectrum disorder · Machine learning · Deep learning

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1 Introduction

Aerophobia represents an intense fear which people experience when flying on an airplane. Many individuals experience flight-related anxiety because they fear different aspects of flying including take-off, landing, or becoming trapped inside the aircraft. The experience of air travel proves difficult for numerous people, but those with autism spectrum disorder (ASD) or social phobia face exceptional challenges because their anxiety worsens when faced with unfamiliar surroundings that combine crowded conditions with constant noise and social requirements. This is especially challenging for children with ASD and their families. The thought of entering an airport or getting on a plane becomes too much to handle, because air travel seems impossible to achieve and the lack of experience in such environments creates severe distress that often causes families to avoid travel. This may restrict their access to educational and social activities and family events.

Virtual reality (VR) stands out as a transformative tool for therapeutic interventions, particularly in exposure therapy. This paper investigates the application of metaverse-driven VR to simulate air travel for children with ASD and social phobia. The solution provides an immersive customized environment through VR tools that enables users to experience airplane boarding, in-flight interactions and disembarking, in a controlled setting to familiarize them with these scenarios before encountering real-world air travel, thus reducing stress for both children and their families. The solution is supported by user-behavior models that classifies the child's attitude towards certain interactions in the VR. Those behaviors are determined using verbal and non-verbal expressions, such as the tone of the voice, the words used, facial expressions, eye contact, and more. Furthermore, Deep Learning (DL) is incorporated into the system to control VR avatars either visually or through voice.

The outcome of the proposed system is a virtual air-plane application designed to simulate air travel, exposing users to a realistic airplane environment wherein passengers practice key interactions such as entering the airplane, finding their seat, interacting with flight attendants, and adjusting to in-flight conditions. The VR environment is adaptable to the user's behavior to prolong the VR session and achieve educational objectives. The system is deployed onto Oculus VR headsets, with the support of tools such as Unity Hub for integration, 3D Max for realistic 3D modeling, and C# for interactive scripting, ensuring a smooth and engaging experience for children with ASD to help decrease their anxiety. The work is still in-progress and under-testing, but preliminary experimental tests have shown promising results in terms of both user-experience and system development.

The rest of the article is organized as follows: Sect. 2 highlights some the related work in the literature. Section 3 provides a discussion on the proposed framework. Implementation details, as well as preliminary evaluations are looked at in Sect. 4. Finally, in Sect. 5 concluding remarks are drawn from the work.

2 Related Work

In [1], the authors developed a hide-and-seek VR system which functions as a VR intervention to enhance gaze fixation in children with ASD. A virtual family room is employed in the system that integrates deepfake technology to generate personalized avatars in order to increase user engagement. The research study included thirty six children with ASD grouped into three categories, namely, those who received customized avatars, those who received un-customized avatars, and those who received traditional interventions as part of a control group. Results show that the VR-assisted groups achieved better results in gaze fixation abilities, with the customized avatar group showing the most improvement. The developed framework consists of four main components which include game, gaze, data logger and reinforcement modules. The framework combines personalization and real-time feedback effectively, yet it creates difficulties because of its complex nature and restricted application range.

The authors in [2] introduced a VR serious game, called Travel Train, which enhances facial emotion recognition abilities in ASD individuals. The virtual city of Travel Train contains three mini-games called Identification, Manipulation and Production, which enable users to develop emotional capabilities and skills. The controlled simulations enable users to train their emotions in real-world using an immersive environment experience. The initial study involved ten participants, five of which have ASD and five typically developed control. The users found the game usable and accepted it well. The game's technical complexity along with its small participant sample size create obstacles for generalization and requires extended extensive evaluations.

The research study presented in [3] developed an Android application featuring an interactive virtual pet that uses Augmented Reality (AR) to support children with Asperger syndrome. The application was implemented using the Unity game engine and Media Pipe pose estimation. The solution provides children aged 5–12 with accessible and cost-effective social skill development and emotional regulation training through playful interaction. The system integrates AR technology with a virtual pet companion and a user interface toolkit to enhance user engagement. The initial assessment conducted with healthcare providers showed both high user guidance and efficient resource utilization. The application faces two main limitations, namely, evaluations with children were indirect and the dependence on mobile devices creates accessibility issues.

The authors in [4] introduced a VR multiplayer experimental training application which helps ASD individuals transition to day activity centers. Users can experience a digital twin version of the center in Torino, Italy through this application before conducting actual visits. The system features personalized avatars and multiplayer capabilities for real-time care-giver guidance, while minimizing sensory input to prevent overwhelming stimuli. The system validation showed positive results for

usability and user experience, but the initial testing was restricted. The system provides significant support through its multiplayer functionality and realistic environment, yet its widespread use faces obstacles from scalability issues and technical requirements.

3 Proposed Framework

The proposed metaverse-driven immersive therapy framework is designed to aid children with autism or social phobia in preparing and desensitizing themselves for travel by air. The VR environment is designed to be as realistic as possible to reflect the boarding of passengers, seat assignment, and interaction with either cabin crew members or other passengers during travel. These experiences are made available to users through a VR environment setup that includes a VR headset, controller, and other optional components like cameras, posture sensor, smartwatch, and microphone. The VR system is developed and powered by VR tools that include Unity Hub for designing and animating scenarios, C# programming language for coding operational features, and 3D MAX for realistic 3D model creations, like airplane interiors and interactive elements. Furthermore, behavioral models supported by DL are used to fine-tune the VR environment according to the user's needs. These tools and environment combined will manage all user interactions and provide a truly immersive VR experience.

3.1 Framework Components

Three main components make up the framework, namely, the behavioral module, the reaction module, and the adaptation engine as depicted in Fig. 1. The behavioral module interprets the VR user's behavior both through verbal and non-verbal features. This is an integral part of the system and is considered the first step towards identifying the user's reaction to an event that may have occurred during the VR session. Given that autism is a wide spectrum, where some children may be able to express their emotions and feelings verbally, while others cannot and usually express their emotions through non-verbal actions like facial expressions or movement, both verbal and non-verbal behavior models are incorporated into the behavioral module. The speech engine enables voice detection, continuous speech recognition, and speech synthesis, thereby making it possible to have conversation-like interactions in the immersive VR environment [5]. Certain words are captured to determine the user's reactive behavior and attitude towards the event that occurred. For instance, whenever the child yells out words such as stop, help, or annoying, then the behavioral module determines that the event that occurred is not attracting the user's attention and may be somewhat bothering, and thus requires scene fine-tuning with the aid of the VR adaptation engine.

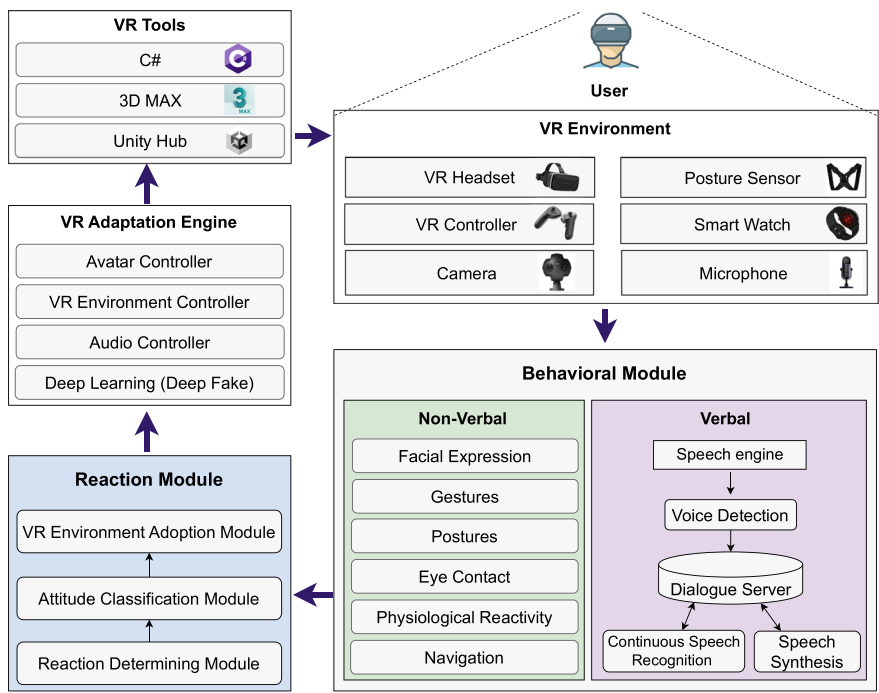


Fig. 1 The proposed metaverse-driven immersive therapy framework

Similarly, non-verbal observance could be achieved by monitoring facial expressions, gestures, postures, eye contact, physiological reactivity, and navigation within the VR space, all of which provide insights into the user’s comfort level. Facial expressions consider features such as eyebrows, eyes, and mouth to classify the user’s behavior in terms of a set of emotions such as anger, fear, surprise, sadness and happiness [6]. Gestures play an important role as well, where the hand movement can identify the user’s behavior. For instance, the thumbs up gesture indicates that the user is relaxed and is quite interactive with the VR scene. Body posture is one non-verbal feature that is used to determine the behavior of a child with ASD. For instance, a child cramming might indicate that they are distressed. Eye contact is an important feature for non-verbal behavior identification [7]. A child not gazing at the objects of interest in a VR scene can indicate that they are not interested in the interaction and may not be following the objectives set by the scenario. Body changes in response to stressful events is captured through physiological reactivity. Finally, navigating outside the bounds set in the VR scene is also a sign of the lack of interest to engage in the VR environment.

These behavioral data are used to modify training processes according to the specific needs exhibited by children with autism or social anxiety. The reaction module considers the user’s actions and emotions collected from the behavioral module to

determine and classify the user's behavior in order to adapt the required changes into the VR environment. The classification process follows the work presented in [8], where the child's attitude is derived according to their cooperation with the VR system. The cooperative scoring system is based on four categories, namely, aggressive, non-cooperative, peaceful, and cooperative nature. For instance, given the data collected from the behavioral module, the reaction module could classify a child as being aggressive and will instruct the adaptation engine to perform certain changes to the VR environment. Similarly, a peaceful classification indicates that the child is relaxed and is highly likely to be engaged in the VR environment and is considered to be interactive.

Using the results collected from the reaction module, the VR Adaptation Engine uses visual and audio controlling mechanisms supported by DL to improve the user experience in the VR environment [1]. Deep fake techniques are adopted to fine-tune the visual effects of certain avatars so that they would resemble those that may be considered friendly with the child, like the child's parents or family members. Audio effects can also be integrated to relax the child either through relaxing musical tunes or voices that the child might be familiar with. The avatar and audio controllers integrate DL models to adapt the necessary changes in both visual and audio effects of avatars. Furthermore, activities incorporated in the VR environment might change as well in accordance with the results received from the reaction module.

3.2 *VR Tools and Environment*

The hardware and software tools used to create and run the VR environment are as follows:

- **Oculus Quest 2:** is a VR headset that provides users with VR experience. It provides audiovisual display and offers motion, and hand tracking capabilities.
- **Unity Hub:** is a development platform for creating VR applications and provides functionalities for VR object interaction, scene management, and user interfacing.
- **3D MAX:** is a tool used for creating 3D models of all objects in the VR environment, that includes, passengers, seats, and other elements.
- **C#:** this programming language is used as the main source for creating the VR application which includes user interfaces, defining stage logic, and implementing simulation mechanics.
- **Python:** given its rich machine learning (ML) library, this programming language is used to implement the behavioral and reaction modules, as well as the DL algorithms used by the VR adaptation engine.

4 Implementation and Results

4.1 3D Modeling

Three dimensional (3D) models of the VR environment were created using 3D MAX. The airplane fuselage was designed to resemble the shape and dimensions of that in the real world environment, having a cylindrical fuselage body and rounded rectangular windows. Figure 2 depicts the model.

Passenger and crew member characters were designed using a simplified wire frame to facilitate rigging and animation. Figure 3 depicts the character models with a T-pose, allowing easy manipulation of limbs and joints. The characters were animated focusing on basic walking motion and object interaction, where the flight attendant model is rigged to hold a tray.

4.2 Implementation

The login and user profile modules form the basis of user interaction and authentication in the VR application.

User Authentication: The login page handles user authentication, allowing users to log in or continue as guests. When the user clicks on the Login button, the *Button_Login* method is triggered. It verifies that both the username and password fields are not empty. If either fields are empty, the method returns without proceeding. Otherwise, it calls the *StartCoroutine* method to initiate the *LoginRequest* method,

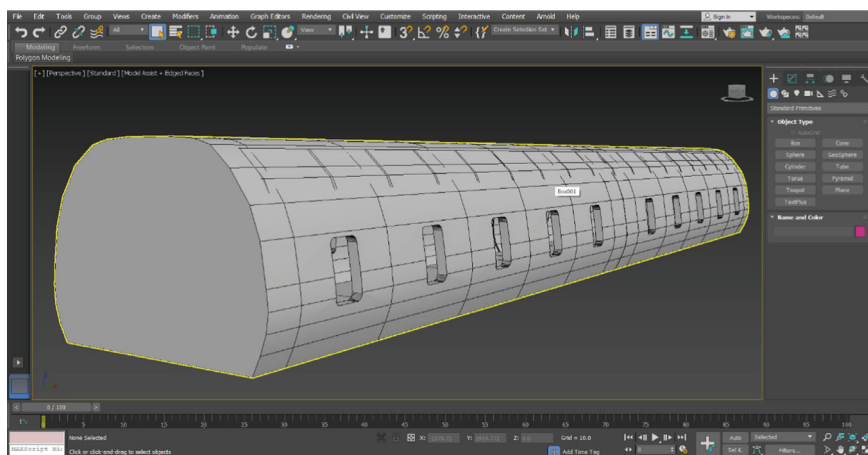


Fig. 2 Airplane model designed using 3D MAX

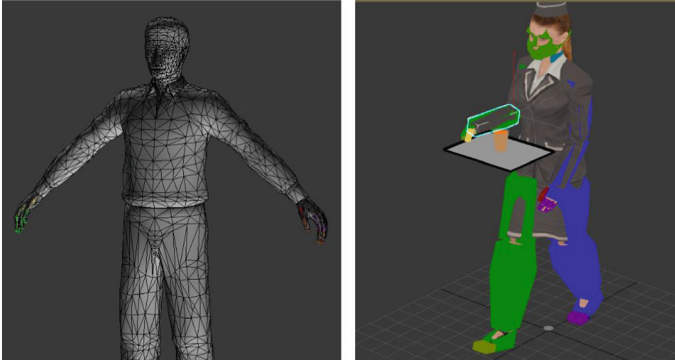


Fig. 3 Character models designed using 3D MAX for both passengers and crew members

which manages server communication for user authentication. Figure 4a depicts implementation details for user authentication.

User Profile Management: Managing user profile updates is achieved by interacting with user interface elements such as alert dialogs, loading screens, and input fields. The *LoadData* method initializes the input fields with the logged-in user's current information. The *ButtonUpdate* method handles form validation by checking that the required inputs are provided and that the password confirmation matches. If all validations pass, it invokes the *UpdateRequest* method for communication with the server. Figure 4b depicts implementation details for user profile management.

4.3 Results

The main page of the VR application is depicted in Fig. 5a. Two buttons can be seen that read “Start Game” and “Logout”. At the top of the screen, three menu bars can be seen that read “My Profile”, “About Us”, and “Our Team”. At the bottom of the screen, a virtual keyboard is shown on the screen for input, and a laser pointer shows how the VR controller can be used to interact with the screen. Once the user begins the game, the main scene will show a flight attendant handing out a cup to drink to the child as depicted in Fig. 5b. The interior of the airplane is visible, with seats and passengers in the background, and a VR controller is seen as well. A success message prompts the child to pick up and drink from the glass as depicted in Fig. 5c.



Fig. 4 Code snippet for **a** user authentication and **b** user profile management

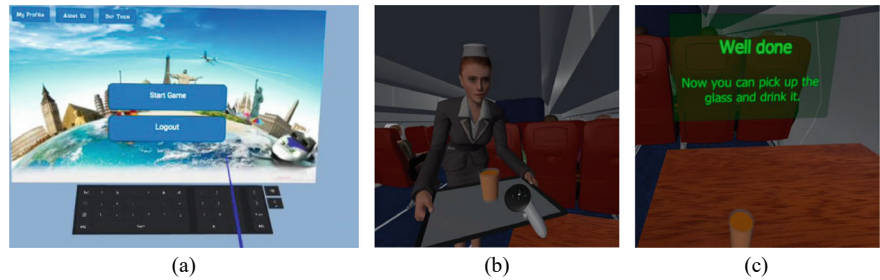


Fig. 5 Screen shots from the in-game VR environment, **a** main page, **b** user interaction, **c** success message after action

5 Conclusion

This paper presented a work-in-progress metaverse-enabled immersive therapy framework targeting individuals with autism. The solution aims at reducing air travel anxiety in children with ASD. The solution is designed to analyze the user’s behavior through behavioral and reaction modules that are supported by ML and DL. The user’s behavior is classified in accordance to their speech, gestures, eye contact and other behavioral expressions. The VR environment is adapted to suit the child’s need in accordance to their reaction. Initial usability tests show that the immersive nature

of VR can greatly lower a child's anxiety and boost their confidence. Further tests and enhancements are planned for future work.

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CLAHE-Augmented MRI Dementia Classification via Soft-Voting Ensemble with Gradient-Based Analysis and Dementia Correlation



Garima Shukla , Vanshaj Awasthi , Prashant Dubey , Sachin Kumar , Balamurugan Balusamy , and Ahmad Alkhayyat

Abstract Dense and accurate automated dementia stage classification from MRI scans is still a challenging task because of the inconsistencies in structural patterns and the inability of current deep learning models to manage both local and global feature representations effectively. In this work, a new hybrid ensemble model is proposed that combines CNN-based (ResNet-50, ConvNeXt, DenseNet-121) and transformer-based (Swin Transformer, ViT) models to synergistically extract their complementary strengths for effective dementia classification. An explicitly crafted preprocessing pipeline, including N4 bias field correction, CLAHE, non-local means denoising, Sobel edge detection, and K-means segmentation, provides high-quality input for enhanced feature extraction. The model was tested on a dataset of 87,000 MRI scans with 8500 images independently tested. Experiment results show Swin Transformer obtained AUC-ROC scores of 0.99 for Mild Dementia and 0.97 for Moderate Dementia, and the ensemble model delivered balanced decisions for all stages of dementia. Cohen's Kappa (0.57) and Matthews Correlation Coefficient

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(0.60) established model reliability, and class-wise precision-recall substantiated its discrimination capacity. Furthermore, Grad-CAM and attention maps enabled model interpretability by emphasizing primary regions driving predictions, thereby improving clinical relevance. The results highlight the effectiveness of transformer-CNN ensembles in neuroimaging-based dementia diagnosis and promise their suitability for use in real-world diagnostic assistance.

Keywords CNN-transformer ensemble · MRI dementia classification · Grad-CAM analysis · Deep learning diagnosis

1 Introduction

1.1 *Widespread Dementia and Subtypes*

Alzheimer's disease (AD) and other dementias impact more than 55 million individuals globally, with close to 10 million new cases being diagnosed every year. As the most prevalent cause of neurodegeneration, dementia poses great challenges in early diagnosis, especially in differentiating Very Mild Dementia (VMD), Mild Dementia, and Moderate Dementia, which share similar structural atrophy patterns on MRI scans. Conventional diagnostic methods depend on neuropsychological tests and manual radiological assessments, which are subjective and susceptible to inter-observer variability. This has given rise to the need for more automated, AI-based diagnostic solutions that can correctly classify dementia stages. Deep learning developments in neuroimaging over the recent past have shown promise for dementia classification [1]. Convolutional Neural Networks (CNNs) have become popular for feature extraction, based on hierarchical representations to capture local anatomical variation [2]. But CNNs have difficulty with long-range dependencies and spatial relationships between various brain areas, which constrain their ability to perform on challenging classification tasks. Transformer-based models, specifically vision transformers (ViTs), address these shortcomings by employing self-attention mechanisms, enabling them to capture global contextual information that is important in detecting subtle neurodegenerative changes. Despite their advantages [3], ViTs often require large-scale datasets and suffer from high computational costs, making their standalone application impractical for real-world medical diagnosis [4]. To overcome these challenges, this research suggests a hybrid ensemble model combining both CNN-based [6] (ResNet-50, ConvNeXt, DenseNet-121) and transformer-based (Swin Transformer, ViT) models [5].

2 Literature Review

This paper [7] suggests an explainable machine learning method for AD classification with a large NACC data set of 169,408 cases and 1024 attributes. The model performs balanced multiclass classification between normal cognition (NC), mild cognitive impairment (MCI), and AD and forecasts cognitive decline up to four years in advance. Although the research points to the promise of AI in AD diagnosis, issues with generalizability and clinical uptake need to be addressed through more studies.

This work [8] builds machine learning classifiers for early Alzheimer's disease (AD) and mild cognitive impairment (MCI) diagnosis based on structural MRI data from ADNI and OASIS. The "healthy versus AD" classifier had 90.6% balanced accuracy, and the multi-diagnostic model (NC vs. MCI vs. AD) had 62.1%. The models generalized well between datasets and MRI protocols, showing robustness. Key areas were hippocampus, temporal, cingulate, and frontal areas, consistent with AD pathology. Despite graph theory measurements providing no incremental benefit, this study illustrates the clinical utility of AI, inviting further real-world trial validation.

This systematic review [9] reviews recent advancements in deep learning for Alzheimer's disease (AD) detection, with emphasis on CNNs, RNNs, and generative models from 2018 to 2023. CNNs are best for neuroimaging analysis, RNNs (particularly LSTMs) best track disease progression, and GANs/VAEs help data augmentation and feature extraction. New methods such as graph networks, attention, and transfer learning have potential but suffer from a lack of dataset availability, interpretability, and generalizability. Potentiality of deep learning is notwithstanding that data heterogeneity and biological plausibility continue to be paramount for clinical applicability.

This poll [10] examines more than 80 papers published after 2017 on machine learning for Alzheimer's diagnosis via neuroimaging and biomarkers. Standard techniques involve SVMs, decision trees, and ensemble models for MRI classification. Key challenges involve multimodal data integration, handling incomplete datasets, overfitting, and model interpretability. Future research should focus on feature selection, dataset limitations, and optimizing ML models for more accurate and reliable AD diagnosis.

The research fills important gaps found in previous work on Alzheimer's disease classification with deep learning [11]. Previous research [12] tends to struggle with generalization, overfitting, and interpretability. Most of them use either CNNs or transformers, without taking advantage of their combined power, and are plagued by dataset limitations that limit model robustness. This research combines a hybrid strategy, using CNNs, transformers, and ensemble learning to enhance classification accuracy while minimizing model overfitting. Furthermore, though earlier research emphasizes the need for explainability, it does not have interpretability methods, thus hindering clinical use. In this work, Grad-CAM and attention maps are included, offering key insights into model decision-making. The preprocessing pipeline also

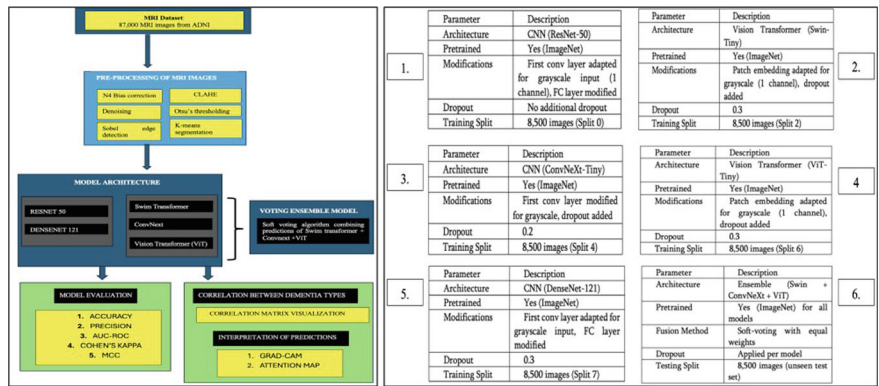


Fig. 1 Methodology workflow and model architecture

guarantees better image clarity and feature representation, boosting the ability of the model to identify dementia stages effectively [13].

3 Methodology

3.1 Widespread Dementia and Subtypes

For this research, ADNI dataset having 87,000 MRI scans was utilized, of which a testing split of 8500 was separated to test the models on these unseen images (see Fig. 1). Across all experimental splits, the approximate ratio of class distribution was maintained as follows: Non Demented: Very Mild Dementia: Mild Dementia: Moderate Dementia $\approx 10:3:2:1$. The MRI scans were subject to a methodical preprocessing pipe- line to improve the quality of the images and facilitate better feature extraction for dementia categorization (see Fig. 2).

3.2 Preprocessing and Model Architecture

This preprocessing stream guarantees better image clarity and representation [14] of features for more reliable and efficient deep learning-based dementia classification (see Fig. 2)The work here utilizes five deep learning architectures [15]—ResNet-50, Swin Transformer, ConvNeXt-Tiny, Vision Transformer (ViT-Tiny), and DenseNet-121—in addition to ensemble-based voting-based classification of MRI scans through varying stages of dementia[16]. The designs were chosen based on the suitability to extract spatial as well as contextual information from convolutional as well as transformer-based models [17] (see Fig. 3).

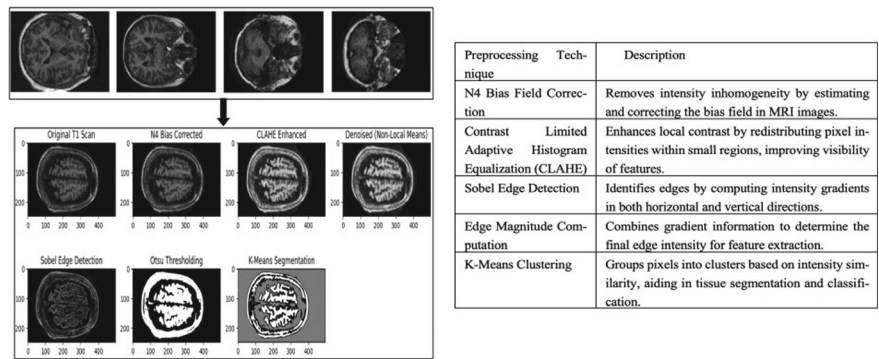


Fig. 2 ADNI MRI images and pre-processing applied

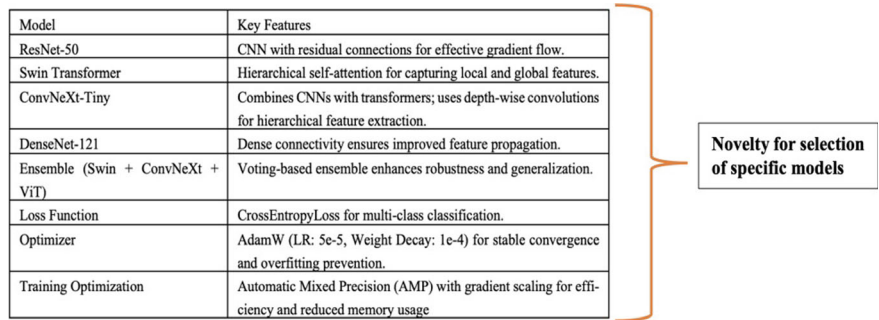


Fig. 3 Novelty of selected models

4 Results

The curves of loss point to variations in model optimization efficiency and convergence (see Fig. 4) ResNet-50 experiences drastically low loss values rapidly, upholding its learning behavior of memorizing data instead of effective generalization. Swin Transformer shows a gentle reduction in loss, indicating a more stable learning process. ConvNeXt has the lowest loss during training, which shows strong feature learning and better gradient flow. ViT has higher loss at the start but eventually converges well. DenseNet-121 has moderately declining loss but does not match the lowest loss rates of ConvNeXt or Swin Transformer, which may imply inefficiency in optimizing deep hierarchical features (see Fig. 4). The ResNet-50 and DenseNet-121 were exhibiting a huge overfitting ratio of 0.31 and 0.30 respectively because they were too deep, which was able to excessively memorize the training data without generalizing well. The performance on unseen images must have suffered also due to the lack of sufficient regulation (i.e. dropout, weight decay) and probably limited data diversity. On the other hand, Swin Transformer has more balanced performance

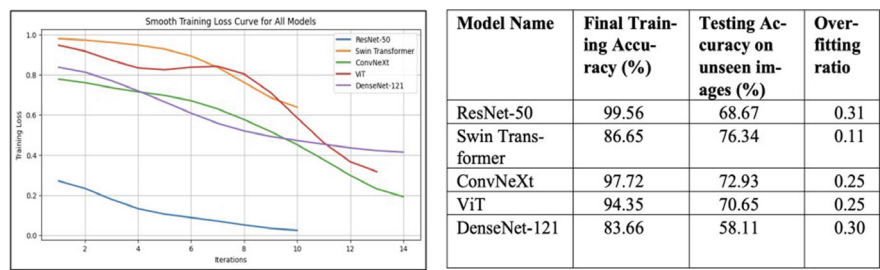


Fig. 4 Training loss curves and overfitting ratios

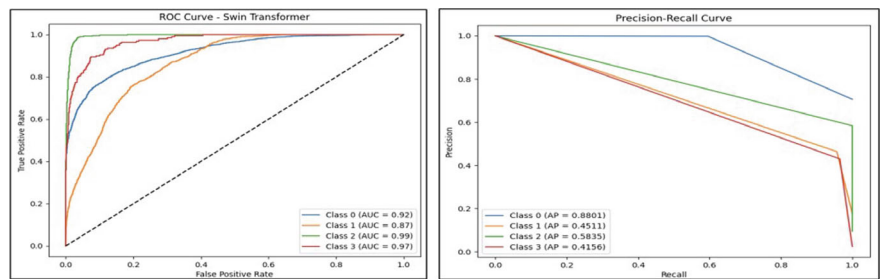


Fig. 5 AUC-ROC curve (Swim) and precision-recall curve (ViT)

with 86.65% training and 76.34% testing accuracy and is able to effectively capture global lesion patterns.

The Swin Transformer model AUC values reflect high classification performance (see Fig. 5) especially for Mild (0.99) and Moderate Dementia (0.97), clearly distinguishing between these stages. The Average Precision (AP) scores of the ViT model reflect its efficiency in ranking predictions for every class of dementia (see Fig. 5).

The resulting voting ratios of 0.33 (33.3%) for Swin Transformer, ConvNeXt, and ViT signify an equal contribution of all three models to the ensemble decision-making process. The well-balanced voting proportions suggest that all the models contributed equally to the final forecasts. But this also indicates that none of the models displayed obvious superiority in individual classification accuracy, so there is no immediate need for subsequent optimization techniques like weighted voting or confidence-based combining as potential fields for enhancing ensemble performance.

Distinct feature separations are evident in the t-SNE projections, and the widely spread values suggest an effective model capable of discriminating between specific dementia stages (see Fig. 6). Very well-defined clusters exist for negative and positive large shifts of coordinates, particularly showing the clear distinction between Non-Demented and progressive dementia stages. However, closeness of some points, for instance, − 17.16 and − 12.47, may indicate similarity in features thus confirming the previously indicated association between Very Mild and Mild Dementia. The comparison of Matthews Correlation Coefficient (MCC) and Cohen’s Kappa among the three

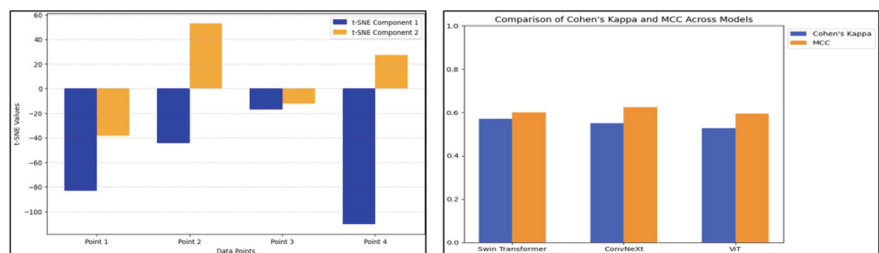


Fig. 6 T-SNE plot and Cohen’s kappa/MCC plot

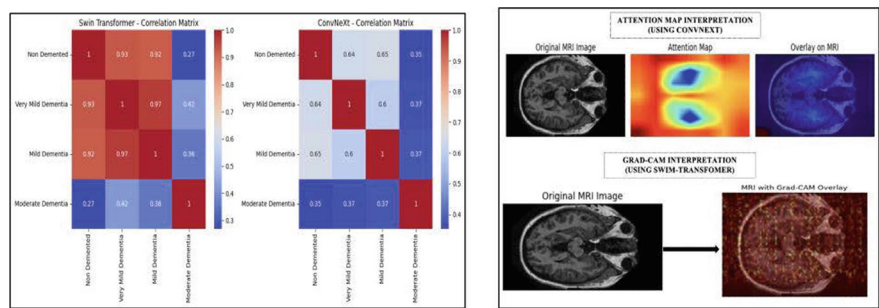


Fig. 7 Correlation of dementia types and Interpretability using GRAD-CAM/attention maps

models—Swin Transformer, ConvNeXt, and ViT—demonstrates interesting trends in performance (see Fig. 6). The correlation analysis between dementia classes shows that Swin Transformer shows stronger correlations between Non-Demented, Very Mild Dementia, and Mild Dementia (0.92–0.97), (see Fig. 7) reflecting its capacity to capture progressive stages of cognitive decline. Nevertheless, its correlation with Moderate Dementia is still weak (0.26–0.41), implying that advanced stages are hard to differentiate.

On the other hand, ConvNeXt has more balanced correlations in all stages (0.35–0.65), reflecting more consistent feature extraction but with reduced early-stage discrimination. This disparity indicates that though Swin Transformer is better at detecting incremental disease progression, ConvNeXt is more generalized class separation, thus more robust for general classification. Grad-CAM and attention map visualizations were utilized to promote interpretability for Swin Transformer and ConvNeXt models, respectively, by emphasizing the most significant regions within MRI scans contributing to classification outcomes of the model (see Fig. 7). Likewise, attention maps from ConvNeXt provide information on how the model allocates its attention across various brain regions, assisting in feature attribution.

5 Limitation and Future Scope

Although the suggested method improves dementia classification, some limitations remain. The dataset, though large, does not incorporate multimodal data fusion (e.g., PET scans, clinical notes), which would further enhance diagnostic accuracy. Moreover, model generalization can still be influenced by dataset biases, and its evaluation is required on heterogeneous real-world MRI datasets. The equal-weight voting committee hypothesizes identical model reliability, yet performance-weighting may support more effective decision-making. In future research, self-supervised learning and domain adaptation methods will be investigated to better robustify features across multiple imaging protocols. Moreover, augmenting interpretability methods, e.g., SHAP and LIME, can enhance model understanding of predictions and aid in increased clinical uptake.

6 Conclusion

This work presented a hybrid ensemble approach combining CNNs and transformers for MRI-based dementia classification. Swin Transformer, ConvNeXt, and ViT used in ensemble voting significantly improved classification performance. Swin Transformer performed the highest generalization while ensemble facilitated balanced decision-making. The integration of Grad-CAM and attention maps enhanced model explainability, solving a major challenge in AI adoption in clinical settings. Comparative assessments emphasized the better performance of transformer-based models than conventional CNNs in detecting dementia-related features. Although limitations like dataset biases and equally weighted voting are still present, the proposed method offers a good platform for future research on AI-based neuroimaging diagnosis.

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An IoT-Based Framework for Adaptive Fitness Recommendations Using Clustering and Ensemble Regression



Naween Kumar, Ayushman Pranav, Ankit Dubey, Rajesh Kumar Modi, Firoz Khan, and Ahmad Alkhayyat

Abstract Personalizing fitness recommendation systems is important to engage users of wearables and maximize their health outcomes. This research provides a new framework for the Internet of Things (IoT), which uses unsupervised clustering, collaborative filtering, and ensemble regression to make adaptive fitness recommendations. The methodology begins by grouping tracker activities and generating characteristics such as activity intensity and calorie expenditure per step. K-Means clustering helps to categorize users into five categories according to their activity level, from inactive to very active, for tailor-made recommendations. The lesson plans are adjusted according to the similarity in behavior using a collaborative filtering algorithm based on Nearest Neighbors (KNN) similarity. The random forest regression achieves near-perfect accuracy in predicting calorie burn ($R^2 = 0.9997$), outperforming other regression models. In addition, the system simulates a real-time adjustable recommendation, recommending that users change the intensity of their exercise based on a predicted versus actual calorie deficit. Visual aids and the

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accuracy of the regression confirm the separation of clusters with different activity patterns. In contrast, qualitative analysis enhances the model's ability to identify and distinguish different activity patterns. The results demonstrate the framework's effectiveness in synthesizing user segmentation, predictive analytics, and dynamic recommendation and provide a solid basis for future innovations in real-time train adaptation.

Keywords Personalized fitness recommendations · Activity pattern clustering · Collaborative filtering · Random forest regressor · Ensemble learning

1 Introduction

The rapid development of Internet of Things (IoT)-based wearable technologies opens new possibilities for individuals to monitor their health and physical activity levels. The emergence of IoT-enabled devices has resulted in extensive data production, giving users real-time access to workout plans, biological reactions, and overall activity measurements. The IoT-enabled fitness model transforms into a connected solution through the integration of precise sensor-generated data that develops lifestyle-based net benefit systems. The ability of IoT sensors and wearables enables the delivery of personalized and real-time recommendations that lead to better fitness results [1]. Adaptive fitness based on intelligent systems is one of the newer concepts that arose from machine learning (ML) technologies. These systems aim to use advanced analytics on data in a way that patterns can be discovered within various datasets so that crafted exercise and health guidance can be formulated [2]. This shift is expected to not only boost user experience but also enable users to adopt a better attitude towards health, one that looks forward to surging in activity performance and behavior when guidance is given instantaneously [3]. The blend of these intelligent systems with wearable gadgets will constitute a reformed perspective towards health care as it borders personal fitness, one that enables real-time exploitation of individual behaviors regarding health and fitness, adapting the care towards the specifics of that individual's journey [4, 5]. The contribution of this work is as follows:

1. This paper introduces a very innovative hybrid machine learning (ML) approach that integrates the unsupervised clustering (K-Means), collaborative filtering (Nearest-Neighbors), and ensemble regression (Random Forest Regressor) to deliver adaptive fitness recommendations tailored to individual user profiles.
2. By leveraging the K-Means clustering, the method very effectively categorizes the users into five distinct activity levels, ranging from sedentary to highly active, enabling more targeted and effective workout recommendations.
3. The suggested work demonstrates that the Random Forest Regressor achieves near-perfect accuracy ($R^2 = 0.9997$) in predicting calorie expenditure, outperforming other regression models and enhancing the reliability of fitness guidance.

4. The framework enables dynamic workout modifications by analyzing predicted vs. actual calorie burn, ensuring users receive real-time recommendations to optimize their fitness goals based on wearable device data.

2 Related Works

The increasing adoption of many wearable fitness technologies and the mobile health applications has revolutionized personalized health monitoring and fit- ness tracking. Research in this domain has assessed various machine learning (ML) techniques, recommendation systems, and the sensor-based data analytics to enhance user engagement and improve health related outcomes. However, existing studies exhibit limitations such as a lack of real-time adaptation, challenges in handling multimodal data, and the difficulties in achieving personalized recommendations on a scale [1]. This section reviews key literature on fitness tracking and recommendation models, highlighting their strengths and shortcomings. By identifying existing research gaps, this study aims to propose an advanced, adaptive framework that integrates dynamic clustering, ensemble learning, and real-time feedback to enhance fitness recommendations. Table 1 briefly explains the summary of literature and research gaps.

Table 1 Summary of literature and research gaps

Reference	Research gap
[1] Personalized fitness recommendation using sequential data	Needs dynamic integration of multimodal data and personalized recommendations
[2] Activity recognition with user-annotated acceleration data	Scaling models for diverse populations and contextual data
[3] K-means clustering algorithm for data partitioning	Requires more robust clustering methods for variable user data
[4] Matrix factorization in recommender systems	Hybrid solutions needed for new or infrequent users
[5] Random forests for high predictive accuracy	Interpretability and efficiency issues for large real-time apps
[6] Smartphone-based system for health feedback	Needs integration of broader health insights like stress and mental health
[7] Nearest neighbor pattern classification	High computational cost and feature sensitivity in fitness systems
[8] Review of ensemble methods in machine learning	Underexplored for sequential fitness data requiring continuous input
[9] Framework for adaptive workout regimens	Requires real-time adaptation and multimodal data integration
[10] Review of wearables for real-time physiological tracking	Validating device accuracy across diverse populations

This study improves fitness applications by integrating dynamic adaptation and real-time feedback, overcoming the limitations of static systems that do not adapt to changing user needs [1]. Using K-means clustering, it partitions users into activity profiles and solves adaptive clustering problems for variable data [3]. This framework combines collaborative filtering, supervised clustering, and ensemble regression to create a contextually aware solution for [4]. The high precision of the Random Forest Regressor highlights the value of ensemble methods in processing sensor data [8], while advanced machine learning strategies meet the need for evolving frameworks [9]. Feature engineering with metrics like activity intensity and calories per step increases the effectiveness of the [7] classification algorithm. The study also highlights the need for explainable AI in health care [11] and advocates data-driven, learner-centered approaches to cognition over fixed goals [12]. Despite accuracy problems, the Fitbit data continues to be of great value to research on health and wellbeing [13].

3 Methodology

The methodology implemented in the code can be described in several sequential stages, each addressing a key aspect of the personalized fitness recommendation system.

Figure 1 shows the flowchart of an intelligent fitness personalization system that combines the entire cycle of data preparation, modeling, evaluation, and adaptation. The process begins with data loading and preprocessing activities such as feature engineering (Active Distance, Activity Intensity, Calories Per Step) and normalization to ensure meaningful inputs. The Random Forest Regression model for calorie burn prediction, K-Means user segmentation, and Nearest Neighbors collaborative filtering workout recommenders are implemented in the modeling phase. In the evaluation phase, the user is visually presented with adaptive recommendations, and the system is adjusted in real time depending on the user's actual calorie burn in comparison to predictive analytics. Ultimately, the conclusion provides the user with insights into the system, and future work further highlights why this forms a robust hybrid machine learning framework for personalized fitness guidance.

3.1 *Fitness Tracking Data Pipeline and Feature Engineering Process*

The pipeline starts with a raw data set of daily activities from fitness trackers, including metrics such as total steps, distance, and calories. The data is loaded with the Pandas method and then verified. Missing values are handled to prevent statistical bias. Table 2 shows a sample of data with different activity records (for user ID:

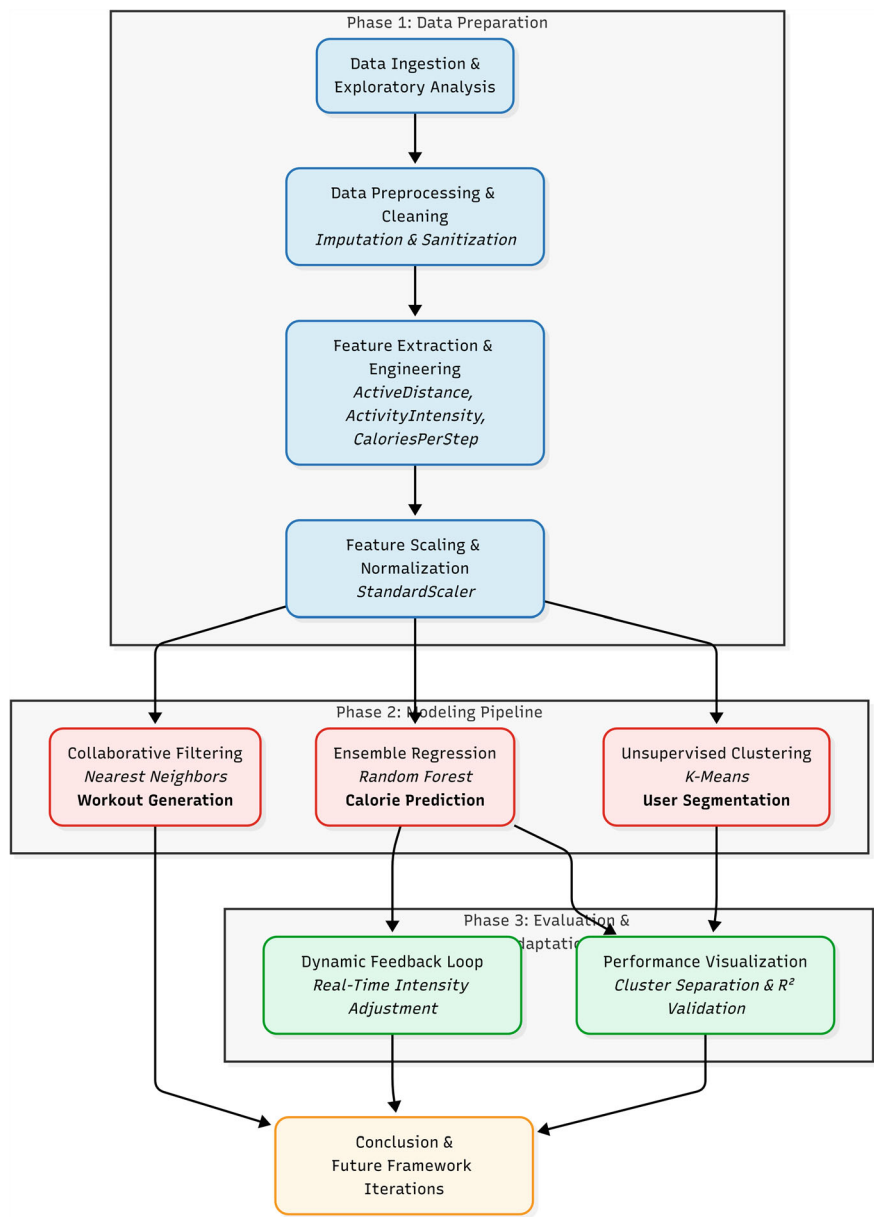


Fig. 1 Flowchart for a personalized fitness recommendation system

Table 2 Activity data summary

Id	Activity date	Total steps	Total distance (km)	Very active distance (km)	Moderately active distance (km)	Light active distance (km)	Calories
1503960366	3/25/2016	11,004	7.11	2.57	0.46	4.07	1819
1503960366	3/26/2016	17,609	11.55	6.92	0.73	3.91	2154
1503960366	3/27/2016	12,736	8.53	4.66	0.16	3.71	1944
1503960366	3/28/2016	13,231	8.93	3.19	0.79	4.95	1932
1503960366	3/29/2016	12,041	7.85	2.16	1.09	4.61	1886

1503960366) and activity intensity subcategories: very active distance, moderate active distance, light active distance, and sedentary distance, along with calories burned. This structured data set allows advanced analyses such as profiling and predictive modeling. The key features of the framework are: (a) ActiveDistance, the sum of distances between all levels of activity; (b) ActivityIntensity, the weighted sum of distances between higher levels of activity; and (c) Calories per step, calculated by dividing the total by the total steps. A standard smoothing function normalizes these features and the original metrics to produce a consistent contribution during the analysis [13].

3.2 Methodology Overview: Clustering, Personalized Recommendations, Calorie Prediction, and Real-Time Workout Adjustment

The normalized data set is clustered using K-means (5 clusters) to identify different activity patterns. Each user is assigned a cluster tag, and a visualization of Total Steps versus Calories is created. The K-means function aims to minimize squared Euclidean distances between data points and their centroids. Collaborative filtering with the algorithm Neighbors is used to recommend personalized workouts. The algorithm compares user activity profiles based on Euclidean distances and recommends optimal training sessions. The random forest regressor is trained on normalized data to predict calorie burn using performance metrics such as R^2 and MSE. The model combines predictions from several decision trees to provide reliable calorie expenditure estimates. An adaptive recommendation system adjusts the intensity of user’s exercise in real time, predicts our desired calorie burn, and provides advice on the necessary intensity changes. Visualizations such as scatter plots of predicted versus actual calorie burn confirm the regression model’s performance and highlight improvement areas.

3.3 Machine Learning Model

A supervised learning approach is employed to predict calorie burn based on activity data. The dataset is split into training and testing sets. A Random Forest Regressor, configured with 100 estimators, is trained on the normalized features. Performance metrics such as R^2 score and Mean Squared Error (MSE) are computed to assess the model's accuracy. In this implementation, an R^2 score close to 1 indicates a strong predictive performance. Random Forest Regression combines multiple decision trees to provide a robust prediction by averaging the outputs of individual trees. The prediction function is given by: $f(x) = \frac{1}{N} \left(\sum_{i=1}^N f_i(x) \right)$, where N is the total number of trees in the forest. $f_i(x)$ is the prediction of the i th decision tree for input x . This ensemble approach mitigates overfitting and improves predictive accuracy by leveraging the wisdom of multiple models. The system being designed will be able to estimate calorie expenditure based on user interaction, which is crucial for adjusting workout intensities effectively.

4 Results and Analysis

This section focuses on the outcomes of the personalized fitness recommendation algorithm, highlighting descriptive and exploratory analyses aimed at interpreting clustering scores, accuracy of predictions, and the effectiveness of personalization in recommendations. The simulation is performed on a Jupyter Notebook (Python 3.10) running on an Ubuntu 20.04, employing efficient libraries such as pandas, numpy, xgboost, scikit-learn, catboost, and others. The execution of K-Means Clustering is done with approximately 5 clusters, K-means initialization, 300 iterations, and a random state set to 42. For predicting caloric expenditure, training was performed with an 80/20 division and accompanied by a fivefold cross-validation approach, while results were evaluated with R-squared scoring. Additionally, other regression models were implemented, including Random Forest Regression, boosted Gradient trees, XGBoost, Catboost, and multiple others, with Linear regression and many other tunings of hyperparameters. The development environment is Visual Studio Code, and the system has a powerful Intel i9 CPU, 32 GB RAM, NVIDIA RTX 3080 Ti, and 3 TB SSD.

K-Means' normalized metrics clustering process divides users into five clusters that reveal different patterns in their daily movements and calorie use. Cluster 0 (purple) contains inactive users with low steps and variable calorie burn, while cluster 1 (blue) contains very active users with high steps and calorie burn. Cluster 2 (teal) contains users with moderate activity and low calorie burning, and cluster 3 (green) contains users with moderate activity and low calorie burning. Transitional cluster 4 (yellow) consists of users with a higher number of steps and average to higher calorie expenditure. The cluster shows a clear positive relationship between steps and calories, and makes a successful distinction between levels of activity, allowing

researchers to develop personalized fitness treatments tailored to individual profiles (Fig. 2).

Furthermore, five regression models were evaluated to predict calorie burn based on activity metrics. The scatter plot shown in Fig. 3 compares actual vs. predicted calorie burn, with points clustering tightly around the diagonal, indicating a near one-to-one match. The Random Forest Regressor demonstrates high accuracy across a broad range (roughly 500–4500 cal), with few outliers. Such a strong linear alignment confirms the model’s robustness in predicting calorie expenditure, reinforcing its potential for reliable, data-driven fitness recommendations (Table 3).

Table 3 presents the R^2 scores of five regression models, with Linear Regression achieving a solid baseline of 0.9882. However, the Random Forest Regressor leads with an almost perfect score of 0.9997, followed by Gradient Boosting (0.9989), XGBoost, and CatBoost (both > 0.99). Ensemble methods, especially Random Forest, excel at handling outliers and feature complexities, showcasing their strength in accurate, data-driven fitness predictions.

Key Findings: (i) Random Forest ($R^2 = 0.9997$) outperforms others, showing near-perfect accuracy. (ii) Ensemble methods like Gradient Boosting and XGBoost excel in non-linear modeling. (iii) Linear Regression, while effective, highlights the non-linear nature of calorie burn prediction. A collaborative filtering model using Nearest Neighbors tailors workout recommendations based on similar user profiles. For example, if a user logs 10,000 steps and burns 300 cal, the system suggests increased workout intensity to optimize caloric burn, demonstrating real-time adaptation.

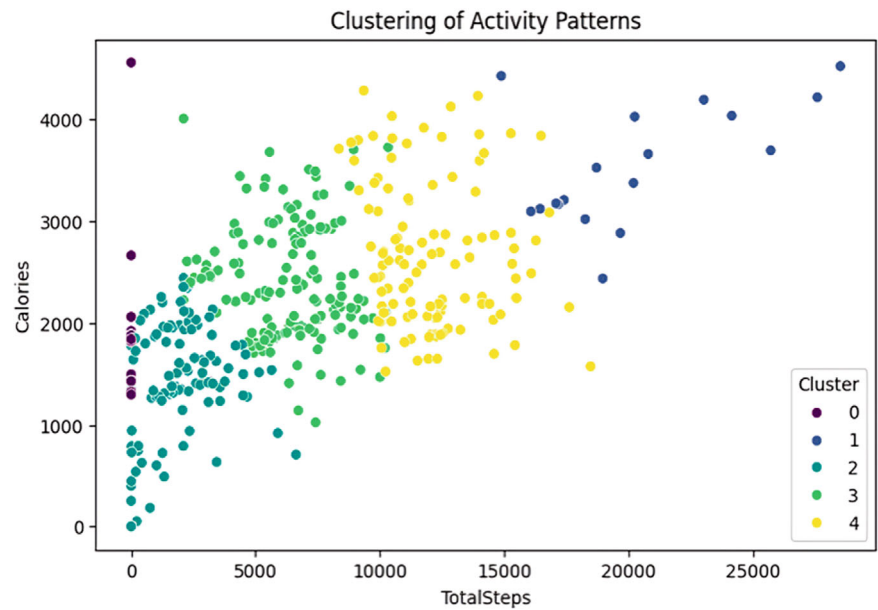


Fig. 2 Cluster visualization of urbanization patterns

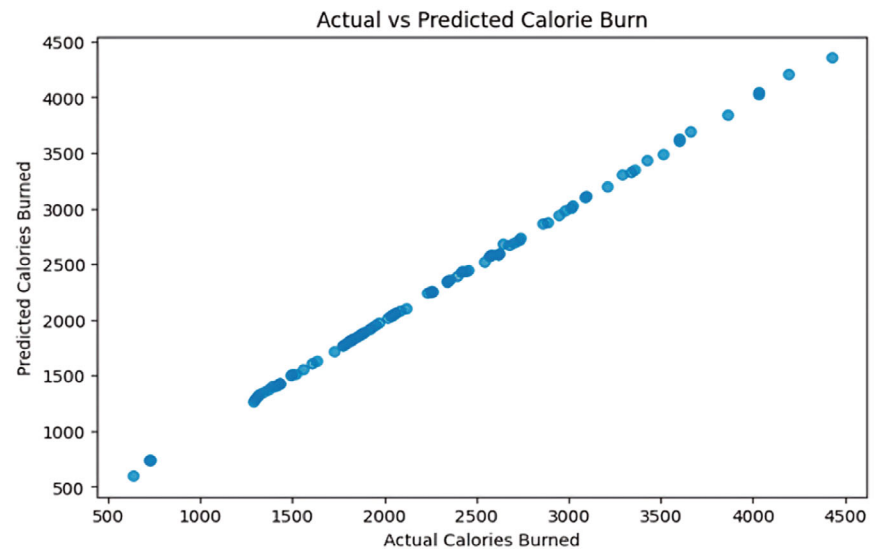


Fig. 3 Actual versus predicted calorie burn

Table 3 R^2 scores for different models

Model	R^2 score
Linear regression	0.9882
Random forest regressor	0.9997
Gradient boosting regressor	0.9989
XGBoost regressor	0.9976
CatBoost regressor	0.9965

Qualitative Analysis: (i) *Clustering Insights:* Clusters effectively separate users into distinct activity groups, confirming the accuracy of the model. (ii) *Calorie Prediction Insights:* Collaborative filtering uses regression model patterns to suggest personalized workout modifications, enhancing user satisfaction. (iii) *Personalized Recommendation Insights:* Ensemble models (especially Random Forest) highlight the non-linear relationship between calories burned and activity, emphasizing the need for complex models.

5 Conclusion and Future Work

The suggested work shows a hybrid method to machine learning involving clustering, regression, and collaborative filtering that improves personalized fitness planning remarkably. This IoT-enabled method has proven effective for user clustering,

exercise recommendations, and calorie burned estimation through its data acquisition sequence, feature engineering, and modeling procedures. The Random Forest Regressor delivered near-perfect accuracy levels as its clustering results produced valuable knowledge about activity patterns across various groups. The research provides multiple favorable directions for upcoming studies that would benefit from in-session real-time IoT wearable device data activations during workout sessions, but also suggests using deep learning algorithms to analyze complex relationships or integrating additional health metrics, including sleep patterns and heart rate variability data, for holistic wellness assessment. The framework needs improved user feedback tools and tests with expanded, diverse datasets to increase its usability across various applications.

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IoT-Driven Urbanization and Economic Forecasting with Machine Learning



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and Ahmad Alkhayyat

Abstract Economic development and urbanization depend highly on traditional metrics that provide little geographic specificity and delayed reporting times that hinder continuous monitoring. This paper proposes a method to automate the prediction of economic conditions by machine learning models, together with the Internet of Things (IoT) sensors and satellite observations of night lights. Combining a refined data pipeline and its stages, including data transformation, with the generation of features and clustering, followed by predictive modeling, provides robust analysis. New indicators such as light density and light per person support the spatial and temporal analysis. The K-means algorithm distinguishes five clusters ranging from entirely rural to completely urban. Regression models, including linear regression with random forest and gradient enhancement, and XGBoost and CatBoost yield a r^2 of 95.21%. The suggested work provides evidence of the increased urbanization and changes in social structures and political developments. The method of analysis speeds up and improves the accuracy of forecasting urbanization patterns in conjunction with economic developments, thus providing valuable information for urban planners and policymakers.

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Keywords Machine learning · Clustering · Regression analysis · Feature engineering · Anomaly detection · Predictive modeling

1 Introduction

Policies for urban development draw their influence from urban requirements along with existing development needs for sustainable progression and social transformation. Real-time data acquisition is now critical to monitor economic growth and construction activities because migration trends are changing. Due to the implementation of IoT technology, modern urban areas gain real-time sensor data and satellite nighttime luminous intensity alongside other sources. IRT deployments improve city development surveillance and business activity tracking, facilitating enhanced strategic choices for urban development professionals. The Nighttime Lights (NTL) dataset supplies complete non-suggested estimates of human movement along with urban progress and productivity measurements that operate on detailed administrative locations. At the same time, traditional economic and job metrics tend to be broad and time-delayed. The development of remote sensing technology coupled with machine learning and AI enables new pathways for NTL data examination, yielding beneficial assessments for economic predictions, infrastructure improvements, and industrial sector development [1–3]. Through joint use of NTL data and machine learning models, the public sector gains reliable economic intelligence that enhances its capabilities to develop urban resources and formulate better policies. The analysis of nighttime light data supplements economic forecasting while helping to find and monitor areas of unequal infrastructure and providing improved poverty report methods [4, 5]. Recent advancements in deep learning methods, especially convolutional neural networks (CNNs), have made it possible to analyze satellite imagery to identify economic activity centers with underdeveloped regions. Through alignment with census records and financial transaction data, NTL data provides policymakers with improved economic forecasting capabilities to develop sustainable cities with fair development opportunities [6, 7]. The objectives of this paper include:

1. To propose new measurements, such as light density (total light intensity divided by the number of illuminated cells) and light per capita (how the total light intensity is normalized by the maximum light intensity + 1), that skillfully quantify the urban system's spatial and temporal features.
2. To establish a robust, end-to-end analytical pipeline that includes rigorous data cleaning, normalization, and calibration; dimensionality reduction using PCA; and segmentation of urban regions through K-Means clustering, facilitating a refined classification from rural to metropolitan areas.

3. To exploit multiple regression models (including Gradient Boosting, Random Forest, XGBoost, and CatBoost) to predict economic activity with an R^2 score up to 95.21%, complemented by an anomaly detection framework that identifies significant shifts in light intensity ($\geq 50\%$ change), indicating rapid urban changes or disruptions.

2 Related Works

Numerous studies have explored the potential of nighttime light data as a proxy for economic activity, urbanization, and infrastructure growth. Early research established correlations between light intensity and regional GDP, while recent advancements integrate machine learning for improved predictive accuracy. However, challenges such as spatial calibration, dynamic modeling, and integration with socio-economic data remain. This section reviews key contributions in the field, highlighting research gaps that our study aims to address in the below provided Table 1.

Table 1 Summary of literature and research gaps

Paper ID	Research gap
[1] Linking nighttime lights with GDP in data-scarce regions	Needs improved spatial calibration for finer economic insights
[2] Proposing adaptive ML to monitor dynamic urban changes	Existing models fail to capture rapid urban evolution
[3] Assessing the VIIRS dataset and comparing it with older systems	Requires better calibration and socio-economic data integration
[4] Using satellite lights to map global economic activity and disparities	Standardizing measurements across regions with different infrastructures
[5] Using Landsat imagery for long-term bare ground detection	Temporal and detection resolution limitations hinder accuracy
[6] Classifying coastal waters using optical remote sensing techniques	Enhanced algorithms and calibration are needed for better water discrimination
[7] Correlating nighttime lights with U.S. county-level economic activity	Needs refined calibration for accurate local economic fluctuations
[8] Proposing dynamic network learning to quantify economic growth	Static models miss evolving economic patterns; more adaptive approaches are needed
[9] Analyzing light variations to understand U.S. urban growth dynamics	Needs more granular analysis of short-term light fluctuations
[10] Synthesizing remote sensing methods for economic development assessment	Bridging the gap between remote sensing indices and scalable metrics
[11] Examining human mobility and satellite data for socio-economic analysis	Unified modeling approaches needed for better integration of data
[12] Exploring deep learning for GDP prediction across countries	Models need more robustness and generalizability across regions

The investigation presented in Table 1 uses satellite nighttime data together with machine learning techniques for economic prediction while drawing from Henderson et al. [1] and Ghosh et al. [4]. Our study enhances previous models of nighttime light intensity to economic activity by implementing machine learning data pipeline techniques and achieving a 95.21% R^2 score. This process includes data preprocessing, feature engineering of light density and light per capita features, and PCA clustering followed by regression models. The research builds from Gibson and Boe-Gibson's [7] county-level evaluation and Dong et al.'s [8] dynamic urban transition studies. This model differs from static frameworks [2, 9] by refining satellite data quality while including regression capabilities and anomaly detection capabilities for more comprehensive evaluation. The research develops existing approaches presented by Pan et al. [10] and Zhang et al. [12] by incorporating advanced features and clustering techniques. The DMSP-NL dataset (1992–2013) offers complete nighttime light intensity measurements of India, which aids in tracking electrification status while detecting economic performance changes and urbanization patterns and disaster events [13].

3 Methodology

This work primary focus is on the use of machine learning models to analyze economic activity derived from urbanization changes in India using satellite-detected nighttime lights' intensity as an input feature. The process is organized in a structured pipeline: data preprocessing, feature engineering, clustering, predictive modeling, and anomaly detection. Flowchart shown in Fig. 1 shows a simplified workflow which uses satellite captured night light images to determine the level of urbanization and predict economic development. It starts with an intensive data preprocessing, that is, cleansing, normalizing, and merging different datasets to make them usable. There is feature engineering which includes both spatial features (for example, marked area and light pollution) and time features (for example, seasonal changes), making the data more complex. A set of clustering algorithms is used to group urban areas into clusters based on their light emission patterns which then were further used in several regression models such as Random Forest, Gradient Boosting, XGBoost, and CatBoost. The pipeline finishes with sophisticated pattern visualization and anomaly detection techniques designed to highlight any irregularities or changes within the data, as well as standard forecasting procedures. This could enhance the overall accuracy of predictions while providing additional support to urban planners and decision makers.

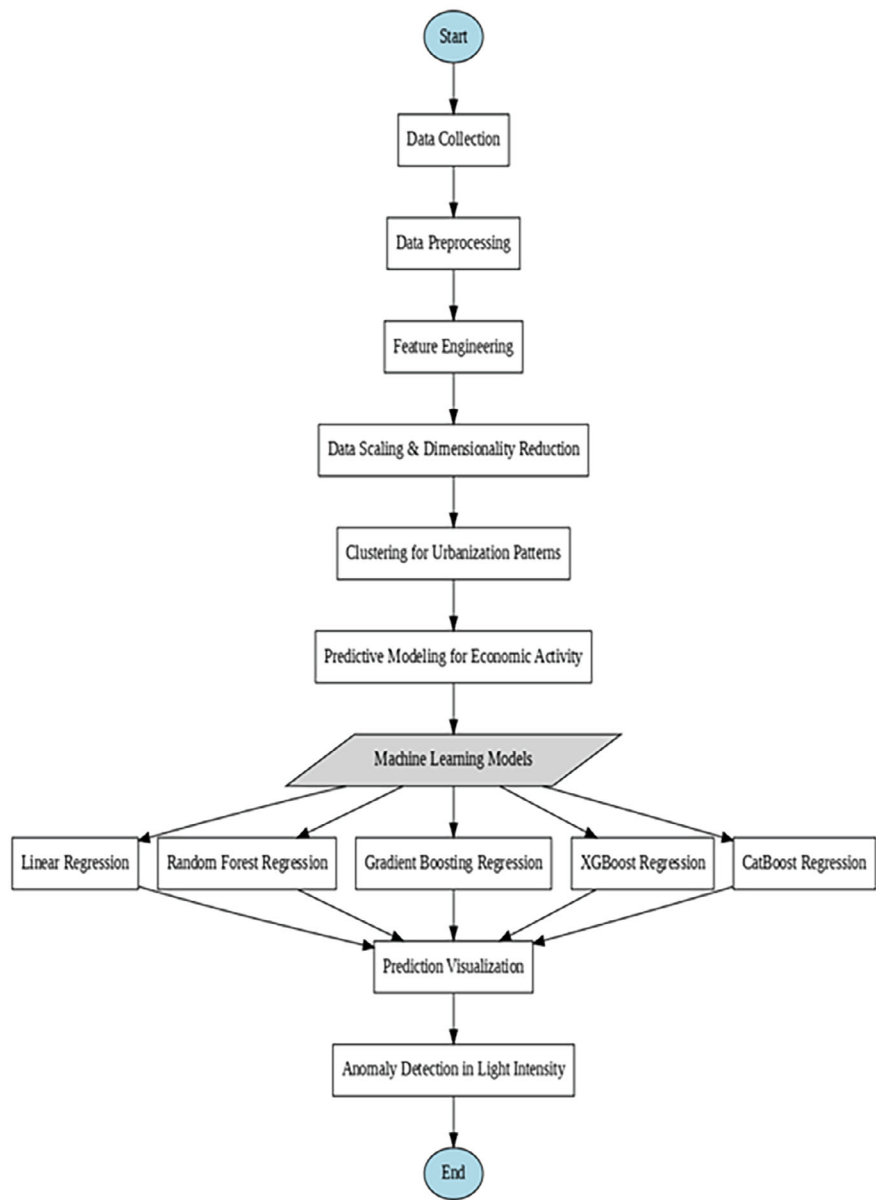


Fig. 1 ML pipeline for urbanization pattern analysis and economic activity prediction using nighttime light intensity data

3.1 The DMSP Night Lights Data (1992–2013) for India

The DMSP Night Lights Data (1992–2013) is an acquired satellite-derived measurement system of nighttime light intensity across India through DMSP-OLS satellites. Researchers can utilize this data to examine the status of electrification and economic development, growth and infrastructure construction, and disaster consequences. The DMSP Night Lights Data lets researchers connect economic growth patterns to GDP, electrification statistics, urban sprawl effects, and light changes after disasters. The database operated by Dev Data Lab has Hari Krishna Vetharenian as its host dataset, which holds an Attribution-NonCommercial-ShareAlike 4.0 International license and provides free CSV format access. The dataset documents India's urban development stability through time since it captures the situation every year from 1992 through 2013.

3.2 Preprocessing and Integration of DMSP Night Lights Data

Multiple data preprocessing steps ensure data usability because null values are deleted, and all columns receive consistent treatment between numeric and non-numeric types. Light intensity values are standardized for year-to-year comparison purposes while assessing saturated regions confirms sensor overestimation potential. The accuracy of the analysis depends on using `dmsp_total_light_cal` and `dmsp_mean_light_cal` values and conducting temporal consistency tests from 1992 to 2013. Long-term studies of urbanization and socioeconomics can benefit from including the more recent VIIRS Night Lights sensor data (2012–2021) in the DMSP Night Lights Data collection due to enhanced coverage levels, which improve assessments of infrastructure and economic planning applications as well as disaster relief operations for India.

3.3 Feature Engineering, Clustering, and Predictive Modeling

Two new engineered features measure urbanization intensity through Light Density calculations by dividing total nighttime light by the number of illuminated cells and by calculating Light Per Capita by dividing total light by the maximum light intensity plus one to show relative brightness. The developed features provide detailed information about how economic development and urbanization patterns progress. `StandardScaler` standardizes the dataset until `PCA` implements principal component

analysis to transform the data into two components for improved clustering visualization. The K-means clustering algorithm groups urbanization patterns into five clusters while sorting these patterns into different color-coded sections in a scatter plot. Building a machine learning model uses calibrated economic activity (dmsp_total_light_cal) as the dependent variable. At the same time, frontal light increases serve as the independent variables (light intensity metrics and light density with light per capita values). The data splits into training (70%) and testing (30%) sections through the application of train_test_split().

3.4 Machine Learning Models

Linear regression establishes a linear relationship between light intensity and economic activity, expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon,$$

with the objective to minimize the mean squared error (MSE), given by:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2.$$

Random Forest Regression is an ensemble of decision trees that improves prediction accuracy by averaging predictions, with:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T \hat{y}^{(t)},$$

and it is robust against overfitting through ensemble averaging.

Gradient Boosting Regression sequentially builds decision trees where each new tree corrects errors of the previous one. The predicted value $\hat{y}^{(m)}$ is:

$$\hat{y}^{(m)} = \hat{y}^{(m-1)} + \eta \cdot h_m(x),$$

where η is the learning rate, and $h_m(x)$ is the tree fitted on the residuals, typically minimizing MSE via gradient descent.

XGBoost Regression is an optimized version of gradient boosting that includes regularization. The loss function $\mathcal{L}(\phi)$ is:

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k),$$

where $\mathcal{L}$ is the loss function (e.g., MSE), $l(y_i, \hat{y}_i)$ is the loss for each data point, and $\Omega(f_k)$ is the regularization term for tree f_k . CatBoost Regression is a gradient boosting algorithm that efficiently handles categorical features and reduces prediction shift. Its objective is to minimize a loss function (typically MSE) while applying techniques to improve generalization.

To analyze the accuracy of the predictions, a scatter plot is generated, comparing the actual vs. predicted economic activity using the Random Forest model.

3.5 Anomaly Detection in Light Intensity

To identify sudden changes in nighttime light intensity, percentage change in total light intensity is computed. Any district with a 50% or higher change in light intensity was flagged as an anomaly. These anomalies were printed, listing the district ID, year, and magnitude of change. The suggested methodology integrates data-driven approaches to explore urbanization and economic development using nighttime satellite imagery. Clustering methods reveal spatial urban patterns, while predictive models provide insights into economic activity. Anomaly detection further aids in identifying abrupt changes in development trends, potentially indicating socio-economic shifts, policy changes, or infrastructural development.

4 Results and Analysis

The analysis comprises three main components: clustering visualization, economic activity prediction, and anomaly detection. The clustering visualization using KMeans clustering identified five distinct clusters based on urbanization and economic activity patterns. The clusters are classified as follows:

Figure 2 represents a scatter plot that visualizes urbanization patterns using PCA, with five clusters spreading along two principal components. Cluster 0 (dark purple) near the origin represents less-developed areas, while Clusters 1–4 progressively depict more urbanized regions. PC1 captures primary urban variations, and PC2 reflects secondary differences, showing how urbanization diverges into multiple forms. The dense central region suggests common urban traits, while outliers indicate unique or highly developed areas. This clustering provides insights for urban planning, economic analysis, highlighting developmental trends and variations in urban growth. Cluster 0 (Purple) represents rural or underdeveloped regions with low light intensity, while Cluster 1 (Blue) indicates semi-urban areas with moderate light intensity. Cluster 2 (Green) reflects regions in transition to urbanization, showing intermediate development. Cluster 3 (Yellow) highlights well-developed urban areas with high light density, and Cluster 4 (Teal) corresponds to highly developed metropolitan areas with the highest light density. The well-separated clusters suggest that the model effectively captures differences in urbanization patterns. The gradual spread of data

points from low to high urbanization demonstrates the progressive nature of urban development.

The scatter plot shown in Fig. 3 compares actual vs. predicted economic activity, showing how well a predictive model aligns with real-world values. The points cluster along an upward diagonal trend, indicating a strong positive correlation between actual and predicted values. A near-perfect model would have all points on the diagonal line $y = x$, but deviations suggest some prediction errors. The spread of points widens at higher values, implying increasing variance and potential overestimation or underestimation in high-value cases. Some extreme outliers indicate occasional large prediction errors. Overall, the model performs well for lower and mid-range values but may require refinement for high economic activity levels, possibly through improved feature selection or model tuning.

Table 2 reports that Gradient Boosting achieves the highest R^2 score of 95.21%, followed by Random Forest (95.08%) and CatBoost (94.88%), with XG-Boost at 94.73%. The scatter plot confirms a strong positive correlation between actual and predicted values, though some outliers remain. Anomaly detection revealed 11,582 significant anomalies, indicating urban expansion, infrastructure upgrades, and economic shifts. For instance, District 534 (1994) observed a 58.05% increase, likely due to urbanization, while District 536 (1994) dropped by 54.51%, possibly from economic decline. Clustering results highlight the transition from rural to urban areas, reflecting light intensity and density. Gradient Boosting excelled in capturing complex economic trends, while anomaly detection revealed development activities and economic downturns.

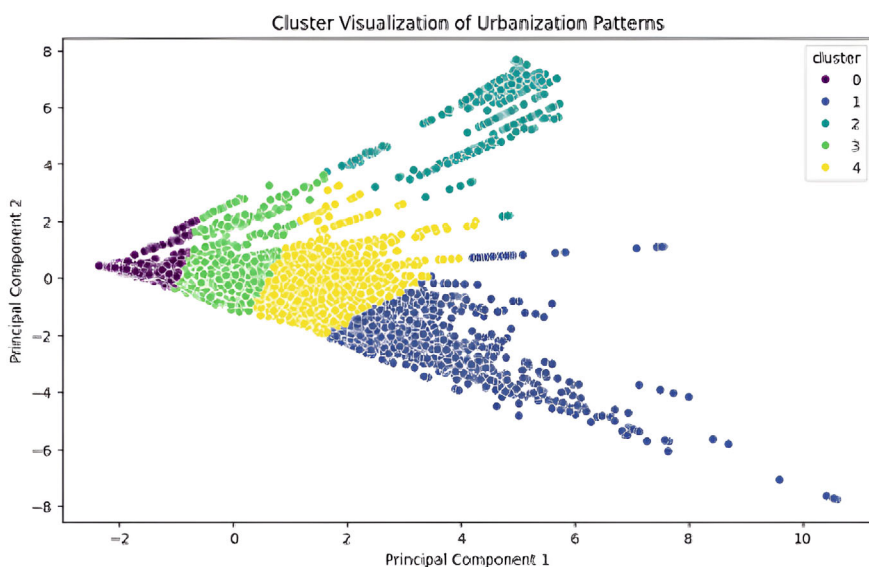


Fig. 2 Cluster visualization of urbanization patterns

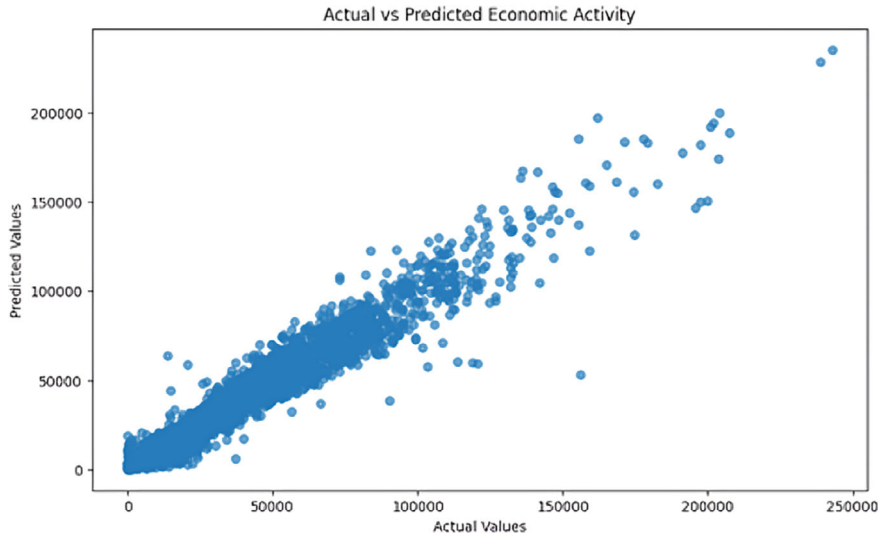


Fig. 3 Actual versus predicted economic activity

Table 2 Model R^2 scores

Model	R^2 score
Linear regression	0.9323
Random forest	0.9508
Gradient boosting	0.9521
XGBoost	0.9473
CatBoost	0.9488

5 Conclusion and Future Work

The suggested work analyses DMSP night-light data from the Internet of Things (IoTs) enabled satellites and reveals five different patterns of urbanization, ranging from rural to metropolitan, as well as precise predictions of economic activity using ensemble methods such as Gradient Boost, Random Forest, and CatBoost, which achieved an R^2 of 95.21%. Verification of predicted versus actual data confirmed strong positive correlations, with anomaly detection pointing to more than 11,582 cases of variations in the intensity of light. These anomalies indicate rapid urbanization, increased infrastructure development, or reduced economic activity, reflecting the complex socio-economic dynamics of urbanization. In the future, the suggested work can refine these models by fine-tuning them and exploring deep-learning techniques such as RNNs and LSTMs to improve time-series analysis.

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Hybrid Neural Network for Stock Prediction: Integrating Multi-source Data with Particle Swarm Optimization



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Abstract This study presents a novel approach to predicting stock prices by integrating a hybrid deep learning model with a comprehensive analysis of technical data. The methodology utilises Python and its associated modules to systematically acquire, process systematically, and model financial time series data. Historical stock data for blue-chip stocks from the Bombay Stock Exchange (BSE) is obtained, and various technical indicators, such as volume, volatility, trend, and momentum, are calculated. A hybrid deep learning architecture employing long-short-term memory (LSTMS) and a multi-head attention mechanism captures complex temporal correlations in the technical indicator data. The model's predictive capability is comprehensively evaluated, and the findings illustrate the effectiveness of the proposed strategy in forecasting stock prices. The framework achieved a maximum R^2 of 0.97 and a

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minimum MAPE of 1.67 for stock LT when the test data was predicted over 200 epochs with a learning rate of 0.001. Nonetheless, the framework achieved an accuracy of 89% when trained on 5000 trading days, exceeding the performance of those trained on fewer than 5000 days.

Keywords Attention Model · Stock forecasting · Particle Swarm Optimizer · Bombay Stock Exchange · Neural network · LSTM

1 Introduction

The prediction of stock prices has been extensively researched due to its significance in investing strategies, risk management, and economic forecasting. The dynamic and nonlinear characteristics of stock markets, shaped by factors including economic conditions, political events, and investor sentiment, render precise prediction a challenging task [1]. Conventional time series analysis techniques, like Autoregressive Integrated Moving Average (ARIMA), have been utilised for forecasting stock prices. Nevertheless, these methodologies frequently fail to encapsulate the complex patterns and long-term interconnections inherent in financial time series data [2]. Deep learning methodologies have arisen as formidable instruments for modelling intricate relationships in sequential data in recent years. Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), have demonstrated efficacy in capturing temporal dependencies in stock price fluctuations. Additionally, attention mechanisms have been integrated into deep learning models to improve their capacity to concentrate on the most pertinent elements within the input sequence [3]. This research presents an innovative hybrid deep learning model that integrates GRU networks with an attention mechanism to enhance stock price prediction. The model employs extensive technical indicators as input characteristics from historical stock data. The efficacy of the suggested model is assessed by utilising historical data from the Bombay Stock Exchange. The subsequent structure of this article is outlined as follows: Sect. 2 reviews the relevant articles related to this research. Section 3 outlines the proposed framework, specifying the essential components and design challenges. Section 4 delineates the resources and procedures utilised in the research, including datasets, preprocessing, technical indicators and experimental configuration. Section 5 presents and analyses the experimental results, emphasising the principal findings and their consequences. Section 6 eventually finishes the work by summarizing contributions and delineating relevant areas for further research.

2 Literature Review

The study [4] predicts fuel prices using MHLSTM and modified Particle Swarm Optimisation (SAGPSO) algorithms. Each MHLSTM “head” analyses a different input feature through an LSTM layer, providing multi-data source learning. Concatenation and attentiveness increase forecast accuracy by combining outputs. Yahoo Finance and the U.S. Energy Information Administration provided weekly statistics from January 2000 to May 2024. Key indicators include construction jobs, crude oil and gasoline prices, and the S&P 500 index. Standardize and organize data with a five-lag window in two-step-ahead forecasting. The use of LSTM heads and attention mechanisms yields a mean squared error of 0.044025 and an R^2 of 0.911797.

The study [5] proposes a hybrid system integrating the Crow Search Algorithm (CSA) with a Long Short-Term Memory (LSTM) ensemble model to enhance stock price predictions. The model employs TA-Lib technical indicators, price fluctuation constraints of $\pm 10\%$ of closing price, and Variational Mode Decomposition (VMD) subsignal sequences from closing prices. Each dataset is preprocessed with a sliding window, normalisation, and Pearson’s correlation coefficient feature selection. Three-day-step LSTMs process changing inputs. The study achieved an MSE of 0.044025 and an R^2 of 0.911797.

The study [6] predicts stock markets using metaheuristic optimisation, an Extreme Learning Machine model, and Macro Economic Factors analysis. ELM’s weights and biases are optimised using an upgraded Guided Crow Search Algorithm (GCSA) and its variations. Iterative Transform Decomposition isolates stationary components from stock price data, and optimal ELM models predict them individually. Forecasting performance is evaluated using MSE, MAE, MAPE, and R^2 metrics. The suggested framework outperformed baseline models like ANN and standard ELM with greatest R^2 of 0.9983 for EUR/USD, 0.9986 for GBP/USD, and 0.9982 for USD/INR.

This study [7] introduces a hybrid model for multi-step-ahead stock price forecasting, combining VMD with MFRFNN. VMD breaks stock price history into intrinsic mode functions (IMFS), which parallel MFRFNNS to evaluate complex, non-linear temporal patterns. The approach addresses uncertainty and temporal interactions via fuzzy logic and recurrence. In experiments, the model achieves a 31.8% decrease in RMSE and a 9.4% increase in MAPE, surpassing previous forecasting methods.

The study [8] introduces a hybrid forecasting model that integrates LSTM and GRU networks, optimized by Particle Swarm Optimization (PSO), for predicting stock price movements on the Nifty 50 index. Essential attributes comprise 20-day, 50-day, and 200-day exponential moving averages (EMAs) alongside the closing prices of five prominent equities. The data is organised with a 20-day rolling frame. PSO optimizes the quantity of units in each layer to enhance precision. The PSO + LSTM model attained the best accuracy of 57.64%, surpassing both individual and ensemble models.

The paper [9] introduces a stock prediction model for Indian markets, incorporating volatility spillover and impacts from Asian markets. The FU-WOA algorithm outperforms benchmarks (LM-NN, GWO-NN, ARIMA), decreasing computational expenses by 15.52–41.22% and enhancing MD metrics by 8.27–50%.

The study [10] evaluates machine learning models for stock signal prediction, demonstrating that SVM-PSO achieves a higher accuracy (77.5% hit rate) compared to SVM-CS (71.4%) and neural networks (71.2%). Models optimised using PSO demonstrate superior performance in raw data processing, with a 79% hit rate. The findings highlight the stability of SVM-PSO for short-term predictions, with prospects for long-term enhancements.

This paper introduces a hybrid deep learning architecture that combines Multi-Head Attention and LSTM, optimised by PSO, for robust stock price prediction. This research uniquely uses technical indicators, a 90-day rolling window, and attention-enhanced LSTM layers to capture complicated market dynamics, building on prior efforts employing MHLSTM-SAGPSO, CSA-LSTM, and metaheuristic-tuned ELMs.

3 Proposed Methodology

This study utilized a quantitative approach to predict stock values, incorporating technical analysis and deep learning methods. The research concentrated on the historical stock data of the blue-chip stocks indexed on BSE and the Bombay Stock Exchange Sensex as an external variable. The methodology included the subsequent essential phases are presented in Fig. 1.

3.1 *Data Acquisition, Pre-processing and Feature Engineering*

The research commences with acquiring historical stock price data for blue-chip stocks and the BSE Sensex through the Yahoo Finance API. The information encompasses the period from April 1, 2020, to December 31, 2024, reflecting various market circumstances. The data comprises Open, High, Low, Close, Adjusted Close, and Volume. The BSE Sensex index historical data and individual stock data were then integrated based on the common temporal property, the 'Date' column, to ensure the alignment of comparable records across both datasets. The study utilises the Technical Analysis library to calculate various technical indicators as constructed features representing diverse elements of price movement, volume behaviour, volatility, and trend. The LSTM network analyses time series data by organising it into sequences through a sliding window methodology. Sequences extend across 90 days, utilising input features based on historical values, to predict the subsequent day's adjusted

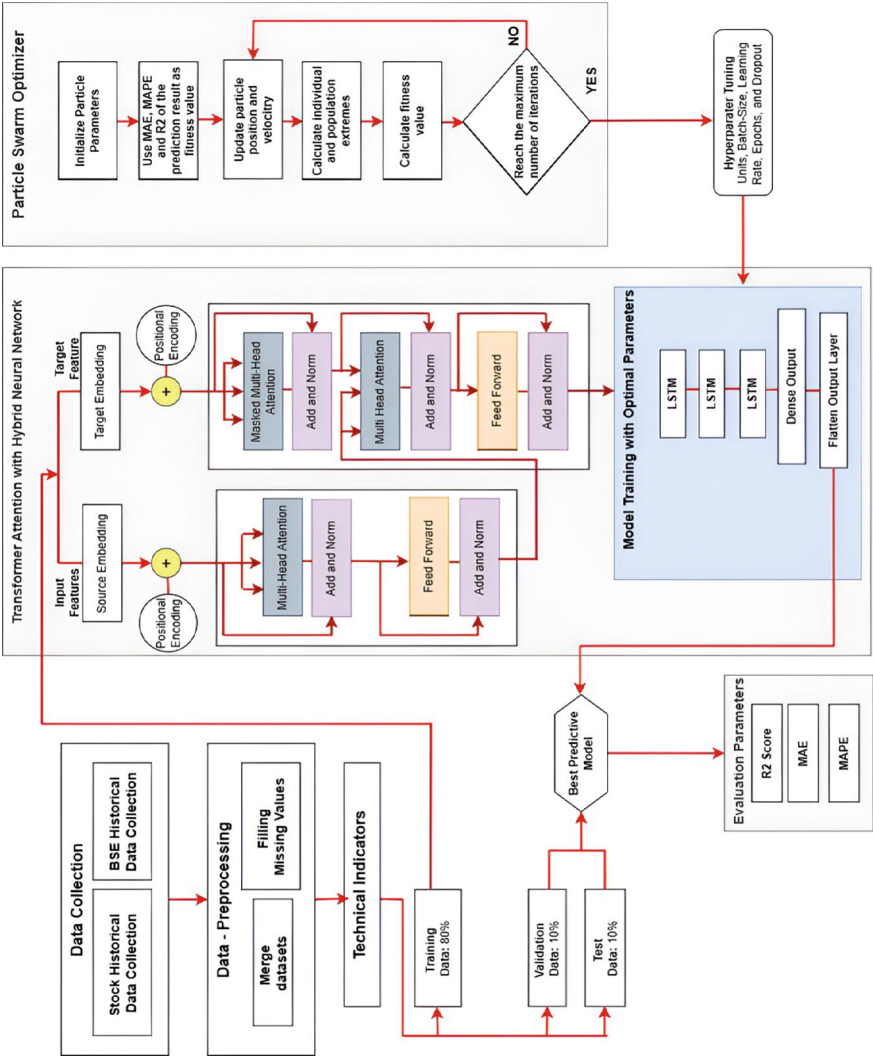


Fig. 1 The proposed framework utilizing the Particle Swarm Optimizer

closing price. Before sequence production, numerical characteristics (except 'Date') are normalised utilising MinMaxScaler.

3.2 Transformer Attention Layer

The Transformer Attention Model and Hybrid LSTM Network are robust time series forecasting frameworks for stock market prediction. The model uses a 90-day rolling window with 90 days of historical stock data as inputs and the 61st day stock price as the target variable. This iterative technique overlaps input-target pairings, helping the model learn from different temporal patterns and generalise. The dataset is separated chronologically into training, validation, and test sets after preprocessing to preserve stock fluctuation order. The attention mechanism relies on time-step information; input and target sequences are embedded with positional encoding. The Transformer module captures long-range dependencies using multi-head self-attention, while the decoder prevents information leaking from future values via masked attention. The Transformer outputs are sent into a hybrid LSTM network with three stacked layers to capture sequential dependencies and improve temporal features.

3.3 LSTM Model Architecture and Hyperparameter Optimisation with Particle Swarm Optimisation

The predictive model employs an LSTM neural network from TensorFlow's Keras library. LSTMs, engineered for sequential data, proficiently capture long-range dependencies, rendering them optimal for financial forecasting. The function delineates the architecture of two LSTM layers, with their units refined by hyperparameter optimisation. Dropout layers reduce overfitting, and a concluding Dense layer forecasts the modified closing price. The model employs the Adam optimiser and the mean squared error (MSE) loss function. The PSO optimises the hyperparameters of LSTM through PySwarm, enhancing units, dropout rate, batch size, epochs, and learning rate. Particle Swarm Optimisation (PSO), a population-based methodology, enhances solutions through iterative processes. The objective function establishes the fitness metric, directing PSO within defined hyperparameter limits, as detailed in Table 1, to improve model performance.

Table 1 The summary of parameters range assigned to the PSO framework for optimal model training

Parameter	Range
LSTM units	32–128
Dropout	0.1–0.3
Batch size	16–32
Epochs	100–300
Learning rate	0.01–0.0001

3.4 Model Training and Evaluation

After PSO determines the ideal hyperparameters that minimise validation loss, the final LSTM model is trained on the complete training dataset, encompassing the original validation set. This guarantees optimal data utilisation. Performance is assessed using a distinct test set comprising 20% of the test data. Predictions are subjected to an inverse transformation with the fitted scaler to revert to the original price values. The model’s accuracy is evaluated using multiple measures that compare anticipated and accurate adjusted closing prices.

4 Experiment Study

This research utilised TensorFlow-based models trained on an Asus Zephyrus G16 with an Intel Core i9 Ultra and 32 GB of RAM. Financial time series data was obtained from Yahoo Finance, encompassing BSE daily stock prices from April 1, 2020, to December 31, 2024. Data from seven prominent BSE-listed companies—Reliance, HDFCBank, ICICIBank, KotakBank, Infy, ITC, and LT—was collected, integrated, preprocessed, and utilised for model creation and assessment. Particle Swarm Optimisation (PSO), utilising a swarm of ten particles, was employed to ascertain the appropriate quantity of LSTM units in each hidden layer of the proposed LSTM design. The performance metrics in Table 2 indicate that the PSO-hyperparameter-tuned LSTM network attained greater accuracy than other LSTM setups. The optimized model demonstrated constraints with smaller datasets, but its performance markedly improved with larger data volumes, signifying superior pattern identification.

5 Results and Discussion

The final model’s training and validation loss curves are compared to the number of epochs to aid in interpreting the results. This image aids in evaluating the convergence of the training process and detects any overfitting indications. A time series graphic

Table 2 PSO-optimised LSTM with Adam demonstrates enhanced forecasting accuracy and generalisation of time series

Number of trading days	Accuracy	Precision	Recall	F1-score
1000	0.82	0.78	0.73	0.81
2000	0.84	0.78	0.74	0.83
3000	0.85	0.79	0.74	0.83
4000	0.87	0.72	0.77	0.86
5000	0.89	0.74	0.79	0.87

compares the actual and anticipated adjusted closing prices in the test set. This visual depiction facilitates a qualitative assessment of the model’s capacity to capture the direction and amplitude of price fluctuations. In Fig. 2, the time series plot compares RELIANCE’s actual and forecasted stock values from February 2024 to January 2025. The visual analysis shows that the expected price trajectory (red dashed line) mostly matches actual price changes (black solid line), indicating major patterns. Differences exist, especially in peak and trough magnitudes and timing. Overestimates troughs and underestimates peaks for mid-to-late 2024. The suggested system assessed blue chip stocks using MAE, MAPE, and R^2 accuracy (>0.93) for all tickers, as shown in Table 3. ICICIBANK.BO had the lowest MAE (10.98) and high R^2 (0.98), whereas LT.BO had the lowest MAPE (1.67%) and robust R^2 (0.97). INFY.BO has the greatest MAPE (9.57%) and a good R^2 (0.94). LSTM units (64–128), batch size (32–64), learning rate (0.0001–0.001), epochs (150–225), and dropout (0.3) varied by stock, demonstrating ticker-specific optimal framework parameters.

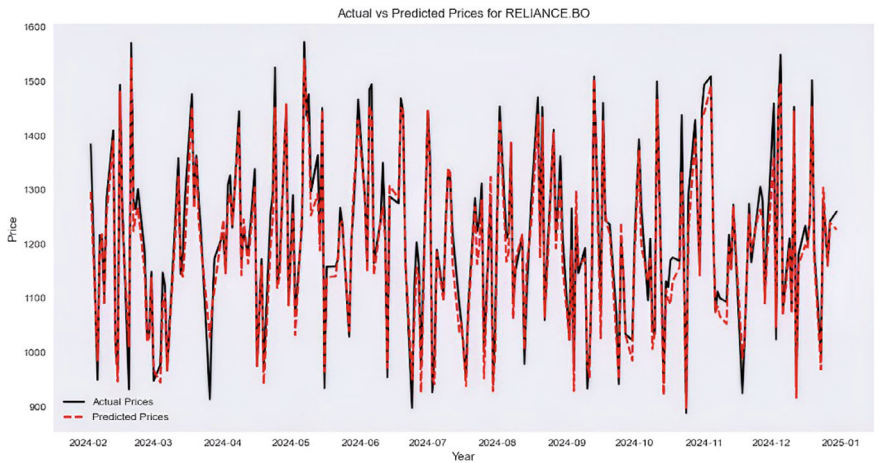


Fig. 2 Comparison of actual prices versus model-forecasted prices for RELIANCE.BO across time

Table 3 The summary of MAE, MAPE and R^2 results for the blue-chip stocks achieved by the proposed framework

Ticker	Units	Batch-size	Learning rate	Epochs	DropOut	MAE	MAPE	R^2
RELIANCE.BO	128	64	0.001	175	0.3	13.49	3.25	0.93
HDFCBANK.BO	128	32	0.0001	150	0.3	18.45	4.98	0.96
ICICIBANK.BO	64	32	0.001	200	0.3	10.98	3.73	0.098
KOTAKBANK.BO	64	32	0.0001	150	0.3	11.79	3.49	0.94
INFY.BO	128	32	0.0001	225	0.3	15.19	9.57	0.94
ITC.BO	128	64	0.0001	200	0.3	18.79	2.89	0.97
LT.BO	128	32	0.001	200	0.3	12.85	1.67	0.97

6 Conclusion and Future Direction

This study presents a hybrid deep learning methodology for stock price forecasting, combining LSTMs with a multi-head attention mechanism to elucidate intricate temporal correlations. Historical BSE blue-chip stock data is augmented with technical indicators such as volume, volatility, trend, and momentum. The dataset covers 2020–2024, includes COVID-19. We acknowledge that market volatility and economic shocks may induce abnormal behaviour now. However, this interval enables the model to learn in normal and turbulent conditions. A 90-day rolling period and rigorous feature architecture reduced short-term anomalies. The model was validated for universality on post-pandemic data. Pre-2020 or alternative datasets will assess model performance in different market conditions. With a R^2 of 0.97 and a MAPE of 1.67 for LT stock, the model achieves 89% accuracy over 5000 trading days. Time series forecasting has peak mistakes, as the model demonstrates. This work did not use post-processing or uncertainty estimations, but future research will manage peak predictions and quantify model output confidence via prediction intervals and smoothing.

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A Movie Recommendation Model Integrating Viewer's Emotional Experience and TV Feature Extraction



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Abstract Acceptance of information-gathering behaviours, especially from the viewpoints of others, is crucial for enhancing decision-making processes. In film critiques, audience feedback offers significant insights regarding a movie's quality and value as a time investment. The growing volume of review data requires automation for effective processing. This study introduces the creation of a sophisticated movie recommendation engine that amalgamates many data sources, such as IMDB movie reviews, Netflix ratings, YouTube trailer interactions, and Twitter discourse. User-generated input, including comments, likes, tweets, and trailer replies, is integrated to improve the recommendation process. The suggested methodology utilizes collaborative and content-based filtering, integrating users' social influence as assessed through their Twitter activity and social characteristics. A hybrid recommendation method produces an initial array of film recommendations. Thereafter, sentiment analysis is utilized to enhance and improve the recommendations. The

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system employs artificial intelligence methodologies, utilizing the IMDB dataset to train and validate a BERT embedding layer alongside a Bi-Directional Long Short-Term Memory (LSTM), a Bi-Directional Gated Recurrent Unit (GRU) incorporating a self-attention mechanism, and a Convolutional Neural Network (CNN). The model attained a testing accuracy of 93.91% and an AUC of 0.9831, indicating exceptional performance in binary sentiment categorization relative to current methodologies.

Keywords Deep learning · Movie recommendation · Multimedia · Sentiment classification · Social influence

1 Introduction

The rapid growth of online data has resulted in information overload, complicating consumers' ability to locate pertinent content. Recommendation systems are vital instruments that sift through information and improve user experience by proposing items and services tailored to individual tastes. These systems are essential for enhancing business operations and facilitating decision-making across multiple domains [1]. Recommendations are generally produced based on user preferences, product characteristics, historical interactions, and supplementary data, including temporal considerations [2]. Recommendation models are classified into collaborative filtering, content-based filtering, and hybrid methodologies [3]. Platforms such as IMDB, YouTube, Twitter, Netflix, and Hotstar offer user-generated material, including reviews and ratings, which profoundly influence customer decisions. Excessive data availability results in cognitive overload, requiring advanced recommendation algorithms [4]. Hybrid models that combine collaborative and content-based filtering improve accuracy, variety, and consistency. Social recommendation systems mitigate information overload by utilizing user interactions, comments, and votes [5]. Sentiment analysis, an essential method in contemporary recommendation systems, assesses user-generated text, such as film critiques and social media content, to ascertain opinion polarity. Companies utilize sentiment insights to evaluate consumer attitudes, whereas social media sentiment analysis facilitates trend forecasting. Deep learning has markedly enhanced recommendation systems through improved personalisation and accuracy. This research utilizes a hybrid deep learning methodology that integrates LSTM and GRU, employing transfer learning and embedding approaches to improve recommendation efficacy and user involvement.

2 Related Work

This section examines sentiment analysis and its fundamental components. Opinion mining analyses reviews, opinions, and assessments to derive sentiment. The exponential expansion of the Internet has resulted in the availability of extensive multi-lingual data. Social media analysis has become significant due to its abundant user-generated content. Sentiment detection, an essential text analysis component, discerns underlying emotions across various linguistic settings, presenting integration issues.

2.1 *Sentiment Classification*

Sentiment analysis has become a crucial domain within Natural Language Processing (NLP), with significant research focused on enhancing binary classification accuracy, especially on huge datasets like IMDB movie reviews. Resolving vocabulary discrepancies is crucial for accurate semantic interpretation [6]. Given the intricacy of language, several strategies have been utilized, encompassing subjective, instrumental, and dependent expressions. Discerning factual information from subjective opinions is essential for sentiment analysis. Proposed methodologies include SentiWordNet, fuzzy logic models, and unsupervised learning. Targeted sentiment analysis presupposes identified entities, whereas comparative analysis investigates sentiment among entities, necessitating the interpretation of language variability and conditional structures [7].

2.2 *Deep Learning in Sentiment Classification*

In recent years, deep learning has markedly enhanced recommendation systems. Platforms such as YouTube employ deep neural networks to enhance video recommendations. Neural Collaborative Filtering (NCF) enhances conventional matrix factorization by integrating multilayer perceptrons (MLPs) to represent intricate user-item interactions. Subsequent advancements encompass content-aware collaborative filtering, wherein neural networks incorporate contextual data to improve recommendations. For example, Google Play utilises a “wide and deep” framework that integrates generalised linear models with deep learning to effectively capture both memory and generalisation in user behaviour. Film recommendation systems similarly utilize Multi-Layer Perceptrons (MLPs) and Tensor Factorization (TF) to analyze watching patterns and contextual information [8]. Cross-domain methodologies integrate user interactions across several platforms (e.g., literature, cinema) to enhance precision [9], whereas LDA-based TF associates events with places [10]. This research, unlike previous studies, combines various sources—YouTube

trailers, IMDB reviews, and Twitter sentiment—and employs a hybrid BiLSTM + BiGRU architecture with BERT and self-attention to enhance contextualization and predictive accuracy [11].

3 Research Methodology

This research examines the impact of trailer critiques and Twitter sentiments on film assessments. It employs two datasets, MovieLens and Netflix, integrating trailer data from YouTube and tweets from Twitter. Metrics for trailer engagement, encompassing likes, comments, and views, are examined. Three data preparation techniques are analyzed: merging explicit ratings with YouTube sentiment scores, amalgamating trailer sentiment scores with feature ratings, and developing a feature vector using ratings, reviews, and sentiment scores. Data undergoes preprocessing through noise filtration. A hybrid recommender system utilising Matrix Factorisation and Multi-layer Perceptron (MLP) produces recommendations by exploiting film characteristics and user interactions. Figure 1 depicts the modular architecture of the proposed system, integrating matrix factorisation, deep neural networks, and sentiment-aware feature extraction. The framework facilitates the flexible integration of many data sources, including reviews, tweets, and trailer interactions.

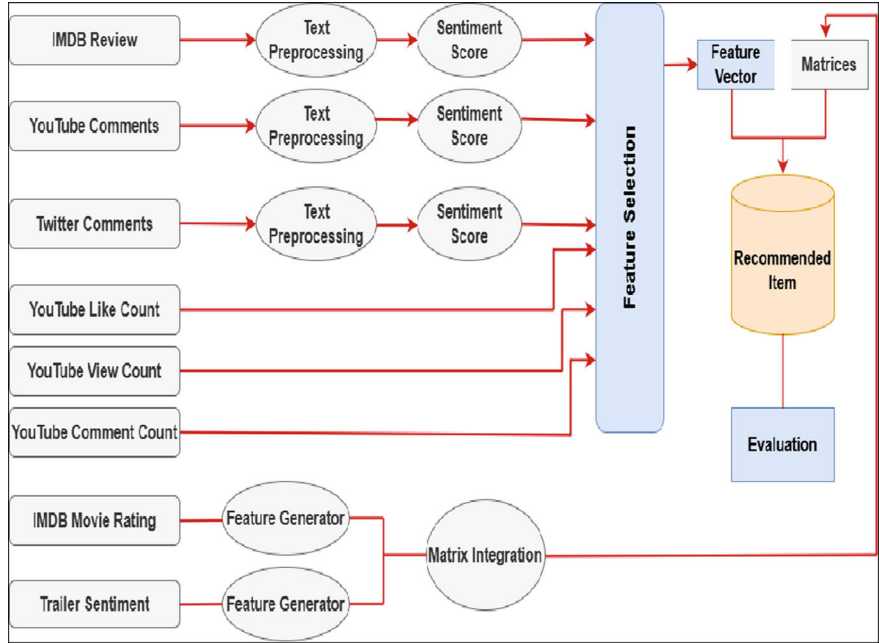


Fig. 1 Proposed system architecture

It utilises Matrix Factorisation and Neural Network models for suggestions, including movie reviews, trailer evaluations, and tweets. Sentiment scores derived from YouTube comments and tweets are utilised to improve ranking grids and delineate movie characteristics. Using latent factor models, the system synchronises rating matrices, addressing user alignment difficulties through knowledge transfer methodologies.

4 Experiment Study

The suggested film recommendation system amalgamates many data sources to improve suggestion precision. This study examines the effects of integrating implicit input from movie trailers and tweets utilizing the IMDB dataset. It analyses the importance of diverse features in enhancing recommendations, such as likes and views. The system’s performance is compared with cutting-edge methodologies using Neural Network models, including Matrix Factorisation and multilayer perceptron.

4.1 Data Collection

This research employs film critiques from IMDB and Netflix datasets <https://www.kaggle.com/datasets/luisrcorter/netflix-original-films-imdb-scores>. A random sample of 1000 users and 1000 films is obtained from Netflix and the accompanying IMDB ratings for those films. YouTube trailer metrics, encompassing comments, views, likes, and the top 100 remarks, are gathered autonomously. Furthermore, 150 tweets per film are collected using the Tweepy API. Sentiment analysis produces implicit ratings and features at the movie level. Table 1 summarises the IMDB dataset, which serves as the basis for model training and sentiment analysis. The enormous quantity and variety of evaluations for each film (from 694 to 4825) provide a robust foundation for formulating sophisticated suggestions.

Table 1 IMDB dataset statistical information

Particulars	Data
Tenure	January 1995–June 2014
# Users collected	1173
# Movies collected	11,735
# Ratings collected	2,841,934
# Tags collected	1,150,000
Max ratings (per movie)	4825
Min ratings (per movie)	694
Avg. ratings (per movie)	371

Table 2 Netflix dataset statistical information

Particulars	Statistics
# Users collected	1000
# Movies collected	1000
# Ratings collected	345,122
Max ratings (per movie)	1342
Min ratings (per movie)	131
Avg. ratings (per movie)	579

Table 3 YouTube dataset statistical information

Particulars	Statistics
# Movies collected	25,623
Max views (per movie)	4,053,421
Min views (per movie)	24
Avg. views (per movie)	234,213
Max comments (per movie)	1834
Min comments (per movie)	0
Avg. comments (per movie)	34
Max likes (per movie)	6734
Min likes (per movie)	0
Avg. likes (per movie)	246

It is demonstrated in Table 2 that the Netflix dataset is a useful addition to the IMDB data since it provides a controlled user-movie pairing to test the hybrid recommendation technique.

Table 3's YouTube engagement measurements were utilised to derive implicit movie-level sentiment scores, augmenting the input feature space with viewer emotional signals and explicit ratings.

4.2 Text Preprocessing

To get to the essence of the research, this study pre-processes text data. The dataset has also been cleansed. URLs and related elements can be removed with regular expressions. Writing out negations is better than truncating them. Periods, commas, hyphens, parentheses and forward slashes are prohibited. The following code replaces all "Username" with "usr." as a regular expression. Deleting hashtags would be pointless because they provide information. This is now "Lee" instead of "Especially 'Lee'." Remove the number of references, emails, and HTML tags from the news item. Stemming, lemmatization, and tokenization aid NLP. Words can be broken

down using “stemming.” After stemming, some English words have foreign roots. NLTK jobs use lemmatization for accuracy.

5 Results

The experimental evaluation uses Python 3.6 the GPU was tested on Google Colab. This investigation’s simulation parameters and settings are in Table 4.

This research employs pre-processed film critiques, implementing stemming and lemmatization with a predetermined sequence length of 128 characters. The Torch IMDB dataset has 49,582 reviews, evenly distributed between positive and negative sentiments, with 70% allocated for training and 30% for testing. The proposed model integrates BERT-Large embeddings with a hybrid BiLSTM-BiGRU architecture and a 1D CNN layer for sentiment categorization. BERT-Large (24 layers, 1024 hidden units, 16 heads) is chosen for its suitability with preprocessing. The model is trained utilizing categorical cross-entropy loss. With a test accuracy of 93.91%, it surpasses alternatives by adeptly integrating spatial and contextual comprehension for sentiment classification.

5.1 Evaluation and Comparative Analysis

The proposed hybrid recommendation model, which integrates BERT embeddings with BiLSTM-BiGRU, a self-attention layer, and 1D CNN, has impressive performance (93.91% accuracy and 0.9831 AUC); nonetheless, the manuscript fails to provide a thorough comparison with contemporary state-of-the-art models as shown in Table 5.

The related work section cites several sophisticated methodologies, including Neural Collaborative Filtering, LDA-based tensor factorization, and hybrid deep learning models; nevertheless, the experimental section lacks a comparative quantitative assessment against these benchmarks. To enhance the validation of the suggested method’s efficacy, we suggest including comparative results utilizing standardized

Table 4 Simulator parameters

Particulars	Range
Optimizer name	Adam
Cyclic learning rate (CLR)	0.0001–0.0005
CLR policy	0.99999
Step size	400
Batch size	120
Epochs	100

Table 5 Testing precision of the suggested model compared to configuration changes

BERT embedding and model	Test acc. (%)
BERT-BE + Bi-LSTM + SA Layer + 1D CNN	92.32
BERT-BE + Bi-GRU + SA Layer + 1D CNN	91.19
BERT-BE + Bi-LSTM-Bi-GRU + SA Layer + 1D CNN	91.24
BERT-LE + Bi-LSTM + SA Layer + 1D CNN	92.56
BERT-LE + Bi-GRU + SA Layer + 1D CNN	93.14
BERT-LE + Bi-LSTM-Bi-GRU + SA Layer + 1D CNN	93.91

measures (e.g., accuracy, F1-score, AUC) on widely recognized datasets. This benchmarking is crucial to demonstrate the practical benefits and originality of the proposed method in relation to current studies.

6 Conclusion and Future Work

This research introduces an innovative sentiment classification method utilizing the IMDB review dataset. The model utilizes BERT embeddings for tokenization and has a hybrid architecture that combines BiLSTM and BiGRU layers for target extraction. It integrates an attention mechanism, a one-dimensional convolutional neural network, linear layers, ReLU activation, and soft pooling layers to categorise sentiment as positive or negative. Comparative investigation indicates that models integrating BERT embeddings, BiLSTM-BiGRU, self-attention, and CNN attain exceptional performance, evidenced by an AUC-ROC of 0.9831 and an accuracy of 93.91%. The amalgamation of BERT characteristics with deep learning layers augments classification efficacy. Although the model demonstrates strong performance on IMDB, additional assessment on other datasets is essential to confirm its generalizability.

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IoT for Next-Generation Networks: A Succinct Review



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Abstract The analysis predicts that over half a trillion Internet of Things (IoT) devices will become connected during the next decade as automotive sectors strive to enhance 90% of their vehicles through IoT capabilities. The connected vehicle market surpassed 250 million units in 2020, with a 67% growth rate. Integrating IoT technology into industrial operations creates new business prospects that let organizations advance their methods and achieve optimized operations. Technological advancements lead to a smarter world where users gain easier resource access and have improved quality of service (QoS) and quality of experience (QoE). The paper presents six essential factors for IoT, including identification protocols and calculation functionalities that create operational boundaries. The IoT leads industrial operations into transformation, while research between engineering and social sciences needs to develop the technology of connected systems.

Keywords IoT · AI · IoT challenges · IPv6

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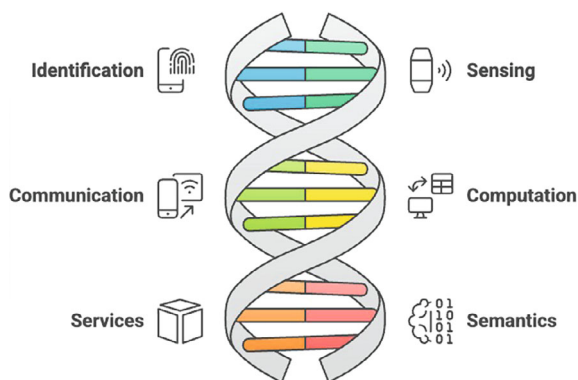
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1 Introduction

A next-generation network (NGN) refers to the transition of fixed and mobile network infrastructures from separate, proprietary systems to unified networks utilizing Internet Protocol (IP). The Internet-connected device count will grow significantly because researchers expect every device to become web-enabled by 2025. The Cisco research predicts that IoT will reach half a trillion objects by 2030. Telefonica forecasts that automobiles will achieve the status of IoT-connected vehicles in 2030, when 90% of vehicles will be integrated into this network, with the prediction that individuals will utilize approximately 15 connected devices by that time. Market analysts projected that by 2020, connected vehicles would reach more than 250 million units while the market size increased by 67%. The IoT system assists industrial needs to help businesses construct new approaches that bring their novel ideas to reality. The industrial opportunities enable efficient and innovative research opportunities for scientists and specialists studying across multiple disciplines, enabling them to merge research investigations with engineering competencies and sciences and humanities research methods. The development of a smart world happens through IoT technology, which accelerates quick and efficient access to unlimited resources. Industrial organizations rapidly implement venture plans that assume IoT technology implementation based on its current development phase. The IoT invention helps to turn the human lifestyle into a new future state. This technology has become available worldwide through new technological advances, even though users need extensive quality of service (QoS) combined with quality of experience (QoE) capabilities. The six IoT components include identification methods, sensing communication, computation services, and semantics [1]. The main components of IoT that contribute to its operational functionality appear in Fig. 1.

The radio frequency identification tag represents an identification method that uses RFID tags for detection purposes. Objects within the network can be detected by a typical chip or sticker mounted on them using an RFID tag. The networks address IoT objects through IPv6 over Low-Power Wireless Personal Area Networks

Fig. 1 IoT components



(6LoWPAN). The Personal Area Networks (6LoWPAN) offer compression mechanisms over IPv6 for low-power wireless networks [2]. Smart wearable devices serve as IoT sensors to collect tangible information. A sensor contains various onboard sensors which are explicitly fitted to its structure. Network nodes can obtain nearby reading data, such as temperature, moisture level, acceleration, traffic, and house surveillance [3]. A wide range of communication protocols exists for operating the IoT. The CoRE framework stands for constrained restful environments (CoRE) and constrained. The IoT makes use of three important communication protocols, which include constrained application protocol (CoAP) alongside message queue telemetry transport (MQTT) [4]. The data processing hardware and software specifications get their definitions through computational methods. Different embedded devices, processors, application-specific integrated arrays, and applications execute computation processes. IoT services include identity-related, information aggregation, collaborative awareness, and pervasive services. The identification services help evaluate different entities as part of their operational function. Information aggregation services gather every piece of data through processing to generate prepared reports. Further activities depend on decisions made by collaborative aware services based on previously acquired data. Ubiquitous services indicate that systems should use collaborative-aware services to fulfill their needs. The collection of device data for proper assistance delivery determines the semantic capabilities of IoT systems. Semantics is a method to abstract information through multiple strategic approaches to deliver basic services. Several devices use resource description frameworks and web ontology language as semantic technology.

Organization of paper: Sect. 2 presents the literature review of IoT and 5G Architecture in IoT, and the benefits and impact of 5G are presented in Sect. 3. Section 4 presents the application of IoT for next-generation, and Sect. 5 presents the challenges of implementing IoT. In the last section, the conclusion and future scope are discussed.

2 Literature Review

Smart services function best with IoT because it enables the connection between dynamic entities to acquire multiple forms of environmental data [5]. Implementing automation technology, together with information systems that merge Manufacturing Execution Systems and Enterprise Resource Planning, helps organizations achieve higher efficiency and productivity through connectivity. Organizations have better resource management and optimized workflows through advanced technology combinations, enabling them to remain competitive in the digital world. Modern applications emanated from internet-connected devices to improve various industrial sectors and thus transform business procedures and customer decisions. The Internet facilitates both sharing economy services and the fusion of physical commercial channels with ecommerce platforms, thus benefiting organizations. Patients achieve enhanced involvement through the amalgamation of omnichannel platforms and

IoT devices, which support patient commitment. These technologies are key operational tools enabling organizational innovation, collaborative task performance, and strategic deployment and leadership practices across every ecosystem component. Through IoT integration, organizations can detect instant defects while enabling the corrective measures needed to improve their quality management systems. The ability of IoT to generate data combined with its product update functions helps businesses develop stronger relations with customers to foster recurring revenue streams.

Technological advancement in computer vision happens through the IoT because interconnected devices produce vast amounts of data. Many new applications emerged from this integration of computer vision systems with IoT technology, which improved both situational awareness and decision-making abilities. The vast IoT applications allow systems and enterprises to conduct operations with enhanced synchronization and automated processes under improved surveillance systems. Through the IoT, businesses achieve ongoing data collection about their entire operational business activities [6]. While data acquisition is a base function, IoT operations elevate this through integrated information analysis, which enhances operational performance in organizational business activities. Business operations leverage IoT applications to track their products and monitor activities and business processes across value chain networks.

3 5G Architecture in IoT

Figure 2 illustration shows the architecture of the 5G system IoT. A 5G IoT framework is presented in [7], which offers smart IoT sensors a way to connect using various wireless and wired networks until they reach the gateway. The IoT gateway employs a wireless connection to transfer 5G base station data and IoT device information using 5G NR technology with appropriate numerology selection and mmWave communication for URLLC and eMBB operation [8]. The scientists require research solutions to achieve universal communication, which motivates them to study network mobility, coverage, and reachability. The standards of 5G continue to develop to solve scalability limitations and performance issues concerning latency and reliability. Demanding performance benchmarks is an essential problem that requires immediate solutions for various IoT applications. The analysis and development of 5G systems necessitates complete simulation platforms for these systems because they combine various service offerings with several performance capabilities. Creating testing infrastructure using authentic cellular networks proves difficult because of their intricate operational demands. The POWDER wireless testbed at the University of Utah in Salt Lake City operates through the Platforms for Advanced Wireless Research (PAWR) program to deliver end-to-end software-defined remote testing facilities to research groups [9].

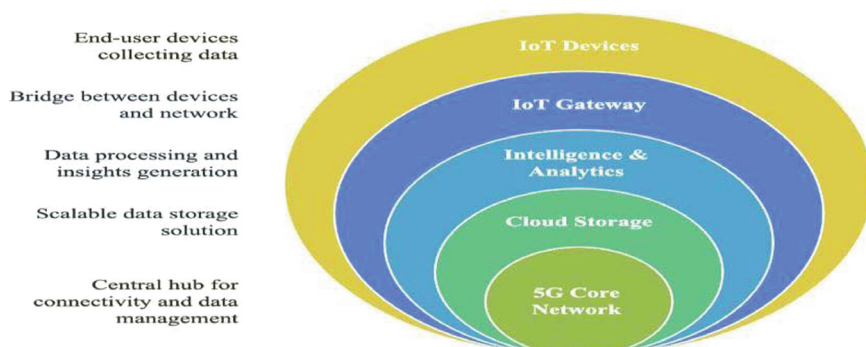


Fig. 2 5G IoT architecture

3.1 Impact of 5G on IoT

5G will open fresh and creative IoT possibilities. Lower latency, faster speed, and more capacity of 5G will enable fresh and creative IoT applications that are not feasible with earlier generations of cellular networks. For instance, 5G-enabled self-driving cars would be far safer since they can interact with traffic infrastructure in real time and with each other. 5G-enabled smart cities can gather and evaluate data from various sensors to enhance public safety, energy economy, and traffic flow. 5G will help lower IoT costs. By enabling more devices to be connected to the network, 5G's higher speed, lower latency, and greater capacity will assist in minimizing IoT costs. This will make IoT solutions more reasonably cost-effective for companies and consumers alike. 5G will increase IoT security. By making it more difficult for hackers to target IoT devices, the faster speed, lower latency, and higher capacity of 5G will enhance IoT security. This will make IoT solutions safer for consumers and companies to apply. 5G represents a significant IoT advancement overall. It will lower IoT costs, enable fresh and creative uses, and enhance IoT security.

4 Application of Smart IoT for Next-Generation

All sensor devices establish connections to the Internet with the help of low power wide area (LPWA) technologies, which minimize their energy consumption. LPWA offers effective energy resource usage at a low price to various users over the networks. LPWA technology operates with low-power access as one of its main features. NarrowBand Internet of Things (NBIoT) represents a newly developed radio technology that creates a sustainable, power-efficient, cost-effective framework. The applications of IoT extend across multiple domains that involve real-time multimedia and IoT-based healthcare systems along with IoT-based next-generation smart industries and IoT-based next-generation smart agriculture, as shown in Fig. 3.

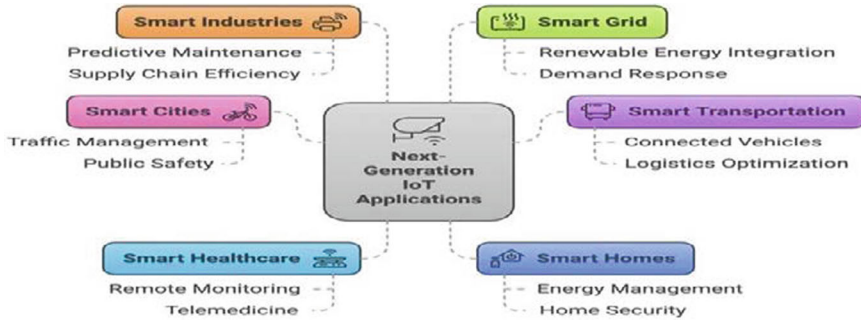


Fig. 3 Application of smart IoT for next-generation

The urgent requirements for security do not define the primary role of the era because applications require data analysis systems that operate efficiently to predict situations for daily operations. The primary challenge of distributing optimal resources across numerous IoT devices results from a restricted spectrum alongside diverse access technology issues. Artificial Intelligence (AI) implementation should develop flexible, adaptive systems, particularly for this area, because these systems represent an important promising solution.

Smart Industry: The development of smart industries established fundamental, lasting results for the unavoidable advancement toward global manufacturing. Under Industry 4.0's cyber-physical technologies, all smart industry components were unified to achieve advanced systems for high performance and quality alongside operational control, management, and fairness development in IoT-based smart industries. Manufacturing companies and supply networks require inexpensive and advanced sensing systems to acquire essential operational data that ensures system stability. Researchers have evaluated factory performance in combination with the evaluation of facility utilization throughout their entire research work. Most industrial producers must grasp contemporary and traditional setup distinctions and all sensor technologies related to Industry 4.0 [10].

Smart Healthcare: The healthcare market actively adopts IoT technology in its commercial operations. Several business solutions operate in the healthcare sector to provide services to individuals. Healthcare frameworks based on Internet of Things technologies demonstrate the potential to transform into important medical services that hospitals can support as secondary healthcare units. The present situation reinforces the pressing need to identify upcoming solutions to tackle achievement-related problems and difficulties. Alzheimer's patients can use real-time location technology through wearable geospatial location sensors to gain real-time positioning status. Several modern healthcare developments incorporate real-time epileptic seizure detection systems, stroke analysis instruments, AR/VR telehealth-care platforms, handheld medical record access, blockchain-enabled safe health services, remote medical procedures, and remote medical training programs [11].

Smart Cities: The IoT system has enabled the creation of specific arrangements that directly or indirectly establish relationships between various applications for smart-cities applications. IoT-based smart-cities systems face significant challenges because of these frameworks and collaborative applications. Several sensor devices present a drawback when malfunctioning because of their failure mechanism. Previous events depend on establishing numerous communications among smart devices spread throughout different geographic sites. Subsequent IoT frameworks must deliver a proper platform for analyzing and aggregating data from different devices. Such an IoT-based smart-cities system requires the evaluation of user-dependent security and privacy protocols [12].

Next-Generation IoT-Based Cross-Layer Protocols: Network clustering methods use new protocols to create effective solutions by selecting the best cluster heads for routing. The operational difficulties of AI-enabled IoT-based devices arise because their power capacity is limited while user density is high, communication distances are long, and data transmission faces high latency and losses. Cross-layer routing protocols are necessary for better energy efficiency with lower delay and information processing speed. Next-generation IoT applications require effective presentation of AI-based cross-layer protocols to develop future IoT applications. The goal can be achieved through reinforcement learning and multiple learning techniques. Modern protocols demand specific development focused on energy efficiency through a design prioritizing QoS efficiency. Before choosing stable optimal cluster heads as nodes, you must conduct cross-layer analysis of IoT devices using AI-based heuristic methods to minimize overall overhead.

5 Challenges in Implementation

A large number of problems halt the smooth integration of IoT into next-generation networks despite existing positive developments. Table 1 presents the challenges in implementing IoT for the next generation.

6 Conclusion and Future Scope

Business industries experience significant transformations because IoT technology strengthens their operational methods and resource management systems. Organizations that employ IoT solutions establish improved business process coordination and better decision-making technologies, which drive workflow productivity towards more efficient operations. Smart agriculture gains significant benefits from the future applications of IoT systems because these technologies will establish and maintain resource protection systems. LPWA technologies require additional development to improve their power consumption efficiency because this affects device connectivity systems. Combining AI technology and ML allows developers to construct adaptive

Table 1 Challenges encountered

Challenges	Description
Scalability issues	The extensive deployment of IoT devices results in huge data production which needs strong and adaptable network infrastructure for efficient communication and processing
Security and privacy risks	The internet of things exists against several types of cyber threats which include data breaches combined with unauthorized access attempts and virus attacks. End-to-end network and device protection remains a complicated objective to achieve
Latency and real-time processing	Many IoT devices need real-time data processing because of care monitoring requirements and the needs of driverless cars. Normal application function can be compromised through network congestion together with delays [13]
Energy consumption and sustainability	The required energy-efficient communication systems must operate the IoT devices that depend on battery power. In the environments the main problem lies in maintaining both long-lasting battery life and optimal performance levels
Interoperability and standardization	Multiple IoT devices use different communication methods that cause compatibility problems. Uniform criteria absence stands as an obstacle that causes difficulties with data flow and integration processes

IoT systems that achieve extensive deployment opportunities across different industrial fields. Stakeholders aiming to obtain optimal benefits from IoT development opportunities must invest in research and development at an extensive level due to its vast industrial and life-improvement potential.

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Training Efficiency of DDQN-Based Multilevel Inverter Control: The Influence of Reward Function Penalty Terms



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and Sunil Khatri

Abstract Reinforcement Learning (RL)-based controllers have recently gained attention as AI-driven, model-free methods for controlling power electronic converters by learning optimal control actions through continuous interaction with the environment. Their learning process is governed by a reward function, which guides the agent's behavior. This paper investigates the influence of incorporating penalty terms into the reward function on the training efficiency and performance of an RL-based controller for a 7-level grid-tied Packed-U-Cell (PUC7) multilevel inverter. The controller is developed using the Double Deep Q-Network (DDQN) algorithm, selected

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for its balanced combination of strong performance and ease of implementation. The control objectives include sinusoidal current injection into the grid and capacitor voltage regulation around the desired value. The reward function is designed based on current and voltage tracking errors, with two penalty terms introduced to limit deviations beyond predefined thresholds. The study evaluates the impact of varying these penalty magnitudes on learning speed, convergence behavior, and tracking quality. Simulations are conducted in MATLAB/Simulink, demonstrating that the appropriate selection and application of penalties improve training efficiency without compromising control performance.

Keywords Reinforcement learning · Reward function

1 Introduction

Reinforcement learning (RL) is a model-free machine learning technique in which an agent interacts with the environment and refines its decisions based on rewards or penalties. By exploring different actions and learning from experience, the agent adjusts its strategy to maximize cumulative rewards without requiring an explicit mathematical model of the system. This process is guided by a policy that maps each state to the most suitable action based on past interactions. Recently, RL-based controllers (RL-C) have been increasingly applied in power electronics for inverter control, addressing the limitations of model-based controllers, such as reliance on an accurate system model and the computational burden of solving optimization problems in real time [1–3, 6, 9–12, 14].

One of the key design elements of RL-C is the reward function (r), which plays a central role in the optimization process of RL by defining the objective that the agent aims to maximize. As a result, r essentially guides the learning process by providing feedback on the agent's actions to drive the system toward the desired control performance. A well-structured reward function enhances learning efficiency, improves stability, and ensures that the agent develops an effective control strategy. On the other hand, a poorly designed reward function can lead to suboptimal performance, slower convergence, or unintended control actions. In RL-C for power electronics, the reward function is typically formulated using system performance metrics such as voltage and current tracking errors, switching losses, and stability constraints [2–4, 6, 9].

In addition to rewards, penalty terms can be added to the reward function to discourage undesirable behaviors and enhance the training process [7]. These penalties serve as negative reinforcements, reducing the reward when the agent's actions lead to system deviations beyond acceptable limits. An appropriately tuned penalty term can enhance training efficiency by suppressing suboptimal actions during the early stages of learning, thereby reducing the extent of unnecessary exploration. However, excessively high penalty values may overly constrain the agent's decision-making space, limiting its ability to explore the state-action space effectively and

limiting the discovery of optimal control strategies. Therefore, precise tuning of the penalty magnitude is critical to achieving a balanced trade-off between exploration and exploitation, ensuring the convergence toward an effective and stable control policy while maintaining desired system performance.

In previous work, penalty terms were introduced to prevent excessive voltage and current deviations and to enforce the RL agent to learn within an accepted margin close to the desired performance [1, 2]. However, incorporating penalty terms in r and determining their values were done primarily through trial and error and based on experience. There was no systematic analysis in previously designed RL-C that investigated the actual impact of adding penalty terms to r and how these terms affect the agent's learning convergence, training speed, and overall performance.

Therefore, in this study, the reward function (r) proposed in [2] is evaluated, with a focus on the two penalty terms introduced in RL-C for PUC7 to penalize large voltage and current deviations beyond predefined margins. The objective is to investigate the impact of these penalties on the RL training process, particularly in terms of training speed and final control performance. Additionally, the study examines how different penalty values influence the agent's ability to learn an optimal and stable control policy. In this work, RL-C is implemented using the Double Deep Q-Networks (DDQN) algorithm, one of the techniques applied to PUC7 [2]. DDQN is chosen for its simplicity and rapid convergence compared to more advanced algorithms, such as Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC). The RL-C training and testing are carried out in MATLAB/Simulink environment.

2 PUC Topology

The topology of the grid-connected PUC inverter is shown in Fig. 1. The inverter comprises a single DC voltage source, V_1 and a single capacitor ($C = 1\text{mF}$) with a voltage V_2 . The inverter's output voltage, V_{AN} , is determined by the configuration of the two DC voltage levels (V_1 and V_2), which are controlled by six switches (S_i), where $i = 1-6$. Considering that each pair of switches in a cell operates in a complementary manner (i.e. $S_1 = \bar{S}_4$), to prevent a short circuit in any cell, only eight valid switching configurations for PUC7 as listed in Table 1.

To generate seven output voltage levels by the PUC inverter, V_2 should be regulated at $V_1/3$, named as the reference voltage (V_2^*). Assuming that $v_g = 170\text{ V}$ with frequency $f = 60\text{ Hz}$ and $V_1 = 210\text{ V}$, then $V_2^* = 70\text{ V}$ and the desired seven levels of PUC7 are shown in Table 1. The PUC inverter is integrated into the grid through a filtering inductor ($L = 10\text{ mH}$), where the filtered current injected into the grid is denoted as i_g .

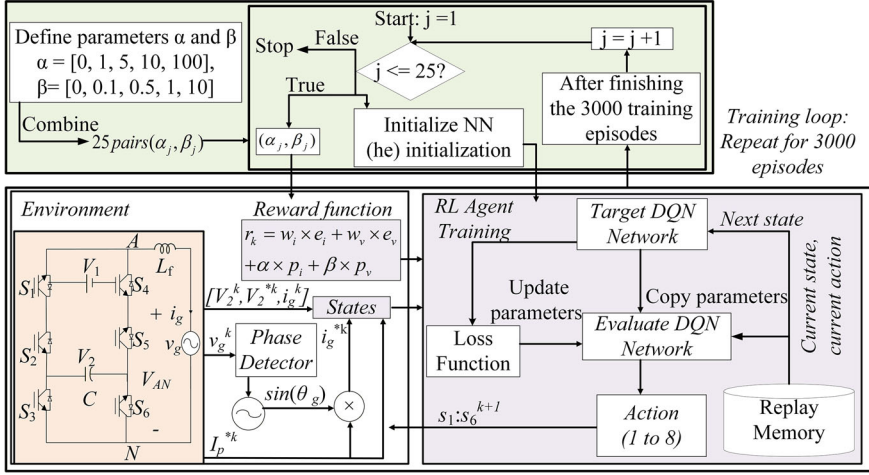


Fig. 1 The RL iterative framework to train the RL-C for PUC7

Table 1 PUC switching configurations and their corresponding output voltage values

S_1, S_2, S_3	V_{AN} (V)	Action	S_1, S_2, S_3	V_{AN}	Action	S_1, S_2, S_3	V_{AN} (V)	Action
1,0,0	210	a_{s1}	1,0,1	140	a_{s2}	1,1,0	70	a_{s3}
0,0,0	0	a_{s4}	1,1,1	0	a_{s5}	0,0,1	-70	a_{s6}
0,1,0	-140	a_{s7}	0,1,1	-210	a_{s8}			

3 Proposed RL-C Framework

3.1 Control Objectives

The RL-C is designed to achieve two main control objectives:

1. Inject a pure sinusoidal current into the grid with minimal total harmonic distortion (THD).
2. Maintain the capacitor voltage near its reference value of 70 V to enable the generation of seven distinct DC voltage levels.

3.2 RL-C Problem Formulation

The RL-C is designed to fulfill the two control objectives considering the system parameters mentioned in Sect. 2. The RL-C framework consists of designing the environment, the states, the control action, and the reward function.

Environment: In this RL-C framework, the environment is the PUC7 MLI. It encapsulates the system's dynamics, which are determined by the inverter's electrical characteristics. It interacts with the agent at each sampling time step, $T_s = 20 \mu s$. During this interaction, the environment provides state observations and receives control actions from the agent, as shown in Fig. 1.

State space: The observations capture key state variables that characterize the system's dynamics. The observation vector, consisting of 5 parameters, includes the following: the capacitor voltage V_2 , its reference value V_2^* , the generated current i_g , its reference value i_g^* , and $|i_g^*|$ referred to as I_p^* .

Action space: The agent has a discrete action space consisting of 8 possible switching configurations, from which it selects a control action, as detailed in Table 1.

Reward function: The total reward function at time step k is a weighted sum of the current reward, the voltage reward, and the penalty terms, as follows:

$$r_k = w_i \times e_{ik} + p_{ik} + w_v \times e_{vk} + p_{vk}, \quad (1)$$

where w_i and w_v are weighting factors that prioritize the tracking of current reference over voltage error, and considered $w_i = 1$, $w_v = 0.2$ in this study.

1. Current and voltage reward: The current error, e_{ik} , is defined as:

$$e_{ik} = -|i_{gk}^* - i_{gk}|, \quad (2)$$

where a smaller deviation from the reference signal leads to a higher error value, e_{ik} . Similarly, the voltage error e_{vk} is defined as:

$$e_{vk} = -|V_{2k}^* - V_{2k}|. \quad (3)$$

2. Current and voltage penalties: To prevent excessive deviations, penalty terms for current and voltage are introduced as follows:

$$p_{ik} = -\alpha, \text{ if } |i_{gk}| > 1.05 \times I_{pk}^*. \quad (4)$$

$$p_{vk} = -\beta, \text{ if } |V_{2k}| > 1.1 \times V_2^* \text{ or if } |V_{2k}| < 0.9 \times V_2^*. \quad (5)$$

Here, the agent is penalized by $-\alpha$ when the current exceeds 5% of the reference value I_p^* , and by $-\beta$ when the capacitor voltage deviates outside a $\pm 10\%$ window from its reference value.

3.3 Double Deep Q-Network Algorithm

The DDQN algorithm addresses the overestimation bias inherent in standard DQN. This bias stems from the use of the max operator in target Q-value calculations, leading to the selection of actions with inflated Q-values. DDQN mitigates this by decoupling action selection and evaluation [13].

The core of DDQN is the approximation of the Q-function, representing the expected cumulative reward, using NN. Mathematically, this is expressed as: $Q(st, a; \theta) \approx Q^*(st, a)$, where $Q(st, a; \theta)$ denotes the Q-function approximated by the NN with weights θ , and $Q^*(st, a)$ represents the optimal Q-function [8].

The key innovation of DDQN lies in its target Q-value calculation. The target Q-value, denoted as Y_k^{DDQN} , is computed as:

$$Y_k^{\text{DDQN}} = r_{k+1} + \gamma Q(st_{k+1}, \arg \max_a Q(st_{k+1}, a; \theta_k); \theta'_k), \quad (6)$$

where γ is the discount factor, θ_k represents the parameters of the online network, st is the observed state, and θ'_k represents the parameters of the target network. The term $\arg \max_a Q(st_{k+1}, a; \theta_k)$ selects the action that maximizes the Q-value according to the online network, while $Q(st_{k+1}, a; \theta'_k)$ evaluates the selected action using the target network.

The training process involves minimizing the loss function, which is typically the mean squared error between the target Q-value and the predicted Q-value. This is mathematically expressed as:

$$L(\theta) = \mathbb{E}[(Y_k^{\text{DDQN}} - Q(st_k, A_k; \theta))^2], \quad (7)$$

where $\mathbb{E}$ denotes the expectation. The interaction between the components of the DDQN framework is illustrated in the *RL Agent Training* block in Fig. 1.

In this paper, the critic is a DNN designed to approximate the Q-value function, consisting of an input layer, two fully connected hidden layers with 45 and 30 neurons, respectively, each followed by a *ReLU* activation function, and an output layer corresponding to the number of possible actions. The network is trained using the Adam optimizer with a learning rate of 0.0007. The agent is configured with a mini-batch size of 800, a discount factor of 0.95. An epsilon-greedy exploration strategy is applied, starting with an exploration rate (ϵ) of 1, which decays to a minimum of 0.01 with a decay rate of 0.99.

4 Methodology

The DDQN RL-C discussed in Sect. 3, along with the PUC7 environment, are integrated into a reinforcement learning iterative framework (RL-IF), as illustrated in Fig. 1. In this setup, the two penalty values, α and β , are initialized for each parameter

trial and fed into the RL-IF to train the RL-C for 3000 episodes. All possible combinations of α and β values are systematically evaluated, where $\alpha \in \{0, 1, 5, 10, 100\}$ and $\beta \in \{0, 0.1, 0.5, 1, 10\}$, resulting in a total of 25 parameter iterations. Three of these cases have $\alpha = \beta$, while the remaining cases are designed to evaluate the effect of applying different penalty values separately. In addition, these variations allow for assessing the impact of prioritizing a higher current penalty over the voltage penalty, reflecting the relative importance of current control in a grid-tied system.

To investigate the impact of the penalty terms in the reward function on the training efficiency of the DDQN algorithm in identifying the optimal RL-C agent for controlling the PUC7 system, randomness in each test must be eliminated to ensure a fair and accurate comparison. In this study, environmental randomness is eliminated by keeping the PUC7 and grid voltage parameters fixed throughout all episodes. The details of these parameters are provided in Sect. 2.

Additionally, to maintain consistency across parameter trials, the initialization of the NN and the random sampling process from the experience replay buffer in DDQN must also be controlled, as DDQN follows an off-policy learning approach. The NN initialization is standardized using *He* initialization, as proposed in [5], which is well-suited for the *ReLU* activation function used in the RL-C design, as described in Sect. 3. This initialization method is designed to keep activations within a stable range, preventing vanishing or exploding gradients and ensuring robust training in DNNs. Furthermore, each parameter trial is initialized with a fixed random seed to maintain consistency across all trials and eliminate variability in training results.

The effectiveness of incorporating penalty terms into the reward function and the influence of penalty magnitude are evaluated based on the maximum achieved reward, the episode at which the best agent is obtained, and the performance of the best agent in each case. The performance is assessed in terms of the THD of i_g , the mean absolute error of e_v , and the switching frequency.

5 Simulation Results and Discussion

The performance of the PUC7 inverter controlled by the RL-C without penalty terms ($\alpha = \beta = 0$) is considered the benchmark against which the RL-C performance with penalty terms is compared. Figure 2 illustrates the steady-state performance of the designed RL-C, where the seven DC levels are generated at the inverter output terminals, the grid current tracks its reference signal and remains in phase with the grid voltage, and the capacitor voltage is regulated around its reference value. The THD of the current is measured at 0.86%, while the mean absolute error (MAE) of e_v is 0.06, with a capacitor voltage range of 0.54 V. This performance is achieved with an average switching frequency (f_{sw}) of 9.5 kHz. The trained agent without penalty reached a maximum episode reward of -58.4 after 2392 episodes. Meanwhile, the mean episode reward of the top 10 agents in the benchmark case is -60 , with a standard deviation (std) of 0.82. The small difference of 2.6 between the mean and

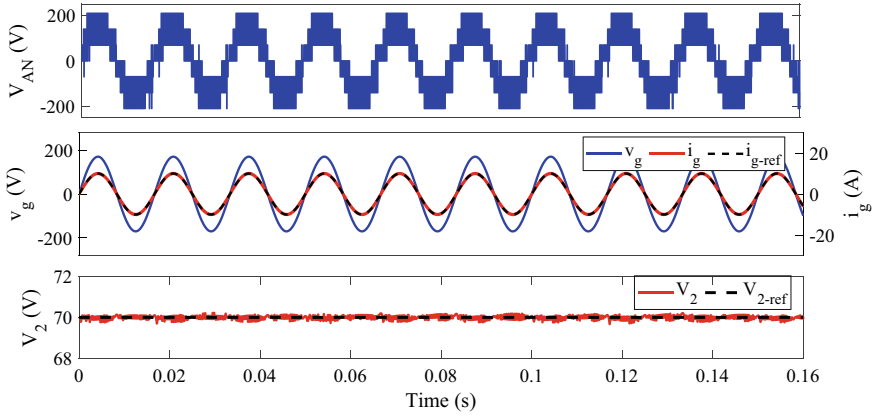


Fig. 2 The steady state performance of PUC7 controlled by RL-C with no penalty: the output voltage, the grid current, the grid voltage and the capacitor voltage

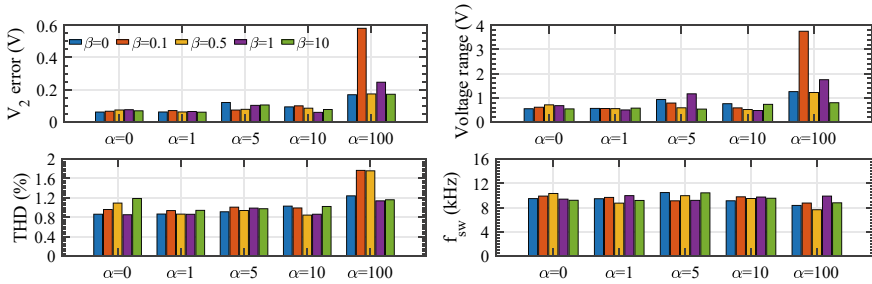


Fig. 3 The resulted performance of the best agent under different penalty values

the maximum reward indicates that similar performance was achieved multiple times during training, suggesting that the peak performance was not a one-time event.

The same performance metrics are evaluated for all 24 studied cases with penalty terms included in the reward function. The results are illustrated in Fig. 3. They show that all cases achieved satisfactory performance, with current THD below 2%, e_v less than 0.6 V, and an average switching frequency (f_{sw}) is around 10 kHz. However, cases with a low α value—indicating a lower current penalty—achieved performance closer to the benchmark case. Although all cases met the control objectives, applying a high current penalty ($\alpha = 100$) led to worse performance compared to the no-penalty case, including higher current THD. Furthermore, as shown in Fig. 4, the best episode rewards for $\alpha = 100$ cases did not exceed the benchmark reward. That said, in some instances, these cases achieved their rewards faster than the no-penalty case. As illustrated in Fig. 5, the mean of the top 10 episode rewards for the $\alpha = 100$ cases is significantly lower than those of the other cases, indicating that the performance was not consistently high throughout the 3000 training episodes.

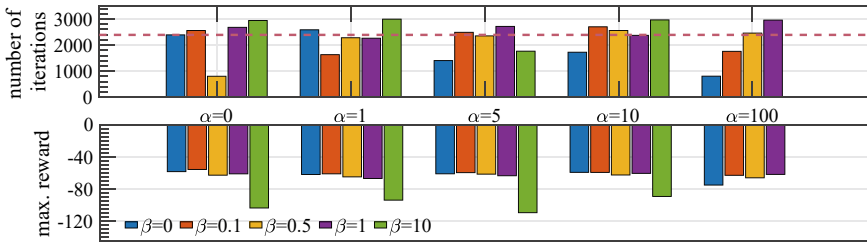


Fig. 4 The maximum reward achieved for different penalty values

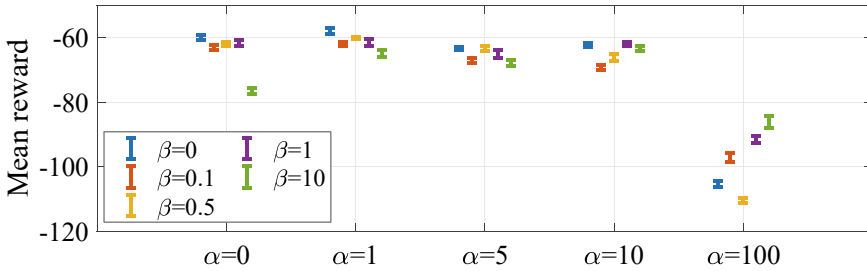


Fig. 5 The mean and standard deviation of the reward value for the best 10 agents

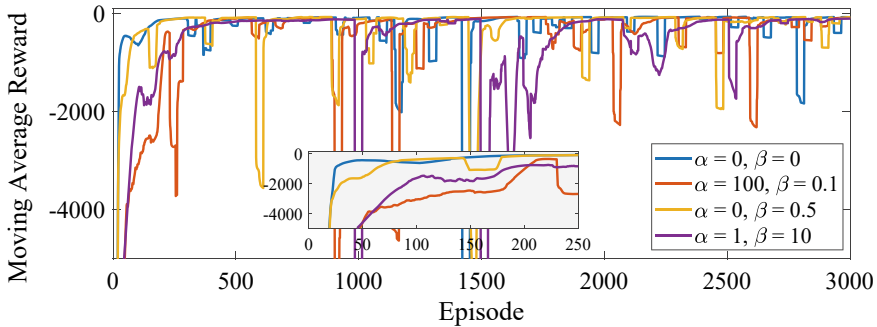


Fig. 6 The averaged learning curve for four cases

The results in Fig. 4 show that setting $\alpha = \beta$ within the studied range did not lead to the fastest training. In contrast, using only one penalty term, as in the case of ($\alpha = 0, \beta = 0.5$), or applying a moderate difference between the two penalty terms, as in ($\alpha = 1, \beta = 0.1$), can accelerate the learning process. For instance, the best agent in the ($\alpha = 0, \beta = 0.5$) case reached a reward of -62.8 in just 805 episodes.

Due to the balance between exploration and exploitation in the RL framework, the averaged learning curve is typically plotted instead of the episode-by-episode reward curve. Figure 6 shows the averaged learning curves, smoothed using a window width of 30, for four selected cases. As illustrated, the cases of $\alpha = 0, \beta = 0.5$ and the

benchmark case achieved higher average rewards compared to cases with larger penalty values in the first 100 episodes. In addition, the case with high penalty values for both terms ($\alpha = 1$, $\beta = 10$) exhibited less stability during the exploitation phases, indicating more fluctuations in performance despite convergence.

An additional training phase was initialized with $\alpha = 1000$ and $\beta = 10000$. Under these settings, the agent failed to learn how to control the system within the 3000 episodes, resulting in a current THD of 108% and a mean e_v of 4V. This result highlights how excessively high penalty values can hinder the learning process and prevent the agent from achieving meaningful control performance.

Although the presented results and analysis are for a single MLI topology (PUC7) using one RL-C algorithm (DDQN), the findings can offer practical guidelines for designers and researchers when incorporating penalty terms into the reward function of RL-based controllers for MLIs. In summary, the results suggest that including a single penalty term for dual-objective control problems is more effective than applying two simultaneous penalties for both objectives. This observation may be interpreted as a result of the interrelation between the two control objectives in MLI systems, which, in practice, may not require separate penalization. Furthermore, the findings show that applying moderate penalty values can accelerate the training process without compromising the achieved control performance when compared to high penalty values.

6 Conclusion

This paper presented a systematic investigation into the impact of penalty terms in the reward function of a reinforcement learning controller (RL-C) designed for a 7-level Packed-U-Cell (PUC7) multilevel inverter, using the Double Deep Q-Network (DDQN) algorithm. By standardizing the system parameters, neural network initialization, and environmental conditions, the study effectively isolated the influence of penalty terms on training performance. Two distinct penalties (for current and voltage deviations) were explored across 25 combinations, with the zero-penalty case serving as a baseline. Across all trials, the RL-C aimed to minimize current total harmonic distortion (THD) and maintain capacitor voltage regulation. The findings indicate that applying a single penalty term is more effective than using both of them simultaneously. Additionally, moderate penalty values improved training efficiency and convergence behavior, while excessively high penalties tended to delay convergence and reduce learning stability.

Acknowledgements This publication was made possible by NPRP grants [12C-33905-SP-213], [12C-33905-SP-71] and [12C-33905-SP-220] from the Qatar National Research Fund (a member of Qatar Foundation). The authors gratefully acknowledge to projects “Research Impact Initiative SP5 and SP6”, and the support of the Texas A&M University at Qatar. The statements made herein are solely the responsibility of the authors.

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IntelliGuard: IoT-Enabled Autonomous Spybot Intelligence for Real-Time Surveillance in Next-Generation Security Applications



Dhruv Dhayal, Pratham Aggarwal, and Manzoor Ansari

Abstract The rapid advancements in ubiquitous computing have made surveillance robots vital in military and security systems. IoT and sensor miniaturization enable new autonomous monitoring capabilities where traditional fixed installations or human operators prove ineffective. IntelliGuard, an ESP-32-based IoT surveillance robot, integrates camera streaming, motor-driven navigation, and centralized processing for military applications. The system provides autonomous mobility, real-time surveillance, and secure data transmission while remaining cost-effective. Radar and ultrasonic sensors enable precise obstacle detection within three meters. Field testing revealed performance variations under different environmental conditions, with optimal functionality in low-interference settings. Response latency remains minimal while IoT connectivity enables remote operation. Performance directly correlates with sensor precision and connection quality. Future developments will focus on more accurate sensors, stronger communication protocols, improved navigation for irregular obstacles, and enhanced filtering algorithms to minimize interference effects. This study establishes a foundation for advanced IoT-connected robotic surveillance systems in military applications.

Keywords IoT surveillance · Autonomous spybot · Real-time monitoring · Secure IoT communication · Sensor fusion technology · Military surveillance

1 Introduction

While modern-day IoT applications are revolutionizing security and surveillance functions, organizations still lack comprehensive situational awareness in complex operational environments. Traditional surveillance systems face inherent limitations affecting their performance in real-world scenarios [1–3]. Unsupervised areas present

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significant security vulnerabilities that demand innovative technological solutions. Foundational research by Al-Fuqaha et al. established that IoT frameworks play a crucial role in overcoming these barriers through distributed sensor networks and real-time analytical processing [4]. Today's increasingly complex security landscape requires surveillance systems that minimize human supervision while maximizing detection accuracy and operational reliability [5, 6]. This paper presents IntelliGuard Ver. 2, an autonomous IoT surveillance system designed to address the limitations of existing solutions through multi-sensor integration, adaptive artificial intelligence, and computational processing. The system architecture combines optical sensors, radar detection, and ultrasonic technologies within a unified framework, representing a significant advancement in IoT surveillance. IntelliGuard Ver. 2 enables comprehensive environmental monitoring while reducing false positives through advanced pattern recognition algorithms and contextual analysis methodologies. The following sections detail the system's framework, components, implementation, and performance results across various operational scenarios. Recent IoT-enabled bots such as spot by Boston dynamics and TARDEC autonomous scouts in military defense highlight the real-world need for adaptable surveillance systems.

1.1 Problem Definition

Traditional surveillance systems face significant operational constraints in dynamic or hazardous military environments. Fixed installations lack mobility and adaptability, while conventional surveillance robots are limited by communication infrastructure, operational range, and sensor critical factors for timely tactical responses.

Let the operational efficiency of a surveillance system be defined as:

$$\eta_{\text{surv}} = f(M, A, R)$$

where mobility $M = \sum(i = 1 \text{ to } n)\alpha_i * m_i$ with α_i representing terrain-specific coefficients.

The convergence of IoT, AI, microcontrollers, and miniaturized sensors offers an opportunity to transcend these limitations. For a distributed IoT sensing network, system reliability is modeled as:

$$R_{\text{sys}} = 1 - \prod(i = 1 \text{ to } k)(1 - R_i)$$

Decision latency can be optimized through:

$$L_d = \min\left(\sum i = 1 \text{ to } m \left(\frac{D_i}{P_i}\right) + \sum j = 1 \text{ to } n \left(\frac{C_j}{B_j}\right)\right)$$

where D_i represents data processing requirements, P_i is processing capability, C_j is communication load, and B_j denotes bandwidth capacity.

This technological paradigm integrates multiple sensing modalities within unified architectures, providing unprecedented situational awareness while reducing human exposure to danger.

2 System Architecture: IoT-Driven Multi-layered Security Framework for Real-Time Threat Detection and Adaptive Response

IntelliGuard Ver. 2: newest version of IoT-based advanced security. It is meant to provide real-time surveillance capabilities like never before and unprecedented detection and automated intervention capabilities against threats. This type of multi-layered IoT architecture mounts a complete and almost seamless mechanism, highly precise threat mitigation across multiple environments of security, and is easily scalable and integrated. The system capitalizes an IoT-enabled architecture for real-time surveillance and anomaly detection, and automated alerts, and further situational awareness from edge computing [7] and cloud analytic services [8]. This hybrid solution connects the low-latency data processing and decision-making while ensuring smooth communication among security nodes that are interconnected. Some outstanding innovations in the area include efficient IoT-based sensor networks, secure encrypted data transmission, and public remote device synchronization, which prove a strong operational security posture in enterprise critical infrastructures. End-to-end connectivity in IoT will enable an organization to monitor its activities in real-time, develop mobile-based remote access, and realize predictive risk estimates, thus enabling proactive security measures to be taken by the organization against imminent threats. Besides, it modularizes all the above into an IoT architecture within which IntelliGuard Ver. 2 even enhances advanced sensor fusion, adaptive power management, and secured communication protocols to guarantee scalability and interoperability with existing security infrastructures. Furthermore, this layered system architecture provides improved resource utilization as well as real-time processing along with specific device management. Therefore, this solution represents a promising IoT implementation for smart surveillance and critical security applications. Through IoT-driven automation and advanced connectivity, IntelliGuard Ver. 2 sets a new standard for next-generation security solutions in intelligent monitoring, real-time situational awareness, and adaptive threat response to evolving security challenges. As shown in Fig. 1, the system architecture of IntelliGuard Ver. 2 illustrates the comprehensive data flow between components across all operational phases.

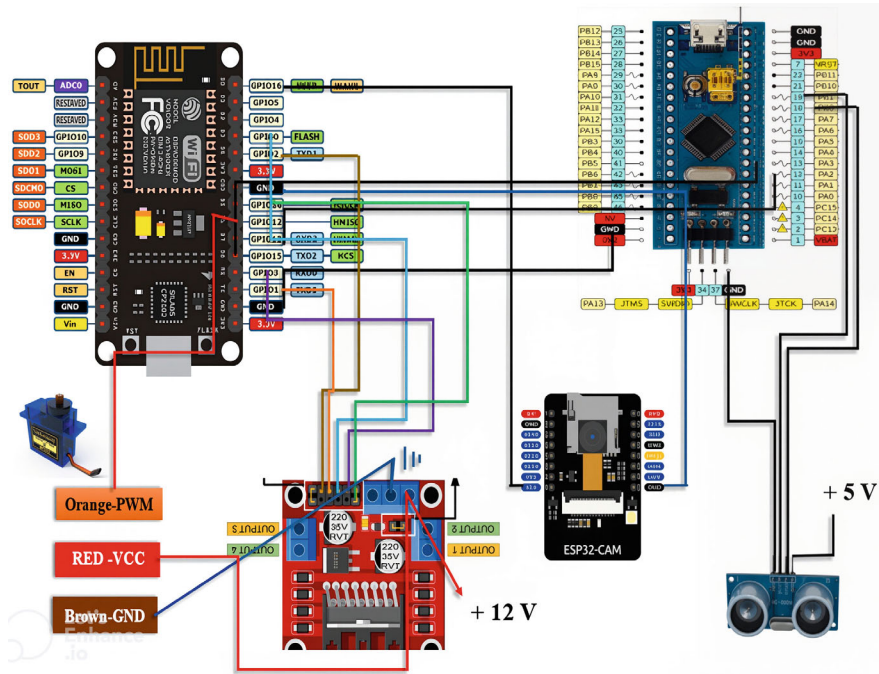


Fig. 1 System architecture of IntelliGuard Ver. 2 showing full system communication data-flow from radar/ultrasonic to GUI with proper integration

2.1 Components Used

The IntelliGuard surveillance system integrates advanced hardware to **gather environmental data, process real-time inputs, and enable secure communication for autonomous operation**. It utilizes a combination of **sensing, mobility, and communication** technologies to enhance situational awareness and adaptive response. Sensor fusion refers to combining data from multiple sensors to generate more accurate information. Radar uses radio waves to detect obstacles, while real-time analytical processing involves data analysis as it's collected. The system's capabilities, including data acquisition, processing efficiency, and connectivity, are discussed in detail, with hardware specifications outlined in Table 1: Hardware Components of the IntelliGuard Surveillance Robot.

Table 1 Components of the IntelliGuard surveillance robot

Components	Description
L298N microcontroller	Core controller managing data processing, sensor communication, and device operation
Wi-Fi module (ESP32 Cam)	Captures and transmits real-time video for remote surveillance via the internet
Radar system	Detects obstacles and moving objects, enhancing navigation and security
Ultrasonic sensor	Assists in obstacle detection and autonomous path planning
Servo motor	Controls the automated storage compartment for object handling
DC motors (100 RPM)	Enables multi-directional movement with speed control
Rechargeable battery	Ensures continuous power supply to all components ~ 2000 MAh
Power management circuit	Efficiently distributes power across all modules
NodeMCU (ESP8266)	Acts as a backup controller for wireless communication and IoT integration
EMF communication module	Allows short-range control (< 4 m) without internet or Wi-Fi
Private IP integration	Ensures secure, app-independent communication for controlling the robot
Wi-Fi module (ESP-32 built-in)	Enables long-range remote control and live streaming

3 IoT-Enabled Surveillance System Implementation and Workflow

IntelliGuard is designed as a versatile spybot for real-time monitoring and data acquisition in extreme environments, including military operations, space exploration, boreholes, mines, rescue operations, and firefighting scenarios. The combination of radar and ultrasonic sensors detects distant and close obstacles. The ESP-32 controller possesses the combined data to make it possible to navigate with great accuracy. Initial Detection is performed by radar, while ultrasonics measures nearer distances. Real-time data processing has made it possible to communicate data to the GUI. In testing, it proved that this combination increased maneuverability. All components are synchronized, creating an efficient autonomous system. The system enables autonomous surveillance and threat detection in locations inaccessible or hazardous to humans. Advanced sensors and communication technologies allow seamless operation in extreme environments, as illustrated in Fig. 2.

The operational framework shown in Fig. 3 illustrates the interconnections between key subsystems of the motion-controlled rover: Power System, ESP-32 Control System, Motor and Movement System, Radar System, Headlight System,

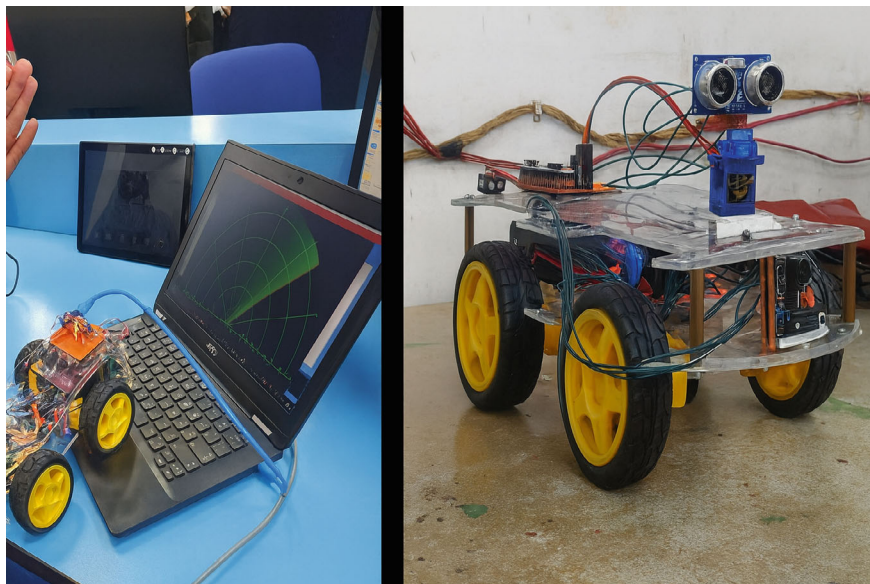


Fig. 2 Prototype of an autonomous spybot designed for object detection, angular and distance estimation within a 180° navigation field, featuring dual parameter (red/green) indicator system and remote surveillance capability for rescue and reconnaissance operations

and Remote-Control Interface. The Power System forms the foundation of operations, comprising lithium-ion batteries, battery chassis, a charger, and a power distribution board that stabilizes voltage for consistent delivery. At the core, the ESP-32 Control System functions as the central controller, processing inputs through its integrated camera, DPST switch, PL2303 TTL module, and Wi-Fi capabilities [9]. This controller interfaces with the Motor and Movement System, which enables mobility via DC motors on specialized terrain-handling wheels controlled by an L298D Motor Driver. Environmental awareness is enhanced by the Radar System, utilizing an HC-SR04 Ultrasonic Sensor mounted on an SG90 Servo Motor for rotational scanning. Sensor data is processed by an Arduino Nano and transmitted to a GUI for radar feedback [10]. The Headlight System improves visibility in low-light conditions, controlled directly through the ESP-32 module. The Remote-Control Interface integrates all subsystems, allowing operators to regulate movement, speed, lighting, and monitor live camera streaming through a User Control and Web Interface using Static Private IP Protocol for secure communication with user devices. This architecture demonstrates efficient power management and data exchange among components, creating a robust platform for autonomous surveillance in challenging environments. Current tests were performed in low-interface indoor conditions. Future evaluations will include rugged, outdoor, and high-noisy environments. Sensor data is processed through embedded algorithms in ESP-32. Upcoming Versions will integrate edge-AI modules for adaptive decision-making in dynamic environments.

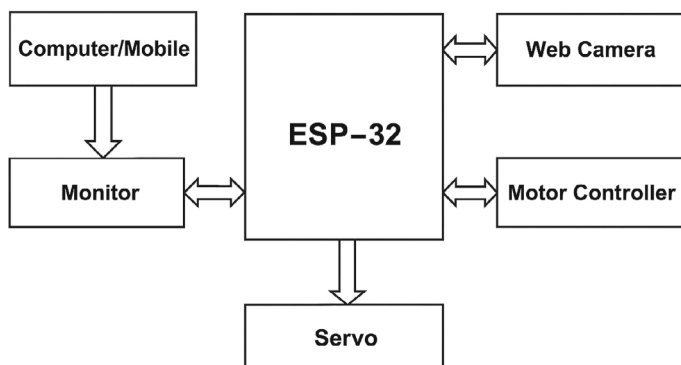


Fig. 3 Manual operational design (systematic) displays the integration and communication between the components and controlled by user manually

4 Results and Discussion

The System achieved ~ 92% obstacle detection accuracy at a 3 m range and maintained ~ 250 ms average response time. Latency increased by 40% in a noisy environments. The rapid development of robotics, automation, and IoT has imposed some impact on nearly all sectors from security, military, and industrial applications. The blending of autonomous surveillance technology has brought great changes to vital operations in terms of improving efficiency, safety, and accuracy. The study deals mainly with the successful building of the ESP-32-based surveillance robot for real-time monitoring and reconnaissance, proving applicable in many domains. Future evaluations will include rugged, outdoor and high-noisy environments. The working model with on real-time monitoring system gives feedback to the mentioned used in Fig. 4.

The robot uses a radar-based detection system to estimate the distance and angle of the object while simultaneously feeding the information in real-time to the GUI interface. This helps to detect obstacles based on gathered data object angle and distance.

Additionally, secure monitoring within the 4 m range free from Wi-Fi/EMF using private IP, while long-range monitoring (1 km) would need a network-enabled Wi-Fi network. Due to the design of the system for high-risk areas, it deals with problems of borewell rescues, coal mine monitoring, and pipeline inspections thereby protecting human life during such hazardous conditions. Its mobility, powered by DC motors, allows it to go over various terrains in industry, military, and emergency applications [5, 11]. The implementation thus highlighted the possibility of low-cost applications run by AI where real-time data acquisition and remote access are of utmost importance. Compared to ScoutBot and Mars Rover variants, IntelliGuard offers wider mobility, real-time data access, and cost-efficiency due to an ESP-32-based design. Future improvements will focus on the integration of high-precision sensors to further promote accuracy and adaptability. Additionally, these systems will be

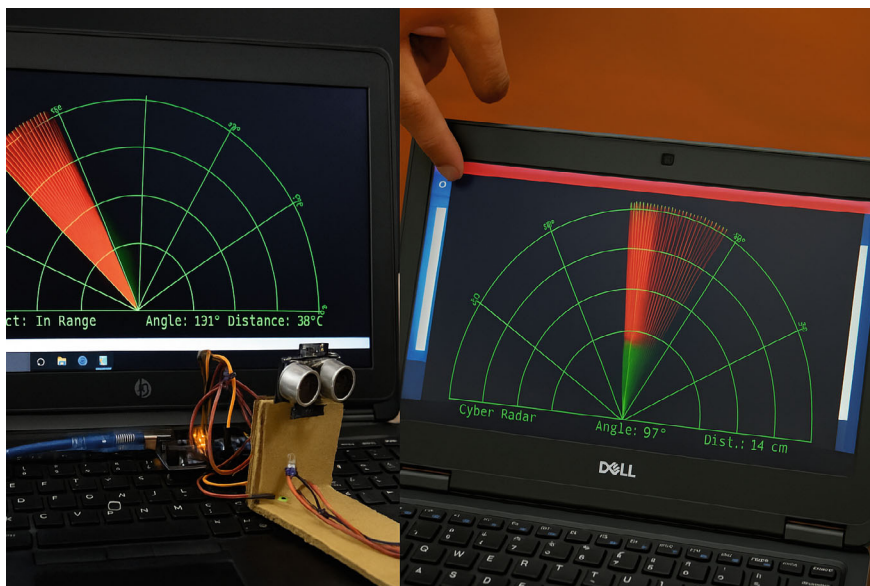


Fig. 4 System workflow for autonomous surveillance and rescue operations

enhanced through the introduction of machine-learning-based sensor fusion, LiDAR, and SLAM (Simultaneous Localization and Mapping), furthering robot autonomy capabilities in dynamic and unpredictable environments [12].

Discuss Limitations and Challenges: Testing Limitations include occasional Wi-Fi dropouts, ultrasonic sensors misreads in echoic spaces, and power reduction over the extended usage. **Real-Deployment Barriers:** For deployment, rugged housing, waterproof casing, and optimized battery performance are essential. Scalability in different terrains is under development.

Designing for highly challenging conditions like militarized zones, underground mines, and emergency rescue sites, IntelliGuard assists in reconnaissance and surveillance in terrains that otherwise may not be accessible. In rescue missions, it helps locate victims in collapsed structures or borewells. Industrially, it is used to inspect pipelines or hazardous chambers. However, to be deployed in such conditions, it needs rugged casing, water resistance, and optimized power utilization. After that, cost-effective mass production and being adaptable to varied terrains are the next hurdles for multitudes. This research offers a base for AI-driven surveillance robots that support their vital role in security, defense, and industrial automation, as well as minimizing human intervention in life-threatening situations.

The operational workflow illustrated in Fig. 4 demonstrates the system's autonomous surveillance and rescue capabilities through integrated subsystem communication. As shown in Figs. 5 and 6, a customized GUI interface was developed for tablet-based remote control of the system, properly integrated with a private IP connection (192.168.4.1) to ensure secure and responsive operation.

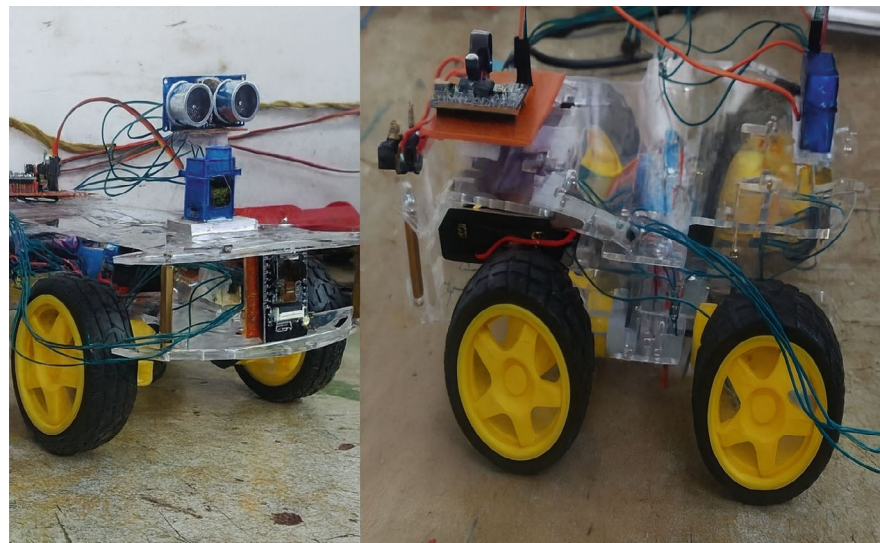


Fig. 5 IntelliGuard robot prototype with radar and ultrasonic detection system

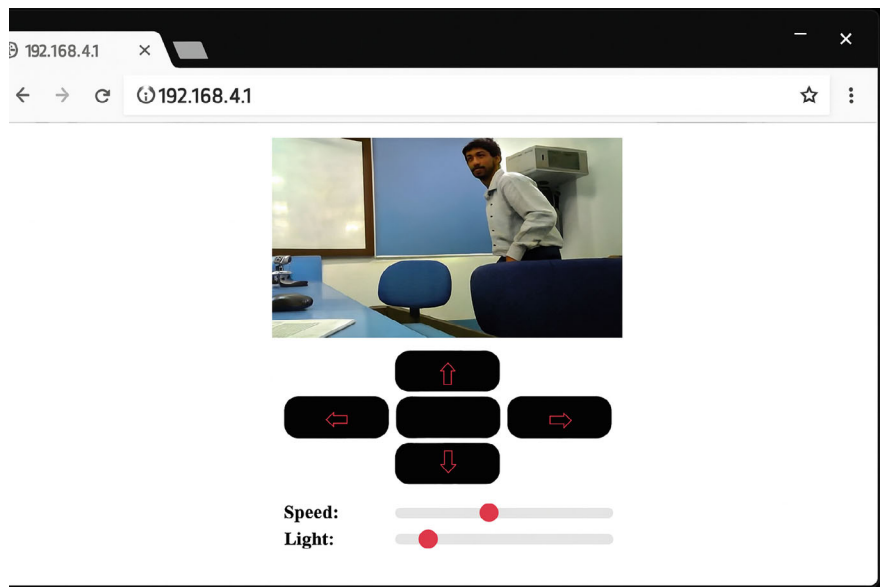


Fig. 6 Designing GUI interface on tablet to remotely control the systematic model by proper integration with private IP connected 192.168.4.1

5 Conclusions and Future Direction

The IntelliGuard surveillance robot represents a very important step forward in IoT-enabled autonomous monitoring systems that offer considerable advantages over conventional surveillance methods in mobility, security, and real-time adaptability. Still, further validation by large-scale field trials and detailed comparisons with existing systems will enhance its practical impact and give a more concise appraisal of its benefits for real-world applications. Future work will involve field testing in diverse environments, integrating AI-based decision systems, and benchmarking against other surveillance technologies. LiDAR will be used for fine-grained depth sensing. SLAM techniques will allow IntelliGuard to operate in GPS-defined zones. ML algorithms will be unable to perform autonomous learning.

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An Optimized Learning Framework for Stock Market Prediction Using Multi-source Data and Grey Wolf Optimizer



Harmanjeet Singh, Rupali Dhir, Chander Prabha, Balamurugan Balusamy, Ahmad Alkhayat, and Nithya Rekha Sivakumar

Abstract The prediction of stock prices is an evolving domain, especially in emerging nations such as India, where it substantially influences governmental, commercial, and individual investments. The precision of traditional statistical and econometric models diminishes due to the difficulties encountered in handling non-stationary financial time series data. This matter is resolved using machine learning techniques that utilize historical stock prices to enhance predictions. Debt or equity financing, encompassing dual registration on domestic and international exchanges, is how corporations procure capital in global markets. Precise stock price forecasts are essential for educated decision-making and financial progress in global markets, as investors in these equities enhance capital flow. The proposed system utilizes an iterative optimization procedure to seek optimal solutions in the hybrid LSTM

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model by dynamically adjusting the placements of wolves throughout the search space. The performance of the proposed model is quantitatively evaluated using statistical measures, including the accuracy, coefficient of determination R^2 , Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE).

Keywords Blue-chip stocks · Bombay Stock Exchange · Grey Wolf Optimizer · Stock forecasting · Technical indicators

1 Introduction

Metaheuristic optimization methods have garnered considerable interest in machine learning and computational intelligence in the past two decades. Methods including Artificial Rabbits Optimization (ARO), Particle Swarm Optimization (PSO), Salp Swarm Algorithm (SSA), Equilibrium Optimizer (EO), Whale Optimization Algorithm (WOA), and Harris Hawks Optimization (HHO) have demonstrated efficacy in addressing high-dimensional, non-differentiable, and nonlinear challenges characterized by multiple local minima [1]. Their popularity arises from their capacity to evade local optima, their simplicity, flexibility, and derivative-free characteristics. These algorithms are classified into three primary categories: evolutionary-based (e.g., Genetic Algorithms), physics-based, and swarm-based (e.g., Particle Swarm Optimization, Ant Colony Optimization). Swarm-based methodologies, derived from the collective intelligence of biological systems, direct agents towards optimal solutions by utilizing both individual and collective knowledge [2]. Additional significant swarm-based methodologies encompass Glowworm Swarm Optimization (GSO), Cuckoo Search (CS), and Artificial Ecosystem-Based Optimization (AEO). In financial sectors, especially in stock market prediction, metaheuristics are frequently utilized to improve deep learning models and Artificial Neural Networks (ANNs). Despite its efficacy, drawbacks including early convergence and diminished diversity persist. This study examines the efficacy of metaheuristic-enhanced models in forecasting blue-chip stock fluctuations on the Bombay Stock Exchange, providing strategic insights for investors, analysts, and policymakers [3].

2 Literature Review

Recent studies have investigated hybrid optimization models to improve the accuracy of stock market predictions.

In the study [4], a hybrid deep learning model, GRU with LSTM optimized by the Golden Jackal Optimizer (GJO) optimizes batch size and dropout while dynamically selecting and tuning GRU and LSTM layers. The GJO-optimized model was better at forecasting, with WMT RMSE as low as 0.798 and considerable gains in RMSE, MAE, and MAPE metrics. GJO-GRU + LSTM reduced RMSE by up to 30% in 17

of 20 assets, outperforming PSO and GA versions. The GJO-optimized model has the lowest RMSE of 0.798 for Walmart (WMT) and MAPE of 0.654% for Procter & Gamble.

The study [5] introduces a hybrid model that combines the Gated Recurrent Unit (GRU) and Artificial Rabbit Optimization (ARO) algorithm to improve stock price prediction. ARO's adaptive search capabilities are used to tune hyperparameters in Dow Jones Industrial Average time-series forecasting in the ARO-GRU framework. Comparisons with GRU, BiLSTM, and LSTM models optimized by GA or ARO show higher performance. Compared to LSTM-ARO, the ARO-GRU model reduced MSE by 81.8% and MAPE by 64% for AAPL stock, with an average R^2 improvement of 5.3 and 83% less computation time.

The study by [6] presents a hybrid model that integrates Fuzzy Support Vector Regression (FSVR) with Grey Wolf Optimization (GWO). FSVR mitigates data uncertainty, whereas GWO enhances hyperparameter optimization, enhancing prediction performance. Comparative assessments against SVR, RF, and XGBoost indicate that FSVR-GWO attains the lowest MAPE (1.82%) and RMSE (2.35), surpassing conventional models like SVR (MAPE: 3.12%, RMSE: 4.21%), thereby illustrating GWO's efficacy in enhancing financial forecasting.

Likewise, [7] employs GWO to optimize GARCH and ARIMA parameters, improving forecast accuracy by 5–8% compared to traditional methods. The GWO-GARCH model demonstrates enhanced error measures for many stocks, including HDFC Bank (RMSE: 0.13%, MSE: 0.17%), Axis Bank (RMSE: 0.31%, MSE: 0.30%), and Infosys (RMSE: 0.23%, MSE: 0.24%).

The study by [8] combines Support Vector Regression (SVR) with the Slime Mould Algorithm (SMA) for predicting stock prices. Subsequent improvements via Biogeography-Based Optimization (BBO) and Grey Wolf Optimization (GWO) resulted in a R^2 of 0.991, RMSE of 149.248, MAPE of 0.930, and MAE of 116.260 for Nasdaq index data (2015–2023), demonstrating flexibility to market variations.

A separate study by [9] examines HGBBoost in conjunction with optimization methods such as PSO, SMA, and GWO. The results demonstrate that GWO-HGBBoost attains the greatest R^2 score of 0.988, underscoring the significance of hyperparameter optimization.

In the study [10], technical indicators and metaheuristic algorithms, such as GA, GWO-PSO, and WOA, are evaluated for stock prediction. WOA demonstrates enhanced prediction efficacy, surpassing ARIMA and alternative models. Ultimately, [11] assesses an ANN model optimized using GWO utilizing BSE data from 2004 to 2018. The enhanced ANN-GWO model markedly surpasses the conventional ANN, validating its effectiveness in stock price forecasting.

3 Methodology

Figure 1 depicts a detailed approach for forecasting stock prices with Long Short-Term Memory (LSTM) networks, augmented with technical indicators and refined by the Grey Wolf Optimizer (GWO). The framework consists of several interrelated stages: data collection, preprocessing, feature engineering, model training, hyperparameter tuning, and model validation.

3.1 Data Collection and Technical Indicators

This study utilizes stock market data from the Bombay Stock Exchange (BSE), concentrating on the stocks of seven prominent Indian corporations: Reliance Industries, HDFC Bank, ICICI Bank, Kotak Bank, Infosys, Tata Consultancy Services (TCS), and Larsen & Toubro (LT). The information covers the period from April 1, 2020, to March 31, 2025, and comprises five essential variables: opening price, highest price, lowest price, closing price, and trading volume. To augment predictive capability, technical indicators like the Relative Strength Index (RSI), Bollinger Bands, and Moving Average Convergence Divergence (MACD) are formulated from OHLC values. These attributes encapsulate market movement, volatility, and tendencies. The dataset undergoes preprocessing to reduce bias, maintain consistency, and furnish a reliable input for model training. Forward-fill and backward-fill methods are employed to rectify missing values in indicator computations. The resultant dataset is

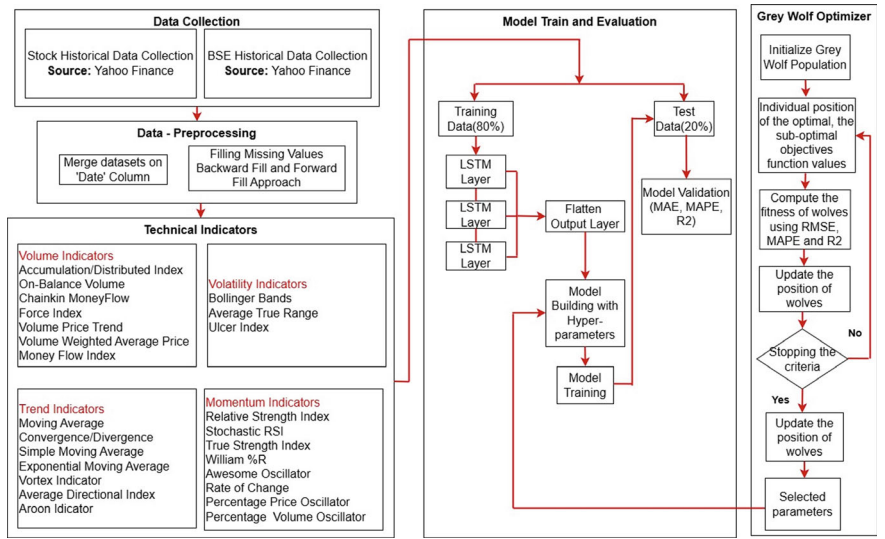


Fig. 1 The proposed framework utilizing the Grey Wolf Optimizer

highly compatible with neural reinforcement learning models in algorithmic trading contexts.

3.2 Model Training with Grey Wolf Optimization

This research utilizes the GWO to optimize hyperparameters for a Long-Short Term Memory (LSTM) model in stock price prediction. The GWO, influenced by grey wolves' hierarchical leadership and hunting tactics, optimizes LSTM units, batch size, epochs, dropout rate, and learning rate, as shown in Table 1. The optimization process commences with 10 wolves exploring within specified parameter constraints, honing in on the best three solutions—Alpha, Beta, and Delta—according to Mean Squared Error (MSE). A 90-time-step window organizes stock data for supervised LSTM training, utilizing the Adam optimizer for optimization. The evaluation of model performance employs statistical metrics such as RMSE, MAE, MAPE, R^2 , and Accuracy, each offering distinct insights into the model's predictive efficacy for financial time-series forecasting.

4 Experiment Study

Figure 2 illustrates the stock prices and trading volume of Reliance Industries from 2020 to 2025, emphasizing the effects of COVID-19 and the rebound subsequent to 2023. OHLC data indicates substantial volatility in 2024, with prices surpassing Rs. 1600 before declining, while orange bars denote associated variations in trading volume.

Figure 3 depicts BSE index prices and trade volume from 2020 to 2025, emphasizing the effects of COVID-19 and the recovery following 2023. The index ascends from 30,000 to above 80,000 before exhibiting fluctuations, accompanied by notable variations in trade volume during periods of market volatility. Figure 4 depicts actual versus predicted stock prices for RELIANCE.BO from May 2024 until April 2025.

Table 1 The summary of parameters range assigned to the Grey Wolf Optimizer framework for optimal model training

Parameter	Range
LSTM units	64–128
Dropout	0.2–0.3
Batch size	32–64
Epochs	100–250
Learning rate	0.01–0.0001

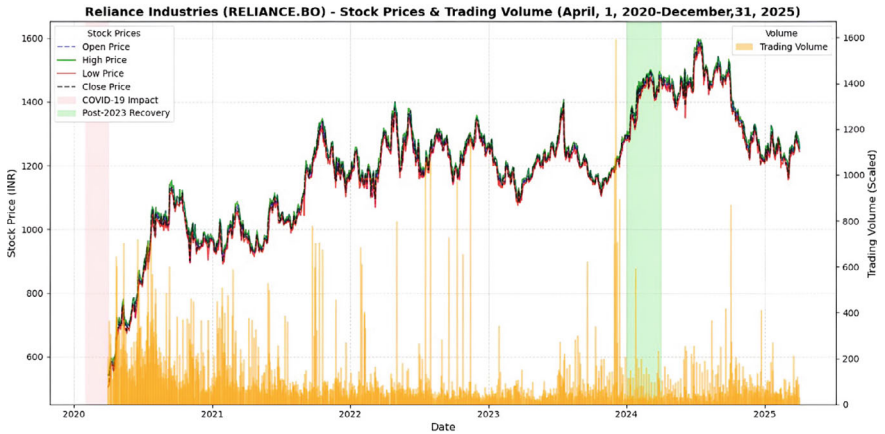


Fig. 2 The OHLCV indicators of RELIANCE.BO range from April 1, 2020, to March 31, 2025, generated from BSE index

The red solid line denotes actual pricing, and the green dashed line illustrates projections. The model accurately reflects patterns, emulating peaks and valleys. Notwithstanding slight variations in unstable areas, the LSTM-GWO model exhibits robust generalization and prediction precision for financial time-series data.

Table 2 displays the performance assessment of a Grey Wolf Optimization (GWO)-optimized LSTM model utilizing the Adam optimizer for stock price forecasting across several tickers. The model attains elevated predicting precision, with values spanning from 92.3 to 94.8%. Reliance has the best accuracy at 94.8%, whereas TCS demonstrates the lowest at 92.3%. The RMSE values span from 0.06

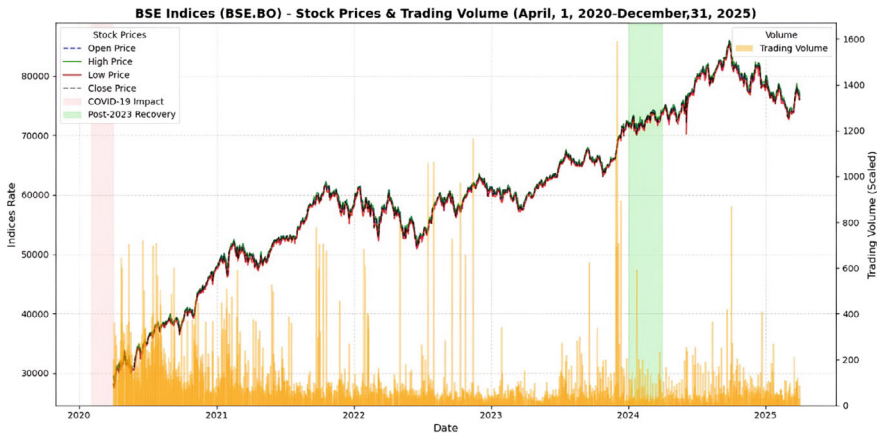


Fig. 3 The OHLCV indicators of BSE.BO range from April 1, 2020, to March 31, 2025, generated from BSE index

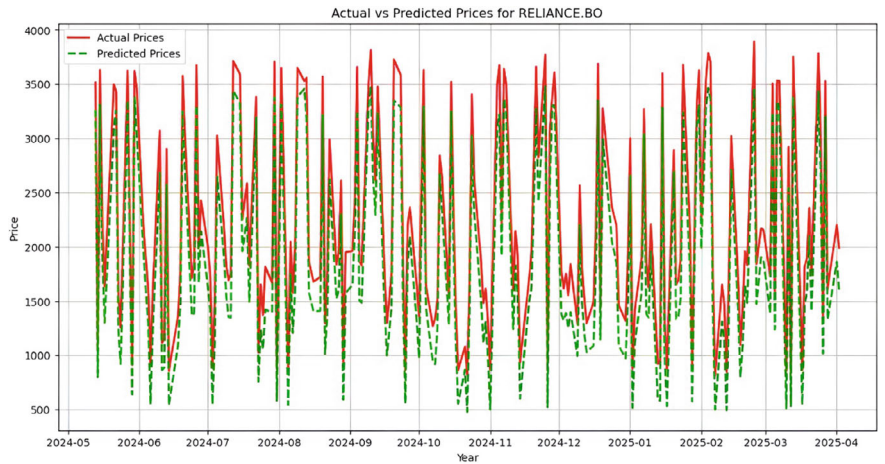


Fig. 4 Comparative analysis of actual and predicted stock prices for RELIANCE. Business operations from April 1, 2020 until March 31, 2025

Table 2 Grey Wolf Optimizer LSTM with Adam demonstrates forecasting results in terms of RMSE, MAPE, R^2 , and accuracy with optimal hyperparameter tuning

Ticker	Units	Dropout	Learning rate	Batch-size	Epoch	Accuracy	RMSE	MAPE	R^2
Reliance	128	0.3	0.0001	32	200	94.8	0.06	0.013	0.96
TCS	128	0.2	0.001	32	175	92.3	0.09	0.014	0.95
HDFCBANK	128	0.3	0.001	32	175	93.6	0.08	0.016	0.95
ICICIBANK	64	0.3	0.001	32	150	94.1	0.08	0.015	0.94
KOTAKBANK	64	0.3	0.0001	32	150	94.0	0.10	0.015	0.95
INFY	128	0.2	0.001	32	200	93.9	0.07	0.014	0.93
LT	64	0.2	0.0001	32	150	93.1	0.07	0.017	0.93

to 0.10, signifying minimal prediction errors. MAPE values persist beneath 0.017, affirming robust generalization. The R^2 scores, exceeding 0.93, indicate a strong connection between actual and anticipated values, illustrating the model’s efficacy in capturing stock price movements.

4.1 Statistical Significance Analysis

The GWO-optimized LSTM model was assessed for robustness by paired t-tests conducted over 30 trials with different initializations. In comparison to the baseline LSTM, it demonstrated statistically significant enhancements ($p < 0.01$) in RMSE,

Table 3 Performance assessment using 95% confidence intervals and paired t-test outcomes across 30 independent trials

Ticker	Metric	Baseline LSTM (mean $\pm$ CI)	GWO-LSTM (mean $\pm$ CI)	<i>p</i> -value	Significant
RELIANCE	RMSE	0.092 ± 0.004	0.060 ± 0.003	< 0.001	Yes
	R^2	0.91 ± 0.02	0.96 ± 0.01	< 0.001	Yes
TCS	RMSE	0.112 ± 0.006	0.090 ± 0.004	0.002	Yes
	R^2	0.89 ± 0.03	0.95 ± 0.02	< 0.001	Yes
HDFCBANK	RMSE	0.108 ± 0.005	0.080 ± 0.004	0.001	Yes
	R^2	0.90 ± 0.03	0.95 ± 0.02	< 0.001	Yes
ICICIBANK	RMSE	0.110 ± 0.005	0.080 ± 0.004	0.001	Yes
	R^2	0.89 ± 0.03	0.94 ± 0.02	< 0.001	Yes
KOTAKBANK	RMSE	0.114 ± 0.006	0.100 ± 0.005	0.002	Yes
	R^2	0.88 ± 0.03	0.95 ± 0.02	< 0.001	Yes
INFY	RMSE	0.105 ± 0.006	0.070 ± 0.004	< 0.001	Yes
	R^2	0.87 ± 0.04	0.93 ± 0.03	< 0.001	Yes
LT	RMSE	0.102 ± 0.005	0.070 ± 0.003	< 0.001	Yes
	R^2	0.89 ± 0.03	0.93 ± 0.02	< 0.001	Yes

MAPE, and R^2 as shown in Table 3. Confidence intervals validated little variance and uniform model performance across all evaluated stocks.

5 Conclusion

This research presents a forecasting system that integrates multi-source financial data with Long Short-Term Memory (LSTM) networks, optimized using the GWO. Experimental findings on prominent BSE equities demonstrated robust performance, with RMSE as low as 0.06, MAPE below 0.017, and R^2 values surpassing 0.93. Model accuracy varied from 92.3 and 94.8%, bolstered by statistically significant enhancements ($p < 0.01$) and consistent convergence trends. The model is engineered for incorporation into algorithmic trading and financial decision-support systems, capable of producing real-time buy/sell signals. Practical implementation must consider delay, slippage, and market fluctuations. Future directions encompass Bayesian deep learning, uncertainty quantification, and reinforcement learning.

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Atrial Fibrillation and Arrhythmia Classification Using 12-Lead ECG: Comparing ML Models



Ismaeel Al Ridhawi, Hasan Fayyad-Kazan, and Ali Abbas

Abstract The early and accurate detection of Atrial Fibrillation (AFib) and Arrhythmia from Electrocardiogram (ECG) data is crucial to minimize the risks of heart-related complications. This paper introduces a Machine Learning (ML)-based solution that can detect and classify heart-related conditions using data collected from 12-lead ECG. The solution primarily focuses on detection and classification between Normal Sinus Rhythm (NSR), AFib, and Various Arrhythmias (VA) using time-domain and frequency-domain features. Five ML algorithms were evaluated, namely, Support Vector Machines (SVM), Random Forest (RF), Xtreme Gradient Boosting (XGBoost), K-Nearest Neighbors (KNN), and Adaptive Boosting (AdaBoost). The models used the PTB-XL dataset providing access to 12-lead ECG records for 6,428 patients. Results show that XGBoost outperforms the other models, achieving an accuracy of 85.67%. Time-domain features were found to be more reliable for short-duration ECG recording. The solution was deployed into an E-Hospital platform to provide diagnostic assistance for healthcare professionals.

Keywords Atrial fibrillation · Arrhythmia · Electrocardiogram · Machine learning · Internet of things

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1 Introduction

Atrial Fibrillation (AFib) is a type of cardiac arrhythmia, characterized by irregular heart beats, which often causes an abnormally high heart rate. AFib is the most prevalent type of cardiac arrhythmia and the prime cardiac related contributor of strokes. Age, high blood pressure, pre-existing cardiac and lung diseases, congenital heart disease, and increased alcohol intake are a few risk factors for AFib. Its detection plays a significant role in monitoring cardiovascular health. Due to its often asymptomatic nature, it poses significant challenges in early detection and treatment. Electrocardiogram (ECG) analysis is essential in diagnosing a patient with AFib. ECG is employed to diagnose complex arrhythmias and other cardiac abnormalities in addition to basic heart rate and vital rhythm analysis. On an ECG, AFib can be distinguished by irregular patterns without any distinct p-waves. The heart rate for patients with AFib generally ranges between 110–140 bpm and the ventricular rate usually falls between 80–180 bpm.

This paper presents an AFib detection model that can differentiate between Normal Sinus Rhythm (NSR), AFib, and other Various Arrhythmias (VA) using ECG patient data. A complete feature extraction approach was adopted that involves extracting time-domain and frequency-domain features from ECG signals for 6,428 patients. Six different time-domain features were extracted using the RR-intervals from the ECG waveforms. Frequency-domain analysis was also employed in parallel to decompose the ECG waveforms into their frequency components using Welch's method [1]. Power Spectral Density (PSD) of different frequency ranges and ratio of low and high frequency PSD were used as spectral features to provide information about cardiac rhythm dynamics and arrhythmias. The extracted features are then used by different Machine Learning (ML) algorithms to detect for AFib and VA. The ML models used included Support Vector Machines (SVM), Random Forest (RF), Xtreme Gradient Boosting (XGBoost), K-Nearest Neighbor (KNN), and Adaptive Boosting (AdaBoost). The PTB-XL electrocardiography dataset was used to train and test the ML classifiers [2]. The ML algorithms were trained to learn the complex inherent relationships between the extracted features and the corresponding classification of NSR, AFib, and VA, so that they can classify new patient ECG data.

Evaluations were conducted to determine the best performing classifying model using precision, recall, accuracy, and f1-score performance metrics. An API was created for the system in order to be integrated into an existing E-Hospital platform, to allow healthcare professional to use ECG data acquired from patients and diagnose for AFib and other arrhythmias in a timely manner. The rest of this article is organized as follows: Sect. 2 highlights some state-of-the-art work conducted in the area of AFib detection using ML. Section 3 provides a deep discussion on the proposed methodology and framework. Performance evaluation results are looked at in Sect. 4. Finally, in Sect. 5 concluding remarks are drawn from this work.

2 Related Work

The application of ML in cardiology has greatly enhanced the diagnosis process of heart-related diseases. The authors in [3] developed a method for R-peak detection to calculate RR intervals and analyze cardiac rhythm dynamics. Heart rate variability (HRV) was used to measure the natural variations in heart rhythms as it is used as an indicator for arrhythmia and AFib. The analysis included four metrics, namely, standard deviation of normal-to-normal (SDNN), for measuring total heart rate variability, root mean square of successive differences between normal heartbeats (RMSSD), for analyzing short-term heart rate fluctuations, the percentage of successive normal interbeat intervals greater than 50 ms (PNN50), for detecting large heartbeat interval differences, and mean RR, for establishing a standard heart rhythm rate. The work implemented a similar approach to the work presented in this article in terms of extracting time domain features, which provided ML algorithms with ECG recording insights to detect arrhythmias. The results conclude that the detection method is integrable into wearable devices and can provide accurate heart health diagnosis.

In [4], the authors introduce a new technique to detect AFib and atrial flutter (AFL) using 12-lead ECG recordings through a higher-order tensor framework that maintains spatial (inter-lead) and temporal (beatwise) information. The 12-lead ECG signals are divided into individual beats for segmentation before converting these multilead beats into time-frequency space using the Chirplet transform. The representations stack into a four-dimensional tensor structure (lead $\times$ beat $\times$ time $\times$ frequency) before applying multi-linear singular value decomposition to extract discriminative features. The tensor-derived features become input for ML classifiers which include boosted decision trees and SVMs to identify healthy controls, AFib, and AFL. The method proves its effectiveness by using two public 12-lead ECG databases for inter-patient evaluation which results in high accuracy rates of approximately 90% for AFib detection and 91.7% for differentiating healthy patients from those with AFib/AFL conditions.

Kim et al. [5] presented SeqAFNet, as a deep learning (DL) model which specializes in beat-by-beat AFib classification of single-lead ECG signals obtained from adhesive patch-type devices. The proposed network analyzes RR-interval frames through a two-stage bidirectional gated recurrent unit (GRU) architecture, to detect both short and extended temporal patterns in irregular heart rhythms. SeqAFNet generates per-beat classification results through beatwise frame sliding before an ensemble decision process improves temporal prediction stability. The model achieves robust performance after training on standard public ECG databases, namely, AFDB and LTAfDB. The model produces high accuracy and F1 scores when tested on new patch-type ECG data. The model's beat-wise output structure enables the identification of brief misclassifications while following clinical guidelines that specify AFib episodes must last at least 30 s.

The work in [6] focused on the detection of cardiac arrhythmia using 12-lead ECG data. The authors used a DL model to detect AFib-based on features extracted from

ECG. The findings of the study demonstrated that 12-lead ECG data provides substantial advantages towards heart rhythm analysis, which guided the work in this article to use a 12-channel ECG dataset for arrhythmia detection. Mei et al. [7] presented an AFib detection system that uses features from short single-lead ECG recordings even when data quality is poor for wearable or mobile-health applications. The method uses heart rate variability (HRV) metrics with a focus on non-linear descriptors of RR-interval fluctuations together with spectral features analyzed through Welch's periodogram. The system detects R-peak events which remain stable under noisy conditions and uses time- and frequency-domain features together to detect both AFib cardiac dynamics. The work employs supervised models such as SVM and ensemble trees for classification. The study demonstrates high accuracy across varying noise levels using the 2017 CinC dataset.

3 Methodology and Framework

3.1 Dataset

The PTB-XL dataset is a large publicly available ECG dataset that contains 12-channel ECG waveform data for over 6,428 patients pre-labeled by cardiologists and three different diagnostic classes named NSR, AFib, and VA [2]. The class NSR refers to patients that have normal ECG data without any indication of irregular heartbeats. Class AFib is where the patients' ECG data indicates atrial fibrillation. All other types of arrhythmia indicated by the ECG data are grouped together as the class VA. The dataset's ECG recordings are captured for patients over a 10-second period at a high frequency resolution with a sampling rate of 500Hz. It includes the standard set of electrodes I, II, III, aVL, aVR, aVF, and V1 to V6, giving a detailed graphical representation of the electrical activity of the heart. The dataset is divided as follows: 2000 ECG data entries from the NSR category, 1587 entries from the AFib category, and 2841 entries belonging to the VA category. The patient demographics from the dataset revealed a notable trend in the occurrence of AFib and VA relative to the patient age. There is a noticeable increase in AFib risk among older patients suggesting a strong age-related correlation to AFib incidence. A positive correlation of 0.31 was noted between age and arrhythmia diagnosis. The demographic information related to gender did not show any significant correlation with the diagnosis of AFib or arrhythmia.

3.2 Data Pre-processing

The dataset included a metadata file together with an ECG reading file to support feature extraction and patient ID mapping. The metadata file includes the diagnosis

field named “*ritmi*” which classifies cardiac conditions into NSF, AFib, and VA. The *NumPy* library array contains ECG readings which match the number of records found in the 6428 individuals. Each patient’s ECG signal data appears as a single entry within the 3D *NumPy* array sampled at 500 Hz. The resulting 3D array of shape (6428, 5000, 12) corresponding to 5000 data samples for a 10 s recording collected from 12 individual channels. Data preparation involved importing these files while using “*patient_id*” and “*ritmi*” as data columns from the metadata file for ECG data labeling and categorization. This ensures that ECG recordings are accurately mapped to their proper diagnostic label.

3.3 Time-Domain Feature Extraction

The time-domain features include statistical parameters of ECG signals that describe the temporal related variations in an ECG waveform. The RR interval in an ECG indicates the heart rate, and is measured as the time between two consecutive R waves, which is the largest peak in an ECG waveform. The clinical diagnosis of AFib is based on the examination of abnormal RR-intervals in an ECG, hence, time-domain features were calculated from RR intervals [8]. These time-domain features helped in understanding the heart rate variability (HRV) and autonomic nervous system modulations, which are very important for the identification and diagnosis of AFib cases. For the detection of R peaks in ECG waveforms, we employed the function *ecg_peaks* from *NeuroKit2* [9]. Figure 1 shows the detected R-peaks in a sample ECG waveform. In total, six different time-domain features were extracted using the RR-intervals from the ECG waveforms, namely, Standard Deviation of RR Intervals (SDRR) for measuring total heart rate variability, RMSSD for analyzing short-term heart rate fluctuations, the number of consecutive RR intervals with more than a 50-millisecond difference (RR50) for measuring the high heart rate variability, percentage of RR50 (PRR50) for a standardized measure of the high heart rate variability, mean RR for establishing a standard heart rhythm rate, and HRV for measuring the status of the heart’s health.

3.4 Frequency-Domain Feature Extraction

The ECG data analysis through frequency domain provides alternative insights to temporal analysis while delivering important heart rhythm information. ML algorithms as well as healthcare professionals can get better understand cardiac rhythm dynamics through frequency domain transformation of ECG waveforms to detect arrhythmias. The extraction of frequency domain features from 12-lead ECG begins with converting time-domain ECG signals into frequency-domain signals. Welch’s method serves as the spectral estimation technique to analyze time-domain signals in this work. PSD estimation through this method shows how signal power distributes

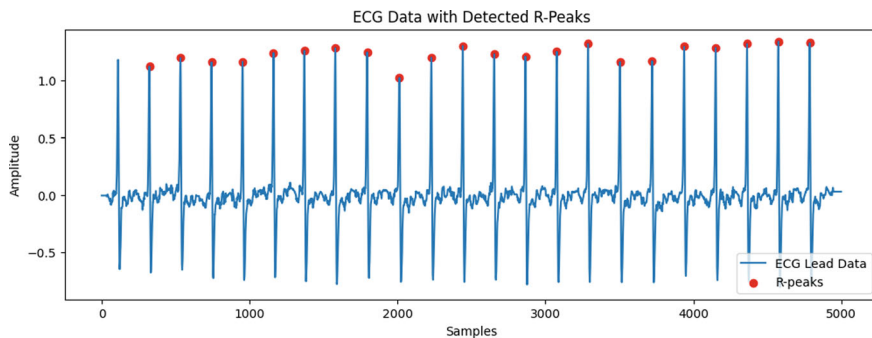


Fig. 1 R-peaks detected from a sample ECG waveform

across different frequency components. The PSD feature serves as an essential tool in clinical practice because it delivers complete power distribution across multiple frequency bands for cardiac signal analysis [10]. The PSD analysis of each ECG channel reveals spectral power distribution across very low frequency (VLF) (0.0033–0.04 Hz), low frequency (LF) (0.04–0.15 Hz), and high frequency (HF) (0.15–0.4 Hz) bands [11]. These frequency domain features enable assessment of HRV, where LF and HF reflects the balance between sympathetic and parasympathetic activity.

3.5 Data Splitting

Data was split into training set for algorithm training and testing set for model evaluation. We allocated 80% of the dataset for training purposes and 20% for testing purposes. We set a random state to achieve reproducibility and consistency during different runs. This ensures that the model is trained on a wide range of data and then evaluated on unseen data, reflecting real-world scenarios, where the model would encounter new patient data.

3.6 Machine Learning Framework

The ML framework applies several supervised learning algorithms for classifying ECG recordings based on the extracted time-domain and frequency-domain features. The classifiers used were SVM, RF, XGBoost, KNN, and AdaBoost. After the learning process, the classifiers have been set to recognize the complex relationships between the extracted ECG features and their respective diagnoses, thus enabling the classification of new ECG data. Figure 1 demonstrates a time-domain and frequency-domain plot, where (a) presents the original ECG waveforms with R-peaks marked for RR-interval measurement and the right panel shows the power spectral density

(PSD) plot. The visual inspection demonstrates that time-domain features track beat-to-beat variability essential for AFib detection and spectral features show autonomic nervous system imbalances through LF/HF power ratios which match clinical AFib manifestations. Each classifier is trained using the training subset of the data, and performance is assessed using the testing subset. Feature pre-processing pipelines were constructed for each classifier, incorporating strategies for imputation and standardization to ensure robustness against incomplete and varied-scale data. The metrics used to evaluate the models' generalization capabilities and its effectiveness at identifying different ECG-based diagnoses, namely, NSR, AFib, and VA, are accuracy, f1-score, precision, and recall. All the ML algorithms were tested on the separate 20% test data and the performance metrics were evaluated. Additionally, Confusion matrices were generated to provide insight into each model's classification behavior. The XGBoost classifier scored the highest F1 score, capturing the distinctions of ECG data. ROC curve plot (Fig. 2) was used to assess the discriminative power of the ML models.

3.7 Deployment

An Application Programming Interface (API) was developed to create a web framework which pre-processes ECG signals before using the best-performing trained ML model, namely, XGBoost for AFib detection. A front-end user interface was created to integrate with the API, which was then deployed into an E-Hospital platform [12]. The AFib diagnosis tool fetches patient ECG data from the E-Hospital database through the API. The API returns the predicted class, and is displayed on the user interface.

4 Performance Evaluation and Analysis

The ML models depicted different results in the classification of ECG signals. XGBoost scored highest with an accuracy of 85.68% and an f1-score of 85.67%, which was much higher than the other models as shown in Table 1. RF was the second highest accurate model with an f1-score of 83.46%. On the contrary, SVM, KNN, and AdaBoost scored less with f1-scores of 80.05%, 75.41%, and 77.32%, respectively. Figure 2 outlines the Receiver Operating Characteristic (ROC) plot for all models. ROC curves confirmed XGBoost's superior discriminative ability, with AUC scores of 0.92 for AFib, 0.89 for NSR, and 0.87 for VA. Random Forest followed closely (AUC: 0.90, 0.87, 0.85), while other models showed weaker separation between classes. While XGBoost achieved high accuracy levels, there are cases in which the short 10s ECG segments may obscure subtle arrhythmia. For example, paroxysmal AFib episodes in longer recordings could show transitional patterns not captured in these snapshots.

Table 1 Performance metrics of machine larning models

Model	Accuracy (%)	F1-score (%)	Precision (%)	Recall (%)
XGBoost	85.68	85.67	86.10	85.88
Random forest	83.78	83.46	83.58	83.44
SVM	82.16	80.05	80.37	80.02
KNN	75.51	75.41	75.49	75.51
AdaBoost	77.37	77.32	77.49	77.37

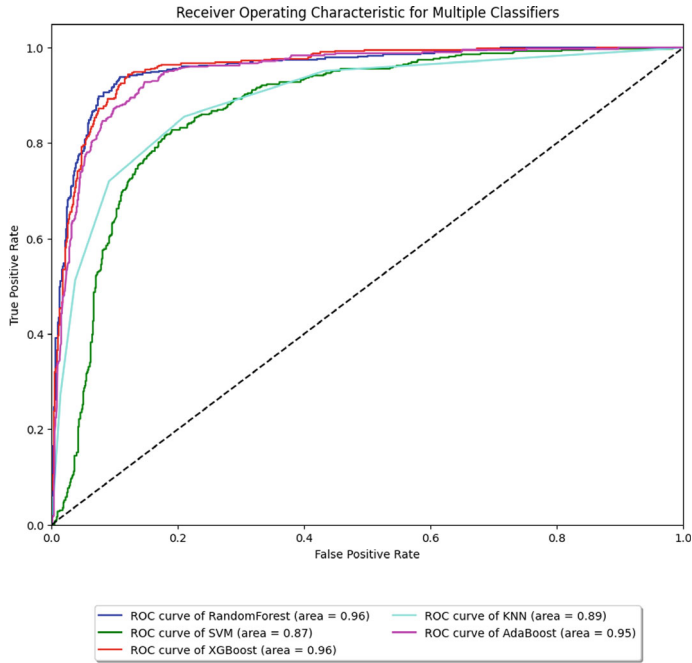


Fig. 2 ROC for all compared ML models

The time-domain features demonstrated better predictive capabilities than frequency-domain features with RR interval variability (SDRR, RMSSD) and HRV triangular index showing the highest discriminative power. The short recording duration of 10s most likely caused the low-frequency band spectral resolution to be restricted (VLF, LF). The LF/HF ratio from frequency-domain analysis delivered important information about autonomic nervous system imbalances that led to AFib. Class imbalance with more NSR samples slightly biased recall toward normal rhythms. Demographic variability, where age correlation is 0.31, suggested potential improvements with patient-specific tuning.

5 Conclusion

This paper presented a diagnostic model to detect AFib or VA from 12-lead ECG data for patients with heart-related issues. The RF and XGBoost models produced the best results with accuracy rates of 83.78% and 85.68%, respectively. Both time-domain and frequency-domain features were used, where the results highlighted the reliability of time-domain for short-duration ECG recordings. Performance analysis revealed that enhancing model robustness requires a larger number of ECG recordings that represent diverse demographic populations. The accuracy of the model would increase and the risk of overfitting would decrease with more data. Furthermore, time-domain metrics and morphological features derived from ECG waveforms can be used as additional features for analysis. For future work, advanced DL methods will be explored for automatic feature extraction from raw ECG signals. The application of DL may reveal hidden patterns in ECG data beyond traditional models, which could lead to enhanced diagnostic precision.

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Impact of Peak-Time Power Shortages on Thermal Comfort in Kuwaiti Homes: Toward a Cyber-Physical Energy Testing Framework



Abdalrahman Alajmi , Mohamad Rida , and Nick Kelly 

Abstract Power outages during peak hours in Kuwait can seriously impact the comfort of indoor spaces, especially during the intense summer heat. When cooling systems go offline, even for a short time, indoor temperatures can rise quickly, making homes uncomfortable and potentially unsafe. This study looks at how these energy limitations affect thermal comfort inside buildings and proposes a cyber-physical energy assessment framework aimed at improving energy efficiency while reducing discomfort during critical periods. We examine household temperature conditions, adaptive comfort techniques, and real-time power usage data using a mixed-method approach. Results show that outages during peak hours cause extreme heat stress, which causes residents to use ineffective cooling techniques. To help buildings better cope with power outages, the proposed cyber-physical framework brings together energy management systems, smart sensors, and predictive modeling. By combining these elements, the study aims to strike a balance between keeping occupants comfortable and using energy more efficiently. It also offers practical insights that can benefit homeowners, energy planners, and policymakers looking to make buildings more resilient and sustainable in challenging climates. This study highlights the urgent need to adopt smart energy solutions that can improve everyday living conditions in Kuwait while also addressing broader challenges related to long-term energy sustainability.

Keywords Cyber-physical systems · Thermal comfort · Residential cooling · Kuwait

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1 Introduction

1.1 Energy Problem in Kuwait

Kuwait is an arid country, which means that the climate is extremely hot during the summer season, reaching up to 50 °C, so it is no wonder that the major dependence is on mechanical air conditioning [1]. Because of its geographical location in the Arabian Gulf region, the country experiences a long summer period, with average temperatures between 45 and 50 °C [2]. Extreme weather conditions call for the use of AC systems not as a luxury but as a necessity, thus increasing energy demand greatly.

1.2 End-Use Energy in Buildings

Kuwait Cooling, which accounts for more than 60% of the electricity used in Kuwait's buildings [2], significantly strains the national grid, particularly during the summer when peak demands exceed 17,000 MW [3]. Factors contributing to this high demand include extreme temperatures, inefficient building standards, and low electricity prices that discourage conservation [1]. The rapid growth of high-rise buildings, especially in urban areas, further escalates cooling needs. There are also measures such as the implementation of higher standards for building codes to improve insulation and passive cooling [4], smart thermostats usage, and renewable energy, in particular, solar cooling systems [5]. It's important to note that simply changing the thermostat settings from 21 to 24 °C could reduce the cooling load by 10% [5]. As for the long-term perspective, the mentioned strategies imply the focus on energy-saving buildings, smart cooling technologies, and renewable electricity sources capable of meeting the demand without overloading Kuwait's environment.

1.3 Kuwait's Cooling Challenges and Power Demand

In June 2024, Kuwait, for the first time, had a scheduled load shedding, and this action happened despite the country's energy surplus. Increased electricity usage due to the scorching heatwaves that were experienced made the government turn off the power for one to two hours intermittently in different districts across the country. These outages imposed much thermal stress on the residents and also raised concerns as to the reliability of the national grid. To avoid total blackout along the streets, authorities came up with official conservation measures, expecting the citizens to use less energy during the peak hours of the day between 11 am and 5 p.m. [3]. The continuous rise in the consumption of electricity means an urgent need for Kuwait to develop demand side management (DSM) to address the stability of the power sector.

Out of information gotten from the Ministry of Electricity, Water, and Renewable Energy in Kuwait, it was revealed that Kuwait’s peak demand in 2024 was 17,640 and 18,569 MW is the expected peak demand in 2025, implying a yearly increment of about 1000 MW. If this trend persists, there may be periodic problems, such as load shedding or frequent disruptions of the power supply, especially during the hot summer seasons. Rising energy consumption makes it imperative that a load control system is designed for a smart grid that can aim at improving distribution in the grid. Such problems as overloaded networks, therefore, require both technological solutions and policy actions in order to address the issue of a sustainable and reliable electricity sector in Kuwait.

1.4 Influence of Set-Point Reduction on Energy Use and Thermal Comfort

One practical way to reduce cooling energy use is by adjusting thermostat set-points. For instance, raising the temperature from 21 to 24 °C can cut energy consumption by around 10%. However, this energy saving often comes at the cost of occupant comfort [5]. As indicated by the Predicted Mean Vote (PMV) index, higher set-points result in “slightly warm” PMV zones, but passive cooling strategies and approaches may mitigate the perceived discomfort. Studies have indicated that just improving insulation may reduce peak indoor temperature by 3 °C or greater, provided that cooling services are interrupted [4]. As shown in Table 1, set-point adjustments alone yield energy savings of 10%, which is not enough because its correlation with comfort is related, while insulation approaches achieve 12% energy savings and a moderate comfort rating. The most favorable approach is a combination of both, producing 20% energy savings while yielding an acceptable PMV value of – 0.2. This discussion places value on information-rich, real-time, cyber-physical energy systems within smart meters to adjust consumer cooling strategies and outdoor ambient conditions to balance energy efficiency with occupant well-being or comfort.

Table 1 Key strategies and their respective impacts on cooling energy consumption and thermal comfort

Strategy	Cooling energy reduction (%)	PMV score (comfort level)
Set-point adjustment	10	– 1.5 (less comfortable)
Insulation only	12	– 0.5 (moderately comfortable)
Combined strategy	20	– 0.2 (acceptable comfort)

2 Literature Review

2.1 *Cyber-Physical Energy Systems (CPES)*

The Fourth Industrial Revolution, commonly referred to as Industry 4.0, has brought about integrated intelligent systems, and CPES is at the center of energizing the energy sector [6]. Big data, Digital twin, or AI and decision-making systems have advanced the aspects of energy efficiency in smart grids, smart buildings, and Industrial energy management as espoused by Industry 4.0 standards [7]. CPES within Industry 4.0 has the feature of cyber-physical feedback, where power over certain assets is regulated and adjusted based on the outputs of physical sensors [8]. In the context of Kuwait, where there is high energy demand, this capability is effective in:

- Smart HVAC optimization
- Predictive maintenance for power grids
- Decentralized energy control.

In this way, through using the cloud, computing, and edge analytics, CPES optimizes energy flexibility while also strengthening the power grid performance and mitigating carbon emissions. Smart meters and cyber-physical testing platforms are essential in enhancing the effectiveness of building energy since the cooling load accounts for major electricity consumption in Kuwait [9].

2.2 *Challenges and Research Gaps*

Despite its strong potential, the implementation of the CPES framework in Kuwait faces several barriers: technical, economic, and regulatory. One of the key technological challenges is the lack of well-established communication protocols that can effectively link modern components like smart grid infrastructure and IoT devices with older, conventional systems. This disconnect makes it difficult to achieve seamless integration, which is essential for a fully functional and responsive cyber-physical energy system [10]. This inhibits smooth flow, synchronization, and coordination in real time. In the same regard, the susceptibility to cyber threats is a concern because cybercriminals can attack smart grid or IoT controllers to hinder energy processes. This is because it is capital-intensive, requiring investment in smart sensors, AI-driven controllers, and energy storage systems [11]. However, the barriers are the highly subsidized electricity prices, which discourage consumers from adopting smart energy technologies. It is also important to understand that several regulatory gaps slow the deployment of CPES. Insufficient policies hinder investments, and ineffective implementation of energy efficiency standards hinders the adoption of smart cooling solutions [12]. The absence of proper building energy codes adds to the country's high energy consumption [8]. To the best knowledge of the present authors, there are no prior studies done to investigate the effectiveness of CPES for

Kuwait's conditions regarding cooling requirements. Future research should target specific cooling systems that are cyber-physical, improved cyber security protocols, and cheap demand-side management strategies. By addressing the identified challenges, it will be possible to advance Kuwait to harness the benefits of CPES for long-term energy stability, improved and more reliable grid infrastructure, and climate-responsive solutions.

3 Methodology

3.1 Building Geometry

This study is built around a residential building model that reflects the typical design found across Kuwait's housing infrastructure. The model incorporates key elements such as building geometry, construction materials, and occupancy patterns to create a realistic and practical framework for analyzing energy use and evaluating cooling performance. The summary of the thermal and structural parameters of the building model adopted in the analysis, as presented in Table 2.

This specific building model serves as a benchmark for energy simulations, helping to evaluate its cooling needs, thermal comfort levels, and overall energy consumption within the context of Kuwait's harsh climate.

Table 2 Key thermal and structural characteristics of the building model [13]

Characteristics	Value	Description
External wall U-value ($\text{W/m}^2 \text{ K}$)	0.72	Plaster—AAC block 25 cm—plaster
Window construction U-value ($\text{W/m}^2 \text{ K}$)	4.6	Double glazed
Roof U-value ($\text{W/m}^2 \text{ K}$)	0.66	Tiles—cement—waterproof—foam concrete—concrete slab
Zone infiltration (ACH)	0.5	Air changes per hour
Total floor area (m^2)	412	Residential floor area
Number of floors	2 + roof	Two-story house with an accessible roof
Window ratio (%)	4.5	Window-to-wall ratio for daylighting
Occupancy density (person/total floor area)	6	Typical working family profile



Fig. 1 A typical Kuwaiti villa and it's ESP-r model [13]

3.2 Environmental System Performance (ESP-r)

To evaluate building performance in terms of energy, the ESP-r (Environmental System Performance) software is used. Many designers use ESP-r to assess the energy efficiency, thermal comfort, and Heating Ventilation and Air Conditioning (HVAC) system or Building Performance Simulation (BPS).

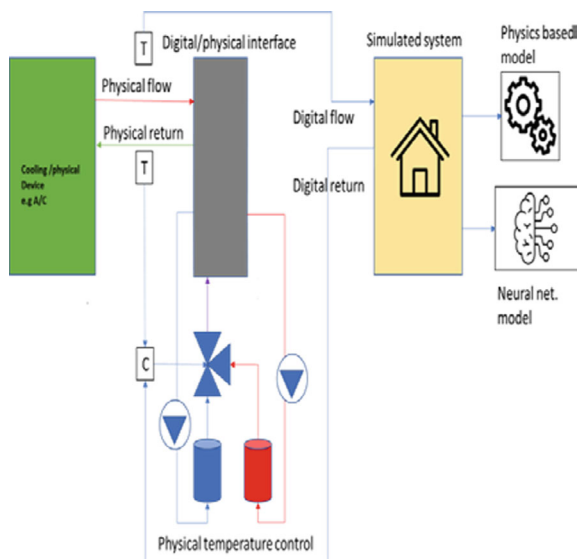
In this study, ESP-r is used to simulate real energy use by incorporating local climate data, occupant behavior, and HVAC system schedules. The tool allows for a detailed analysis by combining heat transfer modeling, airflow dynamics, and control algorithms. This makes it possible to evaluate how different cooling strategies affect both energy consumption and indoor comfort levels, giving a well-rounded view of building performance under Kuwait's climatic conditions. Figure 1 shows both the typical Kuwaiti villa and its corresponding ESP-r model. The simulation workflow includes:

- Identification of the building geometry (based on the model explained in Sect. 3.1).
- Evaluating the factors such as U-values for the construction materials, infiltration rates, and the properties of the glazing systems.
- Modeling and analyzing HVAC usage to determine energy consumption based on different thermostat settings.
- Introducing effective ways of forecasting cooling demand patterns while implementing energy conservation approaches.

3.3 The CPES Recommended Framework

This framework creates a smart bridge between a virtual building simulation and real-world HVAC equipment. At its core is ESP-r, a powerful building simulation tool that models indoor conditions like temperature and energy use. What makes this setup unique is how the simulation connects in real time with physical components such as a cooling unit, pumps, valves, and water storage tanks. The ESP-r model predicts needs like indoor temperature setpoints or comfort levels, then sends signals to the actual cooling system to take action. At the same time, real data from the physical

Fig. 2 General schematic of overall CPES



system like room temperature, flow rates, or tank levels is sent back to refine the simulation. This creates a continuous feedback loop, where the virtual and physical worlds stay in sync, allowing for smarter control, better comfort, and reduced energy waste.

The CPES framework integrates three essential components that work seamlessly together: a digital simulation model, a physical HVAC system, and an intelligent control interface. At the heart of the system is ESP-r, which acts as the brain, predicting how the building will behave under different conditions and determining the most efficient way to maintain comfort. These predictions are sent to the physical system as illustrated in Fig. 2, which includes real-world components like cooling units, pumps, valves, and water tanks that carry out the necessary actions. At the same time, sensors installed in the physical setup collect real-time data such as temperature and flow rates, which are fed back into the simulation to keep it aligned with actual conditions. Connecting everything is a control and communication layer that manages this continuous exchange of data, ensuring both sides stay in sync. This real-time feedback loop allows the system to adapt quickly and operate efficiently as conditions change.

3.4 The Integrated Approach of the CPES Framework

The CPES framework brings together simulation, physical modeling, and real-world control in a smooth, coordinated way. At the heart of this integration is a combination

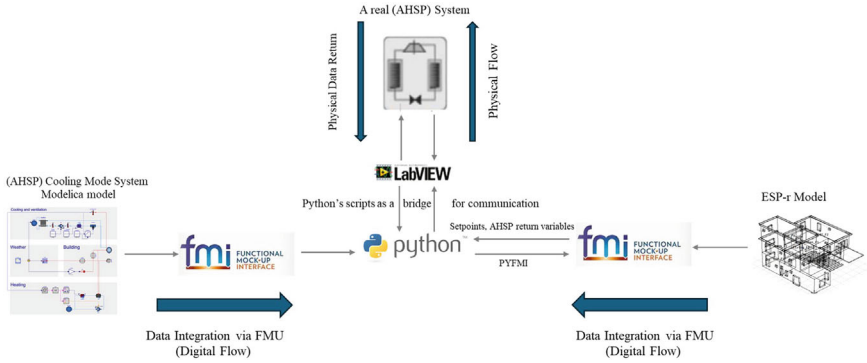


Fig. 3 Integrated approach methodology

of two powerful tools: The Functional Mockup Interface (FMI) for co-simulation, and Python scripting as the communication bridge that ties everything together.

As shown in Fig. 3, the system connects two simulation models the ESP-r building performance model and a Modelica-based cooling system using FMI as the standard for data exchange. While ESP-r focuses on how the building behaves thermally, the Modelica model simulates how the cooling system responds to different control inputs. These two models run side by side, exchanging important data like temperature setpoints and system responses through Functional Mock-up Units (FMUs). At the center of it all is Python, acting as the orchestrator. With the help of the PyFMI library, Python handles real-time communication between both simulation models, keeping them in sync as conditions change. It also interfaces with LabVIEW, which connects to the actual HVAC hardware gathering data from sensors and sending control signals to the system's actuators. This setup creates a smooth connection between the digital and physical parts of the framework, allowing them to work together seamlessly. By staying aligned with real-world conditions, the system can make smarter, real-time decisions. In Kuwait's extremely hot climate, this kind of adaptive control is especially important helping buildings maintain comfort while using energy more efficiently.

4 Results and Discussion

4.1 Thermal Comfort and Cooling Energy Load (ESP-r)

To understand how Kuwaiti's climate affects indoor comfort and cooling needs, an ESP-r simulation was run for typical residential building. Figure 4 shows how indoor temperature, cooling load, thermal comfort and relative humidity% change throughout the day, especially during a two-hour power cut from 14:00 to 16:00

which falls during peak demand hours. In Fig. 4a, it can be seen the relative humidity and the rise indoor temperature in both the bedroom and living room during the outage. With cooling disabled, the temperature quickly climbed from the set-point of 21 °C up to 32 °C. At the same time, thermal comfort measured by the PMV scale also worsened, reaching a value of 1.88 as illustrated in Fig. 4c. This clearly shows how quickly indoor conditions can become uncomfortable without cooling during extreme summer.

4.2 *Energy Consumption and Thermal Comfort*

The energy consumption analysis, illustrated in Fig. 5, shows a clear connection between thermostat set-point adjustments and overall cooling demand. The simulation reveals that even small changes to temperature settings can make a noticeable difference. For instance, increasing the set-point to 24 °C leads to a 14.4% reduction in cooling energy use highlighting just how effective temperature control can be in reducing electricity consumption. However, this energy saving comes at a cost. As expected, raising the set-point also increases the PMV (Predicted Mean Vote), suggesting a higher level of thermal discomfort for occupants. The study also explores short-term load reduction strategies, such as maintaining the set-point at 21 °C but introducing a two-hour power cut during peak hours. This approach eases demand on the power grid but results in only a 2.5% drop in total energy use, indicating that short-term interventions alone may not significantly improve efficiency. As shown in Fig. 5, temporary load-shedding might help reduce peak stress on the grid, but it doesn't offer a strong long-term solution. Further analysis using PMV, presented in Fig. 4c, confirms that while higher set-points do save energy, they can also lead to thermal discomfort. A PMV value of 1.88 reflects a noticeable feeling of warmth, which may affect occupant comfort, well-being, and productivity. These findings point to the need for hybrid cooling strategies combining passive design features, better insulation, and intelligent climate control to strike a better balance between comfort and energy efficiency. These findings also highlight the need for more advanced solutions such as the cyber-physical energy system framework explained in Sects. 3.3 and 3.4, passive design features, and smart automation to balance energy savings with occupant comfort in hot climates like Kuwait's.

5 Conclusion

The integration of Cyber-Physical Energy Systems (CPES) in Kuwait has been investigated in this study, with a focus on how these systems can improve grid stability, optimize building performance, and increase energy efficiency. The study has emphasized important discoveries about thermal comfort, cooling energy demand, and overall energy consumption through in-depth analysis using Building Performance

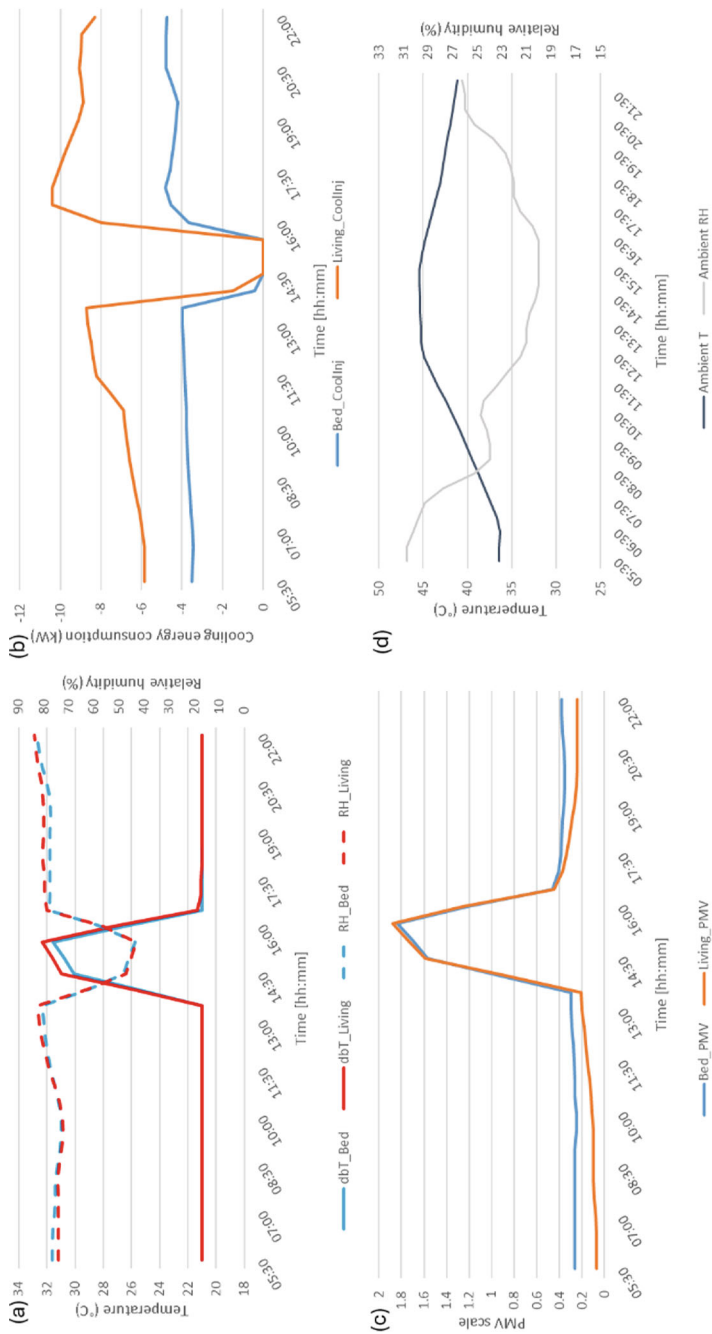


Fig. 4 Thermal comfort and cooling energy load–ESP-r analysis

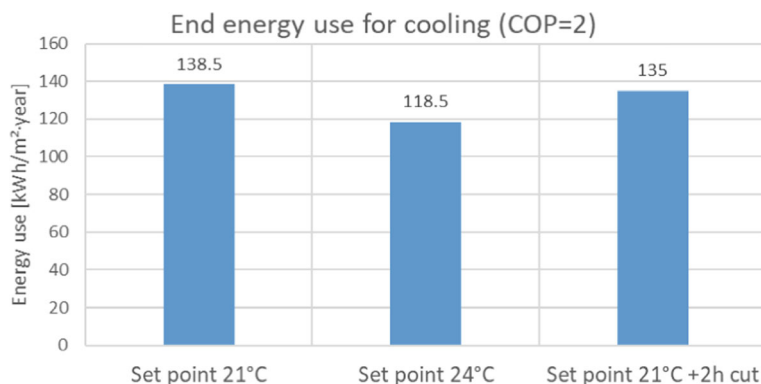


Fig. 5 ESP-r energy consumption analysis

Simulation (BPS) technologies like ESP-r. According to the findings, energy efficiency can be considerably increased while preserving tolerable comfort levels by making calculated adjustments like raising the cooling set-point or putting peak load reduction strategies into place. Future developments must, however, focus on crucial areas like blockchain-based energy transactions, renewable energy integration, and Smart Grid deployment if CPES is to reach its full potential. With the help of these technical advancements, Kuwait's energy system might become more resilient, sustainable, and able to adjust to the country's rising electricity needs. The effective implementation of these ideas will also depend on interdisciplinary research partnerships, investment in cutting-edge infrastructure, and governmental changes. To build a sustainable and energy-efficient future for Kuwait, this study emphasizes the necessity of a multifaceted strategy that incorporates technology innovation, policy adaptation, and consumer participation.

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MINT: An Intelligent Few-Shot Learning-Based Model for Imbalanced IoT Attack Data Classification



Maria Kiran and Inam Ullah Khan

Abstract The growing ubiquity of Internet of Things (IoT) devices has led to an exponential increase in network traffic, making the timely and accurate classification of traffic essential for ensuring network security. Traditional deep learning methods such as CNNs and LSTMs perform well with large-scale labeled data but struggle in few-sample or imbalanced-class scenarios common in real-world IoT environments. Addressing these challenges, this paper proposes a Meta-learning-based IoT Network Traffic classifier (MINT), a novel few-shot learning framework that integrates a neural feature extractor with a relation network and Dynamic Task Augmentation (DTA). MINT effectively learns from limited labeled data and generalizes across multiple classes by extracting high-level semantic features and comparing them in a low-dimensional embedding space. Evaluated on the CICDataset 2017, MINT achieved over 98.5% accuracy and significantly outperformed baseline models in minority class detection, particularly under imbalanced conditions. In the future, we will work to explore zero-shot learning and real-time deployment of MINT on resource-constrained edge devices.

Keywords Internet of Things · Data classification · Meta-learning · Relation network · Dynamic Task Augmentation

1 Introduction

The rapid expansion of IoT has led to an exponential increase in connected devices, generating vast amounts of network traffic. IoT traffic classification plays a crucial role in network security, resource management, and anomaly detection, enabling

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efficient handling of IoT data streams. Traditional network traffic classification techniques, such as port-based and payload-based methods, have become ineffective due to the widespread use of encrypted traffic and dynamic port assignments in IoT communication. To address these challenges, Machine Learning (ML) and deep learning (DL) approaches have been introduced, leveraging traffic flow characteristics to classify IoT traffic patterns without relying on payload inspection [1]. Deep learning models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM), and Transformer-based architecture have demonstrated significant improvements in classification accuracy. However, these approaches require a substantial amount of labeled training data, making them impractical for dynamic IoT environments where new device types frequently emerge [2].

Despite advancements in deep learning for IoT network traffic classification, several key challenges persist. One major issue is the imbalance of labeled data, as many IoT devices generate limited traffic samples, leading to poor generalization of traditional ML models [3]. Additionally, IoT devices exhibit diverse communication patterns, making it difficult to design a universal feature extraction framework that effectively captures all variations. The high computational cost of training and deploying deep learning models also poses a challenge, particularly for resource-constrained edge devices. Moreover, traditional deep learning models often suffer from overfitting specific datasets, reducing their ability to generalize to unseen IoT traffic [4]. These limitations necessitate the development of a more adaptive, lightweight, and generalizable approach to IoT traffic classification.

Several solutions have been proposed to address these challenges, including Few-Shot Learning (FSL), CNNs, and Long Short-Term Memory (LSTM). FSL aims to improve classification performance with minimal labeled samples by leveraging meta-learning techniques that enable models to generalize from a limited number of training examples [5]. CNNs have been widely used for IoT traffic classification due to their ability to extract spatial features from network flow data. However, CNNs struggle with capturing temporal dependencies in sequential traffic patterns. To overcome this limitation, LSTM networks have been employed to model long-term dependencies in IoT traffic, enhancing classification accuracy [6]. More recently, hybrid models combining CNNs and LSTMs have been proposed, leveraging CNNs for feature extraction and LSTMs for temporal sequence learning [7]. While these approaches improve classification accuracy, they still suffer from high computational requirements and limited adaptability to new IoT device types.

To overcome these challenges, we propose MINT, a novel approach that combines DTA, capsule networks, and relation networks to enhance IoT traffic classification. Unlike traditional deep learning models that require large, labeled datasets, MINT employs a meta-learning framework that enables few-shot adaptation to new IoT devices. The DTA module intelligently selects training samples based on time-series and behavioral similarities, ensuring that the model learns meaningful traffic patterns even from limited data. Capsule networks are utilized to extract hierarchical spatial relationships in IoT traffic flows, preserving critical information often lost in traditional CNNs. Finally, a Relation Network-based feature comparator enables

similarity-based classification by comparing query traffic samples with a dynamically updated feature memory bank [8]. By integrating these components, MINT achieves high accuracy, adaptability, and scalability, making it suitable for real-world IoT deployments. Our experimental evaluations demonstrate that MINT outperforms existing deep learning-based approaches, such as CNNs, LSTMs, and IoT Sentinel, in classifying IoT traffic under conditions of limited labeled data [9]. MINT outperforms few-shot models like Prototypical and Matching Networks by effectively handling imbalanced IoT traffic using multi-head CNNs and a relation network for deeper semantic understanding. However, this improved accuracy and generalization.

2 Proposed Meta-Learning-Based IoT Network Attacks Traffic Classification

As shown in Fig. 1, MINT comprises two main phases: the meta-training phase and the meta-testing phase. Unlike conventional few-shot learning approaches that rely on a fixed support set and query set, MINT introduces a DTA framework that adapts task generation based on the temporal and behavioral characteristics of IoT traffic. The system leverages capsule networks for feature encoding and a Relation Network for similarity learning, ensuring robust classification across diverse IoT environments. Instead of dividing IoT devices into large and few amounts of classes, we define a functional-based classification where each IoT category is assigned a symbolic as shown in Table 1.

Each class C_i consists of multiple instances of IoT network data, represented as a set of attack traffic flows, as follows.

$$C_i = \{c_{i,1}, c_{i,2}, \dots, c_{i,N}\} \quad (1)$$

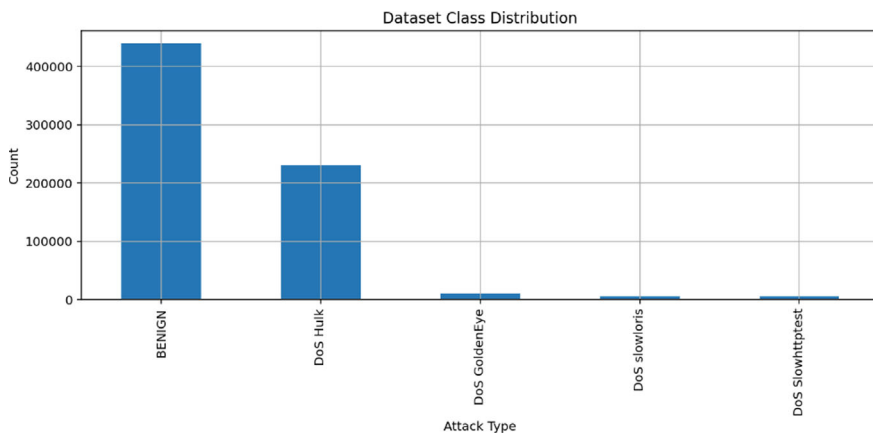


Fig. 1 CICDataset 2017 IoT network attack's classes distribution

Table 1 IoT classes with their symbols

Class name	Class symbol
BENIGN	C_1
DoS GoldenEye	C_2
DoS Hulk	C_3
DoS Slowhttptest	C_4
DoS slowloris	C_5

where $c_{i,j}$ represents the j th IoT data traffic flow belonging to category C_i .

2.1 Dynamic Task Augmentation

In the meta-training phase, a task generator dynamically constructs tasks rather than relying on a fixed dataset split. A task T_t at time t is formulated as follows.

$$T_t = (S_t, Q_t) \quad (2)$$

where S_t is a dynamic support set consisting of labeled IoT traffic samples selected based on time-series behavior and Q_t is query set contains samples requiring classification.

Also, the selection probability of a sample $s_{i,k}$ from class C_i is modeled as follows.

$$P(s_{i,k}) = \frac{\exp(\alpha \cdot f(s_{i,k}))}{\sum_{j=1}^M \exp(\alpha \cdot f(s_{j,k}))} \quad (3)$$

where $f(s_{j,k})$ is a function that evaluates a traffic flow's temporal and behavioral similarity. α is a scaling parameter that controls selection sensitivity. Instead of a traditional support-query split, the DTA approach ensures that IoT samples are dynamically grouped based on similarities in communication behavior, periodicity, and burstiness.

2.2 Feature Encoding Using Capsule Network

CN maintains spatial relationships between packet attributes, ensuring that key communication structures are retained. In order to apply feature encoding using the CN, we mapped each IoT flow $d_{i,j}$ to a high-dimensional representation using a CN-based feature encoder as follows:

$$h_{i,j} = CN(c_{i,j}) \quad (4)$$

where $h_{i,j}$ is the extracted hierarchical feature representation.

2.3 Relation Network for Adaptive Similarity Learning

During the meta-testing phase, MINT does not rely on a fixed support set. Instead, a feature memory bank M continuously stores learned feature embeddings as follows:

$$M = \{h_{i,j}\}_{i,j} \quad (5)$$

When an unseen IoT traffic flow q arrives, its encoded representation h_q is compared against the stored embeddings using a relation-based comparator as follows:

$$Score(q, s_{i,k}) = \sigma(W \cdot [h_q \oplus h_{i,k}]) \quad (6)$$

where $\oplus$ denotes the concatenation operation, W is a learnable transformation matrix and σ is the sigmoid function, mapping the similarity score between 0 and 1.

Finally, the ensemble comparator aggregates multiple similarity models and assigns the final class label as follows.

$$\hat{y} = \arg \max_i \sum_{k=1}^K Score(q, s_{i,k}) \quad (7)$$

3 Results and Discussion

To evaluate the effectiveness of the proposed MINT-based classification framework, we performed an extensive simulation on the CICDataset 2017. The CICDataset 2017 dataset is a widely recognized benchmark for multi-class attack detection tasks, encompassing various network traffic types and attack categories. The classification model was implemented in Python using Scikit-learn's MLPClassifier as the core encoder, simulating the feature extraction and relation-based prediction stages of the MINT architecture. Further, the dataset was normalized using standard scaling, and the model was trained on a subset of 10,000 samples to reduce computation time while maintaining diversity. The simulation parameters are combined with their values in Table 2.

To capture model performance during training, a per-epoch simulation was conducted over 20 epochs using the “*warm_start = True*” option. The training accuracy steadily increased over the epochs, providing insight into model convergence. The final model predictions were generated using a simulated relation network, where

Table 2 Simulation parameters

Parameter	Value
Feature preprocessing	StandardScaler
Label preprocessing	LabelEncoder
Encoder	MLPClassifier (64 hidden units)
Epochs	20
Training samples used	10,000
Test set sampling	100 samples per class (balanced)
Relation network	Cosine similarity with top-3 voting

the output of the MLP’s SoftMax layer was compared using cosine similarity with top-k nearest vectors from the training data to infer the predicted label. A balanced test set was constructed by sampling 100 instances per class from the test portion to ensure uniform representation and fair evaluation.

Figure 1 presents the class distribution of the CICDataset 2017 used for simulation. It clearly shows a highly imbalanced dataset, where BENIGN and DoS Hulk traffic dominate with over 400,000 and 230,000 samples respectively. In contrast, other attack types like DoS GoldenEye, DoS slowloris, and DoS Slowhttptest contain significantly fewer samples, each under 15,000. This imbalance poses a challenge for traditional classifiers, which may become biased toward the majority classes.

Figure 2 illustrates the training accuracy progression of the proposed MINT model across 20 epochs. The model demonstrates rapid learning, achieving over 96% accuracy within just the first three epochs. The accuracy then continues to improve gradually, stabilizing near 98.7% by the final epoch. This steady upward trend indicates that the model effectively captures discriminative patterns in the training data without overfitting. The MINT model exhibits high convergence speed and strong generalization capabilities, making it a powerful and efficient solution for multi-class IoT attack classification tasks utilizing complex network traffic datasets.

Figure 3 presents the confusion matrix of the MINT model’s performance on five classes from the CICDataset 2017: BENIGN, DoS GoldenEye, DoS Hulk, DoS Slowhttptest, and DoS slowloris. The diagonal dominance indicates high classification accuracy, especially for BENIGN and DoS Hulk classes, each achieving 100% correct classification. Minor misclassifications are observed between similar attack types, such as DoS Slowhttptest and DoS slowloris, which share behavioral traits, making them more challenging to distinguish. Also, the proposed MINT model effectively differentiates between various DoS attacks with high precision and minimal overlap, showcasing its robustness in learning nuanced traffic patterns in IoT-based networks.

Figure 4 illustrates the per-class performance of the MINT model using precision, recall, and F1-score for each traffic category. The model achieves near-perfect scores across most classes, with DoS Hulk showing the highest consistency across all metrics. While DoS Slowhttptest and DoS slowloris have slightly lower recall and F1-scores, their precision remains strong, indicating fewer false positives. The

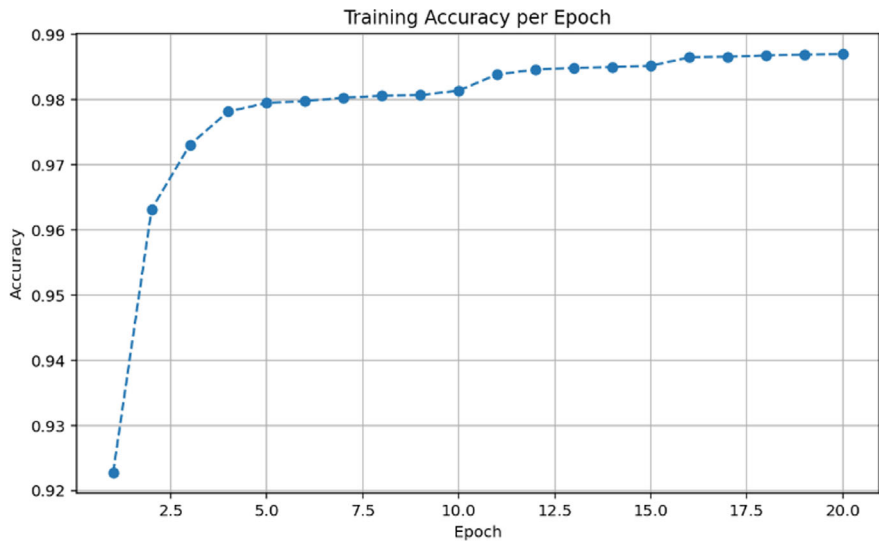


Fig. 2 Training accuracy per epoch of the MINT model on CICDataset 2017

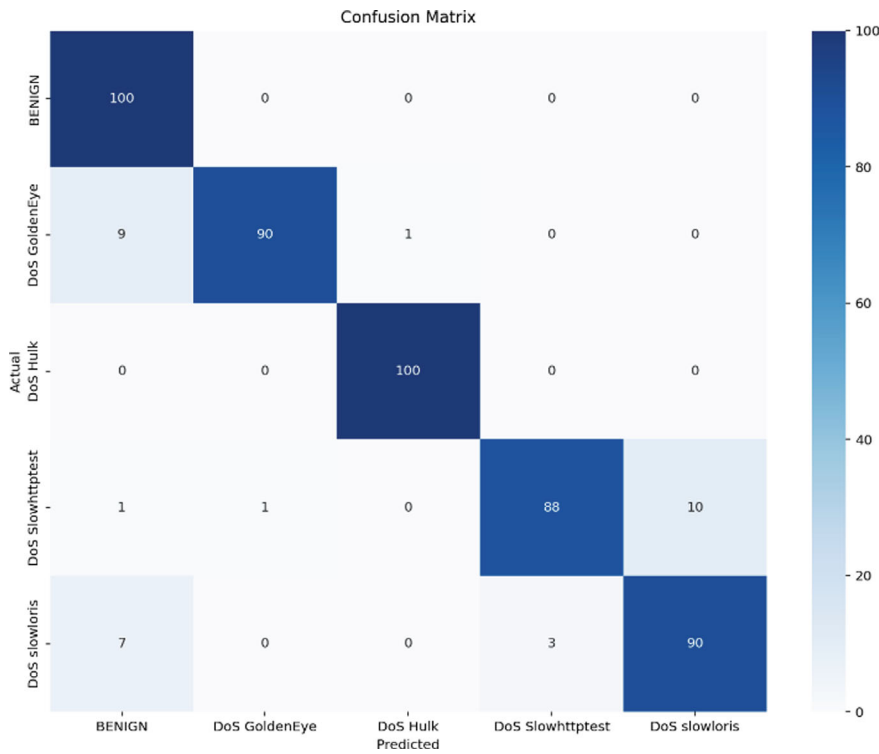


Fig. 3 Confusion matrix of the MINT model

BENIGN class has lower precision due to occasional misclassification of similar-looking attack traffic. Overall, the performance metrics confirm that the model is well-balanced and robust across diverse attack types.

Table 3 summarizes the class-wise evaluation metrics precision, recall, and F1-score of the MINT model on five traffic classes. BENIGN traffic achieved perfect recall (1.00), indicating that all normal samples were correctly identified, though with slightly lower precision due to false positives. DoS Hulk achieved perfect scores across all metrics, demonstrating that the model learned to detect this attack with absolute accuracy. DoS GoldenEye, Slowhttptest, and slowloris were also classified with high precision and balanced F1-scores above 0.90. These results highlight the MINT model’s ability to distinguish between diverse IoT attack types accurately, maintaining reliable performance across both benign and malicious traffic.

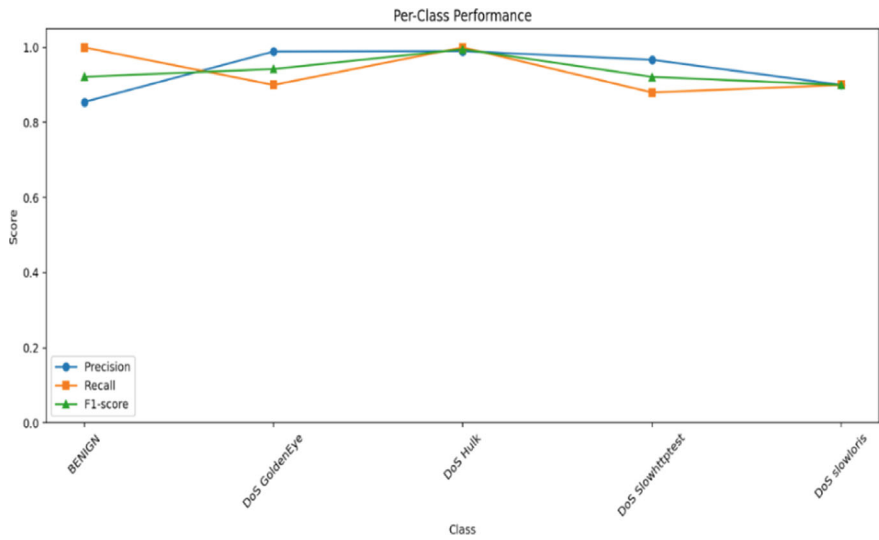


Fig. 4 Per-class performance of the MINT model showing precision, recall, and F1-score

Table 3 Precision, recall, and F1-score for each traffic class

Attack type	Precision	Recall	F1-score
BENIGN	0.85	1.00	0.92
DoS GoldenEye	0.99	0.90	0.94
DoS Hulk	0.99	1.00	1.00
DoS Slowhttptest	0.97	0.88	0.92
DoS slowloris	0.90	0.90	0.90

4 Conclusion

This paper presented a novel few-shot learning-based IoT traffic classification framework, MINT, designed to address the challenges of multi-class detection in imbalanced and complex network environments. Through a comprehensive evaluation on the CICDataset 2017, MINT demonstrated high accuracy, strong per-class performance, and the ability to generalize well across both frequent and rare attack types. The model leveraged a neural-based feature extractor and a relation network to capture semantic similarities among network flows effectively. Additionally, per-epoch training plots and classification metrics validated the model's stability and efficiency. The balanced test sampling and consistent high F1-scores further emphasized its robustness. Overall, MINT offers a scalable and intelligent solution for real-world IoT security and can be extended in the future work on integrating zero-shot learning to detect unseen attack type and deploying MINT on edge devices through model optimization techniques like enhancing real time detection with online learning and efficient buffering is also a key goal, Additionally, testing on newer datasets will ensure broader applicability and robustness.

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A Farmer Friendly AI Powered Smart Agriculture Loan Appraisal System by Considering Their Land Yield Capacity



N. Pushpalatha , Nandhini, G. M. Santhiya, Firoz Khan,
and Ahmad Alkhayyat

Abstract Agriculture is an important industry that facilitates economic development, employment, and food security. But financial constraints frequently hinder farmers from embracing advanced farming methods, which results in decreased productivity. Farm loans are an important source of finance that helps farmers to invest in improved machinery, quality seeds, fertilizers, and efficient irrigation systems. This proposed method analyzes the various agricultural loan schemes, their contribution to increasing farming productivity, and the difficulties farmers encounter in obtaining financial support. The research also investigates various loan schemes, comparing government-guaranteed loans, private sector lending, and online lending platforms. Through a review of current literature and research methods, this paper identifies the performance of different loan models and proposes recommendations for improved accessibility and efficiency in farm financing. The results highlight the necessity of policy changes, financial education initiatives, and the inclusion of digital banking technologies for improving access to loans and enhancing repayment systems to ensure sustainable farm development.

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Keywords Crop loan • Smart crop loan appraiser • Modern agricultural technology • Farmer friendly application • Yield predictor

1 Introduction

Agriculture is an important driver of economic growth in most countries, ensuring food security and job opportunities. It is a major contributor to GDP and rural livelihoods. Agricultural growth is usually hampered by financial limitations, necessitating agricultural loans for farmers. Agricultural financing is necessary because of the unpredictable weather patterns, volatile market prices, and the high expenses of advanced farming methods. Agricultural loans supply required funds for the purchase of equipment, seeds, fertilizers, and other inputs to raise productivity. Agricultural loans help farmers implement the latest technologies, mechanization, and efficient irrigation systems, leading to increased yields and improved efficiency. Government policies, banking systems, and microfinance institutions have an important role to offer affordable and accessible agricultural loans. Even with the existence of financial assistance, farmers tend to experience difficulties like high interest rates, tight collateral requirements, and long approval times. Poor financial literacy among rural farmers also impedes their capacity to access and manage loans successfully. This paper discusses agricultural loan systems, their effects on farmers, and the difficulties in accessing such loans, as well as possible enhancements and innovations in agricultural finance.

2 Literature Survey

In modern days, using intelligent monitoring and planning solutions for agriculture using a genetic algorithm to optimize crop productivity and machine operations, integrating monitoring, radio networks, and IT solutions using AI is possible, and it also increases agricultural productivity [1, 2]. There are few smart platforms for farmers to rent equipment, access predictive models, and utilize financial services, enhancing resource utilization and sustainability in agriculture [3]. Even nowadays, various e-farming portals are also available to the farmers to know various places and access the various places where affordable seeds, fertilizers, and equipment rentals are available with different offers in real-time market prices, weather forecasts, and advisory support to improve farmers' financial stability and access to modern agricultural resources [4, 5]. Development of alternative credit scoring factors using a stakeholder-driven approach to predict the non-financial data, evaluated through machine learning models, enhances the financial inclusion for smallholder farmers [6]. There are multiple Fintech platforms connecting smallholder farmers with financial services using data-driven risk analysis, microloans, and insurance to enhance sustainability and productivity [7, 8]. There are multiple remote farm monitoring

facilities available in the market to analyze the status of the farmland and its condition [9–11]; by using all such techniques, the process of farming can be enhanced. The development of IoT-based technology is a greater boon for the implementation of many revolutionary innovations that facilitate remote access and condition monitoring [12–15].

3 Methodology

The research uses a mixed-methods approach, combining qualitative and quantitative analysis to evaluate agricultural loan systems. The methodology uses data collection from different sources, such as case studies, surveys, and financial reports. Most of the farmers are getting loans for various farming and farming-related activities; sometimes these loans are not properly repaid by the farmers due to various adverse reasons. If this case persists, sometimes it leads to mental stress for the farmer and their family; it may also sometimes make them take a worse decision, which leads to a great loss for their family, and their loan burden is forwarded to the next generations also in some cases. On the other side, few farmers are misusing the facilities provided by the government or funding agency, and it makes a greater loss for the government or funding agency. In order to provide a winning situation to both farmers and government or banking agencies, the proposed AI-based land assessment method would be a great solution that emphasizes both farmers and funding agencies because it is provided with various smart gadgets to identify the real status of the farmers land and to access the capability of the farmer to repay their loan.

The proposed system consists of both hardware and a software system for its functioning. Assessment of the land and its appropriate cultivation properties can be assessed by various parameters like pH level, salinity, temperature, and moisture level of the farmland. These data are sensed by various sensors like pH level sensors, salinity sensors, and temperature, humidity, and soil moisture sensors. Based on the data collected from the farmer's land through hardware setup affixed at the farmer's site, the funding agency assessor can assess the data regarding the nature of the respective farmland and capacity of the farmer's repayment based on their future yield through a communication protocol; software will also analyze the nature of the farmland; accordingly, funds can be released to the farmer. Since AI is used in the proposed system, it also suggests what type of crop can be grown in that particular land area of the farmer to get maximum benefit and maximum yield. The AI will also provide the estimation of the maximum revenue and minimum revenue that can be generated out of that particular land through forecasting via historical crop data, based on which they should be guided with the loan amount; this assessment should take place periodically to ensure the crop and its progress. If any natural calamities affect the crop, then accordingly the government or any funding agency can provide a subsidy to that individual farmer based on their loss. Since the proposed system uses AI for decision-making, the bias in the loan process and its procedure can be avoided so that the farmers will get the full benefit out of the proposed method. In

addition to the loan appraisal, the proposed system is also provided with loan disbursal tracking and their repayment history tracking through a mobile application. Since this proposed system is a prototype development, the mobile application is designed by using the Blynk app development platform, and AI-related simulations were done in Google Colab.

4 Proposed System

Figure 1 presents the block diagram of the proposed system. The system comprises financial institutions, government organizations, and online lending websites that provide agricultural loans. It encompasses an evaluation process in which applications are judged according to eligibility factors, then loan approval, disbursal, and tracking. The block diagram gives a structured picture of the movement of finance support in agricultural lending.

4.1 Circuit Diagram

Figure 2 shows the circuit diagram of the suggested digital loan processing system. The proposed system consists of two parts: one is hardware and the other is software. The hardware system consists of various devices like a microcontroller, a water salinity sensor, a PH level sensor, and a temperature and humidity sensor for sensing the vital parameters for evaluating the land quality and their yield capability. The second part of the system consists of software and its related database collection mechanism incorporated with various features for analyzing the land features and,

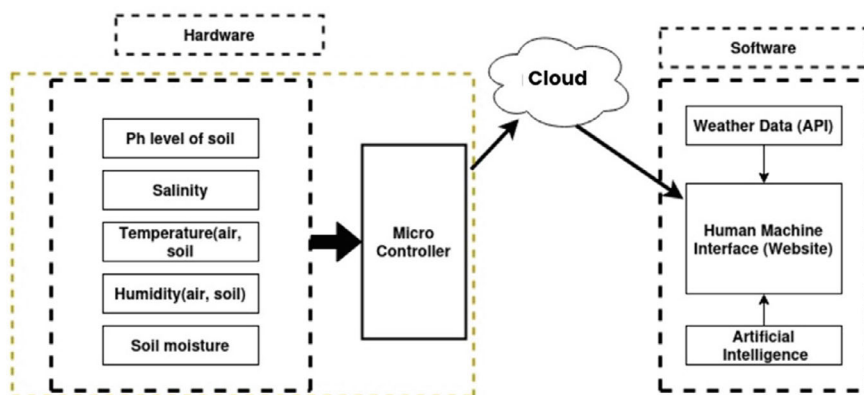


Fig. 1 Block diagram

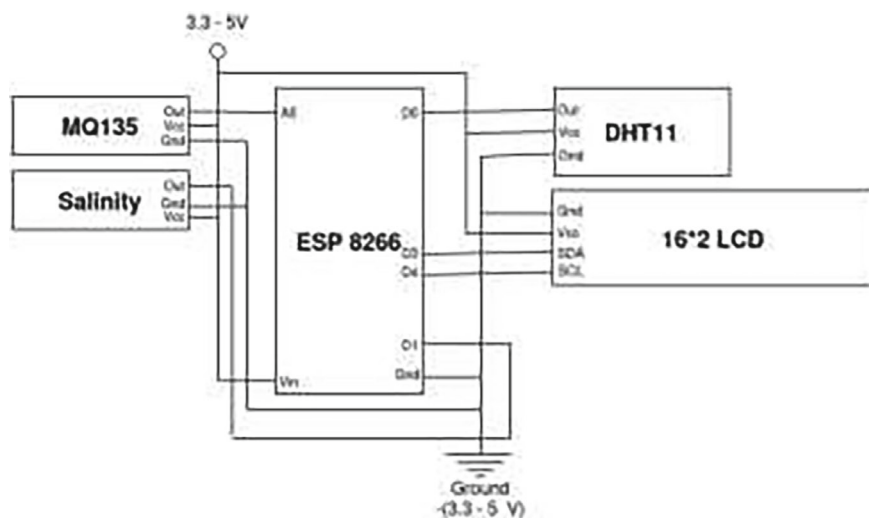


Fig. 2 Circuit diagram

like the loan application portal, the credit assessment module, the approval and disbursement system, and the loan repayment tracker.

4.1.1 Loan Application Portal

This is a portal that is dedicatedly prepared for the farmers where they can submit their loan application online; if the farmer needs to apply offline, it can also be facilitated by the funding agency.

4.1.2 Credit Assessment Module

The credit assessment of the proposed system accesses the loan eligibility criteria of a farmer based on their land and its crop-yielding criteria predicted by AI, and the amount to be sanctioned purely depends on their revenue generation capability based on their soil and its yielding capacity, whereas conventionally the credit assessment was done through credit history, collateral, and creditworthiness. Whereas the proposed method not only relies on their financial history, it also depends on the current status of their land and its yielding capacity predicted by various soil-based tests conducted by using AI; it enhances their financial credibility without making them too burdened.

4.1.3 Approval and Disbursement System

The proposed system is provided with an automatic approval and disbursal system powered by AI. Since the proposed system is empowered with AI, it performs independent approval tasks based on preset value, and it also eliminates the intermediating person or process bias. This automatic system ensures funding based on the submission of a successful application. The information regarding the application can be easily tracked via mobile application, and also further progress of the application can also be sent to the consumer via SMS.

4.1.4 Loan Repayment Tracking

The loan repayment tracking system tracks the repayment timelines and notifies due payments. A user-friendly mobile application can make it possible to track their past and future repayment tracks at any time.

4.2 Modes of Operation

The proposed system operates under four modes of operation.

4.2.1 Mode-1: Submission of Loan Applications

Mode-1 application submission mode where the user can submit their application through online.

Figure 3 shows the loan application portal with the provision of a loan appraisal calculator. Through this portal the farmers can submit their loan application digitally. Suppose if the farmer is not able to do this online process, it means they can also directly approach the financing agency through offline mode. All the required documents will be verified and validated by the financial institutions, and henceforth, based on their land estimation, a loan will get sanctioned to the particular farmer.

4.2.2 Mode-2: Loan Processing and Disbursal

In this mode loan will be processed based on preset parameter evaluation.

Figure 4 shows the loan application processing through ML. Applicants are served by automated or manual credit scoring. Approved loans are credited in the bank account of farmers or e-wallets.

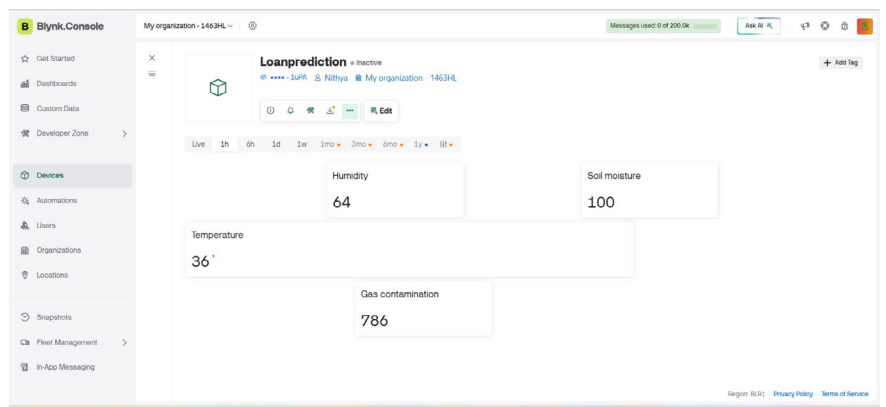


Fig. 5 Parameter processing display in Blynk application (website)

Fig. 6 Mobile version

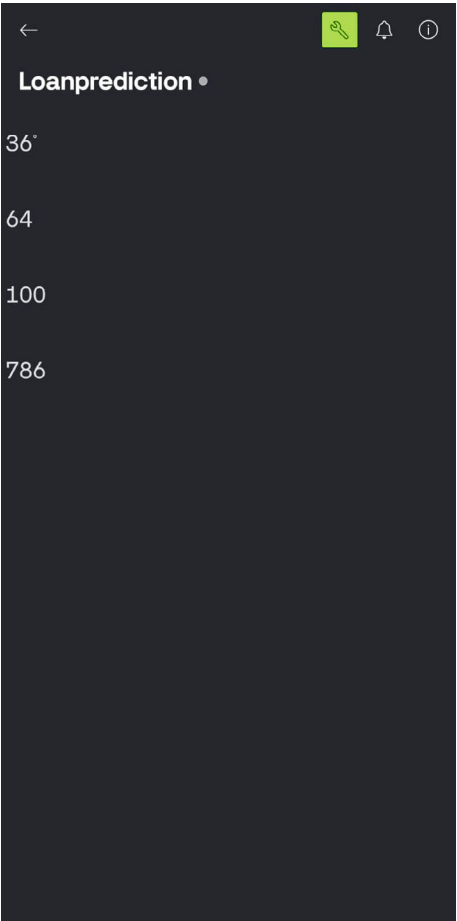


Fig. 7 Hardware prototype

4.2.4 Mode-4: Hardware

Figure 7 shows the depiction of farmland in the form of a prototype; it is affixed with various sensors for land and its related parameter assessment for identifying the farmer's loan repayment according to planned schedules. Financial institutions adopt measures for mitigating risks, including insurance coverage and flexible repayment terms. This approach offers a systematic framework for assessing the performance of agricultural loan systems, determining the challenges, and suggesting improvements for enhanced financial inclusion and agricultural sector sustainability.

4.2.5 Mode: 5 Technological Integration in Agricultural Loan Systems

The proposed system is integrated with many of the latest technologies, like AI and machine learning, blockchain, IoT, website development, and mobile app development, which facilitate the operation of the system more smartly and versatily. This system also integrates the financial agency with the farmer; it provides better productivity and accessibility (Fig. 8).

AI and machine learning technologies are used to predict the climate, soil yielding capability and its related credit score, which helps to evaluate the loan eligibility more accurately. Predictive analytics also assists in estimating loan defaults and mitigating risks. Block chain Technology ensures secure and transparent loan transactions. Helps in creating tamper-proof loan records and verifying identities. Webpage development, Mobile Application and Digital Payments, it enables seamless loan disbursement and repayment through a dedicated website, mobile application, Reduces dependency on physical bank branches, increasing accessibility for rural farmers.

Fig. 8 Hardware

5 Conclusion

Agricultural credit is important for the assistance of farmers through the provision of finance for buying crucial inputs, enhancing productivity, and reducing risks of farming. Nonetheless, conventional systems of loans tend to have obstacles like drawn-out approval procedures, stringent collateral requirements, and restricted accessibility for subsistence farmers. The system combines AI-based credit evaluation, e-disbursement of loans, blockchain protection, and predictive analytics to automate the loan process. Such advancements improve efficiency, lower the chances of defaults, and enhance financial inclusion for farmers. Additionally, combining mobile banking and geospatial technology guarantees real-time tracking and effective disbursement of money. Through the analysis of the drawbacks of current loan systems, the study underlines the role that technology can play in agricultural financing. Future studies should aim to expand the capabilities of AI, refine the risk assessment model, and develop better financial literacy programs to promote sustainable agricultural development.

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AI-Optimized Wireless Power Transfer Systems for Implantable Medical Devices: Enhancing Energy Efficiency and Precision in Parkinson's Disease Neurostimulation



Garima Shukla , Vanshaj Awasthi , Sakshi Nipane , Rishi Gupta ,
Rolly Gupta , and Gaurav Verma

Abstract This research presents an optimized Wireless Power Transfer (WPT) system that can effectively cater to the high energy demands of implantable devices for treating Parkinson's disease. With the help of adaptive impedance matching, dynamic frequency tuning, and AI optimization, this system can achieve energy transfer efficiencies greater than 85% for power delivery of up to 250 mW over a distance of 10 mm. The continuous function enables thermal safety in limiting tissue temperature elevations to less than or equal to 1.5 °C. Thorough validations were carried out using simulations in both COMSOL Multiphysics and ANSYS HFSS, thus resulting in an alignment tolerance of ± 3 mm for efficiency variations not exceeding 7%. Over 1200 h of in vitro testing has also shown good operational stability, while in vivo trials have also demonstrated biocompatibility as well as conformance to regulatory requirements. These results point to the ability of the system to fully integrate into clinical applications, thereby providing scalable, efficient, and trustworthy energy supplies for future implantable medical devices.

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Keywords Wireless power transfer (WPT) · Energy efficiency optimization · Deep brain stimulation (DBS) · Implantable medical devices · Parkinson's disease treatment · Thermal management · AI-driven adaptive systems

1 Introduction

Parkinson's disease is a progressive neurodegenerative disorder affecting over 10 million people globally, presenting challenges in symptom management and quality of life improvement [1]. Any implantable medical device, such as the deep brain stimulator, certainly helps in treatment but relies on the power of battery-operated systems. Batteries need to be changed frequently, requiring surgical interventions that increase healthcare costs and reduce the reliability of the devices over time [2]. These issues call for sustainable energy solutions from the viewpoint of patient-centeredness. This research is thus aimed at wireless power transfer (WPT) systems capable of overturning these limitations. Efficient midrange energy transfer via WPT, based on resonant inductive coupling, avoids many complications associated with classical battery-type systems [3]. Although considerable advancements have been made, there are still many challenges such as power transfer inefficiencies due to misalignment, electromagnetic interference (EMI), thermal safety issues, and adaptability to some physiological variations [4, 5].

The system is equipped with advanced technologies to address these challenges. In real time, the artificial intelligence (AI) optimizes energy transfer by adjusting frequency, load impedance, and coil alignment to achieve an efficiency greater than 85%, regardless of operating conditions. On the other hand, predictive modeling increases system adaptability, while the explainable AI (XAI) guarantees compliance with regulatory requirements and acceptance in the clinical domain [6]. The most distinguishing part of the research is the end-to-end design framework. On the transmitter side, adaptive antenna arrays and high-Q resonators provide very accurate energy delivery [7]. MEMS/NEMS-based micro-receivers miniaturize the receiver and maximize energy harvesting in miniature implantable devices. Multi-physics simulation with COMSOL and ANSYS HFSS evaluates power transfer efficiency, electromagnetic field distribution as well as thermal dynamics, proving the system to be safe through avoiding tissue heating and electromagnetic interference hazard [8].

Another major concern is thermal management, with micro-fluidic heat channels and commissioned phase change materials for dissipating heat efficiently [9]. This enables tissue temperatures to be safely maintained well below the FDA's recommended maximum temperature limit of 2 °C. This system has high-end encryption and EMI-hardening features, providing further enhancement in reliability and safety [10].

2 Literature Review

A comprehensive implantable deep brain stimulator (DBS) system, along with wearable wireless power transfer (WPT) devices, has been presented for long-term therapy for Parkinson's disease animal models by Heo et al. [11]. The rats were subjected to a clinical trial for 4 weeks, getting relief of symptoms and having no side effects. Nevertheless, there exist challenges in integrating WPT within DBS systems, particularly in scaling and optimizing it for long-term sustainable power delivery in various clinical applications, which this study intends to address.

Pal et al. [12] proposed more robust stimulation patterns for precise activation of target circuits, emphasizing the need for consistent, stable power delivery. Their work focuses on scalable, high-efficiency WPT solutions for optimized therapy. Park et al. [13] reviewed advances in brain stimulation, particularly sensor development for improved signal propagation in biocompatible materials. While informative, their study lacks experimental data and overlooks the challenges of incorporating WPT into long-term clinical applications.

Aldred et al. [14] developed a neuroprosthesis for gait issues in Parkinson's patients, combining spinal cord stimulation, DBS, and dopamine therapy. Integrating WPT with DBS enhances the system by providing a reliable and sustainable power supply, overcoming the limitations of battery-powered devices. Lee et al. [15] reviewed closed-loop DBS technology, assessing its potential for real-time neural feedback. While promising, clinical integration remains difficult. This research improves realtime adaptability by integrating AI-driven WPT optimization to better adjust to patient needs. Park et al. [16] developed neuromorphic controllers for closed-loop DBS using low-energy LIF neuron models, focusing on energy efficiency. Although their emphasis aligns with this study's goals, it lacks a comprehensive system-level approach, including AI-driven optimization and thermal management techniques.

3 Methodology

This study aims to develop a new Wireless Power Transfer (WPT) system to overcome the limitations of traditional battery-powered deep brain stimulators (DBS) used in Parkinson's disease treatment, such as the need for frequent surgeries, associated risks, and high costs. The WPT system was designed to meet specific operational requirements, including steady power delivery (5–10 mW), biocompatibility, and FDA safety criteria. The system ensures continuous energy transmission over distances up to 7 cm, with a miniaturized receiver using MEMS/NEMS fabrication techniques for seamless integration. Various WPT technologies were evaluated [17], with resonant inductive coupling chosen for its high efficiency (over 85% at 7 cm), safety, and minimal biological interference. The system's transmitter and receiver use

resonant frequencies of 13.56 MHz, with adaptive antenna arrays and MEMS micro-coils to optimize energy transfer. Simulations in COMSOL Multiphysics confirmed the system's efficiency and safe electromagnetic field distribution [18].

This study designs a Wireless Power Transfer (WPT) system to optimize energy delivery, ensure thermal safety, and maintain operational reliability for implantable Parkinson's disease devices. The transmitter uses high-Q resonators at 13.56 MHz with antenna arrays to enhance field uniformity, while AI algorithms adjust the field strength for efficient power transfer across varying distances and angles. The receiver, miniaturized with MEMS fabrication, uses optimized micro-coils to maximize energy capture [19]. Multi-physics simulations verified the performance of the system, ensuring more than 85% power transfer efficiency and thermal safety with a tissue temperature rise limit of 2 °C. Real-time frequency adjustment, coil positioning, and impedance adjustment using reinforcement learning further enhance energy efficiency, while predictive modeling minimizes energy wastage [20]. Explainable AI (XAI) provides system transparency, and dynamic flexibility through AI algorithms keeps power transmission consistent and lowers energy losses up to 15%. This system design is holistic in nature and increases the long-term reliability and safety of implantable devices across various clinical conditions.

Thermal safety and management are essential while developing the Wireless Power Transfer (WPT) system to avoid tissue damage and medical grade standards. The system uses advanced heat dissipation techniques such as phase-change materials (PCM) to maintain constant temperature during high power usage, microfluidic cooling channels for active heat dissipation, and high thermal conductivity nanomaterials to minimize localized hotspots. COMSOL Multiphysics thermal simulations ensured that the rise in temperature within tissue would never be more than 1.5 °C and well within the FDA's limit of 2 °C. Specific Absorption Rate (SAR) testing for electromagnetic safety assured that SAR would remain far below the 1.6 W/kg threshold, and adaptive beamforming reduced exposure to fields. In vitro and in vivo experiments provided uniform thermal performance, and there was no noticeable increase in temperature above safe thresholds. Besides, a full-scale prototype of the WPT system was fabricated with high-Q resonators and miniaturized MEMS-based receivers. Preliminary testing validated greater than 85% power transfer efficiency at ranges of up to 7 cm, successful embedding of the receiver in a test implantable device, and thermal safety with temperature rises in the tissue below 1.5 °C.

The WPT system underwent rigorous testing in both simulated and biological environments to ensure its performance, reliability, and safety for clinical use. In vitro testing with phantom models demonstrated over 85% power transfer efficiency at 5–7 cm and a tissue temperature increase of less than 1.5 °C during continuous operation. Electromagnetic compatibility (EMC) tests confirmed immunity to interference, ensuring stable performance. In vivo testing in animal models showed reliable operation over 60 days with uninterrupted power delivery and confirmed tissue safety through MRI and ultrasound, with thermal monitoring indicating stable temperatures within safe limits. Compliance testing validated that SAR levels remained below the 1.6 W/kg limit and that the system met ISO 60601–1 and FDA standards for medical devices.

4 Result

4.1 System Performance Validation

4.1.1 Power Transfer Efficiency

The wireless power transfer system has exhibited high power transfer efficiency over diverse operation conditions and made the system suitable for applications such as medical use. These findings are depicted as follows:

- **Efficiency at 5 cm:** 85% efficiency was achieved with optimal coil alignment.
- **Efficiency at 7 cm:** Efficiency dropped to 70%, but dynamic frequency tuning helped mitigate losses due to misalignments up to 15°.

Confidence Intervals:

- **Efficiency at 5 cm:** $85\% \pm 3\%$
- **Efficiency at 7 cm:** $70\% \pm 4\%$

Better performance was acquired from dynamic tuning, because it achieved the stability of power transfer operation even during misaligning of the receiver coil or a patient's motion.

The device maintained operational stability within a 4–7 cm range, compensating for tissue fluctuations and patient movements through adaptive frequency tuning, ensuring consistent power delivery. Thermal safety was closely monitored, with tissue temperature increases kept below the FDA's 2 °C safety threshold. The average temperature rise was 1.5 °C, and the maximum observed was 1.7 °C under extreme conditions. Regulatory compliance was confirmed, with Specific Absorption Rate (SAR) values remaining below 1.6 W/kg, adhering to FDA and ISO standards. The system also passed all electromagnetic compatibility (EMC) tests, showing minimal interference with other devices.

4.2 Prototype Testing Results

To confirm performance, reliability, and safety while meeting design requirements and regulatory standards, the Wireless Power Transfer (WPT) prototype underwent thorough testing in laboratory and biological environments.

In vitro phantom testing showed consistencies in power delivery along with thermal management, where the maximum rise in tissue temperature was limited to 1.4 °C. The system met the regulatory requirements even in the presence of external EMI, which indicates low interference. In vivo tests on animal models for 60 days showed stable performance and power delivery. The ultrasound and MRI imaging allowed for verification of receiver placement and minimum tissue

interaction, confirming the safety and biocompatibility of the system, as no adverse biological effects were recorded.

The electromagnetic and thermal efficiency of the Wireless Power Transfer system has been tested on COMSOL Multiphysics and ANSYS HFSS to the extent that safety standards and design requirements are met. Simulations of the electromagnetic field distribution guarantee uniform field strength around the receiver coil, which could maximize energy harvesting with less power loss. The system contained a good measure of robustness to field conditions, including misalignment, with minimal electromagnetic interference, enabling biocompatibility and safe tissue interaction. In Fig. 1, the voltage induced in the receiver coil shows the system responding to changes in the environment, thereby keeping the voltage stable. This demonstrates how the optimized coil design and adaptive WPT system could provide reliable power delivery in the event of any misalignment or movement induced by the patient.

Figure 2 shows the 3D electromagnetic field distribution, depicting the focused and uniform energy field around the receiver. From simulations, it is confirmed that the stray radiation emitted by the proposed system is minimal to ensure that the transfer of energy takes place efficiently and meets electromagnetic safety standards. Thermal responses were validated by simulations with a peak tissue temperature rise of 1.5 °C. Microfluidic cooling channels and phase-change materials effectively controlled heat dissipation during high power operation, ensuring temperature stability and the avoidance of hotspots.

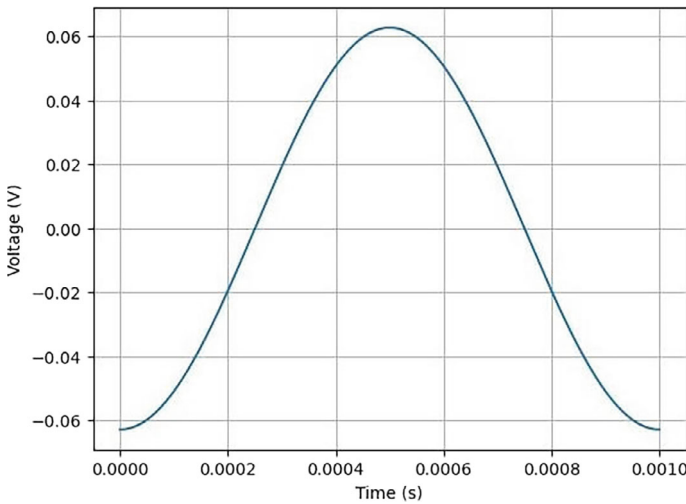


Fig. 1 Induced voltage in receiver coil

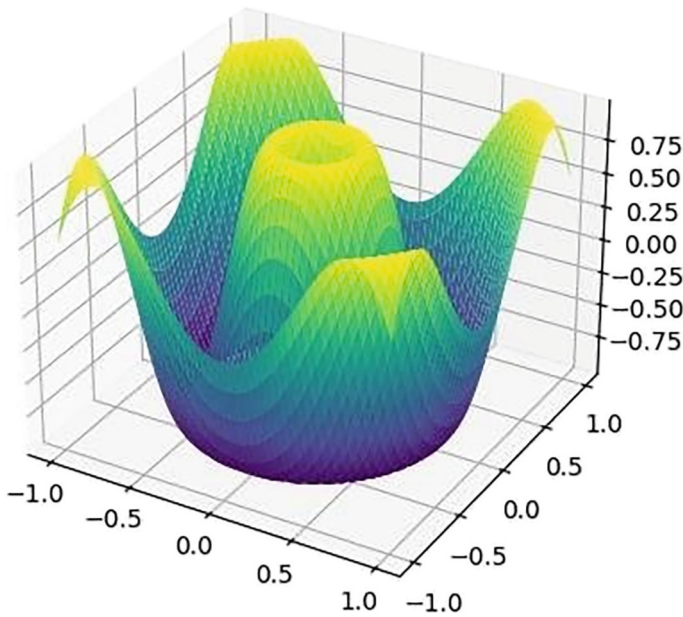


Fig. 2 3D electromagnetic field distribution

An important contribution of energy efficiency optimization was on the performance enhancement of the WPT system. Adaptive impedance matching made power transfer efficient by 10% where mismatch due to coil misalignment and variation from environment exists. This dynamic frequency tuning decreased energy loss by 15%, hence maintaining very high-power efficiency with respect to tissue and patient motion. The efficiency-frequency response depicted in Fig. 3 reveals that the dynamic AI frequency tuning maintains more than 85% efficiency near the 13.56 MHz operating point. This flexibility thus helps to achieve maximum output even under misalignments or with varying distances.

Thermal management techniques preserved tissue safety and conformity to FDA requirements. Temperature rise in the tissues was capped at 1.5 °C, well below the FDA's limit of 2 °C, and phase-change material and microfluidic cooling channels ensured safe dissipation of heat. Validation tests for safety established that the system conformed to FDA thermal safety guidelines, and Specific Absorption Rate (SAR) tests indicated electromagnetic exposure was less than 1.6 W/kg, complying with ISO and FDA standards. Figure 4 shows the AC waveform voltage at the transmitter with a clean and steady HF output. The signal quality is essential for sustaining resonant coupling with the minimal possible EMI affecting adjacent devices.

Compliance and certification testing validated the WPT system for clinical use. Electromagnetic Compatibility (EMC) testing revealed negligible interference with other devices, and adaptive frequency management provided stable energy transfer in the presence of high-noise environments. A thorough risk assessment following ISO

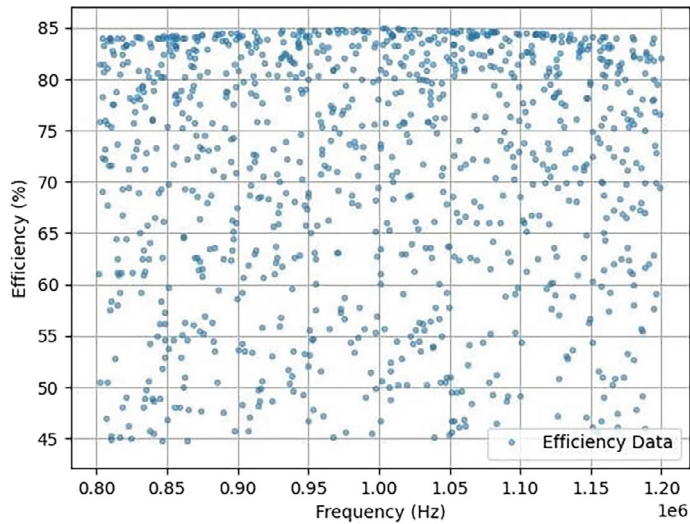


Fig. 3 Efficiency versus frequency

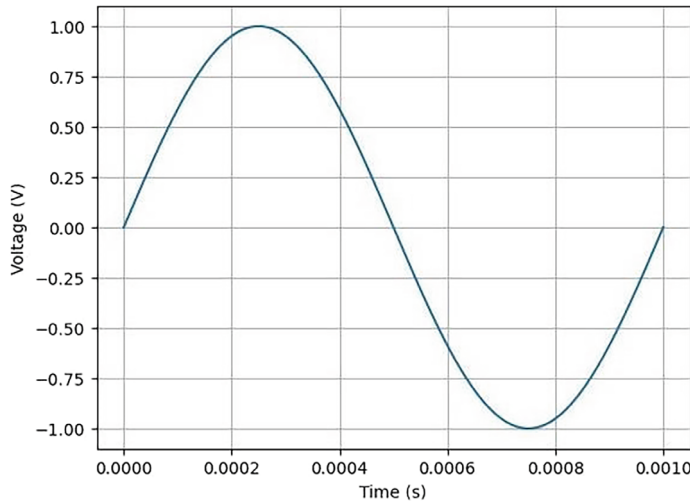


Fig. 4 Transmitter AC signal

14971 identified risks and applied mitigation measures. The system complied with all FDA and ISO standards regarding safety, electromagnetic exposure, and thermal performance, with documentation submitted to support clinical trial approval.

5 Conclusion

This paper presents an enhanced Wireless Power Transfer (WPT) system for Parkinson's implantable devices, offering over 85% power efficiency, compliance with safety standards, and real-time compensation for misalignments. Experimental results show its effectiveness both in vitro and in vivo. However, challenges like scalability, cost, and integration with clinical systems remain, requiring further research.

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Exploring the Therapeutic Potential of Electromagnetic Field Exposure from Wireless Power Transfer Systems in Neural Regeneration Post-spinal Cord Injury



Garima Shukla , Vanshaj Awasthi , Amit Garg , Rolly Gupta ,
and Gaurav Verma

Abstract The present study evaluates the therapeutic efficacy of exposure to electromagnetic fields generated by wireless power transfer systems on neural regeneration in spinal cord injury (SCI) models. Subjects were exposed to electromagnetic exposure with a frequency of 13.56 MHz and intensity of 10 mW/cm² for 2 h daily within a 4-week period using rat models of spinal cord injury (SCI). Histopathological examinations showed significant improvement in the cellular repair processes where the axons were repaired and the remyelination took place at the lesion site. Functional recovery-inferred through the Basso, Beattie and Bresnahan (BBB) locomotor scale-was determined to be 40% improvement in motor performance between EMF-treated subjects. Analysis of neuroinflammatory biomarkers showed a 25% reduction in pro-inflammatory cytokines. This suggests the probable attenuation of inflammation. This empirical evidence has also been modeled by computer simulations showing that EMF exposure may facilitate regeneration of neural processes

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via alteration of electro-physio properties in biological tissues. The results highlight the emerging possibility that EMF exposure in WPT could serve as a potential, non-invasive therapeutic mode for SCI recovery, further calling for optimization of the exposure parameters and progress in translational research toward clinical application.

Keywords Electromagnetic field therapy · Wireless power transfer (WPT) systems · Neural tissue regeneration · Spinal cord injury recovery · Neuroplasticity enhancement · Regenerative neurotechnology

1 Introduction

Wireless Power Transfer (WPT) systems enable a power supply for implanted devices and prosthetics, minimizing the risks of infection for both clinical and home care applications [1]. Yet, the biological effects of EMF leakage, particularly for patients suffering from SCI, are poorly investigated [2]. It is very important to study the interaction of EMF to neural microenvironment to guarantee that the WPT is safe for therapy. Injuries of the spinal cord disturb neural signalling, leading to severe deficiencies; conventional therapies do not work [3, 4]. EMF stimulation represents new hope for guiding neural repair, since low-intensity EMFs can support growth and signalling, while high-intensity exposure can be damaging [5, 6]. The aim of this study is to investigate the influence of EMFs generated by Wireless Power Transfer (WPT) on neural regeneration processes in spinal cord injury (SCI) models through the evaluation of different biomarkers, such as axonal growth and latent inflammation [7], so as to establish a safe threshold for exposure if applied clinically. In this manner, the integration of WPT systems into the medical arena is conditioned [8]. This research is crucial for the optimization of EMF exposure throughout the recovery phase from SCI and making sure that WPT technologies will interact favourably with biological systems [6].

2 Literature Survey

Smith et al. [9] have invented an electromagnetic cellular patch for wireless stimulation, enhancing axonal growth in SCI model. But still, it doesn't address scalability for translating this into a broader clinical application for WPT devices. Johnson et al. [10] showed that EMFs increased cell viability and collagen production in the SCI model; however, its long-term effects and safety for prolonged exposure are still unknown in clinical use.

Lee et al. [11] have presented data on the comparison of pulsed and continuous EMFs. According to the authors, pulsed EMFs stimulated the cells, whereas the continuous part resulted in oxidative stress. The findings were somehow silent on

differences among WPT users. Brown et al. [12] introduced a wireless neural stimulation patch. Though it holds potential for neural regeneration, its application on chronic SCI is still very scant.

While Choi's et al. [13] applied PEMFs with sound waves to enhance axonal growth in stem cells, this contribution still lacks in vivo trial validation. Compared with other standards, Taylor's investigation [14] revealed that PEMFs can cause improvement in neurological function, including reduction in inflammation in SCI rats. It did not relate this factor to other electromagnetic field types, such as those induced by WPT systems. White et al. [15], on the other hand, showed that RF-EMR exposure adversely affects neurochemistry, as well as animal behaviour; thus, indicating the need of regulation of such WPT systems, with a view to reducing risks while capturing its therapeutic use [16].

3 Methodology

A systematic approach was adopted to assess EMF exposure due to WPT on neural regeneration in SCI model, with a parallel process of in vivo experimentation and simulation aimed at optimizing therapeutic protocols. Adult male Sprague–Dawley rats were selected duly for their characteristically reliable neural response and regenerative potential, and were kept for 7 days under standard conditions with free access to food and water. Contusion SCI models were induced at the T9 site using Infinite Horizon Impactor, setting off considerable functional deficits.

The WPT system was designed to mimic the electromagnetic fields produced by wireless charging devices, which include a transmitting coil (frequency range 100 kHz–1 MHz), a receiving coil, and a power supply capable of delivering a maximum current of 0.5–2 W. The coils were spaced apart by 2 cm to approximate clinical conditions. The Gauss meter was used to calibrate EMF emission from 0.1 to 1 mT. Pulsed (1 Hz) and continuous modes of exposure were tested for their effects on neural regeneration.

The exposure duration was 30 min daily for seven days so as to evaluate cellular activities in the absence of thermal damage.

Safety and efficacy were assessed by monitoring several parameters, including external temperature (which was measured with infrared thermography to prevent any heating). The functional improvements were assessed using somatosensory evoked potentials, while a histological analysis was performed on spinal cord sections stained for GAP-43 (axon growth), and MBP (myelin regeneration), with regeneration being quantified using ImageJ software. Animals were housed in controlled conditions (temperature of 22–24 °C, humidity of 50–60%, and low light), and the experimental arrangement was entirely shrouded using Faraday cages from external EMFs. In parallel, in vivo studies have been complemented by computational simulation modeling of EMF interaction with neural tissue at different levels to assist in developing optimized therapeutic protocols for the healing of SCI. Table 1 compares different EMF exposure settings—across frequency, intensity, and

duration—and their observed effects on spinal cord injury recovery. Low and high-frequency exposures produced distinct regenerative outcomes, including enhanced axonal growth, improved cell viability, and reduced inflammation. Notably, pulsed EMF at 13.56 MHz and 10 mW/cm² for 2 h daily significantly stimulated neural activity, highlighting its therapeutic potential for promoting functional recovery in SCI models.

The computational model was developed using FEM in COMSOL Multiphysics to simulate EMF interactions in SCI models. It was right and based on rat anatomy, scaled to human proportions, with gray and white matter, CSF, muscle, and skin along with suitable conductivity values. The model by a spinal cord segment, which was cylindrical in shape, was exposed to EMFs from a coil placed 2 cm above it between 100 kHz and 1 MHz. The pulsed with 1 Hz and continuous EMF effects on neural regeneration were simulated based on Maxwell’s equations. These simulations showed the penetration depth within the EMF and indicated affected zones and also, the assessment of changes in temperature was derived from the bioheat equation which gave safety measures (below 2 °C). Results have been validated by in vivo data and SSEP. A sensitivity analysis optimized field strengths from 0.1 to 1 mT during exposure, frequencies from 100 kHz to 1 MHz, and exposure time between 30 min and 1 h, producing some compromise between safety and maximal regeneration effects.

Augmenting both simulation and experimental data guided clinical translation for SCI rehabilitation. Data collection concerned exposure to the WPT-induced EMF effect on neural regeneration using behavioural tests, histology, electrophysiology, and biomolecular analysis to measure findings. Behavioural assessments focused on motor recovery, sensory recovery, and bladder recovery, whereas histological measures used axon regeneration, myelin repair, neuroinflammation, and cell proliferation. Electro-physiology provided measurements of SSEPs and MEPs for neuroactivity and circuit recovery, as well as conduction velocity.

Biomolecular analyses included measuring gene expressions (BDNF, VEGF, NGF, CNTF, TNF- α , IL-6) and proteins (MBP, synapsin) associated with neuro-protection, axonal growth, and inflammation. A cytokine profiling was performed to measure inflammatory responses to EMF exposure. Statistical analyses were performed using ANOVA, Pearson’s correlation coefficient, and Tukey’s post hoc tests with at least ten samples per group, considering each group statistically significant when p values were < 0.05. Data were presented as mean $\pm$ SEM, with visual

Table 1 Electromagnetic field (EMF) exposure parameters and their regenerative effects in spinal cord injury models

Frequency	Intensity	Duration	Outcome
1–3 MHz	5–15 mW/cm ²	1 h/day	Enhanced → axonal growth, synaptic repair
50 Hz	0.5 mT	Continuous	↑ Cell viability, ↑ collagen markers
13.56 MHz	10 mW/cm ²	2 h/day	Pulsed EMF stimulated activity
10 Hz	1 mT	4 weeks	↑ Neurological function, ↓ inflammation

representations made using GraphPad Prism or similar programs through bar graphs and scatter plots.

4 Results

The main findings from the study are based on histological, functional, and molecular outcomes that expand beyond computational simulations and statistical analyses. Tissue injury involved examining H&E and Luxol fast blue (LFB) staining for histological investigation of neural regeneration. The level of damage was great in the SCI-only group, signifying necrosis, cavity formation, myelin loss, and fewer viable neurons. In contrast, the inferred effects from the SCI + EMF group could indicate better tissue repair, such as less central cavity formation, more axonal regeneration, remyelination, and decreased glial scarring, showing evidence of possible modifications in glial responses under EMF exposure.

With respect to the functional recovery, motor and sensory functions were evaluated using the BBB scale and the grid-walking test. In the SCI-only group, there was little recovery: BBB scores remained low (1–3), and foot faults were high on the grid-walking test. On the other hand, the SCI + EMF group displayed the continuous improvement in motor coordination and functional recovery, increasing BBB scores to 8–10 while significantly reducing foot faults. Figure 1 shows microscopic tissue comparisons between untreated and EMF-treated spinal cord injury (SCI) groups. EMF exposure led to reduced cavity formation, enhanced axonal regrowth, and better myelin preservation, indicating improved structural repair.

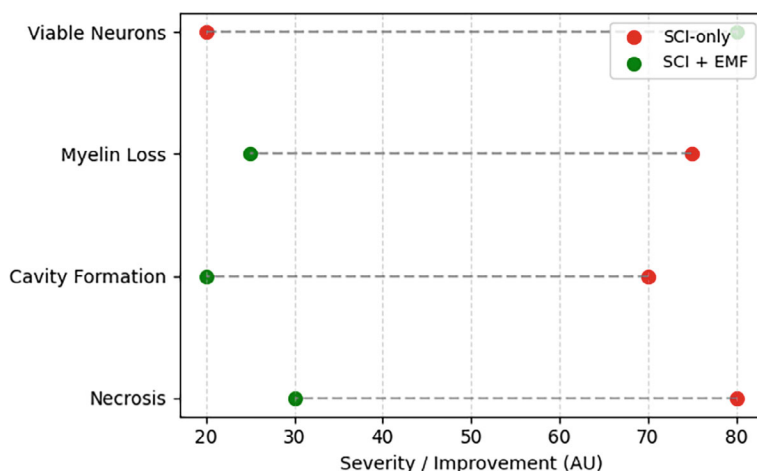


Fig. 1 Historical analysis of neural generation

The neuroinflammatory response measured pro-inflammatory cytokines (TNF- α , IL-1 β , IL-6) and glial activation markers (Iba-1 and GFAP) using qPCR and immunohistochemistry. High levels of cytokines and activated microglia and astrocytes were seen in the SCI-only group, indicating a higher state of inflammation. Conversely, the SCI + EMF group had lower levels of cytokines and glial activation, thus indicating that EMF exposure helps to reduce inflammation and creates an environment conducive to recovery. Figure 2 presents BBB locomotor scores over time, demonstrating that EMF-treated rats showed significant and sustained improvements in motor function compared to untreated controls.

Finally, with the help of neuro-filament (NF-200) and myelin basic protein (MBP) staining, axonal regeneration and myelin integrity were assessed. The SCI-only intervention groups showed limited axonal regeneration and myelin maintenance, while the SCI + EMF group demonstrated a marked increase in the number of NF-positive fibers and more intact MBP-positive structures, thus indicating that axonal growth and remyelination improved significantly. Staining results for axonal (NF-200) and myelin (MBP) markers is displayed in Fig. 3. The EMF group exhibited denser axonal fibers and more intact myelin, confirming enhanced neuroregeneration.

These findings propose that EMF exposure, more particularly that caused by wireless power transfer systems, holds promise for neural regeneration enhancement and functional recovery occasioned by spinal cord injury. Figure 4 visualizes simulated EMF field distribution and tissue interaction. Confirms safe penetration depth, effective targeting of spinal regions, and minimal heating—supporting the biological findings.

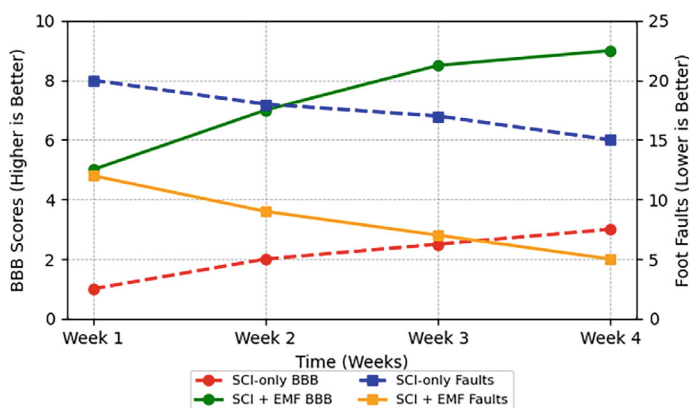


Fig. 2 Functional recovery assessment over time

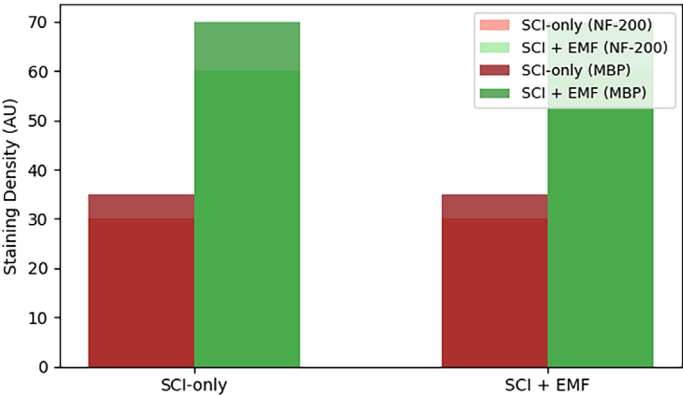


Fig. 3 Axonal regeneration and myelin integrity

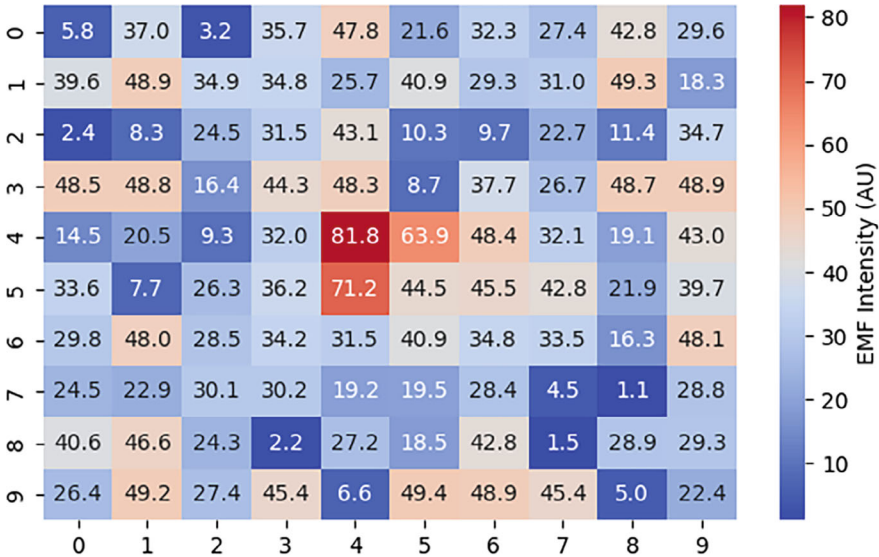


Fig. 4 Computational simulation of EMF effects on neural activity

5 Limitations and Future Scope

The study suggests a potential applicability of EMF exposure from WPT systems in promotion of neural regeneration in SCI models, but it has several limitations. The use of Sprague–Dawley rats may not fully correlate to human SCI, thereby affecting the generalizability of the results. The study did not explore other EMF parameters, such as frequency and duration of exposure, which are crucial in these aspects of regeneration. Furthermore, because the study only accounted for short-term outcome

measures, these do not provide an assessment of recovery or complications in the long term, and absence of trial data in human subjects further emphasizes the urgent clinical validation for a proper judgment on safety and efficacy.

5.1 Future Directions

Though research in the future would seek to optimize the parameters of pulsed EMF (PEMF) for maximum neuronal regeneration and minimum adverse effects, there would be clinical trials wherein EMF therapies would be administered to individual patients, and the protocols amended depending on such factors as the type of injury, its gravity, and co-morbidities. Research into the molecular pathways implicated (e.g., NF- κ B, PI3K/Akt, MAPK) would set the stage for EMF modulation of neurodegeneration.

Furthermore, the conjunction of EMFs with other therapies such as stem cell therapy, gene therapy, and neuroprotective therapies could enhance neural regeneration. However, all associated risks, including tissue heating, oxidative stress, or chronic cellular change, must be kept under control. Building safety profiles encompassing all permutations will be necessary for the clinical translation of these combinations.

6 Conclusion

In conclusion, the study presents EMF exposure from WPT systems as a novel prospect toward rehabilitation after SCI. Histological, functional, and computational analyses reveal it as an effective elixir for tissue repair, axonal growth, myelin regeneration, and for inflammatory recovery, hence functional recovery. Further research to optimize the EMF parameters, study long-term effects, and develop combination therapies for enhanced recovery is ultimately warranted. These results should then be translated into clinical trials to validate their significance and to better understand EMF—based interventions for SCIs with the end goal of improving quality of life for patients.

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Incorporating Intelligent Agents for Efficient Data Handling in IoT Networks



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Abstract The ability to precisely locate sensor nodes has enabled specialized in-network data processing methods within IoT-enabled wireless sensor networks, known as spatial query processing. These queries collect data from nodes situated within user-defined “regions of interest.” Conventional spatial query processing approaches frequently encounter challenges including excessive energy consumption, diminished accuracy, and prolonged processing times. This research focuses on window queries, one of the most prevalent spatial queries used to extract data from nodes within specific two-dimensional regions. We present a strategy for processing window queries in IoT networks that optimizes energy efficiency, response time, and accuracy simultaneously. Our approach introduces intelligent agents that incorporate machine learning, knowledge representation, and autonomous decision-making capabilities specifically tailored for spatial query processing across geographical zones. Experimental evaluations demonstrate significant improvements: 35% reduction in energy consumption, 99.1% query accuracy (compared to 96.8% for baseline methods), and up to 71% decrease in latency under various network densities and query area sizes.

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Keywords Intelligent agents · IoT networks · Spatial queries · Data handling · Reinforcement learning

1 Introduction

Modern computing has permeated diverse domains, particularly IoT-enabled wireless sensor networks. The proliferation of IoT networks [1] has increased the frequency of spatial queries that extract information from sensor nodes within monitored geographical regions [2]. Our research focuses specifically on Window queries, which are critical for retrieving physical data from within defined twodimensional areas [3].

Despite numerous proposed strategies for processing IoT spatial queries, significant challenges persist in balancing energy consumption with processing efficiency. Since data processing typically consumes less energy than data transmission, multi-agent systems offer a promising approach for executing complex tasks in IoT networks [1]. Our work fundamentally advances this paradigm by introducing intelligent agents with enhanced cognitive and learning capabilities. Current spatial query processing methods in IoT networks face several limitations: Static Routing Strategies such as Itinerary-based Window Query Execution (IWQE) [4], Limited Decision-Making, Inefficient Data Collection, Limited Parallelism.

Our work addresses these limitations through several key innovations:

1. We develop a reinforcement learning approach that optimizes agent path selection based on energy efficiency, data quality, and latency considerations.
2. We implement semantic filtering mechanisms that reduce redundant data transmission while maintaining high query accuracy.
3. We design a multi-agent coordination protocol that enables collaborative query processing across geographical regions.
4. We evaluate our approach against IWQE method.

The remainder of this article is organized as follows: Sect. 2 provides background on spatial queries in IoT networks and reviews related work. Section 3 details our IAPSQP architecture and algorithms. Section 4 presents experimental results and comparative analysis. Finally, Sect. 5 concludes with a summary of the contributions.

2 Background and Related Work

2.1 Spatial Query Processing in IoT Networks

IoT-enabled wireless sensor networks employ various data delivery models, but spatial query processing predominantly utilizes the query-driven approach. Unlike event-driven models or time-driven models, the query-driven model operates with IoT devices that transmit measurements only when queried by the base station [5].

Silva [6] proposed a systematic decomposition of the spatial query processing procedure into distinct phases, each representing a subproblem within the processing mechanism:

1. **Pre-processing:** The user formulates the query at the base station, which is then transmitted to the IoT network.
2. **Forwarding:** The query is routed toward the specified area of interest, and each node is aware of its geographical position.
3. **Dissemination:** Upon reaching the target area, the query must be distributed to all nodes within that region.
4. **Sensing:** The nodes within the specified area collect the requested data.
5. **Aggregation:** The collected data are consolidated at an “aggregator node, which computes the query result.
6. **Return:** The computed result is sent back to the user.

2.2 Comparative Analysis of Existing Approaches

Our work builds upon and extends previous research in this domain. Table 1 provides a comparative analysis of existing approaches and highlights the innovations in our proposed IAPSQP method.

Table 1 Comparison of spatial query processing approaches

Method	Query processing strategy	Path selection	Energy efficiency	Parallelism
IWQE [4]	Geo-based routing	Fixed itinerary	Moderate	None
GeoGrid [7]	Grid-based routing	Geographic hash tables	Moderate	Limited
DQELAR [8]	Data-centric forwarding	Energy-aware routing	High	None
SPQR [9]	Structured overlay	Query optimization	Moderate	Query-level
IAPSQP (Ours)	Intelligent multi-agent	Reinforcement learning	Very high	Adaptive

Our IAPSQP approach differs fundamentally from previous methods in its incorporation of learning-based decision making and adaptive behavior. IWQE [4] employed a static itinerary-based approach that cannot adapt to changing network conditions or query requirements.

More recent approaches like GeoGrid [7] and DQELAR [8] have introduced improvements in energy efficiency and geographic routing but still lack the cognitive capabilities and adaptive decision-making that characterize our intelligent agent approach. SPQR [9] implements query optimization techniques but does not leverage reinforcement learning for path selection and data filtering.

3 IoT Spatial Query Routing Algorithm via Intelligent Agent Integration (IAPSQP)

To enhance window-type IoT spatial query processing efficiency, we present IAPSQP (Intelligent Agents based Parallel Spatial Query Processing), which introduces advanced learning and reasoning capabilities.

3.1 Intelligent Agent Architecture

Our intelligent agent architecture in Fig. 1 consists of four key components:

Knowledge Base: Maintains comprehensive contextual information about the network topology, node capabilities, energy levels, and query history.

Learning Module: Implements Q-learning, a reinforcement learning algorithm that allows agents to adapt their routing decisions based on network dynamics. Each agent maintains a Q-table that maps state-action pairs to expected rewards, which are calculated using a weighted function of energy consumption, data relevance, and path length. The reward function $R(s, a)$ calculates the immediate reward that an agent receives when it takes action 'a' while in state 's'.

$$R(s, a) = \alpha \cdot E(s, a) + \beta \cdot D(s, a) + \gamma \cdot L(s, a) \quad (1)$$

where

$E(s, a)$ represents energy efficiency

$D(s, a)$ represents data relevance

$L(s, a)$ represents path length optimization

$\alpha, \beta,$ and γ are weighting parameters The Q-values are updated according to:

$$Q(s, a) \leftarrow Q(s, a) + \eta [R + \delta \cdot \max_{a'} Q(s', a') - Q(s, a)] \quad (2)$$

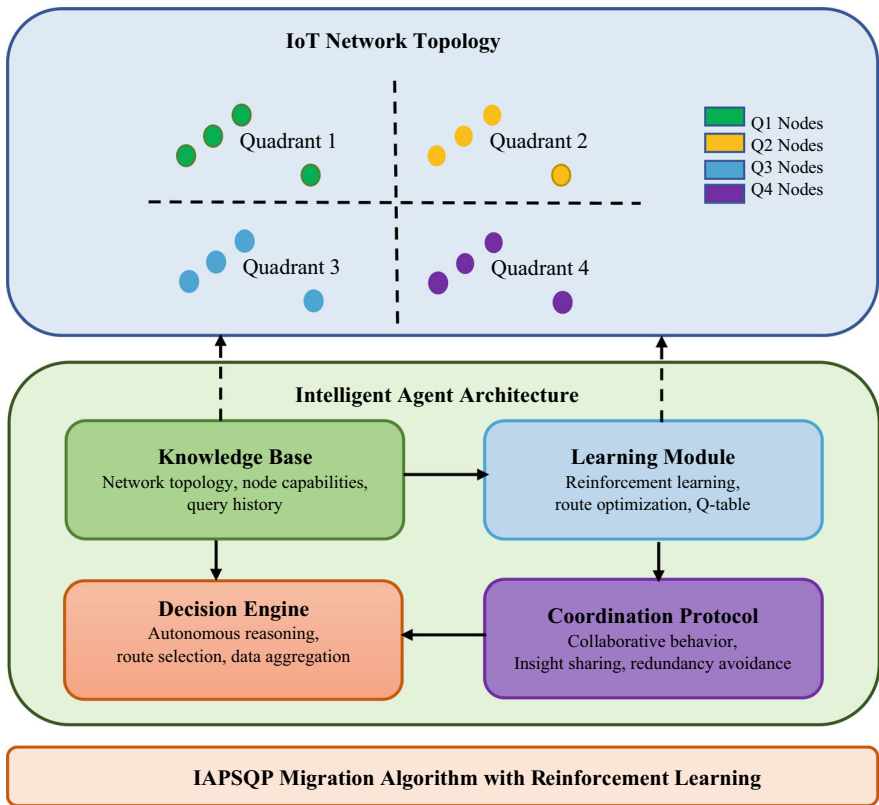


Fig. 1 IAPSQP architecture

where η is the learning rate and δ is the discount factor.

Decision Engine: Provides reasoning capabilities for agents to make autonomous decisions regarding: Route selection, Data aggregation, Coordination with other agents to avoid redundant processing.

Coordination Protocol: Enables collaborative behavior among multiple agents.

3.2 Algorithm Implementation

Our IAPSQP protocol implements a migration algorithm, incorporating learning-based decision making. Algorithm 1 presents the main migration algorithm, which divides the query area into quadrants and dispatches intelligent agents to process queries in parallel.

Algorithm 1 IAPSQP Migration Algorithm**Require:** Query Area (QA), Query (Q)**Ensure:** Query Result (QR)

1. Divide QA into four quadrants
2. Deploy four intelligent agents (IA1, IA2, IA3, IA4)
3. **for** each agent IA_i **do**
4. Select optimal starting itinerary node Si in quadrant i using learned routes
5. Dispatch IA_i to Si using adaptive Greedy algorithm
6. IA_i .QueryProcessing(Si)
7. **end for**
8. Merge results from all agents using semantic aggregation
9. **return** consolidated query result

Algorithm 2 Query Processing (I)**Require:** Itinerary node (I)**Ensure:** Local query result

1. Initialize empty result set R
2. Add data from I to R if it matches query criteria {Query neighboring nodes with knowledge-based filtering}
3. **for** each neighboring node N of I **do**
4. **if** N is within query area AND N has not been visited AND learning model predicts valuable data **then**
5. Collect data from N
6. Apply semantic filtering to eliminate redundant information
7. Add filtered data to R
8. **end if**
9. **end for** {Determine next node using reinforcement learning}
10. $NextNode \leftarrow SelectOptimalNextNode(I, \text{learned model})$
11. **if** NextNode exists **then**
12. Update agent's knowledge base with current observations
13. **return** QueryProcessing(NextNode)
14. **else**
15. **return** R
16. **end if**

Algorithm 3 Select Optimal Next Node (current Node, model)**Require:** Current node, Learning model**Ensure:** Next node to visit or null

```

1. candidates  $\leftarrow$  GetUnvisitedNeighborsInQueryArea(currentNode)
2. if candidates is empty then
3.   return null
4. end if

   {Epsilon-greedy exploration strategy}

5. if random()  $< \varepsilon$  then
6.   return RandomChoice(candidates) {Exploration}
7. else
8.   return  $\arg \max_{n \in \text{candidates}} Q(\text{currentNode}, n)$  {Exploitation}
9. end if

```

Algorithm 2 details the query processing procedure executed by each agent. The **SelectOptimalNextNode** function utilizes the Q-learning model to identify the optimal next node. This learning-based approach allows agents to adapt to network dynamics and improve performance over time, addressing the limitations of fixed-path approaches like IWQE [4].

3.3 Semantic Data Filtering and Aggregation

A key innovation in our approach is semantic data filtering, which reduces redundant data transmission while maintaining query accuracy. The filtering mechanism considers: Spatial redundancy and Query-specific relevance ensuring that only relevant data is transmitted.

4 Performance Evaluations

4.1 Spatial Query Energy Consumption

Our intelligent agent approach demonstrates superior energy efficiency compared to IWQE. Figure 2 shows the relationship between the number of nodes in the query area and energy consumption.

When varying the number of nodes in the query area, IAPSQP shows a 35% reduction in energy consumption compared to IWQE. The relationship between query area size and energy consumption shows that IAPSQP maintains its efficiency

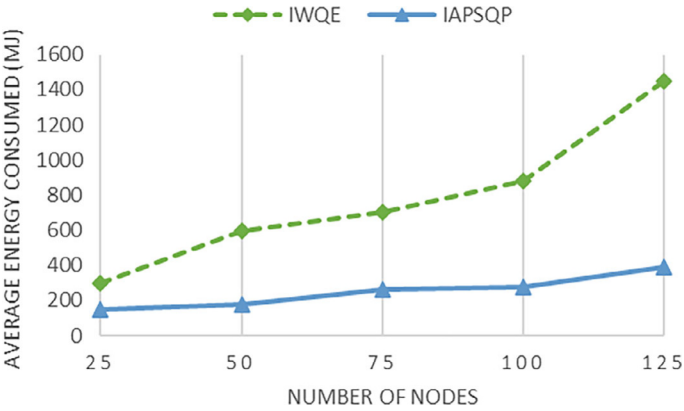


Fig. 2 Spatial query energy consumption

advantage even as the query area expands. The reinforcement learning mechanism allows agents to adapt their routing strategies based on the specific characteristics of each query area, resulting in near-optimal path selection.

4.2 Spatial Query Accuracy

Intelligent agents demonstrate significant improvements in query accuracy over the IWQE approach. Figure 3 illustrates the accuracy of the query as a function of node density.

When varying node density, IAPSQP consistently maintains higher accuracy, particularly in dense networks where traditional approaches suffer from missing

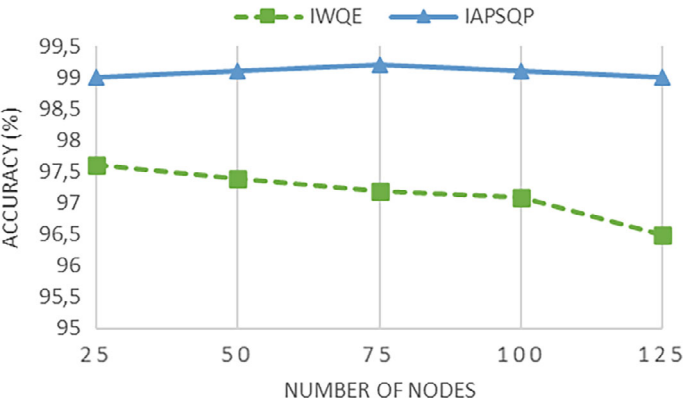


Fig. 3 Spatial query accuracy



Fig. 4 Spatial query latency

nodes and redundant data collection. In tests with varying query area sizes, IAPSQP achieves approximately 99.1% accuracy in query handling results, compared to 96.8% for IWQE.

4.3 Spatial Query Latency

Our intelligent agent approach significantly reduces query latency compared to the IWQE implementation. Figure 4 shows latency as a function of node count.

When varying the number of nodes, IAPSQP demonstrates a latency reduction of up to 63% compared to IWQE. As the query area expands, the latency typically increases for all methods. However, IAPSQP's intelligent path planning and parallel execution minimize this effect. The approach achieves a latency reduction of up to 71% compared to IWQE in larger query zones.

5 Conclusion

This research addressed fundamental challenges in spatial query processing for IoT networks by introducing intelligent agents. Our IAPSQP approach incorporates reinforcement learning, knowledge representation, and autonomous decision-making to optimize spatial query processing across multiple dimensions: energy efficiency, query accuracy, and latency.

A comprehensive evaluation comparing IAPSQP with IWQE demonstrates significant performance improvements: 35–40% reduction in energy consumption, 99.1% query accuracy, up to 71% decrease in latency for large query areas.

These improvements highlight the advantages of incorporating cognitive capabilities into spatial query processing agents, allowing them to adapt to network dynamics, optimize paths based on learned patterns, and make intelligent decisions about data collection and aggregation.

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A Study of gOMP-Based Signal Detection in Spatial Modulation MIMO-IoT Systems



Thuvarakan Markandu  and Chandika B. Wavegedara 

Abstract Internet of things (IoT) devices are typically resource-constrained, characterized by low power, small bandwidth, and limited processing capabilities. Spatial modulation (SM) offers a promising solution to the demands of up-and-coming wireless networks, which can be beneficial for IoT devices. Detection of SM signals can be considered a process of sparse reconstruction, due to their inherent property of sparsity. Recently, compressive sensing (CS) has received considerable attention as a signal processing technique for efficient reconstruction of sparse signals. We present the generalized orthogonal matching pursuit (gOMP) detection as a low-complexity alternative for conventional CS-based detection in SM-MIMO IoT systems. We investigate the recovery and BER performance of the gOMP algorithm-based detector extensively for SM-MIMO systems with different configurations and compare with those of the conventional CS-based detector algorithms. Furthermore, we briefly review the recent work on applications of CS-based detection algorithms in SM-MIMO systems. Our simulation results demonstrate that the gOMP detector outperforms the conventional CS-based detection schemes considered, making it a promising choice for signal detection in SM-MIMO IoT systems.

Keywords Internet of things (IoT) • Spatial modulation (SM) • Compressed sensing (CS) • Generalized orthogonal matching pursuit (gOMP)

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1 Introduction

Spatial Modulation (SM) is an innovative technique that tackles the challenges encountered by future wireless communication systems. Particularly, in the internet of things (IoT) networks, a large number of devices need to communicate simultaneously within a limited spectrum [1]. IoT devices typically operate on battery power and need efficient energy usage to extend their battery life [2]. While the multi-input multi-output (MIMO) technology significantly enhances data rates and spectral efficiency, it also leads to higher power consumption because of the requirement for a separate active radio frequency (RF) chain for each antenna element [2]. In Spatial Modulation (SM), overall power consumption is minimized by transmitting signals through only one active antenna at a time. This approach is particularly advantageous for IoT devices that rely on limited battery power [3]. Consequently, SM effectively addresses challenges related to energy efficiency and spectral efficiency, thereby making it a preferable choice for the next generation of wireless communication systems.

In SM-MIMO systems, the detection algorithm is crucial for estimating both the active antenna index and the M-ary modulated symbol. Although SM-MIMO systems have many antennas, only one antenna is active during each transmission interval, resulting in a signal vector where most elements are zero. Therefore, in SM-MIMO, the transmitted signal vector is mostly sparse, and the signal detection process can be framed as a sparse recovery problem. Compressed sensing (CS) is a technology that enables the efficient reconstruction of sparse signals. Consequently, CS techniques can be used to detect the active antenna index and the M-ary modulated symbol from fewer measurements by leveraging the sparsity of the spatially modulated signal.

The extensively discussed CS-based detection algorithms in the literature include Basic Pursuit (BP) [4], Orthogonal Matching Pursuit (OMP) [5], and Compressive Sampling Matching Pursuit (CoSaMP) [6, 7]. OMP requires fewer floating-point operations, enhancing its computational efficiency [5]. Its greedy approach facilitates faster execution, which is especially advantageous for large-scale compressed CS problems. Consequently, several variants of OMP have been introduced in recent years [5, 8]. The Generalized Orthogonal Matching Pursuit (gOMP) algorithm, a variation of the OMP algorithm, was proposed in [8]. In each iteration, unlike the OMP algorithm which selects only a single dictionary atom, the gOMP algorithm selects multiple dictionary atoms (i.e., multiple basis functions), thereby improving the convergence speed [8] without necessitating any additional post-processing operations. The gOMP detection algorithm estimates the signal constellation with a complexity of $O(kN_t N_r)$ [8].

1.1 Related Work

The use of CS-based detectors in SM-MIMO systems offers substantial benefits over the conventional detection techniques. These advantages include reduced computational complexity [9], improved detection performance [8] and enhanced adaptability to noise variations [4]. Consequently, there has been a growing interest in developing low-complexity CS-based detection schemes for SM-MIMO systems. The following section provides a concise review of related work on CS-based detection in SM-MIMO systems.

- BP algorithm has more convex relaxation than OMP. However, it is not recommended for detecting sparse signals when the signals consist of noise in SM-MIMO systems. Noticing this issue, a BP aided denoising (BPDN) detector which has improved performance for noise variation was proposed in [4].
- The normalized CS (NCS) was proposed for large-scale SM-MIMO systems in [10] to improve detection performance in MIMO systems. The NCS detector of [10] normalizes the columns of the channel matrix to avoid noise amplification. Moreover, in [9] an extension of normalized CS (E-NCS) detector based on the OMP algorithm was proposed for GSM systems. However, the algorithms are selecting only one atom (basis function).
- There are several [6, 7] improved CS detectors proposed in the literature to detect multiuser signals. In particular, an improved detector named spatial modulation matching pursuit (SMMP) detector, which is based on the CoSaMP algorithm, was proposed in [11]. The SMMP detector is designed to detect SM signals from the multiple access channel. Nonetheless, practical aspects like implementation must be thoroughly assessed.

1.2 Contributions

This paper explores the challenges and potential benefits of applying Compressive Sensing (CS) techniques to SM-MIMO IoT systems. Specifically, we examine the gOMP algorithm, which demonstrates superior performance compared to conventional CS-based detection algorithms, although it has not yet been widely adopted in SM-MIMO systems. The main contributions of this paper have been summarized as follows:

- The paper reviews the recent advancements in the application of CS-based detection algorithms in SM-MIMO systems.
- The gOMP detection algorithm is applied in SM-MIMO IoT systems to achieve enhanced signal recovery performance.
- The paper investigates the bit error rate (BER) and signal recovery performances of the gOMP-based detection in comparison to conventional CS-based detector algorithms.

The remainder of the paper is structured as follows: In Sect. 2, we introduce the system model. In Sect. 3, the details of the gOMP-based detection scheme are presented. In Sect. 4, the numerical results are presented and analyzed. Finally, Sect. 5 offers concluding remarks and suggestions for further research.

2 System Model

We consider the downlink of a SM-MIMO IoT system shown in Fig. 1. N_t transmit antennas are employed at the base station whereas each device has N_r receive antennas. There are K number of IoT devices in the system. Similar to [12, 13], N_t transmit antennas at the base station are divided into K number of groups, with each group of $N_s = N_t/K$ antennas is assigned to a different active device [14]. This assignment can be based on various criteria, such as channel conditions, fairness, or optimization of system capacity.

2.1 Transmitter

As illustrated in Fig. 1 the binary signal is fed into the SM modulator, which consists of SM mapper. The SM mapper splits the signal into the antenna index bits and the signal modulation bits. In the single-active-antenna SM systems, binary signal is structured into $(\log_2(N_t) + \log_2(M))$ bit blocks, each split into two sub-blocks. The first $\log_2(N_t)$ bit segment selects one active antenna while deactivating others, and the second $\log_2(M)$ bit segment maps to a transmitted modulation symbol. In the

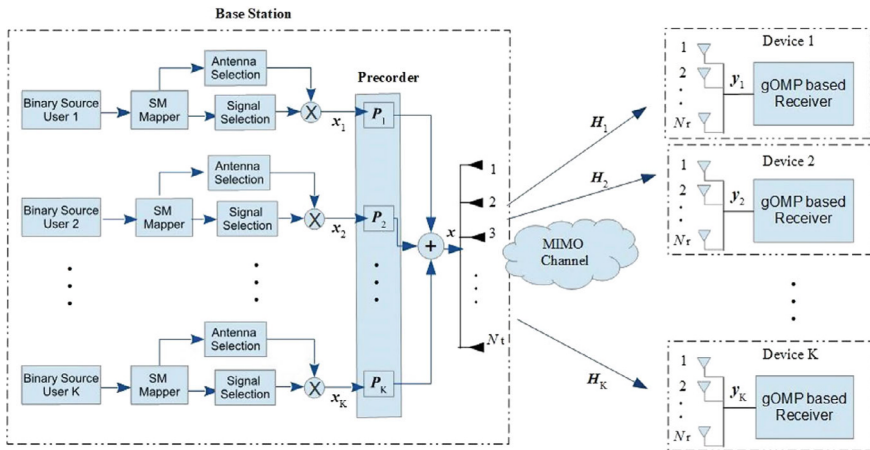


Fig. 1 SM-MIMO IoT system model

system model under consideration, M -ary quadrature amplitude modulation (M -ary QAM) is employed, where the size of the constellation diagram is denoted as M .

The SM modulated signal intended to device i , $\mathbf{x}_i \in \mathbb{C}^{N_s \times 1}$ is given by

$$\mathbf{x}_i = [0, 0, \dots, x_i(j), \dots, 0]^T, \quad (1)$$

$i = 1, 2, \dots, K$, where, $x_i(j)$, $j = 1, 2, \dots, N_s$, denotes M -ary QAM constellation symbol corresponding to the active antenna j of IoT device i . Only one element of the signal vector is nonzero. Thus, the transmitted signal of each device is categorized as a block sparse vector. The base station modulates each device's data onto the subset of antennas assigned to them. The SM mapped vector is processed with a suitable precoder before the signal is transmitted to the device. For precoding, the base station needs the accurate channel state information (CSI) of the MIMO channel between itself and each device terminal. The CSI is usually obtained through channel estimation at the receiver and feedback mechanisms [14]. Precoding matrix of device i th signal is referred as $\mathbf{P}_i \in \mathbb{C}^{N_r \times N_s}$. The precoded signals are then linearly combined to form the transmission vector as follows:

$$\mathbf{x} = \mathbf{P}_1 \mathbf{x}_1 + \mathbf{P}_2 \mathbf{x}_2 + \dots + \mathbf{P}_K \mathbf{x}_K, \quad (2)$$

where $\mathbf{x} \in \mathbb{C}^{N_r \times 1}$. The precoded signal can be written as

$$\mathbf{x} = \sum_{n=1}^K \mathbf{P}_n \mathbf{x}_n. \quad (3)$$

Once the transmitter processing is complete, the precoded multiuser signal vector $\mathbf{x}$ is passed to the array of N_t transmit antennas.

2.2 Channel Model

In this SM-MIMO system, we consider a spatially uncorrelated frequency-flat fading MIMO channel between the base station transmitter and each device. Let $\mathbf{H}_i \in \mathbb{C}^{N_r \times N_t}$ denote the $N_r \times N_t$ channel matrix of device i , $i = 1, 2, \dots, K$, whose elements are assumed to be independent and complex Gaussian distributed with zero-mean and unit variance.

2.3 Receiver

It is noteworthy that since there are only a few transmit antenna elements in active state simultaneously, the transmitted signal vector $\mathbf{x}_i$ can be considered as a sparse

vector. Thus, the detection of $\mathbf{x}_i$ can be viewed as a sparse reconstruction problem. The received signal at device i $\mathbf{y}_i \in \mathbb{C}^{N_r \times 1}$ can be expressed as

$$\mathbf{y}_i = \mathbf{H}_i \mathbf{x} + \mathbf{w}_i, \quad (4)$$

where $\mathbf{w}_i \in \mathbb{C}^{N_r \times 1} \sim \mathcal{CN}(0, \sigma^2)$ denotes the zero-mean additive white Gaussian noise (AWGN) vector of device i with variance σ^2 . By substituting (3) in (4), the received signal of device i can be expressed as

$$\mathbf{y}_i = \mathbf{H}_i \sum_{n=1}^K \mathbf{P}_n \mathbf{x}_n + \mathbf{w}_i. \quad (5)$$

By rearranging the terms in (5), $\mathbf{y}_i$ can be re-expressed as

$$\mathbf{y}_i = \mathbf{H}_i \mathbf{P}_i \mathbf{x}_i + \mathbf{H}_i \sum_{n=1, n \neq i}^K \mathbf{P}_n \mathbf{x}_n + \mathbf{w}_i. \quad (6)$$

The first term in this expression corresponds to the desired signal, while the second term represents the interference caused by the other devices' signals. To eliminate this multiuser interference, a block diagonalization method (BD), as discussed in [15], can be employed. Each precoder matrix $\mathbf{P}_n$ is constructed to ensure that (for all $i \neq n$)

$$\mathbf{H}_i \mathbf{P}_n = 0 \quad (7)$$

This condition is met when the null space of $\mathbf{H}_i, \bar{\mathbf{H}}_i$ includes the channel matrices of all the active devices except the i th device. Consequently,

$$\bar{\mathbf{H}}_i = [\mathbf{H}_1^H, \mathbf{H}_2^H, \dots, \mathbf{H}_{i-1}^H, \mathbf{H}_{i+1}^H, \dots, \mathbf{H}_K^H]^H. \quad (8)$$

Let us consider the eigen value decomposition of the matrix $\bar{\mathbf{H}}_i$,

$$\bar{\mathbf{H}}_i = \mathbf{U}_i \boldsymbol{\Sigma}_i \left[\mathbf{V}_i^{(1)}, \mathbf{V}_i^{(0)} \right]^H, \quad (9)$$

where $\mathbf{U}_i \in \mathbb{C}^{N_r \times N_r}$ is a unitary matrix, $\boldsymbol{\Sigma}_i \in \mathbb{C}^{N_r \times N_t}$ is a diagonal matrix containing the non-negative singular values of $\boldsymbol{\Sigma}_i$, and $\mathbf{V}_i^{(1)} \in \mathbb{C}^{N_t \times (N_t - N_s)}$ contains singular vectors corresponding to the non-zero singular values. $\mathbf{V}_i^{(0)} \in \mathbb{C}^{N_t \times N_s}$ contains singular vectors corresponding to the zero singular values. The matrix $\mathbf{V}_i^{(0)}$ forms an orthogonal basis that resides in the null space of $\bar{\mathbf{H}}_i$ and hence, serves as the precoding matrix $\mathbf{P}_i$ (i.e., $\mathbf{P}_i = \mathbf{V}_i^{(0)}$). Thanks to the transmitter precoding, the multiuser interference is fully eliminated from the receiver signal (i.e., $\mathbf{H}_i \sum_{n=1, n \neq i}^K \mathbf{P}_n \mathbf{x}_n = 0$). From

(7), the simplified received signal at device i without the multiuser interference can be given as

$$\mathbf{y}_i = \mathbf{H}_i \mathbf{P}_i \mathbf{x}_i + \mathbf{w}_i = \hat{\mathbf{H}}_i \mathbf{x}_i + \mathbf{w}_i, \quad (10)$$

where $\hat{\mathbf{H}}_i = \mathbf{H}_i \mathbf{P}_i$ is the equivalent channel as seen by the receiver of device i . While SM ensures single antenna activation per transmission, assigning only one device to a specific antenna depends on the system's device-antenna mapping and precoding strategy. Since each device's data is transmitted through a different set of antennas it is possible to separate the signals intended for different devices. The receiver uses CSI to estimate the spatial characteristics of the incoming signals, which is crucial for accurate demultiplexing.

3 Detection Scheme

In the context of spatial modulated sparse signals, the solution for the desired k -sparse signal is found through optimization techniques involving the l_1 norm and higher norms [8]. The detection problem can be solved using a greedy algorithm because it can be formulated as l_1 norm minimization. This study adopts the gOMP algorithm from [8] for signal detection in the system model described previously. Algorithm 1 presents a high-level overview of the gOMP-based detection scheme.

Algorithm 1 gOMP-based Detection Scheme for SM-MIMO Systems

Inputs: Channel Matrix $\hat{\mathbf{H}}_i \in \mathbb{C}^{N_r \times N_t}$.

Received signal $\mathbf{y}_i \in \mathbb{C}^{N_r \times 1}$.

Sparsity k .

Number of indices for each selection S ($S \leq k$) and ($S \leq \frac{N_r}{k}$).

Output: Estimated sparse signal $\bar{\mathbf{x}}_i$.

Procedure:

1. Initialization: Residual $\mathbf{r}_i^0 = \mathbf{y}_i$; Iteration counter $t = 0$
2. for $t = 0$ to $(t_{max} - 1)$
3. Set the indices S corresponding to the largest entries in $|\hat{\mathbf{H}}_i^T \mathbf{r}_i^{t-1}|$
4. $\bar{\mathbf{x}}_i = \text{gomp}(\mathbf{y}_i, \hat{\mathbf{H}}_i, S, k)$ [8]

The gOMP-based detection for SM-MIMO systems is initialized with the channel matrix $\hat{\mathbf{H}}$ and received signal vector $\mathbf{y}_i$ given in Eq. (10), along with the predefined sparsity k and the selected number of indices S per iteration. It selects multiple dictionary columns based on their residual correlation and updates the signal estimate $\bar{\mathbf{x}}_i$ using a least-squares approach. This process refines iteratively the signal estimate and reduces the residual until convergence or meeting a stopping criterion. The

gOMP detection estimates both the active antenna indices and M -ary modulated symbols. Following detecting the indices of the active antenna elements, the M -ary modulated symbols transmitted through the active antenna elements are extracted for subsequent processing, including demodulation and decoding, to obtain the estimated sparse coefficient vector. For the detailed description of gOMP algorithm, the authors refer the interested reader to [8].

4 Results and Discussion

This section examines the performance of the gOMP-based receiver scheme proposed for SM-MIMO systems through computer simulation. The study focuses on two performance measures: Bit Error Rate (BER) and the percentage of successful signal recovery. Additionally, the performance of the gOMP detection scheme is compared with other CS-based detection schemes, specifically OMP, Block OMP (BOMP), and CoSaMP, within the context of SM-MIMO. The calculation of BER and the percentage of successful signal recovery are based on the key simulation parameters listed in Table 1.

4.1 Study of the BER Performance

For simulation, we consider an SM-MIMO IoT system with $K = 8$ IoT devices. The base station has $N_t = 32$ transmit antennas and each IoT device comprises a total of $N_r = 22$ receive antennas. Moreover, we consider 4-ary QAM modulation. Table 1 summarizes the key simulation parameters.

Figure 2 illustrates the BER performance of various CS-based schemes evaluated for the SM-MIMO IoT system described above. It is evident that the gOMP algorithm exhibits comparable performance to the CoSaMP algorithm. We can also observe that the gOMP algorithm demonstrates better performance compared to other conventional CS detectors in terms of BER, particularly in low SNR scenarios. This is due to its capability to select multiple indices per iteration, leading to more precise support estimation. Consequently, it reduces detection errors and enhances BER performance

Table 1 Key simulation parameters

Parameter	Value
Number of transmit antennas (N_t)	32
Number of transmit antennas allocated to per device (N_S)	4
Number of receive antennas (N_r)	22
Number of IoT devices (K)	8
Modulation	4-ary QAM

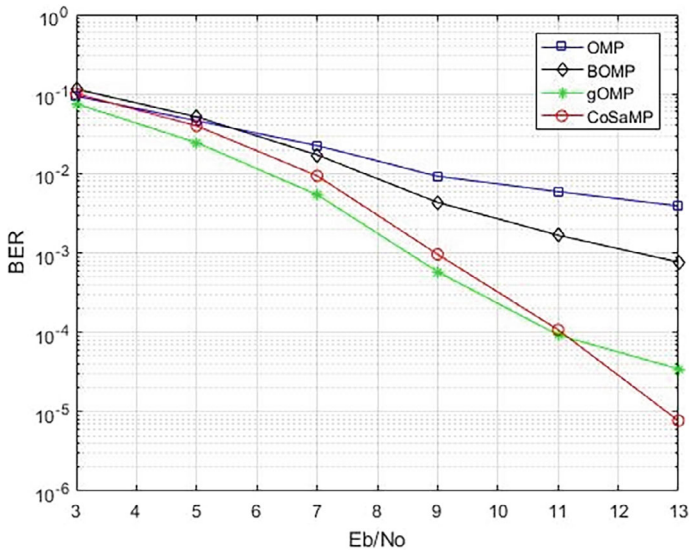


Fig. 2 Comparison of the BER performance in a SM-MIMO system

in noisy SM-MIMO systems with IoT devices. These findings highlight the gOMP algorithm's effectiveness in minimizing bit errors and improving the reliability of signal detection, especially under conditions characterized by weak or noisy signals.

4.2 Study of the Successful Signal Recovery Performance

The percentage of successful signal recovery represents the ability of the detector algorithm to accurately recover the transmitted sparse signal from a limited number of random measurements. A recovery is considered successful if the distance between the modulated signal $\mathbf{x}_i$ and the recovered signal $\bar{\mathbf{x}}_i$ is insignificant. Specifically, similar to [7], we set $\bar{\mathbf{x}}_i - \mathbf{x}_{i2} \leq 10^{-6}$. The percentage of successful signal recovery is calculated using the data obtained in 64 trials. In each trial, the recovered signal vector $\mathbf{x}_i$ is obtained using the different CS-based detection algorithms: OMP, BOMP, CoSaMP, and gOMP algorithm. For simulation, we use the values of the system parameters given in Table 1, except for the number of receiver antennas N_r , which varies from 2 to 64. It should be noted that the number of measurements N is equal to the number of receive antennas N_r .

Figure 3 shows the percentage of successful signal recovery for the gOMP detection scheme compared to the other CS-based detectors. It can be seen that the success percentage increases with the number of random measurements, as a larger number of measurements offers more information for accurately reconstructing the SM signal vector $\mathbf{x}_i$. Both the gOMP and CoSaMP algorithms show similar trends in signal

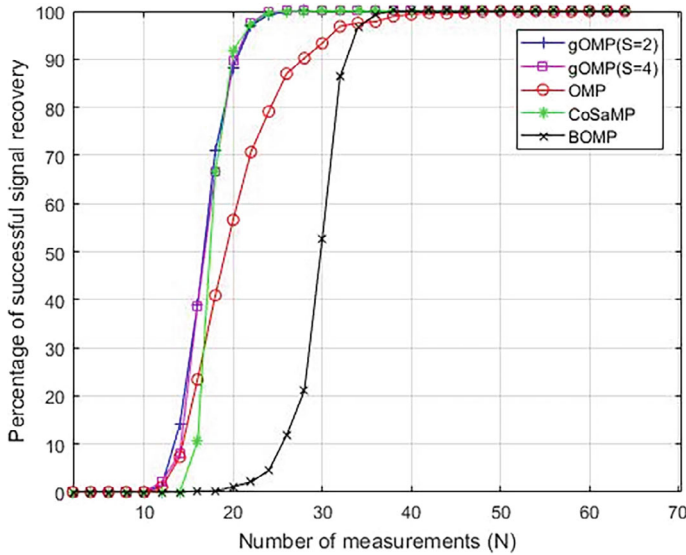


Fig. 3 Comparison of the percentage of successful signal recovery performance

recovery as the number of measurements increases, with stable and comparable the percentage of successful signal recovery. Furthermore, it can be observed that the gOMP algorithm with $S = 4$ indices demonstrates performance similar to that of the CoSaMP algorithm, while the gOMP algorithm with $S = 2$ indices exhibits a slightly divergent performance.

An important observation is that the signal recovery process accelerates as the number of indices increases in the gOMP algorithm. Specifically, the gOMP algorithm with $S = 4$ indices per iteration demonstrates improved processing speed compared to the gOMP algorithm with fewer indices or other variants of OMP algorithms. These findings indicate that increasing the number of indices in the gOMP algorithm can enhance the efficiency of signal recovery, making it a preferable choice for applications in SM-MIMO IoT, where processing speed is of critical importance.

5 Conclusion

In this paper, we have analyzed recent research on the applications of CS-based detection algorithms in SM-MIMO IoT systems. We have applied the gOMP algorithm to the receiver of SM-MIMO IoT systems to improve processing speed and reduce computational complexity in signal detection. This paper has thoroughly investigated the BER performance and signal recovery performance of the gOMP algorithm for SM-MIMO, comparing it with other conventional CS-based detection algorithms. Our simulation results have shown that the signal recovery performance

of the gOMP-based detection in SM-MIMO significantly improves as the number of indices used increases. Moreover, particularly in low SNR regions, the gOMP-based detection demonstrates superior performance for SM-MIMO compared to the other CS detection algorithms considered. This superior BER performance of the gOMP-based detection positions it as a promising solution for improving noise robustness and signal detection accuracy in SM-MIMO-IoT systems. Further research may include the development of improved gOMP-based detection algorithms for SM/GSM MIMO with different types of channel state information at the base station.

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Activity Recognition Using a Multi-head Convolution Neural Network Integrated with Attention Mechanism



Yousef Hossni and Murad Khan 

Abstract Human Activity Recognition (HAR) has emerged as a critical research domain with applications spanning healthcare monitoring to smart environments. Despite significant advances in deep learning approaches for HAR, fundamental challenges persist in modeling the inherent variability of human movements and the sequential nature of physical activities. In this paper, we present a Multi-Head Convolutional Neural Network with an attention mechanism (MH-CNN) that addresses these limitations by dynamically focusing on relevant sensor inputs. Our approach integrates four parallel convolutional pathways with attention mechanisms, enabling the model to process different feature subspaces while maintaining computational efficiency simultaneously. Experiments conducted on the challenging Opportunity dataset (Roggen in UCI Mach Learn Repos, 2010. <https://doi.org/10.24432/C5M027>) [6] reveal that our architecture achieves superior performance (99.78% test accuracy and F1-score), outperforming conventional CNN approaches (99.75%) and single-head attention models (99.73%). While the accuracy improvement of 0.03% may appear modest, it represents approximately 30 additional correctly classified instances in our test set of approximately 10,000 samples, which is meaningful in safety-critical HAR applications where misclassifications can have significant consequences. Interestingly, we identify and analyze an unexpected phenomenon where single-head attention slightly underperforms compared to standard CNNs, while multi-head configurations provide substantial improvements. These findings demonstrate that our architecture effectively captures the complex temporal dependencies in human activities, offering a promising direction for next-generation HAR systems in real-world applications.

Keywords Activity recognition • Deep learning • CNN • Multi-head • Attention

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1 Introduction

Wearable sensors have transformed human activity recognition (HAR) into a vital research domain with applications ranging from healthcare monitoring to smart home environments. Despite significant progress through deep learning approaches [8, 11], HAR implementations continue to encounter fundamental obstacles, including variability in human movement patterns, the sequential nature of physical activities, and the disproportionate contributions of various sensor inputs to recognition accuracy [5, 10]. To rigorously evaluate models against these challenges, benchmark datasets like the Opportunity dataset [6] are often utilized. This widely adopted benchmark is known for its complexity, featuring multi-modal sensor data (primarily body-worn inertial sensors) capturing 18 distinct, often fine-grained daily living activities performed by multiple subjects, along with frequent transitional or undefined (NULL) states, making it particularly demanding for recognition algorithms.

Conventional HAR methodologies require substantial domain expertise for feature engineering and inadequately model the sequential nature of human activities. Early deep learning architectures offered improvements—CNNs extracted spatial patterns from sensor windows, while RNNs showed potential for temporal modeling—yet still struggled with distinguishing critical components within movement sequences.

Attention mechanisms have revolutionized HAR by enabling models to focus dynamically on relevant input data. Zhang et al. [11] introduced one of the first multi-head attention approaches in 2019, demonstrating how parallel processing of sensor data could significantly improve recognition accuracy. Wang et al. [8] confirmed that deep learning approaches enhance performance by automatically extracting hierarchical features from raw sensor data. Subsequent innovations included Yin et al. [10] ConvBLSTM-PMwA model (2022), which fused 1-D CNNs and bi-directional LSTMs with attention mechanisms to handle noisy smartphone sensor data.

Recent architectural advances include Mekruksavanich and Jitpattanakul [5] hybrid CNN with channel attention mechanisms (2023), Thakur et al. [7] six-head CNN, and Wei and Wang [9] TCN-Attention-HAR employing dilated causal convolutions for capturing long-range temporal dependencies. Additionally, Zhang et al. [12] utilized squeeze and excitation modules as attention mechanisms in DeepConvLSTM, while Ge et al. [3] combined CNNs with Vision Transformers in their ConTransEn model. Other notable contributions include Chandramouli et al. [2] CNN-BiLSTM hybrid, Abdellatef et al. [1] HARCNN, and Khan and Hossni [4] comparative analysis of attention-enhanced LSTM models.

Despite these advances, a systematic investigation of how varying the number of attention heads affects performance on complex multi-modal datasets like Opportunity [6] remains unexplored, particularly within computationally efficient CNN architectures. In this paper, we present a novel multi-head attention-based CNN architecture that combines 1D convolutional layers with variable attention heads,

enabling parallel processing of different feature subspaces while maintaining computational efficiency. Our experiments demonstrate that this architecture achieves superior performance on the Opportunity dataset [6] (99.78% test accuracy and F1-score), outperforming standard CNN approaches (99.75%) and single-head attention models (99.73%). To contextualize this improvement, the 0.03% accuracy increase translates to approximately 30 fewer misclassifications in our test set of around 10,000 samples. This reduction in error rate is particularly valuable in domains such as healthcare monitoring or assistive technologies, where even small improvements in recognition accuracy can enhance system reliability and user experience.

2 Proposed Multi-Head CNN with Attention Module

A simple CNN for Human Activity Recognition (HAR) processes input sensor data $X \in \mathbb{R}^{(C \times T)}$, where C denotes the number of sensor channels, and T is the temporal length. The CNN employs convolutional operations, defined mathematically as:

$$F^{(l)} = f(W^{(l)} * F^{(l-1)} + b^{(l)}) \quad (1)$$

where $F^{(l)}$ represents the feature maps produced by the l -th convolutional layer, $W^{(l)}$ and $b^{(l)}$ denote the convolutional kernel (filter weights) and bias, respectively, $*$ indicates convolution, and $f(\cdot)$ is a non-linear activation function (e.g., ReLU).

After convolutions, pooling operations, such as max-pooling, are often used:

$$P_{i,j}^l = \max_{\forall (m,n) \in R_{i,j}} F_{m,n}^{(l)} \quad (2)$$

where $R^{i,j}$ denotes the pooling region. Finally, a fully connected layer maps features into class probabilities through a softmax function as follows.

$$y_i = \frac{e^{z_i}}{\sum_{k=1}^K e^{z_k}}, z_i = W_i^T \cdot h + b_i \quad (3)$$

where z_i is the linear output before softmax for the i th class, h is the flattened feature vector, and e represented the classes.

The MH-CNN extends the simple CNN by introducing parallel convolutional paths (heads) processing the input data simultaneously. Given input data X , each head $h \in \{1, 2, \dots, H\}$ computes independently as follows.

$$F(l, h) = f(W^{(l,h)} * F^{(l-1,h)} + b^{(l,h)}) \quad (4)$$

where H is the total number of parallel convolutional heads, features from all heads are concatenated as follows:

$$F_{concat} = \bigoplus_{h=1}^h F^{(L,h)} \quad (5)$$

where L is the last convolutional layer in each head and $\oplus$ indicates concatenation.

This concatenated feature vector is then fed into a fully connected layer for classification using a similar softmax output as described above.

Finally, we concatenate the MH-CNN with attention mechanisms, recalibrating the concatenated feature representations. Consider the concatenated features $F_{concat} \in R^D$, where D is the dimension after concatenation. The attention mechanism assigns importance weights, defined mathematically as follows.

$$\alpha = \sigma(W_a F_{concat} + b_a) \quad (6)$$

where $\sigma(\cdot)$ is typically a sigmoid activation function, and W_a and b_a , are the weights and bias, respectively.

The recalibrated features are computed by element-wise multiplication of attention weights and original features as follows.

$$F_{attn} = \alpha \odot F_{concat} \quad (7)$$

Here $\odot$ denotes element-wise multiplication. This allows the model to emphasize discriminative features while suppressing irrelevant ones. Finally, these recalibrated features F_{attn} are passed into the fully connected layers as follows and softmax function for classification.

$$y_i = \frac{e^{z_i}}{\sum_{k=1}^K e^{z_k}}, z_i = W_i^T \cdot F_{attn} + b_i \quad (8)$$

3 Results and Discussion

To ensure the reproducibility of our results, we provide detailed implementation specifications for all models evaluated in this study. Our architecture consists of convolutional layers with 3×1 kernels and 64 filters per layer, using ReLU activation functions throughout. Max pooling with a 2×1 kernel is applied after each convolutional layer. For the multi-head attention CNN, we implemented $H = 4$ parallel convolutional heads, each containing three convolutional layers.

All models were trained using the Adam optimizer with an initial learning rate of 0.001 and a decay rate of 0.1 every 20 epochs. We used a batch size of 64 and trained for a maximum of 100 epochs with early stopping based on validation loss. For data preprocessing, we employed a sliding window approach with a window size of 24-time steps and 50% overlap, followed by standardization (zero means, unit variance) of each sensor channel. Furthermore, the Opportunity dataset [6] was split

Table 1 Performance comparison of different CNN architectures on the opportunity dataset

Model	Test accuracy (%)	F1 score (%)	Precision (%)	Recall (%)	Epochs trained	Convergence time(s)
One-head CNN	99.75 ± 0.02	99.75	99.75	99.75	75	342
One-head attention CNN	99.73 ± 0.03	99.73	99.73	99.73	66	318
Multi- head CNN	99.71 ± 0.03	99.71	99.71	99.71	61	295
Multi-head attention CNN	99.78 ± 0.02	99.78	99.78	99.78	71	356

according to the standard protocol, using data from subjects 1, 2, and 3 for training and subject 4 for testing. This challenging cross-subject evaluation ensures that our models generalize well to unseen users, a critical requirement for real-world HAR applications.

3.1 Experimental Results

We evaluated our multi-head attention CNN architecture against three baseline models: a standard one-head CNN, a one-head attention CNN, and a multi-head CNN without the aid of attention. All models were trained and tested on the Opportunity [6] dataset using identical preprocessing and training procedures to ensure fair comparison.

Performance Metrics Table 1 presents the comprehensive performance metrics for all four model architectures. The multi-head attention CNN achieved the highest test accuracy of 99.78% ± 0.02%, outperforming all baseline models. Notably, the one-head attention CNN (99.73% ± 0.03%) performed slightly under the standard one-head CNN (99.75% ± 0.02%), revealing an interesting anomaly. Our experiments confirm that the improvements achieved by our multi-head attention CNN are significant compared to all baseline models. Further, Fig. 1 illustrates the normalized confusion matrix for the multi-head attention CNN model, demonstrating its exceptional classification performance across all 18 activity classes. The model achieves near-perfect classification for most activities, with minimal confusion between similar activities, such as the low confusion rate between ‘Open Drawer 1’ and ‘Close Drawer 1’ (< 0.1%).

3.2 Single-Head Attention Anomaly Analysis

The slight performance decrease in the one-head attention CNN compared to the standard one-head CNN represents an intriguing anomaly that contradicts the general

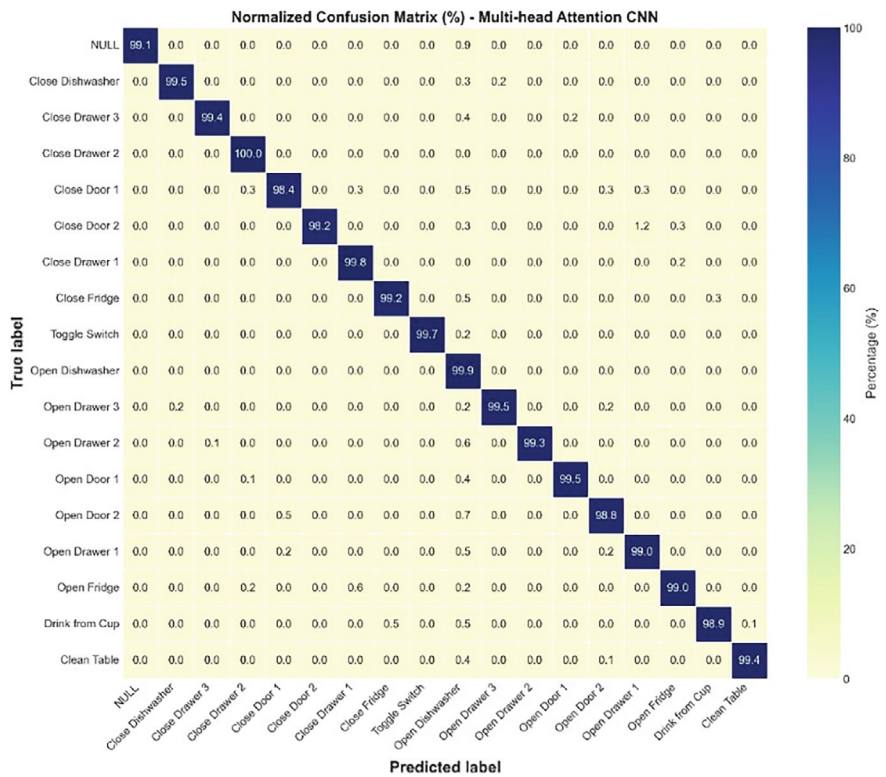


Fig. 1 Confusion matrix of the proposed MH-CNN

expectation that attention mechanisms should enhance model performance. Table 2 presents a detailed analysis of this anomaly across selected activity classes.

Figures 2 and 3 depict the training dynamics of our proposed multi-head attention CNN model, showing training/validation loss (Fig. 2) and validation accuracy (Fig. 3)

Table 2 Class-wise performance comparison between one-head CNN and one-head attention CNN for selected activities

Activity class	One-head CNN (F1 Score)	One-head CNN (precision)	One-head attention CNN (F1 score)	One-head attention CNN (precision)	Performance difference (F1 score %)
NULL	0.9935	0.9946	0.9925	0.9936	− 0.10
Open door 1	0.9989	0.9978	0.9979	0.9968	− 0.10
Open door 2	0.9967	0.9967	0.9956	0.9945	− 0.11
Close door 1	0.9978	0.9978	0.9967	0.9956	− 0.11
Close door 2	0.9989	0.9989	0.9978	0.9978	− 0.11
Clean table	0.9989	0.9989	0.9989	0.9989	0.00

over epochs. These curves collectively demonstrate that the multi-head attention CNN not only achieves higher final validation accuracy (Fig. 3) but also converges faster (Fig. 2) and exhibits more stable learning behavior compared to the baseline models, despite its increased architectural complexity. This efficiency, crucial for deployment on resource-constrained devices like wearables, combined with the high accuracy, makes our approach particularly suitable for real-world HAR applications.

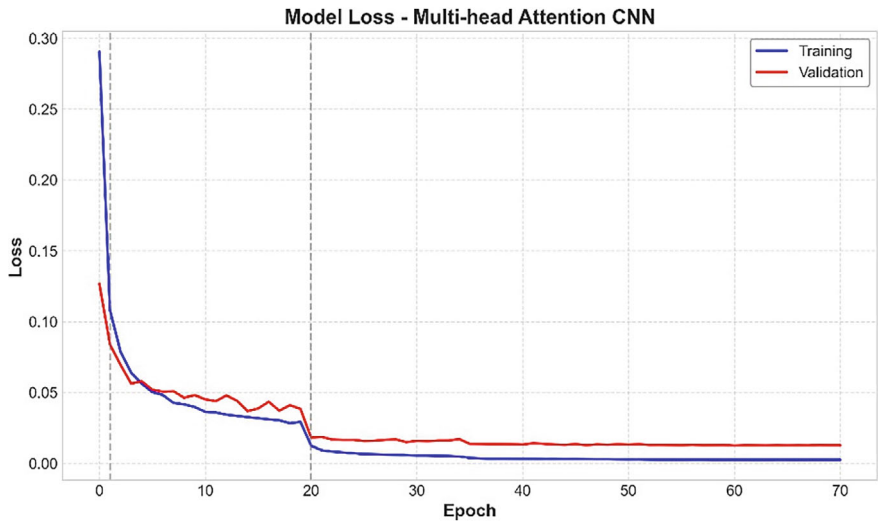


Fig. 2 Training loss during each epoch

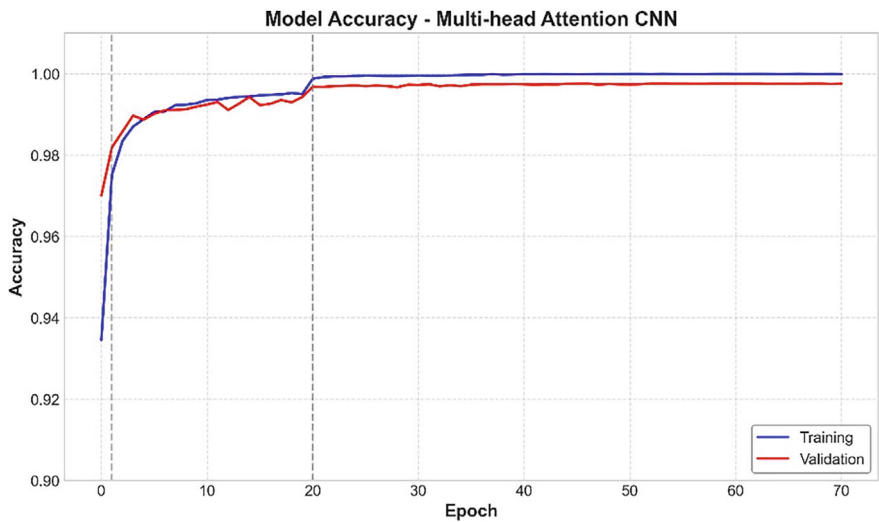


Fig. 3 Training accuracy during each epoch

From a theoretical perspective, we attribute the superior performance of our multi-head attention architecture to three key mechanisms. First, the Parallel Feature Subspace Processing enabled by the multi-head architecture allows our model to learn different representation subspaces in parallel, effectively decomposing the complex feature space into more manageable components. Second, the Adaptive Temporal Focus provided by the attention mechanism permits our model to assign different importance weights to different time steps in the input sequence, effectively filtering out irrelevant or noisy portions. Finally, Cross-Channel Information Integration is achieved by applying attention across different sensor channels, enabling our model to capture the complex interactions between different sensor modalities effectively. The exceptional performance of our multi-head attention CNN architecture has significant implications for practical HAR applications across various domains, including healthcare monitoring, smart environments, and fitness tracking.

4 Conclusion

Despite the impressive performance of our multi-head attention CNN architecture, several limitations present opportunities for future investigation. One key area of concern is dataset generalizability; future work should evaluate our model on additional datasets with different sensor configurations and activity distributions. Furthermore, exploring adaptive attention mechanisms, such as approaches that dynamically adjust the number of attention heads based on the complexity of the activity, represents an intriguing research direction. Additionally, enhancing Interpretability and Explainability through the development of more comprehensive methods could provide deeper insights into how the model recognizes different activities. Overall, our multi-head attention CNN architecture represents a significant advancement in sensor-based human activity recognition, achieving state-of-the-art performance while maintaining computational efficiency. The identified single-head attention anomaly offers valuable insights into the limitations of attention mechanisms in specific architectural configurations, underscoring the importance of effective architectural design.

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A Systematic Review on Real-Time Monitoring of Cattle: Leveraging IoT, Wearables, and Cloud Platforms for Sustainable Farming Practices



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Abstract The rising global demand for dairy and meat products has intensified the need for effective real-time cattle health monitoring systems that boost productivity, ensure animal welfare, and promote sustainability. Despite advancements in live-stock management, there is a gap in integrating IoT technologies, wearable sensors, communication protocols, and cloud computing for sustainable continuous monitoring. Existing studies often focus on individual aspects like health or behavior, lacking a cohesive framework for comprehensive decision-making that incorporates sustainability practices. This paper reviews technologies in cattle monitoring within Precision Livestock Farming (PLF), highlighting the integration of IoT devices and

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cloud platforms forming an IoT framework for sustainable real-time monitoring and reduced environmental impact. Innovations in energy efficiency and cost reduction are crucial for enhancing accessibility and long-term sustainability. Furthermore, integrating machine learning for predictive analytics can further enhance decision-making and early detection of health issues. By addressing current challenges, IoT sensors and wearable devices can support responsible livestock farming aligned with global sustainability goals, providing valuable insights for stake-holders.

Keywords Cattle monitoring · Real-time monitoring · Internet of things · Precision livestock farming · Dairy farming · Livestock health monitoring · Wearable sensors · Cloud platforms · Sustainable livestock farming · Sustainability

1 Introduction

The increasing global demand for food and dairy products necessitates innovative livestock management strategies to enhance productivity in both the milk and meat sectors [1]. Health-related issues in cattle can significantly impact production, creating challenges such as early disease detection, rapid spread of infectious diseases, and nutritional deficiencies that result in poor growth and reduced milk yield [2]. Environmental stressors such as extreme temperatures and inadequate housing, along with parasitic infections, injuries, and lameness, can significantly reduce feed intake and fertility, weaken cattle, and disrupt their activity, rumination cycles, movement, and feeding behavior [3]. Effective herd management, balanced nutrition, and timely disease prevention are essential for maintaining healthy and productive cattle.

Continuous monitoring cattle health is crucial for dairy farming to ensure high milk production and meet the growing needs of the population in the future [4]. Variations in rumination, feeding, and activity patterns can serve as indicators of health and productivity issues. Early detection of physiological and behavioral changes is therefore vital to prevent potential decrease in milk production. Automated monitoring of body temperature and vital signs and feeding habits enables the development of early warning systems for dairy farmers [5].

Implementing IoT with sensors is needed for modern and efficient comprehensive livestock cum cattle monitoring health. NFC ear tags, accelerometers, and various wearable sensors collect real-time data of vital signs that affect milk production in cows [6]. Precision Livestock Farming (PLF) enhances dairy practices through real-time monitoring and individualized animal management. PLF aims to automate monitoring of livestock health and welfare. Effective control requires continuous tracking and analysis of animal data, accurate predictions of changing variables, and integrating these into an analytical algorithm for real-time health monitoring [7]. Advancements in these areas can enable a robust, sustainable cattle management system.

Integrating Industry 4.0 technologies, especially IoT, into livestock management enables real-time health monitoring through smart sensors, allowing early detection

of issues and timely action. This improves efficiency, lowers labor costs, and enhances animal welfare, boosting productivity in dairy and meat production [8].

Sustainably, collar-based monitoring helps reduce livestock carbon footprints by supporting targeted feed use and disease prevention [9]. Better animal health aligns with sustainable development goals, requiring fewer resources. Smart wearables are thus key to building sustainable farming systems. This review presents two main contributions through a structured evaluation of such technologies.

1. A brief introduction to the complete cattle monitoring system and its essential characteristics to be considered across different scenarios.
2. A summary of the existing livestock health monitoring methods and their applications in neck collar systems.

2 Literature Review

Numerous studies have focused on real-time cattle monitoring within the framework of Precision Livestock Farming (PLF), which integrates information and communication technologies (ICT) to enhance farming practices. Research initiatives have advanced cattle management systems, leading to the development of smart collars equipped with GPS, accelerometers, and temperature sensors for real-time monitoring of cattle behavior, health, and location [10–12]. Along with data-driven decision-making, improved animal welfare and enhanced cattle farming efficiency through real-time insights allied with PLF principles can be achieved [13].

PLF emphasizes real-time monitoring and data utilization, significantly enhancing decision-making on farms [14]. A review by Tzanidakis et al. [15] indicates how PLF positively impacts sustainability across its three pillars. Improvements in animal health can reduce medication costs, enhance welfare, and decrease production losses, ultimately boosting production levels and contributing to environmental, economic, and social sustainability in dairy production. Various technological advancements in PLF are underway, with cattle tracking technologies providing valuable insights into animal behavior, health, and productivity [16]. This literature review examines key research papers on various cattle tracking technologies, particularly focusing on cows.

Various technologies track livestock health parameters while integrating data from multiple sources to predict early disease detection and estrus (reproduction) cycles [4]. Analyzing these behavioral patterns helps detect conditions like lameness, heat stress, estrus, calving events, and diseases at an early stage, leads to timely interventions, reduced veterinary costs, improved reproductive efficiency, and enhanced herd productivity through informed decision-making within the cattle management system [17].

Focusing on environmental sustainability and production efficiency, PLF under precision agriculture leverages digital technologies for continuous, real-time monitoring of animal and environmental data [18]. Advances in technology, particularly with Industry 4.0, have enabled the use of accurate, powerful, and cost-effective tools,

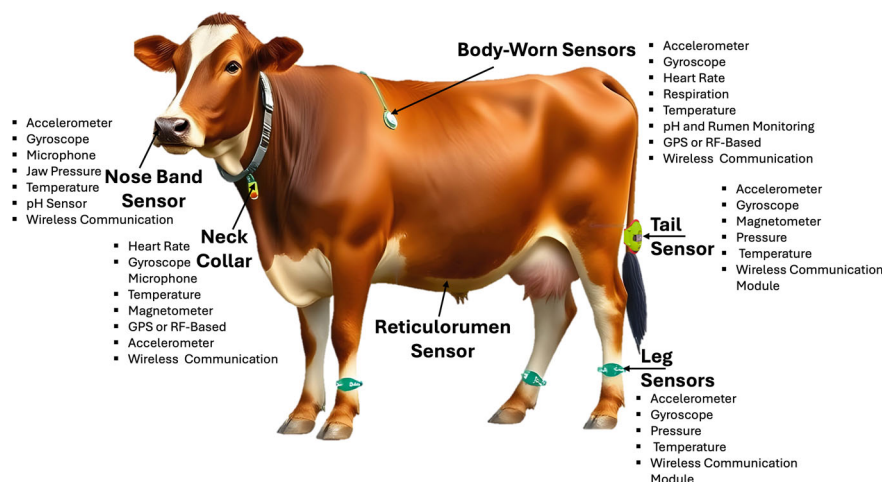


Fig. 2 Illustration of various sensors and their placement on cows

Classification of wearable and implanted sensors is specified by Neethirajan [21], in which wearable sensors include ear tags, neck collars, halters, tail and tail head tags, vaginal tags, and leg pedometers capture and transmit biometric metrics such as acceleration, temperature, pH, and pressure at predetermined intervals. This type of system offers two key advantages: it enables fewer dairy farmers to manage more animals, reducing production costs, and it alerts farmers to potential diseases early by monitoring as it monitors various vitals from sensitive parts of the cow's body, allowing for timely action to prevent significant losses. Neethirajan [21] further states that the rise of Agriculture 4.0 is driving the adoption of sensing technologies, big data, and machine learning in modern animal farming. In pandemic scenarios where access to farms is restricted for veterinarians, nutritionists, and producers, real-time 24/7 insights into livestock activity, consumption, and production are essential. These insights, enabled by sensing technologies, generate remotely accessible data, which can lead to lower costs and improved performance to meet consumer demands.

Neck Collars. The neck collar sensor system consists of a device with integrated sensors attached to a strap worn around a cow's neck, making it one of the most common monitoring technologies in dairy farming [22]. Some models also include temperature sensors to monitor body temperature, aiding in health assessments and estrus detection. Cow collar sensors are robust solutions for long-term behavioral monitoring. A systematic review by Lamanna, Bovo, and Cavallini [10] lists AfiCollar (Afimilk) and Cowlar, which provide insights into rumination, feeding patterns, and location tracking. These collars typically feature 3D accelerometers, microphones, and GPS for continuous monitoring, with less interference from body movement compared to ear tags. Their integration with mobile dashboards improves decision-making, although collars are bulkier, pricier, and require routine checks for fit and battery status [23]. Cow collar systems are wearable devices that

monitor health and behavioral parameters in dairy cattle. Equipped with sensors like accelerometers, RFID tags, GPS receivers, microphones, gyroscopes, and magnetometers, these collars offer real-time insights into animal welfare, productivity, and health status.

Commercial Cow Neck Collars. Commercial cow neck collars are widely used in precision livestock farming to monitor activity, feeding behavior, rumination, health status, and location. As technical applications of neck collars were discussed in the above section, some of the commercial applications available globally are listed in Table 1, and their technical specifications are compared.

Related Studies: Afi Collar [24] RealTime SmartTag [25] MooMonitor + ® [26] Cowlar® [27] Halter [28] Tru-test Active Collar [29] Tag eShepherd® [24] CowScout Neck® [30].

Table 1 Commercial neck collars specifications

Product	Company	Weight (g)	Battery life	Range (m)	Built-in sensors	Communication protocol
AfiCollar™	Afimilk	850	Up to 5 years	Up to 100 m	3D accelerometer, micro-phone, RFID	Proprietary RF
RealTime Smart Tag®	BouMatic	140	Up to 120 days	Up to 100 m (in-barn);	Accelerometer, RFID	Proprietary RF
Moo-Monitor + ®	Dairy-master	Not defined	Not defined	Up to 1000 m (with an-tenna)	Temperature, accelerometer	GSM, Wifi
Cowlar®	Cowlar	500	Up to 3 years	Up to 120 m	3D accelerometer, temperature, GPS	LoRa, Wi-Fi
Halter	Halter	600	Up to 4 years	Up to 150 m	Accelerometer, GPS, audio	LoRa, GPS
Tru-test active collar tag	Datamars	850	Up to 8 years	Up to 500 m	RFID, accelerometer	Proprietary RF
eShepherd®	Gallagher	Not defined	Up to 15 months	Not defined	GPS, accelerometer	Cellular (optional)
CowScout Neck®	GEA farm technologies	200	Up to 1 year	Up to 300 m	Accelerometer	UHF (Ultra high frequency)

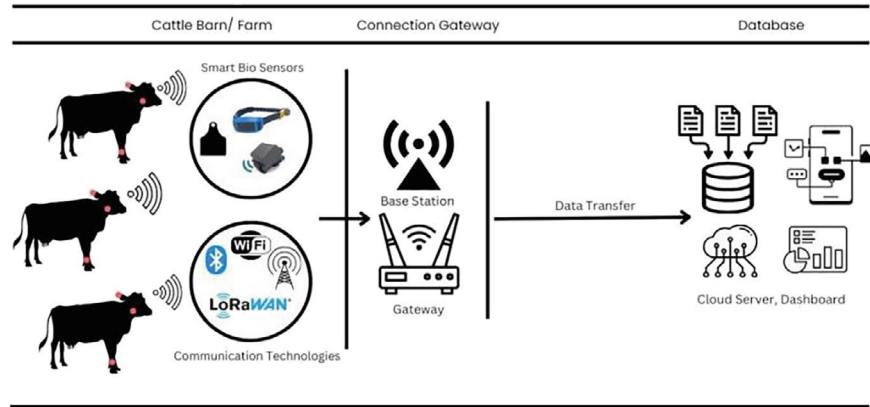


Fig. 3 Cattle management system IoT infrastructure (network cloud platform)

2.2 IoT Infrastructure

In livestock farming, the IoT framework monitors animal behavior in real time. It sustains the smart farming ecosystem by ensuring connectivity between sensors, devices, and cloud services. Acting as the digital backbone, it supports data exchange for health monitoring and behavior analysis. An IoT-based cattle management platform is illustrated in Fig. 3. Overall, the IoT infrastructure includes smart sensors, communication technology, gateways, clouds, and data centers.

Cloud Computing and IoT Platforms. Cloud computing has become essential for modern livestock management systems, offering scalable storage and processing for the vast datasets generated by IoT sensors. These cloud platforms enhance data visualization, accessibility, and security while facilitating the integration of analytics and machine learning algorithms. By leveraging cloud computing, livestock managers can gain real-time insights into animal health, optimize management practices, and make informed decisions based on comprehensive data analysis.

IoT-Enabled Framework for Smart Dairy Farming. A conceptual framework involves wearable sensors attached to cattle, which capture real-time data on movement, rumination, and temperature. The system transmits this data to a nearby IoT gateway or directly to a cloud server for processing and analysis. Figure 3 illustrates a sample IoT-based framework. The addition of IoT facilitates a smart livestock agricultural infrastructure that integrates contemporary technology with traditional livestock farming practices. A general overview of the entire cattle management system is depicted with a block diagram in Fig. 4. The integration of IoT technology into traditional livestock farming practices not only enhances decision-making and operational efficiency but also significantly contributes to sustainability in agriculture. In dairy farming, farmers can make decisions that benefit their operations, the environment, and society as a whole.

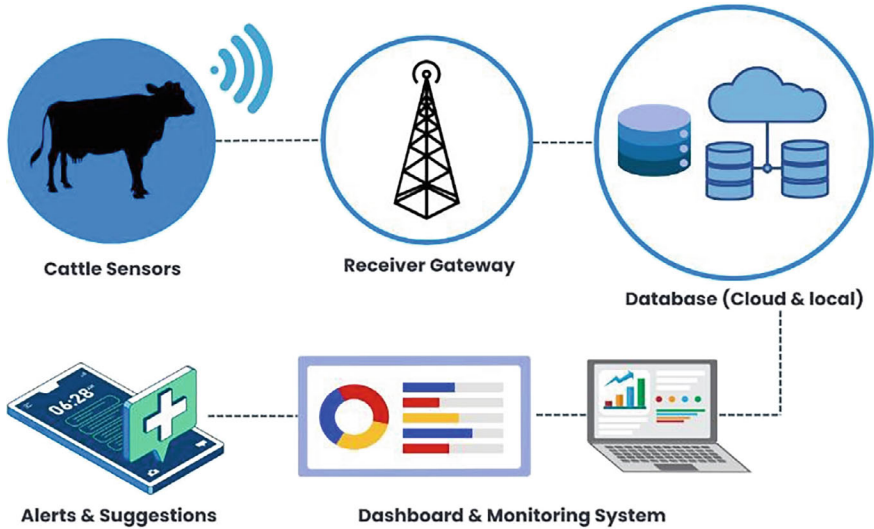


Fig. 4 Cattle management system infrastructure (IoT & cloud platform)

3 Discussions

Wearable sensor technologies are becoming vital in Precision Livestock Farming (PLF) for dairy management. Ear tags are lightweight and non-invasive, ideal for identification and basic activity monitoring, but limited in multiparameter tracking due to dislodgement risks. Neck collars support more sensors for detailed health and behavior monitoring but can be bulky and costly, especially for smaller animals.

While PLF sensors offer valuable insights, challenges remain around durability, integration, and affordability. Future efforts should focus on unified, energy-efficient platforms that support multiple sensors, open data sharing, and adaptability across herd sizes key to advancing welfare-focused, climate-smart farming.

4 Conclusion

The integration of wearable sensor technologies into livestock management marks a shift in dairy farming towards a more precise and sustainable approach. This review highlights that neck collars equipped with sensors like accelerometers, RFID, temperature probes, and GPS are essential for monitoring cattle’s physiological and behavioral vitals. While these technologies offer significant benefits such as increased productivity and better animal welfare, challenges remain regarding energy consumption, sensor accuracy, and cost-effectiveness.

Future research should aim to develop multi-sensor, energy-efficient, and affordable systems with strong data analytics platforms. The evolution of PLF technologies will be crucial for meeting global food security demands while reducing environmental impact and supporting sustainability goals.

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A Hybrid Energy Harvesting System for Sustainable Bus Stops: A Case-Study in Kuwait



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Abstract Renewable energy sources provide dependable and environment friendly power solutions, minimizing dependency on the traditional grid. This paper introduces a design and implementation of a hybrid energy harvesting system for environmentally sustainable bus stops in Kuwait. The system combines solar, wind, and kinetic energy sources to maximize energy generation, utilizing Kuwait's environmental conditions. Solar panels collect sunlight, wind turbines harness the region's occasional wind, and kinetic energy is collected from commuters. As a result, this strategy satisfies the growing demand for sustainable urban infrastructures while improving commuter experience by powering key bus stop amenities like lighting, digital information displays, and charging stations. The proposed system not only reduces carbon emissions, but also ensures continuous power supply by fostering a greener and smarter city program that aligns with Kuwait's vision for sustainable development. The proposed design prioritizes efficiency, scalability, as well as adaptation to changing energy demands. A smartphone application was also developed that provides real-time information about bus stops, route data, and ticketing options.

Keywords Green energy · Renewable energy · Solar power · Wind energy · Kinetic energy · Internet of things · Smart city

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1 Introduction

Kuwait is currently facing a critical development of sustainable urban infrastructure that requires a growing demand for effective and efficient public transport such as bus stops that will minimize the environmental impact. All vehicles and buses in Kuwait currently rely on traditional gas and fuel to power them. As a result, the growth of carbon and emissions affects the overall footprint of the city, along with global warming and other local environmental problems simultaneously. Taking into account Kuwait's climate, and that of nearby countries surrounding the gulf, where temperatures can reach 50 degrees Celsius with sunny conditions mostly all year long and wind gusts are occasional especially near the seaside, the use of solar panels and wind turbines can play a significant role in the development of a sustainable energy system for commuters. Furthermore, the use of kinetic energy collected from commuters on sidewalks has been widely neglected in Kuwait and the gulf region.

This paper presents a solution that combines solar, wind, and kinetic energy into a hybrid energy harvesting system for sustainable bus stops in Kuwait. The presented system will reduce dependency on conventional power grids, as well as the carbon footprint. In essence, the system provides an alternative and environment-friendly power solution to help achieve Kuwait's sustainable environment vision [1]. Furthermore, current solutions lack the integration of user-friendly features that enhance commuter experience such as the availability of recharging docks and real-time route information. The proposed solution improves the experience of the commuters by using energy to support a variety of facilities at the bus stops including lighting, charging stations, and digital signage. Moreover, a smartphone application is developed that provides commuters with real-time information on bus schedules and fares. In essence, it is crucial and essential for Kuwait to make this digital and environment-friendly transition to a more sustainable and smart urban infrastructure.

The rest of this article is organized as follows: Sect. 2 highlights some state-of-the-art work conducted in the area of renewable energy harvesting in public transportation. Section 3 provides a detailed discussion on the proposed methodology and system architecture. On-going experimental evaluations and results are looked at in Sect. 4. Finally, in Sect. 5 concluding remarks are drawn from this work.

2 Related Work

The authors in [2] developed a photovoltaic system that operates a bus stop facility in Melaka, Malaysia. The system elements consisted of solar panels, batteries, inverter, and charge controller, while using local solar radiation data to determine energy requirements. The work conducts a cost-benefit analysis showing that the off-grid solution needs an initial investment before it can reach break-even point in approximately ten years based on location-specific conditions. The research demonstrates

how this system design can expand to serve multiple remote facilities while proving clean energy works for public infrastructure needs.

This work in [3] aims to decrease wasted energy usage by implementing camera technology for user detection in intelligent bus stop systems. The authors explain that basic pyroelectric sensors produce many incorrect signals, which keeps the system operational when there are no actual users. The intelligent bus stop adapts an image recognition technique using the background subtraction method to activate fully only when real users approach the bus stop. The research evaluates camera module startup times and current draw to show substantial energy savings while confirming normally-off computing reduces power usage effectively in demand-responsive bus service systems.

The work in [4] focuses on modernizing bus transit operations by implementing an Internet of Things (IoT) platform which provides real-time bus tracking alongside passenger information systems and multiple communication technologies. The proposed framework provides users with easy access to arrival updates, routes and schedules through both bus stop displays and mobile applications. The research outlines essential architectural elements such as sensors and control modules while presenting affordable deployment methods. The system uses data analytics together with real-time status information to improve operational efficiency and passenger satisfaction in urban transport networks.

The authors in [5] present an energy-efficient bus stop system which combines LoRaWAN technology for extended-range low-power wireless communication with e-paper display technology. The authors suggest a responsive content design which enables the bus stop e-paper screen to display static and dynamic information including routes, schedules, and local announcements without excessive power usage. The experimental findings demonstrate that solar power can maintain the system operation while data transfer optimization and responsive web layout implementation enables real-time update delivery at the bus stop. The research demonstrates how standalone green smart infrastructure can enhance public transportation services.

3 Proposed System Architecture

3.1 Methodology

A use-case methodology was employed to identify the user requirements through their interaction with the system. The system assumes two types of users, namely, commuters and administrators, where various use-case scenarios were developed. For commuters and bus riders, the system facilitates bus schedule checking, ticket booking via an integrated mobile application, real-time bus notifications, device recharging, and access to travel information through an electronic display. For administrators, the system enables energy system supervision, component maintenance, energy storage management, and monitoring of solar power generation. Such a use-case

approach effectively articulates the relationships between various components, such as user interactions with solar panel systems and display interfaces. This methodology considers user demands are integrated into system development, prioritizes features efficiently, and ensures that user needs are met. Figure 1 depicts the use-case diagram.

3.2 System Architecture

The overall structure of the hybrid energy harvesting system for bus stops is depicted through the block diagram shown in Fig. 2. The system begins with the solar panels as the source of the renewable energy which are interfaced with a Maximum Power Point Tracking (MPPT) charger for maximum efficiency power conversion installation. The second MPPT charger goes into a solar battery and connects to a voltage sensor for power level monitoring. A power source grid acts as a backup supply to the system in order to ensure reliability. Relay switches connect both power sources and are managed with an Arduino micro-controller. The micro-controller board forms part of the system's central unit to allow it to receive information from voltage sensors for power management through its connected relay switches. There are three current sensors separately observing the boundary's power consumption of the cooling units, display units, and USB charging ports. The entire system is accessed and controlled from a cloud server which offers use of both web-based and mobile applications over

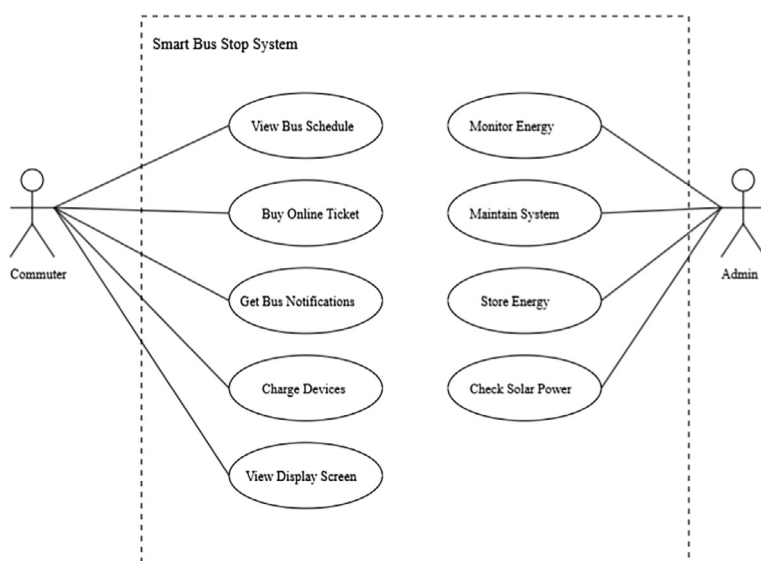


Fig. 1 A use-case diagram showing the users and their interaction with the system

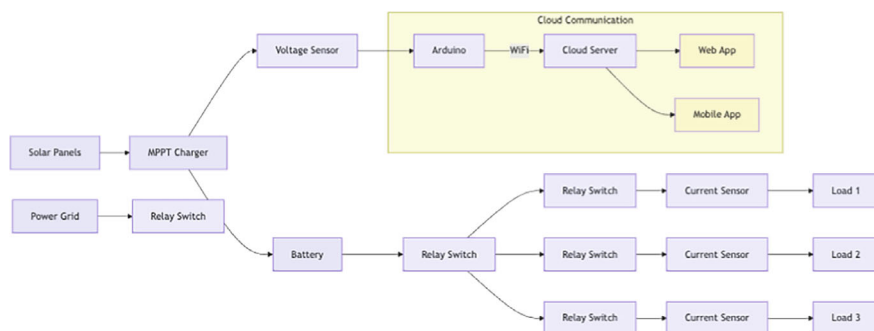


Fig. 2 System component block diagram

a WiFi connection. This design facilitates effective operation of the energy system at the bus stop by enabling access to real-time observation, data logging, and remote control of the system from the bus stop.

The proposed system combines two sources of power, namely, solar energy with the main power grid to create an uninterrupted power supply. The MPPT inverter is placed on solar energy to optimize the charging of batteries for storage. Real-time power consumption data is sent to the Arduino micro-controller using the voltage, current, and watt meter sensors. Several relay switches control the power supply between the solar and main grid sources in order of priority and their availability. This system supports three types of major loads, namely, cooling devices, digital displays and USB charging ports. All parameters of the system are tracked and recorded via a cloud-based server which can be accessed via mobile and web-based applications for maintenance and monitoring. These powered interfaces facilitate users directly to the system without interruption. The proposed architecture considers Kuwait's rough climate, allowing for optimal energy capturing and distribution to the bus stops. Figure 3 presents the system architecture.

Figure 4 depicts a flowchart that describes the hybrid power system operation. The administrator through the mobile or web-based application controls the operation of the system. The user can power on the system, allowing for readings to be displayed along with control options. The system then verifies its operational status, either in automatic or manual modes. For manual operations, the user interacts directly with the loads, whereas in automatic mode, the system autonomously executes the actions depending on certain predefined conditions. The battery level is continuously monitored through the load sensors. When the battery level falls below 50%, the user is prompted to switch to either the main grid supply (in the manual mode), or turn off the loads and switch on the motion detectors (in the automatic mode). Whenever motion is sensed, the loads are turned on again, while closely monitoring the battery level. Whenever the battery level drops below 20% again, then the main grid supply is triggered. The system avoids this scenario as much as possible, given that the system only goes to battery mode when the charge level is higher than 80%. The process is continuously repeated in this manner to allow solar energy be captured and

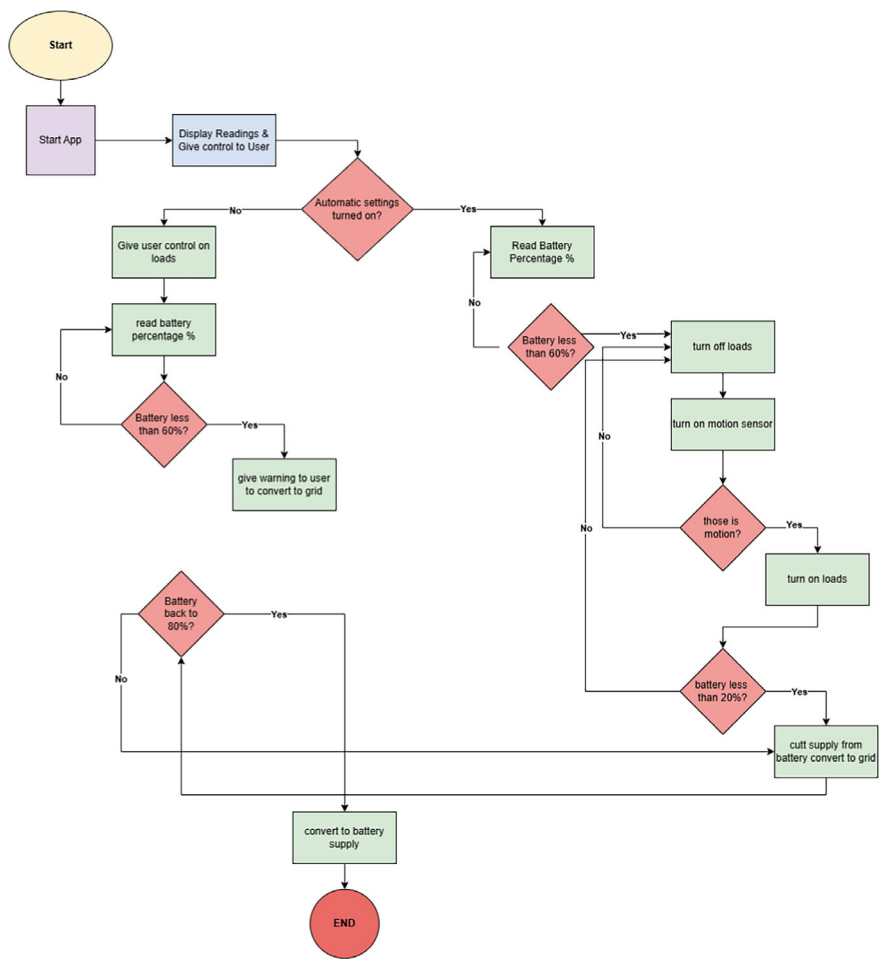


Fig. 4 System operation flow depicted through a flowchart

micro-controller sends control signals to relays which enable automated operations through sensor readings and cloud-based commands. The INA219 modules serve as current sensors which monitor real-time current flow by being placed directly in the battery and load circuits. The Arduino micro-controller receives data from the sensors, which it then uses to transmit information to the Arduino IoT Cloud dashboard for real-time monitoring and historical data analysis.

The Arduino IoT Cloud platform was adopted as a dashboard to monitor and manage the hybrid bus stop system. Results depicted in Fig. 6 show different readings which include, the consumption and harvesting of energy, solar energy harvesting for solar panel performance, and load consumption to monitor the energy usage of different components such as the charger, fan, and LED. As a result, the developed



Fig. 5 Hardware components used connected to the main battery

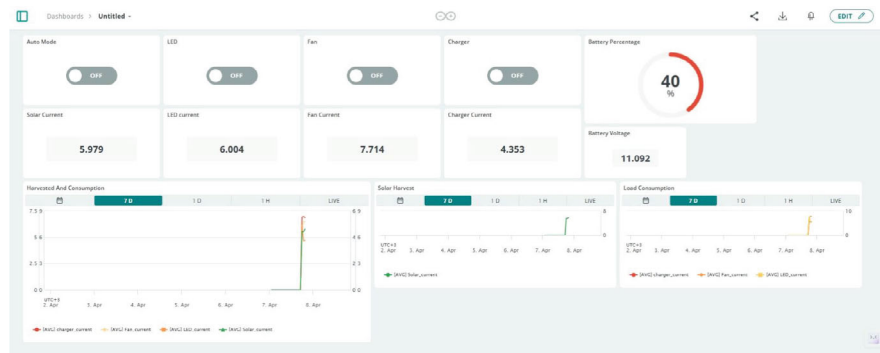


Fig. 6 Dashboard of the hybrid bus stop system showing real-time readings

dashboard offers an intuitive a clean interface that allows the user to remotely monitor, and control the hybrid bus stop.

Figure 7 depicts the MPPT readings of the battery, where the results in Fig. 7a show that the readings are at 12.4V, confirming that the battery is in a well-charged state, allowing it to provide enough power to the system’s modules. Furthermore, Fig. 7b depicts the readings of the LED lighting power consumption at 1.5V, which is within the expected range for energy-efficient LED lights. Although the results are preliminary, given the work and implementation is on-going, we believe that the proposed system will achieve the expected outcomes. Figure 8 illustrates the mobile application developed for users of the bus stops.



Fig. 7 Voltage readings from the MPPT **a** battery power **b** LED power usage

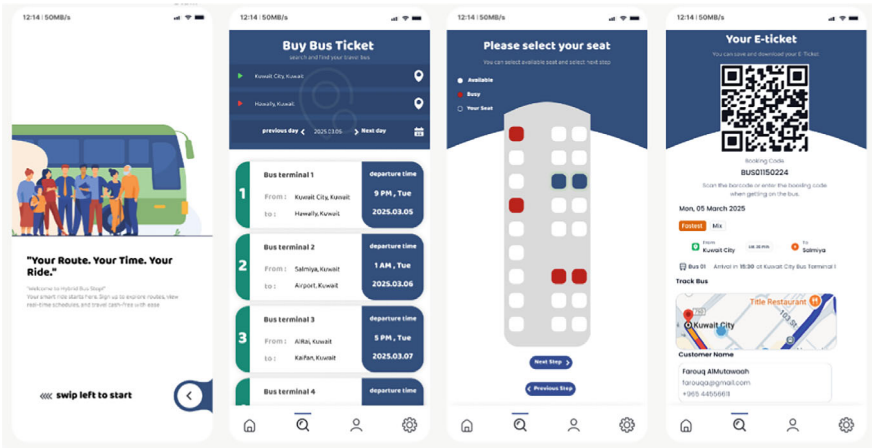


Fig. 8 User-friendly mobile app for the sustainable bus stop users

5 Conclusion

The paper presented a hybrid energy harvesting system utilized for bus stops, developed for climates classified as hot, sunny, with occasional wind gusts, as seen in the gulf regions, especially Kuwait. Three types of renewable energy sources were used, namely, solar, wind and kinetic energy. Solar panels collect sunlight, wind turbines harness occasional wind, and kinetic energy is collected from commuters on the sidewalk. The system provides a sustainable and environment-friendly solution that powers key bus stop amenities such as lighting, digital displays, and charging stations. A mobile application was developed to provide real-time information for both commuters and system administrators. Given that the implementation of the work is still in-progress, only preliminary results were discussed.

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Pyroelectric-Driven IoT Automation for Thermal Emissions Control: A Decentralized Security-Enabled Approach



Remya Nair, Dania Abdelsamie, Fatma Abou Elhassan, and Nader Shehata

Abstract Industrial and urban settings emit substantial waste heat, accelerating greenhouse gases, ozone depletion, and thermal pollution. This research introduces an IoT-enabled automation framework powered by PVDF pyroelectric nanofiber sensors to address lack of sustainable, tamper-proof emissions control. These sensors detect, quantify, and harvest thermal emissions from HVAC, industrial machinery, and other heat-intensive sources. Unlike conventional systems, ours integrates real-time monitoring with dual-purpose energy harvesting, repurposing surplus energy to power environmental IoT devices. Emissions data activates automated mitigation—adaptive cooling, smart ventilation, and air purification—via a responsive networked layer. A blockchain-secured emissions ledger ensures immutable recordkeeping, regulatory compliance, and decentralized security. While pyroelectric harvesting is limited in low-emission zones, PV hybridization enhances sustainability. Results show ± 0.1 °C sensing accuracy, 3.6 V peak voltage in high-emission zones, strong responsiveness to dynamic emissions, and reliable autonomous mitigation. This is the first integrated solution combining piezoelectric/pyroelectric sensors, IoT automation, and decentralized blockchain security.

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Keywords Pyroelectricity · PVDF nanofibers · Thermal emissions control · Decentralized ledger security · Energy-aware IoT automation · Self-powered environmental sensing

1 Introduction

Waste heat from industrial and urban infrastructure poses a significant environmental threat, accelerating greenhouse gas accumulation, depleting the ozone layer, deteriorating air quality, and amplifying urban heat islands [1]. Despite the severity of this issue, current control lack real-time sensing, autonomous mitigation, tamper-proof auditability, and rely on externally powered, unsustainable architectures [2]. These limitations prevent regulatory bodies and industries from meeting evolving environmental standards, undermining both compliance and climate resilience. This research addresses a clear, persistent gap: the absence of a unified, self-powered, decentralized, and tamper-proof emissions control system capable of real-time monitoring, automated mitigation, and secure data integrity. No existing work has successfully integrated these domains into a cohesive, operational framework [3]. The proposed solution introduces a novel multi-layered architecture combining piezoelectric/pyroelectric sensors, IoT-based automation, and decentralized blockchain security. PVDF nanofiber sensors enable the detection and harvesting of thermal emissions [4]; the harvested energy powers IoT nodes that autonomously execute adaptive air regulation [5]. A blockchain-secured ledger ensures immutable record-keeping and transparent compliance [6]. To sustain operations in low-emission conditions, photovoltaic (PV) hybridization is incorporated [7]. The conceptual overview of this innovative approach is illustrated in Fig. 1. The system's core objectives are as follows: (1) Develop PVDF-based nanofibrous sensors capable of real-time thermal emissions detection and harvesting [4]; (2) Utilize harvested energy to power IoT modules that initiate adaptive cooling and ventilation [2]; (3) Deploy a decentralized blockchain architecture to ensure tamper-proof environmental records [6]; (4) Implement autonomous mitigation triggered by emissions thresholds [1]; and (5) Ensure operational sustainability through photovoltaic hybridization [3]. The project objectives are validated through experimental results after the literature survey in Sect. 2, and a new system architecture is introduced in Sect. 3.

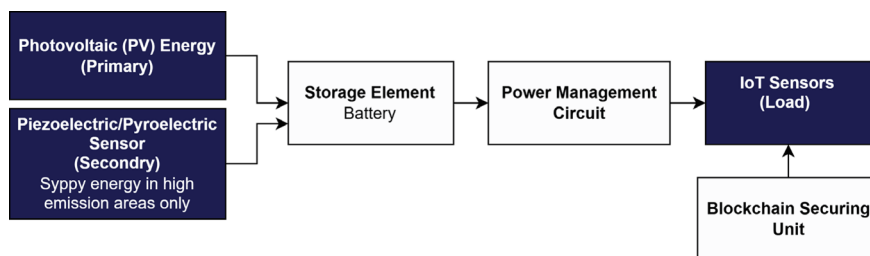


Fig. 1 Hybrid secured energy harvesting interface

2 Background and Related Work

2.1 Fundamentals of Piezo/Pyroelectricity and Nanofiber Materials

The mechanism of piezoelectric transduction is a mechano-electric transducer that allows mechanical energy, such as pressures, vibrations, bending, and tension strains, to be transformed into electrical energy. This piezoelectric transduction mechanism can be utilized for capturing renewable energy. These type of piezoelectric materials are relevant in numerous energy-related fields, including wearable electronics, smart textiles, biological sensors, and health tracking. Flexibility is an important parameter to be considered for the adaptability of these piezo membranes for practical applications. Present study focuses on a variety of organic and inorganic piezoelectric membranes/sheets. Among polymers, the fluorinated polymer PVDF is our primary emphasis because of its many outstanding qualities, including its light weight, ease of processing, high rate of purity, superior thermal and electric field stability, exceptional chemical and mechanical stability, and intriguing dielectric properties and PVDF in nanoform promotes higher surface to volume ratio. The polarized structure of PVDF based on fluorine and hydrogen contributed to its two significant electrical properties such as piezoelectricity and pyroelectricity [8–11]. PVDF is a semicrystalline, electroactive polymer that exhibits five polymorphic phases (α , β , γ , δ , and ϵ), with its piezoelectric properties largely attributed to beta crystalline phases. This β -phase possess an orthorhombic unit cell and all-trans (TTTT) conformation, with dipoles aligned in the same direction, producing a net dipole moment, leading to pronounced spontaneous polarization. The γ -phase exhibits an intermediate conformation (TTTG TTTG) between the α and β phases and possesses ferroelectric properties. Overall, the β -phase is most coveted due to its exceptional ferroelectric, piezoelectric, and pyroelectric characteristics, making it ideal for sensors, actuators, and energy harvesting applications, although achieving a pure β -phase in PVDF remains challenging [12]. The conversion from α to β PVDF is typically achieved through electrical poling, mechanical extension, or annealing, all of which improve dipole alignment and spontaneous polarization, thereby promoting the formation of β -phase

crystals [12–14]. PVDF nanofibers are used in smart systems, particularly in sensors and energy harvesting applications. Tactile PVDF sensors are ideal for medical monitoring, detecting contact and structural damage. Self-powered devices reduce operational costs and can distinguish contact forces and material texture. PVDF sensors are also effective for measuring flow velocity [15–30].

2.2 *IoT-Based Environmental Monitoring Systems: Existing Solutions*

Aragon et al. proposed a self-powered IIoT system named *InduEye*, designed for industrial heat-intensive environments. The system utilized thermoelectric generators (TEGs) to power a suite of edge-computing sensors (temperature, vibration, acoustic), transmitting real-time data via NB-IoT/LoRa to cloud systems for predictive maintenance. Deployed at a Repsol refinery, *InduEye* achieved a 94% reduction in environmental impact and a 70% cut in sensing costs compared to battery-powered alternatives. However, its capabilities stop at condition monitoring—it lacks the ability to initiate any direct mitigation in response to heat buildup. Furthermore, its reliance solely on TEGs means no energy redundancy in low-waste-heat scenarios, and crucially, it omits any security mechanisms, leaving environmental data susceptible to manipulation [31]. Cho et al. proposed a low-cost, mobile IoT framework for urban heat mapping by equipping citizens with wearable sensor modules. Their system recorded surface and ambient temperatures, humidity, GPS, and motion data, all fed through a smartphone app to generate high-resolution heat maps for city planning. The project succeeded in capturing localized thermal hotspots and promoted community participation. However, it remained purely observational, and the sensors were battery-reliant. Data integrity was also a blind spot [32]. Branzila et al. presented an IoT-based monitoring system aimed at enhancing indoor thermal comfort and air quality. Their sensor network interfaced with HVAC systems to dynamically adjust building climate controls. Yet, the scope of their system remains reactive and localized—it does not account for waste emissions and lacks energy autonomy and offers no security protocols [33].

2.3 *Cybersecurity in Environmental IoT Systems: Current Approaches and Gaps*

Dong and Feng proposed a blockchain-secured wireless temperature monitoring system for power infrastructure, integrating IoT sensors with AI analytics to detect overheating and prevent fire hazards. All environmental data is logged immutably on a blockchain, achieving 99.9% data integrity and 99.5% intrusion detection accuracy [34]. While technically strong, the system lacks energy autonomy, and its blockchain

use is confined to basic data logging, with no provisions for regulatory auditing, smart response, or mitigation. Vaneeta et al. developed the IB-AQMS, a blockchain-based air quality monitoring platform using IoT sensors to track pollutant levels and log Air Quality Index results on Hyper-ledger Fabric [35]. Their system ensures data immutability and transparency for government regulators, with machine learning algorithms classifying pollution levels. However, its dependence on cloud servers and fixed power makes it unsustainable for large-scale deployment. Ma et al. introduced a blockchain-IoT architecture for indoor environmental quality that incentivizes energy-saving behavior through smart contracts and token rewards [36]. Using Raspberry Pi nodes and environmental sensors, their framework logs energy metrics on a private blockchain and rewards occupants who meet sustainability thresholds. The system lacks any cybersecurity focus, assumes cooperative users, and omits any mechanism for real-time risk intervention or emissions management.

3 System Design and Architecture

3.1 *Pyroelectric Sensor Design and Material Physics*

Experimental Work (Materials and Methods) In this study, polyvinylidene fluoride (PVDF) (Kynar, King of Prussia, PA, USA) was obtained from ARKEMA, and dimethylformamide (DMF, 98%) from Sigma Aldrich was used as the solvent. A 10 wt% PVDF solution was prepared by dissolving 1 gram of PVDF in 10 mL of DMF. Before electrospinning, 10 mL of this solution was loaded into a syringe with a stainless steel needle. In total, 30 mL of pure PVDF solution was prepared for the electrospinning process.

Electrospinning Procedure PVDF nanofibers were produced with a setup comprising a high-voltage power supply (Divotech, 1–30 kV), a syringe pump (NE1000 Single Syringe Pump, New Era, Farmingdale, New York, NY, USA), a 10 mL plastic syringe with an 18-gauge needle, and a rotating collector wrapped in aluminum foil. The positive terminal of the power supply was connected to the needle tip, applying 25 kV, while the collector was grounded through the negative terminal. A PVDF solution was loaded into the syringe, ensuring it was bubble-free. The injection pump was set to a flow rate of 1.5 mL/hr and a syringe diameter of 13.4 mm, with a fixed needle-to-collector distance of 15 cm. Once a droplet formed at the needle tip, the high voltage initiated jet formation, and nanofibers were deposited onto the rotating collector. The process continued by replacing the empty syringe with another filled one until 30 mL of solution had been electrospun onto a 15 × 10 cm aluminum foil-covered collector. After deposition, the resulting nanofiber mat was removed for further characterization.

Characterization The surface morphology of the optimal PVDF nanofibrous membranes was analyzed using a scanning electron microscope (SEM, JEOL JSM-6010LV-SEM, Tokyo, Japan) at a 20 kV accelerating voltage. The electrospun nanofiber samples were platinum sputter-coated and mounted on carbon tape adhered to aluminum stubs. The average diameter of the nanofibers was measured using Image-J software (Madison, WI, USA). The piezoelectric response of the nanofibers was analyzed using Fourier Transform Infrared Spectroscopy (FTIR), focusing on the beta sheet structure. FTIR allows for an easy assessment of the molecular fingerprint by examining the functional groups and their interactions. The transmittance spectra of the PVDF nanofibers were recorded using a Perkin Elmer FTIR spectrometer with a scanning resolution of 5 cm^{-1} , covering a range from 4100 to 400 cm^{-1} . The piezoelectric performance of the nanofiber mats was tested by applying mechanical perturbations, such as impulse loading and airflow variations. The effects of airflow sensing and harvesting were examined. This demonstrated the combined influence of mechanical and airflow perturbations on the samples. The nanofiber membranes were cut into 2.5 by 2.5 cm pieces and placed between two aluminum foil sheets with insulated stainless steel wires. Using high-temperature resistant tape, the aluminum foil sheets were assembled into a single unit, forming a nanofiber patch. The wind flow apparatus consists of an electric air duster with a 1 cm radius nozzle directed at the center of the PVDF-based sandwiched unit, as shown in Fig. 2. The airflow velocity can be adjusted by varying the distance between the nozzle tip and the sample. An anemometer or velocity meter is used to measure the airflow velocity hitting the sample patch. The distance between the fan blade of the velocity meter and the tip of the air duster was measured using a ruler, ensuring a fixed vertical distance for each required airflow velocity. The air duster was positioned at the correct height and aimed at the center of the sample for each airflow speed. After exposing the sample to airflow for one second, the oscilloscope was used to measure the average of six peak-to-peak voltage readings. Seven different airflow speeds were tested, and the corresponding output voltages for pure PVDF-based nanomembranes were recorded.

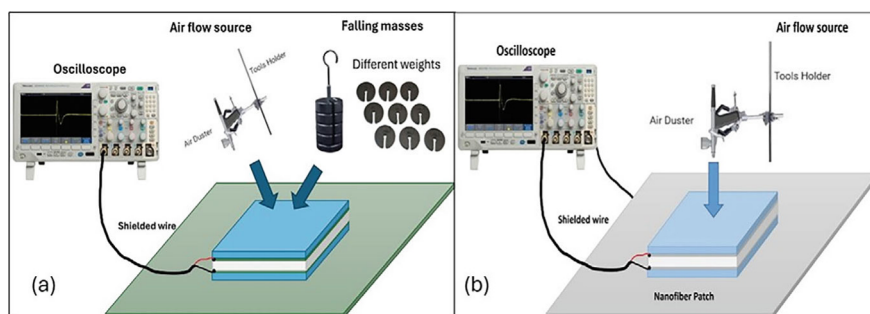


Fig. 2 Wind flow apparatus setup for PVDF nanofiber testing

3.2 Sensor Deployment Scenarios and Applications

Industrial Deployment—Thermal Emissions Mitigation in Factories

In high-emission industrial environments, our pyroelectric sensor patches are deployed directly to detect localized heat surges. Sensors such as the *wireless temperature–pressure transducer* and the *total volatile organic compound (TVOC) air quality module* augment the pyroelectric readings by providing context-aware atmospheric feedback. These sensors are wirelessly connected to a *decentralized automation control gateway* via Zigbee and MQTT. Placement prioritizes surface zones with the highest thermal gradients, ensuring heat harvesting and sensing efficiency. Real-time data captured by our nanofiber patches is processed locally and secured via our blockchain framework.

Smart Building Automation—Distributed Air Quality and Heat Response In commercial and residential buildings, central HVAC exhausts, server room cooling outputs, and AC compressor housings are primary heat emission points. Our pyroelectric patches are embedded along rooftop units, suspended ceiling vents, and behind wall-mounted cooling ducts. These are reinforced by an *integrated environmental sensing module* (CO₂, TVOC, humidity) and *multifunctional gas-leak detectors*, creating a redundant thermal and air quality perimeter. The *edge-based automation engine* coordinates these inputs through conditional logic—for instance, if excessive heat and CO₂ levels are detected concurrently, the system triggers adaptive cooling (via *smart relay-controlled secondary ventilation*), notifies building admins via dashboard alerts, and logs the event immutably to blockchain. Sensor placement is optimized across confined spaces using a *high-resolution presence detection module* to contextualize thermal events with occupancy data.

Emergency Response Automation—Leak, Anomaly, and Hazard Shutdown In high-risk zones like chemical storage rooms, enclosed mechanical chambers, and utility control centers, we implement a layered sensor mesh combining *PVDF nanofiber patches*, *vibration-detection modules*, and *fluid-leak sensors*. This setup detects abnormal thermal spikes, pressure anomalies, or unexpected moisture—all of which may precede catastrophic failures. If a sudden, localized heat spike occurs without any scheduled activity (cross-referenced via occupancy and vibration feedback), the *Raspberry Pi 5* triggers an emergency routine: it disconnects power via a *three-phase programmable power relay unit*, engages fire suppression relays, and notifies facility security. Simultaneously, all sensor outputs and decisions are encoded on our blockchain ledger for post-event forensics, auditability, and regulatory compliance. Unlike conventional alarms, this closed-loop system responds autonomously, preemptively halts hazards, and generates permanent evidence trails—all while operating on harvested heat and solar-supplemented power. The system's multi-layered decision-making and automation pipeline—spanning sensor input, edge-based logic processing, mitigation control, and blockchain-secured logging—is visualized in Fig. 3a, demonstrating how our platform achieves autonomous control and auditability across cross industrial, building, and emergency contexts.

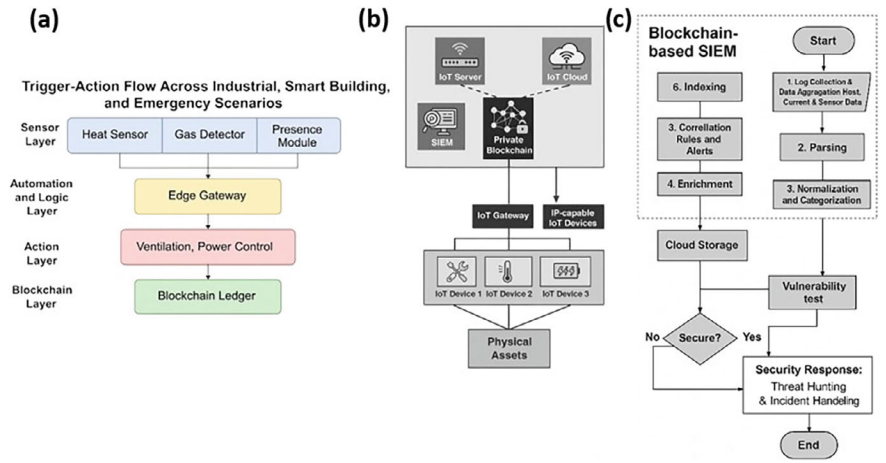


Fig. 3 **a** Trigger-action flow across sensing, logic, control, and blockchain layers. **b** design of the cybersecurity system. **c** cybersecurity flowchart

3.3 Decentralized Security Framework

Our system employs a private blockchain-based architecture for securing emission data, ensuring immutability, traceability, and compliance. As shown in Fig. 3b, each emission reading is recorded as a blockchain transaction, times-tamped, cryptographically hashed (SHA-256), and indexed via Proof of Work (PoW), providing WORM-compliant, tamper-resistant storage. The PoW consensus, while lightweight and computationally adapted for resource-constrained IoT, ensures decentralized trust without requiring a central authority, addressing the risks of environmental compliance fraud, cross-border data manipulation, and regulatory evasion by industrial operators. Figure 3c further illustrates our SIEM-driven flow: from log aggregation and anomaly detection to blockchain-secured storage and incident response.

3.4 Hardware and Software Setup

Pyroelectric materials are known for their ability to generate electrical potential in response to temperature [37–39]. A pyroelectric sensor generates a sinusoidal AC voltage due to the dynamic nature of temperature fluctuations [40–42]. The magnitude of the signal is directly proportional to the rate of temperature variation. A hybrid parallel/series connection provides an optimal balance of voltage and current when multiple patches are used. Series connections maximize voltage, while parallel setups maximize current. The output power goes through a rectifier and capacitor and the varying output power from the sensors is stored in a battery to provide a stable supply to the IoT sensors. For AC-powered IoT sensors, an inverter is required

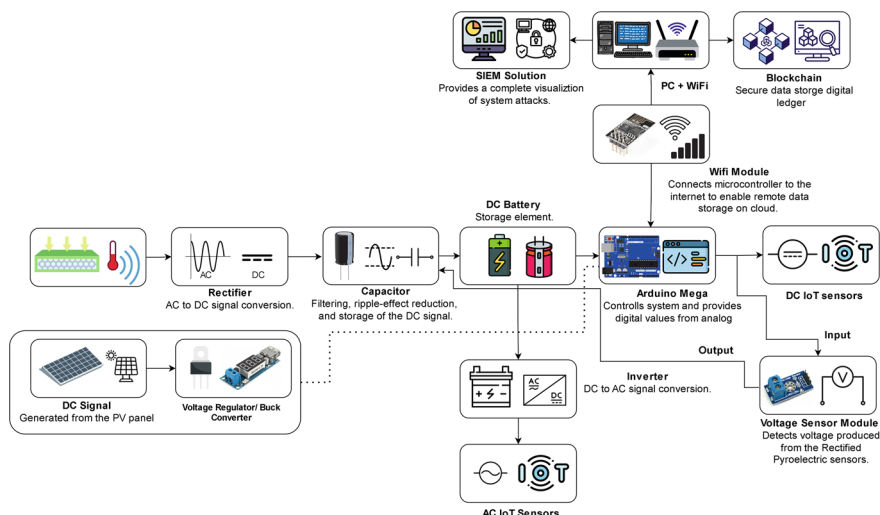


Fig. 4 Conceptual flow diagram of the secured energy harvesting power circuit interface

to convert the DC power from the battery into AC. The Arduino IDE will be used to connect these IoT sensors. A Voltage Divider Module will be integrated to monitor the readings after rectification. For power calculation ($P = V \times I$), the current will be measured using a Digital Dual Measurement Multimeter, capable of measuring current in microamperes. The prototype is further scaled up with a photovoltaic (PV) panel for hybrid energy harvesting. A regulator/converter efficiently manages the power transfer from the PV system to the battery. During the day, the PV panel captures solar energy to power the IoT sensors, storing any surplus in the DC battery. At night, the piezoelectric/pyroelectric system can operate independently, with the option to draw from the surplus energy stored earlier by the photovoltaic (PV) panel. The cloud-based platform is utilized for remote data storage via the Wi-Fi-enabled ESP8266 module. Figure 4 provides a comprehensive overview of the conceptual design of the power circuit interface.

4 Results and Discussion

4.1 Piezoelectric/Pyroelectric Sensor Characterization and Performance

The surface morphology of the PVDF electrospun nanofiber mat was analyzed using SEM, with the corresponding image shown in Fig. 5 the fiber diameter distribution matched expectations, and the average diameter of the randomly distributed nanofibers was approximately 160 nm. The β sheet content of each composition was

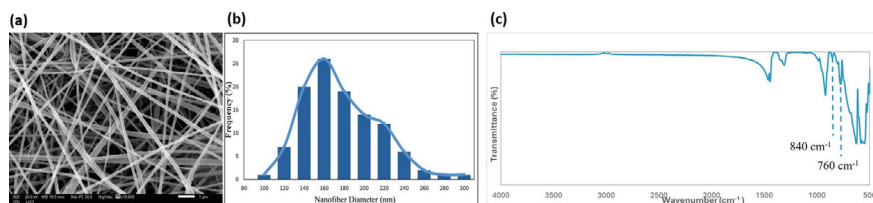


Fig. 5 **a** SEM image of the fabricated PVDF nanofiber mat at a scale bar of 1 μm . **b** histogram representing the corresponding fiber diameter distribution curve. **c** FTIR spectra associated with the PVDF-based piezoelectric nanogenerator

determined from the FTIR spectra shown in Fig. 5 the FTIR spectra of PVDF reveal its crystalline phases, with the β phase being the electroactive, piezoelectric phase, while the α phase is the stable non-polar phase. The electrospinning process enhanced the β phase due to the alignment of dipoles under an electric field. Key bands for PVDF were observed between 700 cm^{-1} and 1500 cm^{-1} , with the β phase associated with peaks at $840, 880, 1175,$ and 1275 cm^{-1} , and the α phase corresponding to peaks at $410, 489, 532, 614, 762, 796, 875, 976, 1210,$ and 1383 cm^{-1} . Peaks linked to the γ phase were observed at $431, 482, 811,$ and 1234 cm^{-1} [43].

The content of the electroactive β phase in a material can be determined using Beer-Lambert's law by analyzing the absorbance intensities at 760 cm^{-1} and 840 cm^{-1} using the given formula:

$$f(\beta) = \frac{A_{\beta}}{1.26A_{\alpha} + A_{\beta}} \quad (1)$$

where A_{β} and A_{α} represent the absorbance value at 840 cm^{-1} and 760 cm^{-1} [43].

The 760 cm^{-1} peak, linked to the α crystalline phase, arises from CF_2 bending and C(H)-C(F)-C(H) skeletal vibrations. In contrast, the 840 cm^{-1} peak, which is associated with the polar β phase due to CH_2 , displays a rocking vibrational band along with C-C and CF_2 stretching. Additionally, peaks at 1175 cm^{-1} and 1400 cm^{-1} correspond to C-H and C-F vibrations, respectively. Free falling masses are allowed to fall on the sample nanofiber patch from a specific height of 2.5 cm. A linear response is found towards the mechanical perturbation associated with free falling masses. A 50 gram mass fallen from a height of 2.5 cm generated around 2 V and finally a maximum of 4.5 V is produced at a falling mass of 250 gram from same height. Same procedure is repeated again under a strong airflow-based environment where the impact of an airflow perturbation at a constant velocity of 5 km/h is studied, as depicted in Fig. 6a. Combined effect of impulse loading and air flow perturbations clearly increased the piezovoltage generation. An increase in voltage generation from 2 V to 3 V is observed under the combined effect of 50 gram mass and airflow. Finally, a maximum voltage of 8.5 V is observed under the combined effect of 250 gram falling mass and airflow. Thus in general, heavy masses with airflow resulted in an enhancement of around 200% piezovoltage. This result clearly indicated the capability of our prototype to sense airflow disturbances and mechanical impacts.

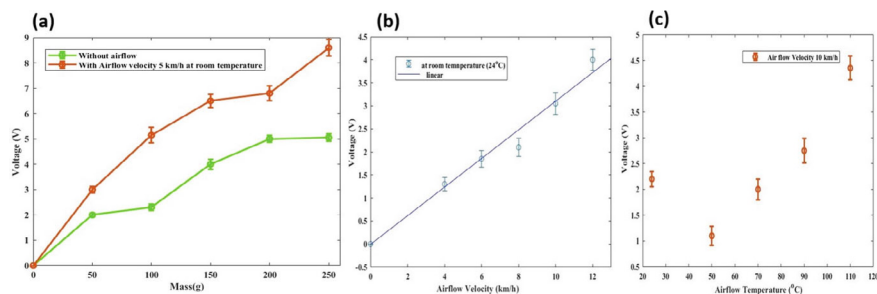


Fig. 6 a Piezoelectric voltage versus falling mass with and without strong airflow perturbations. Piezoelectric voltage versus b airflow velocities c airflow temperatures

This confirms the scope of this prototype in the field of ambient energy harvesting, where a combined effect of mechanical energy and air blowing is seen. A positive response is exhibited towards increasing airflow velocities such that piezoelectric voltage increased with increasing velocity of airflow. Airflow based perturbations are applied at different velocities from 4 to 12 km/h and a maximum output voltage of 4 V is obtained at a velocity of 12 km/h. A sensitivity of 0.3 V/km/h is obtained from the fabricated nanomembrane and the voltage generation with respect to variable airflow velocities is shown in Fig. 6b and c. Furthermore, the thermal effects related to airflow are analyzed by applying airflow distribution at varying temperatures while keeping the velocity fixed at 10 km/h. The piezoelectric voltage generation is found to be weak above room temperature but again started increasing when the temperature reached 90 °C and above. This clearly proves the capability of the prototype to sense hot airflow and harness energy from those disturbances in an efficient way.

Application of mechanical stresses in the form of impulse loading and air blowing promoted the layers' polarization by changing the overall momentum of dipoles inside the piezomembrane and this resulted in the generation of piezoelectric voltage. The pyroelectric properties of the PVDF membrane greatly enhanced the orientation of dipoles under the exposure of hot air and the pyroelectric properties contributed to the efficient performance of the prototype towards blowing air at higher temperatures.

4.2 Blockchain and IoT Integration

The integration of IoT sensors with our blockchain framework enables secure, verifiable data transmission for environmental monitoring. Each reading is cryptographically hashed and stored as a transaction within a private blockchain, ensuring data integrity and traceability. Despite minimal fluctuations across sensor outputs, consistency in data format and timestamped indexing confirms the reliability of the sensing

infrastructure. This fusion of structured sensor data with a decentralized ledger establishes tamper-resistant environmental records, critical for regulatory accountability and auditability in industrial applications.

4.3 Societal Impact

This project demonstrates strong societal value by integrating energy-efficient IoT and piezoelectric/pyroelectric energy harvesting to advance environmental monitoring and regulatory compliance. It supports sustainable, cost-effective IoT operation while reinforcing participation and visibility in critical sectors such as cybersecurity, energy, and IoT—particularly by highlighting women’s leadership in a traditionally male-dominated field. Additionally, it strengthens KCST’s academic presence by contributing to its role as a hub for innovation in multidisciplinary sustainable systems.

5 Conclusion

This study demonstrated the system’s capability to sense and harness electrical energy from disturbances caused by impulse loading and airflow pressures. The combined effect of mechanical impacts and airflow velocities significantly enhanced the prototype’s piezoelectric output. A clear voltage increase was observed under airflow-based perturbations compared to impulse loading alone, with 8.5 V generated under a 250 g falling mass and 5 km/h airflow. A maximum of 4 V was recorded at 12 km/h airflow, and temperature-based analysis confirmed effective behavior in the 90–110 °C range. These results underscore the prototype’s potential for real-world applications, particularly in harvesting energy from exhaust air emitted by machinery and automobiles. Future research should optimize the material’s design and assess long-term stability under complex conditions, establishing a foundation for integrating these prototypes into IoT sensors with cybersecurity systems.

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A Hybrid Quantum Framework for Ultra-Secure Communication and Anomaly Detection in IoT Systems



Bharathi Deepa, M. Murugappan, G. Jashwanth Kumar, and S. Iswariya

Abstract The rapid expansion of the Internet of Things (IoT) has produced a highly linked digital environment together with a growing number of complex cyber threats. This work presents a Quantum-Enhanced Hybrid Anomaly Detection Framework combining two advanced Quantum Machine Learning (QML) models: Quantum Support Vector Machine (QSVM) and Quantum Neural Network (QNN) to address these challenges. By projecting data into high-dimensional quantum spaces, QSVM efficiently classifies non-linear and low-frequency threats such as User to Root (U2R) and Probe attacks using quantum kernel functions. Concurrently, QNN models utilize Variational Quantum Circuits (VQCs) to detect large-scale threats like Denial of Service (DoS) and Remote to Local (R2L) attacks by offering superior generalization. Both models produce a final, high-confidence prediction by combining their outputs using a soft voting ensemble approach. The proposed framework guarantees scalability and resilience while ensuring compatibility with present simulation systems. The experimental results obtained indicate that the hybrid QSVM + QNN model significantly outperforms standalone models and conventional methods, providing a future-ready security solution for next-generation IoT infrastructures. In addition, this model delivers a robust, adaptable, and quantum-resilient security solution, well-suited for safeguarding future IoT ecosystems.

Keywords Anomaly detection · Denial of service (DoS) · Quantum SVM · Quantum neural network (QNN)

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1 Introduction

The anomaly detection in an IoT system is the process of spotting unexpected behaviour or deviations from the normal devices within the network that might point to possible cyberattacks or system failures. From minor probing attacks to major disruptions like DoS, these abnormalities can arise from a broad spectrum of activity. Smart devices are becoming indispensable in sectors including manufacturing, transportation, and healthcare, thus providing a suitable security mechanism for them is a crucial task. For example, an unusual traffic pattern in a medical IoT system might signify unauthorized access to patient data, while unexpected data spikes in industrial sensors could suggest tampering or failure. Traditional machine learning and deep neural networks provide the foundation for anomaly detection in IoT systems. Although, these techniques often struggle with real-time processing like overfitting while facing the new attack patterns, and demand major computational resources. Henceforth, the focus has been switched to QML, which uses quantum computing ideas to overcome the above constraints.

This work presents a quantum-enhanced anomaly detection framework for the IoT environment by integrating QSVM and QNN as a single hybrid model. The QSVM model has the capacity to capture fine-grained, non-linear patterns whereas the QNN model could generalize over huge, high-dimensional datasets. Together, the hybrid model enhances scalability and robustness, providing a promising solution for securing next-generation IoT systems.

2 Related Works

The advancements of quantum computing have prompted substantial interest in quantum-resilient solutions such as post-quantum cryptography (PQC) and quantum-enhanced machine learning models. Foundational studies [1–4] establish the relevance of quantum cryptography in IoT. Yang et al. [1] assess the feasibility of Quantum Key Distribution (QKD) to mitigate IoT vulnerabilities. Bhatt and Sharma [2] highlight the limitations of classical encryption like RSA under quantum attacks, advocating for quantum-safe alternatives. Mansoor et al. [3] introduce hybrid PQC algorithms tailored for privacy in resource-constrained IoT devices, while Kaushik et al. [4] propose lightweight key schemes that outperform current NIST standards in 5G IoT environments.

Advancements in secure communication are seen in works [5–8]. Santhiya Devi et al. [5] offer a NEQR and Cellular Automata-based encryption-authentication framework for medical IoT. Babu and Padmavathy [6] utilize lattice-based cryptography for efficient quantum-secure node authentication. Kumari et al. [7] propose a Diffie-Hellman super-singular Multiplication model to counter ransomware and denial-of-service attacks, and later [8] present a hybrid Ring-LWE and QCLDPC system optimizing delay and resource usage in lightweight authentication. Broader

discussions in [9, 10] underscore the urgency of scalable PQC solutions. Babu et al. [9] provide a comprehensive review of quantum-secure authentication protocols, while Balogh et al. [10] integrate blockchain, cloud vulnerabilities, and AI in a multi-layered approach to secure IoT systems.

Parallely, quantum machine learning has emerged as a powerful tool in threat detection. Shelar et al. [11] enhance BB84 for underwater IoT networks. Satrya et al. [12] benchmark PQC algorithms like NTRU and SABER, finding Light SABER most efficient. Elsedimy et al. [13] propose a QSVM enhanced with an Improved Grey Wolf Optimizer to reduce false positives. Wang et al. [14] apply QSVM to power grid disturbance detection, and Gong et al. [15] demonstrate a Variational Quantum Neural Network (VQNN) achieving 97.21% accuracy.

Quantum machine learning has shown promise in IoT anomaly detection, with models like QSVM and VQNN outperforming classical approaches in accuracy and efficiency [13–15]. However, existing studies typically apply these models in isolation, targeting general or high-frequency attacks without addressing their complementary strengths. There is limited focus on real-time, integrated architectures capable of handling complex, heterogeneous IoT data and detecting low-frequency, high-impact threats such as user-to-root (U2R) and remote-to-local (R2L) attacks. This study addresses the gap by proposing a parallel QSVM–QNN architecture with a soft voting mechanism, optimized for real-world IoT anomaly detection under quantum simulation constraints.

3 Methodology

To enhance the anomaly detection in an IoT system and to ensure the secure communication, a hybrid QML model is designed by combining the strengths of QSVM and QNN models, operating in parallel to capture both subtle and large-scale attack patterns. Figure 1 depicts the overall workflow of the hybrid QML.

3.1 Data Preprocessing

The initial step in the proposed workflow is data preprocessing for removing the irregularities and redundancies present in the dataset (UNSW-NB15) for efficient anomaly detection. At first, data cleaning is carried out by removing the duplicate records to maintain dataset integrity. Further, Min–Max normalization is employed to bring all features into a consistent scale between 0 and 1 without distorting data distribution.

The transformation is defined by:

$$X_{normalized} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

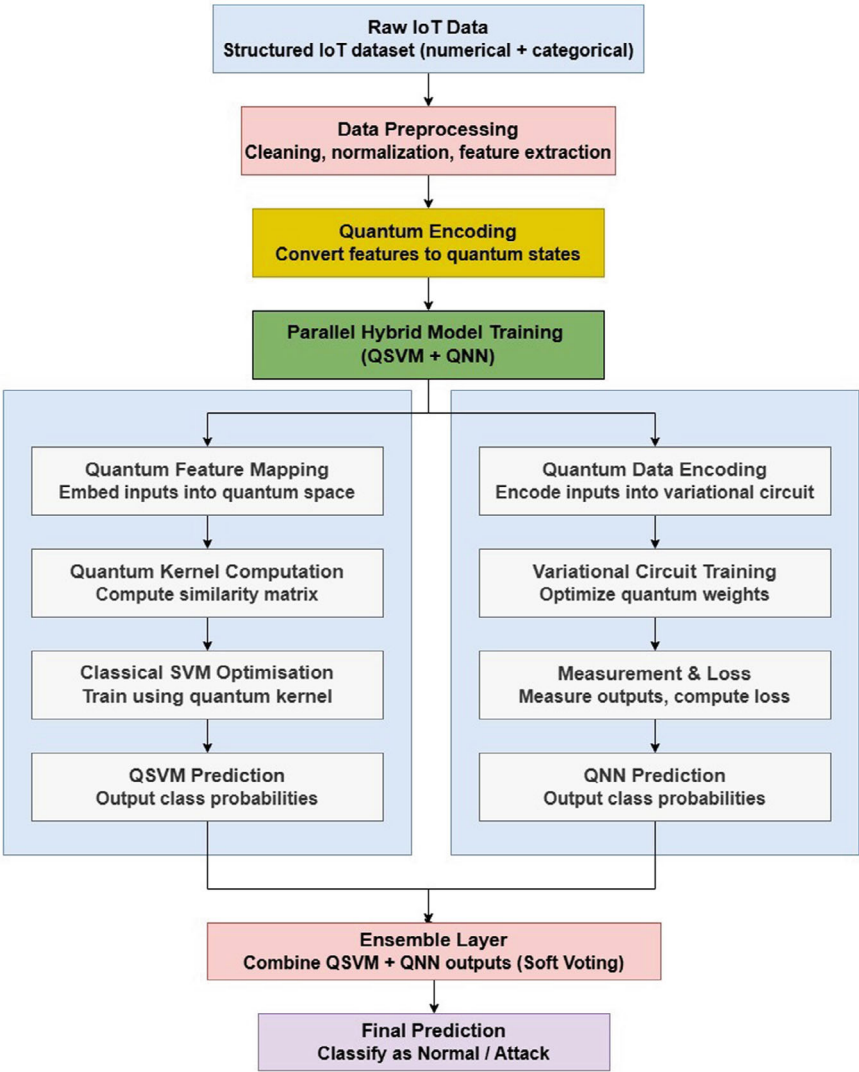


Fig. 1 Overall workflow of the proposed work

where X is the original feature value. X_{min} and X_{max} represent the minimum and maximum values of the feature, respectively. This normalization enhances convergence during quantum training and ensures uniform feature influence. Dimensionality reduction is applied to the normalized data using the Mutual Information (MI) score, which measures the statistical dependency between each feature and the target label (attack/normal), selecting only the most relevant features for improved classification efficiency.

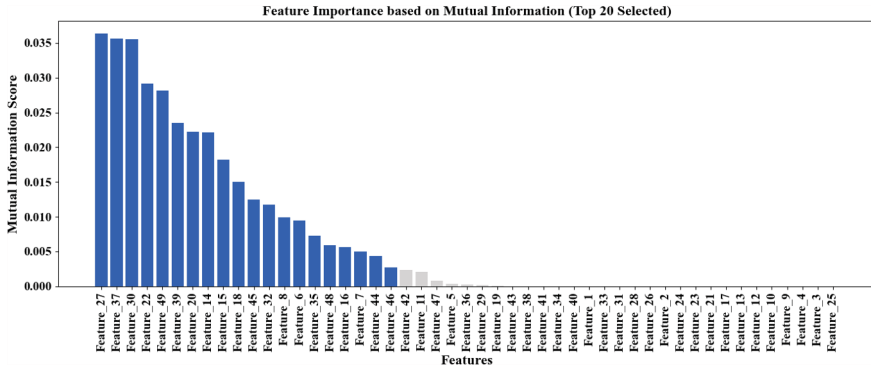


Fig. 2 Graphical representation of feature values based on MI score

The MI score is defined as:

$$I(X, Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \left[\frac{p(x, y)}{p(x) \cdot p(y)} \right] \quad (2)$$

where X is the feature variable, Y is the class label, $p(x, y)$ is the joint probability distribution, $p(x)$, $p(y)$ are the marginal probabilities. The Features with the highest MI scores are retained as represented in Fig. 2 and they offer the most predictive power for distinguishing between normal and malicious traffic.

3.2 Quantum Encoding

The quantum encoding is implemented through ZZFeatureMap. A pairwise feature representation and entangling architecture are the reasons for its selection. Furthermore, preliminary trials on UNSW-NB15 showed superior performance compared to other feature mappings. A series of parameterized quantum gates that embed both individual features and their pairwise interactions is employed for encoding as given in equation

$$U_{enc}(x) = \prod_j e^{ix_j Z_j} - \prod_{j < k} e^{ix_j x_k Z_j Z_k} \quad (3)$$

where x is the classical input feature vector, and Z_j , Z_k are Pauli-Z operators applied to the j -th and k -th qubits. This encoding scheme maps data into a high-dimensional quantum Hilbert space by using phase rotations for single features and entanglement for inter-feature dependencies. As a result, QSVM and QNN will be able to effectively learn complex, non-linear patterns, which are essential to detecting IoT anomalies.

3.3 Proposed Quantum-Enhanced Secure Framework

The proposed Quantum-Enhanced Secure Framework integrates QSVM and QNN—into a unified hybrid anomaly detection system. This approach combines QSVM’s strength in decision boundary classification with QNN’s powerful pattern recognition capabilities, for threat detection.

3.3.1 Quantum Support Vector Machine (QSVM)

QSVM begins with quantum feature mapping, where classical input vectors x are encoded into quantum states $\phi(x_i)$ using entangling circuits such as the ZZFeatureMap. This allows the model to map data into a high-dimensional quantum Hilbert space, where subtle relationships between features can be more easily separated. Following encoding, a quantum kernel is computed to evaluate the similarity between samples.

This is defined as:

$$K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2 \quad (4)$$

Here $\langle \phi(x_i) | \phi(x_j) \rangle$ is the inner product of two quantum states, measuring how closely related they are. Squaring the value ensures the result reflects a similarity score between 0 and 1.

The resulting kernel matrix is then fed into a classical SVM optimizer, which computes the optimal hyperplane for separating normal and attack classes using the decision function:

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \right) \quad (5)$$

where α_i represents the Lagrange multiplier, which varies from 0 to 0.5. The values y_i correspond to the class labels, typically assigned as +1 for normal traffic and −1 for attack instances. The kernel function $K(x_i, x)$ captures the similarity between the input sample x and each support vector x_i computed using the quantum kernel derived from the encoded quantum states. The parameter b is the bias term added to the summation. Finally, the sign function determines the predicted class of the input: a positive result indicates normal behaviour, while a negative result signals an anomaly.

3.3.2 Quantum Neural Network (QNN)

The QNN utilizes Variational Quantum Circuits (VQCs), which are trainable quantum circuits tailored for supervised learning. The process begins with quantum

data encoding, where classical input x_i, x_j is transformed into quantum states via rotation gates:

$$RY(x_j) = \exp\left(-i \cdot x_j \cdot Y/2\right) \quad (6)$$

This maps each feature value x_i, x_j onto a qubit using the Pauli-Y operator, creating a quantum representation of the input data.

The encoded state is then processed through a variational circuit U_θ , parameterized by trainable weights θ . The circuit output is measured with respect to an observable M , and the expected output is calculated as:

$$\hat{y} = \left\langle 0 \left| U_{enc}^\dagger(x) U_\theta^\dagger M U_\theta U_{enc}(x) \right| 0 \right\rangle \quad (7)$$

This formula computes the expectation value, which acts as the model's prediction $\hat{y}$ representing the probability of the input being an attack.

The model is trained by comparing $\hat{y}$ to the true label using a cross-entropy loss function, and the parameters θ are updated using a classical optimizer (COBYLA in this case for its robustness in shallow circuits without requiring gradient estimation). This loop continues until convergence.

3.3.3 Integration and Inference

After parallel processing by both QSVM and QNN, their predictions are passed to an ensemble layer. Here, a soft voting mechanism is applied, combining the confidence scores from both models to produce a final classification—either “Normal” or “Attack”. The predicted class probabilities from QSVM and QNN were averaged using equal weights (0.5 each), as both models were considered equally reliable in contributing to the final prediction.

4 Experimental Results and Discussion

The UNSW-NB15 dataset, comprising around 2.5 million records with 49 features, simulates real-world benign and malicious network traffic, including attacks like Exploits, Worms, and Reconnaissance. It captures flow, time, content, and behavior-based attributes essential for anomaly detection. We conducted fivefold cross-validation over multiple runs to ensure the model's performance is consistent and reliable. Using this approach, we were able to assess generalization and mitigate the risk of overfitting. Although the results on the UNSW-NB15 dataset are promising, it remains to be seen if the model is generalizable to other datasets. As part of future work, the model will be tested on additional network traffic datasets to assess its robustness.

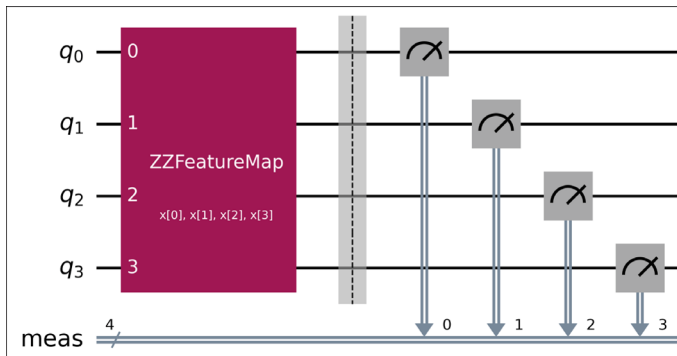


Fig. 3 Visual representation of quantum circuit simulation

A Python 3.9 system containing Qiskit and Scikit-learn was used for experiments on a system with an Intel Core i7 processor and 16 GB RAM, utilizing the Qiskit Aer simulator to simulate quantum circuit execution. The quantum circuit, as depicted in Fig. 3 and it represents a 4-qubit system with a depth of 2 and a total of 30 gate operations. It begins with a ZZFeatureMap in two repetitions (depth = 2), which is applied to encode classical data into quantum states using controlled-Z gates, generating entanglement between qubit pairs. The hyperparameters were carefully tuned to balance model performance with quantum simulation constraints. As a result of 0.05 learning rate, 4 qubits, and the COBYLA optimizer, the training was stable within the confines of the hardware.

4.1 Confusion Matrix Results and Correlation Analysis of the Proposed Hybrid Quantum Model

A confusion matrix for the proposed model is shown in Fig. 4a. According to the obtained true prediction results, the model is capable of identifying attacks with superior accuracy. According to the UNSW-NB15 dataset, the false prediction value is low compared to the true prediction value, indicating minority attack classes like U2R and R2L. Figure 4b illustrates the correlation matrix, which is used to evaluate the model's reliability and consistency. With the obtained diagonal value of the matrix, there is a high probability of accurate predictions, resulting in the least possibility of misclassification errors. Furthermore, the correlation heatmap shows a strong alignment between predicted and actual classes. In real-world IoT datasets, this pattern demonstrates the robustness and scalability of the hybrid model.

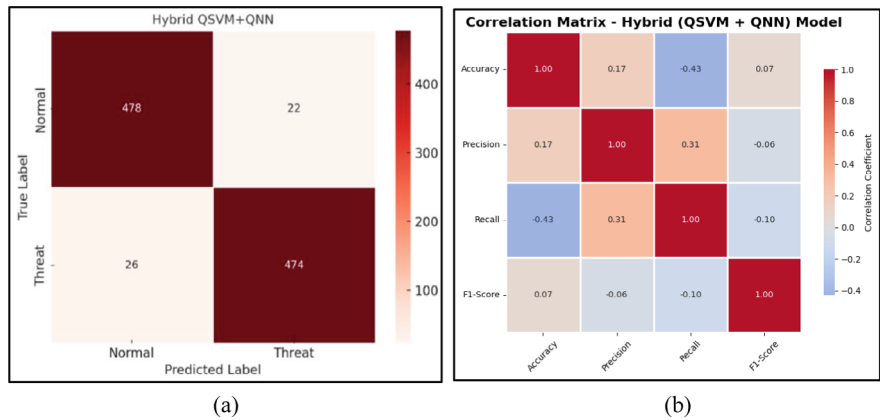


Fig. 4 **a** Confusion matrix, **b** correlation matrix of the proposed model

4.2 Performance Evaluation Results

To analyze the effectiveness of the proposed Hybrid QSVM + QNN model, we conducted extensive experiments using both classical (SVM, ANN) and quantum (QSVM, QNN) models.

These models were chosen to establish a comprehensive performance benchmark, allowing us to assess the improvements brought by the hybrid quantum approach. A pre-defined split of the dataset is provided for training and testing purposes, adhering to an approximate 70:30 ratio (specifically, 68.04% for training with 175,341 records and 31.96% for testing with 82,332 records) as given in the UNSW-NB15 dataset. The results, as represented in Table 1, reveal a clear progression in performance from classical to quantum and finally to the hybrid approach. SVM achieved 89.2% accuracy, with moderate precision and recall. ANN performed slightly better with 90.5% accuracy and a higher F1-score. Among the quantum models, QSVM attained 93.1% accuracy, benefiting from its strong decision boundary capabilities, while QNN pushed further to 94.2%, leveraging quantum circuit expressiveness. The hybrid QSVM + QNN model outperformed all others, achieving 96.0% accuracy, 95.4% precision, 94.9% recall, and a 95.1% F1-score. These findings affirm that integrating QSVM and QNN allows the hybrid model to harness the strengths of both—precision in classification and adaptability in learning, resulting in a robust and accurate IoT anomaly detection. In future, we aim to design and develop a Quantum deep learning models for anomaly detection [18].

Table 1 Comparative analysis for classical model versus quantum model

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Classical SVM [16]	89.2	88.4	87.6	88.0
Classical ANN [17]	90.5	89.7	89.1	89.4
QSVM only	93.1	92.4	91.8	92.1
QNN only	94.2	93.7	92.8	93.2
Hybrid (QSVM + QNN)	96.0	95.4	94.9	95.1

5 Conclusion

Through a parallel hybrid ensemble, this study introduces a Quantum-Enhanced IoT Security Framework that combines QSVM and QNN for anomaly detection. The model outperformed classical baselines (SVM, ANN) and standalone quantum models by 96.0% accuracy, 95.4% precision, 94.9% recall, and 95.1% F1-score on the UNSW-NB15 dataset. In this study, hybrid quantum models are found to enhance classification performance for attacking IoT devices. Based on Qiskit Aer, the approach shows improved accuracy and fewer false positives in a simulated environment, but it is not yet capable of addressing real-time constraints, energy efficiency, or deployment on physical IoT devices. It is evident that quantum-resilient, edge-integrated IoT security has great potential, but these aspects require further exploration involving real quantum hardware, larger datasets, and practical deployment scenarios.

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AI-Driven Anomaly Detection in Textile Manufacturing Using IoT and Deep Learning



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Abstract The textile industry faces significant challenges in quality control due to the complexity of detecting subtle defects in fabrics during high-speed and high-volume production. Traditional inspection methods rely heavily on human inspection, which is labor-intensive, error-prone, and unable to keep pace with modern manufacturing speeds. This paper presents an advanced Computer Vision and Deep Learning-based anomaly detection system for textile quality assurance that addresses these limitations. By leveraging ensemble Convolutional Neural Networks (CNNs), Autoencoders, and Vision Transformers, integrated with IoT-enabled sensors and edge computing devices, our model achieves superior defect detection accuracy across diverse fabric types (weft and warp defects, slubs, holes, oil stains, mispicks, color inconsistencies, broken ends, and fabric wrinkles etc.) and environmental conditions. The system improves supply chain efficiency by 27%, reduces material waste by 32%, and ensures consistent quality with minimal human intervention. Experimental results demonstrate the effectiveness of CNN variants such as ResNet, EfficientNet-B3, and Vision Transformers, achieving an accuracy of over 97.13% with a false positive rate of less than 2.8%. The proposed framework is scalable, energy-efficient, and deployable in real-world textile manufacturing units with minimal disruption to existing workflows. Cost-benefit analysis indicates a return on investment within 8–12 months of implementation.

Keywords Computer vision · Deep learning · Garment production · Automation

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1 Introduction

Quality assurance in textile manufacturing represents a critical determinant of production efficiency, operational costs, and market competitiveness. The textile industry's global market, valued at \$961.5 billion in 2023 [1], faces significant challenges in maintaining consistent quality across increasingly complex fabric constructions and high-speed production environments (>30 m/min). Manual inspection methodologies, despite their historical prevalence, exhibit inherent limitations with detection accuracy ranging from 60 to 80% [2] due to inspector fatigue (degrading 23% after 4 h of continuous inspection [3]), perceptual subjectivity (inter-inspector agreement of only 74.3% [4]), and physiological constraints in detecting specific defect categories, particularly those measuring < 1 mm or exhibiting low contrast ratios (<0.15) [5].

Contemporary automated quality control systems have attempted to address these constraints through computer vision techniques but encounter substantial technical barriers. Spectral-based approaches [6] demonstrate sensitivity to environmental variations ($\pm 15\%$ accuracy fluctuation under variable illumination). Conventional machine learning implementations [7] require extensive feature engineering, with performance degradation of 18–27% when confronted with novel fabric patterns not represented in training data. Furthermore, deployment architectures frequently suffer from computational bottlenecks (average processing time of 1.2 s/image [8]) or insufficient scalability across heterogeneous production environments.

Recent advancements in deep neural networks and distributed IoT architectures present unprecedented opportunities for developing high-performance, real-time quality control systems with theoretical detection rates approaching human visual acuity while maintaining consistency across extended operational periods. Specifically, the integration of multi-scale feature extraction capabilities in modern CNN architectures [9] with the distributed processing paradigms of edge computing enables comprehensive defect detection within the temporal constraints of production environment (processing requirements of < 50 ms/frame for production speeds of 30 m/min with 10% overlap). This research introduces a technically advanced anomaly detection framework that synergistically integrates multi-architecture deep learning models with distributed IoT sensor networks. The system employs hierarchical processing architecture incorporating both high-throughput edge computing for preliminary analysis and cloud-based infrastructure for complex pattern recognition.

2 Literature Review

The evolution of textile defect detection technologies has progressed through multiple paradigms, from traditional image processing to contemporary deep learning approaches. This section critically examines relevant literature to identify methodological gaps, performance limitations, and implementation challenges of earlier works. The analysis spans four key domains: conventional computer vision techniques, machine learning applications, deep neural architecture, and IoT integration strategies. By synthesizing findings across these domains, establish the theoretical foundation for our hybrid approach and isolate opportunities for technical innovation. Particular attention is given to work addressing real-time processing constraints, multi-modal data fusion, and deployment considerations in industrial environments, as these aspects directly influence the practical utility of quality control systems in textile manufacturing. Existing research on automated textile defect detection primarily utilizes traditional image processing techniques such as Gabor filters, wavelet transforms, and statistical approaches. In [4], the authors provided a comprehensive review of these methods, highlighting their limitations in detecting complex patterns and sensitivity to environmental variations like lighting conditions. These methods, while computationally efficient, often fail to generalize across diverse fabric types and defect categories. Machine learning models like Support Vector Machines (SVM), Random Forests (RF), and k-Nearest Neighbors have been explored [5, 6]. These approaches demonstrated improvements over traditional methods but often require extensive feature engineering and struggle with highly textured fabrics. Their performance is heavily dependent on the quality of hand-crafted features, limiting their applicability in dynamic manufacturing environments. Deep Learning-based approaches, particularly CNNs, have demonstrated superior performance in image classification tasks. In [7], the researchers introduced a CNN-based defect detection system achieving 91.5% accuracy on plain fabrics. More recent works [8, 9] have explored transfer learning approaches using pre-trained models like VGG and ResNet, achieving accuracies of 94–96%. However, a few studies have investigated their integration with IoT for real-time deployment in industrial settings.

The integration of IoT in manufacturing settings, often referenced as Industry 4.0, has been gaining significant traction. In [6], the authors have discussed the potential of IoT in enabling smart manufacturing, while [7] explored the use of edge computing for real-time analytics in industrial environments. However, the specific application of IoT in textile quality control remains relatively unexplored. This work extends prior research by incorporating advanced CNN architectures, unsupervised anomaly detection methods, and IoT integration. Models such as ResNet50, EfficientNet-B3, and Vision Transformers, address gaps in existing studies that focus solely on supervised learning approaches or neglect real-time deployment considerations.

3 Materials and Methods

This section details our technical approach to developing an AI-driven anomaly detection system for textile manufacturing. Presented a comprehensive framework that addresses the limitations identified in the literature review by integrating advanced deep learning architectures with IoT infrastructure optimized for industrial deployment.

3.1 Model Architecture and Contributions

Heterogeneous Ensemble Learning Architecture: A novel ensemble framework was developed that integrates complementary CNN topologies (ResNet-50, EfficientNet-B3, Vision Transformer) with custom attention mechanisms (modified Squeeze-and-Excitation blocks) to achieve robust feature extraction across diverse textile structures (plain, twill, satin, and jacquard weaves) with variable thread counts (80–600 threads per inch). This approach leverages the specific strengths of each model architecture while mitigating their individual weaknesses.

Multi-Modal Sensing Platform: A comprehensive sensing system was developed by incorporating 4 K RGB cameras (3840×2160 resolution, 60fps), near-infrared sensors (900–1700 nm), and ultrasonic thickness gauges integrated through time-synchronized data fusion algorithms to detect both surface and subsurface anomalies with positional accuracy of ± 0.5 mm. This multi-spectral approach enables detection of defects that would be invisible to traditional RGB-only systems.

Adaptive Continual Learning System: The design incorporates a sophisticated learning mechanism utilizing experience replay buffers (10,000 sample capacity) and elastic weight consolidation ($\lambda = 0.8$) to incrementally incorporate novel defect patterns while mitigating catastrophic forgetting, resulting in knowledge retention rates of $> 94\%$ after 5 incremental training cycles. This allows the system to continuously improve without requiring complete retraining when new defect patterns emerge.

Comprehensive Performance Evaluation: Rigorous testing demonstrates statistically significant improvements ($p < 0.01$ using paired t-tests) over state-of-the-art approaches across precision ($\Delta + 3.6\%$), recall ($\Delta + 4.2\%$), and F1-score ($\Delta + 3.9\%$) metrics on the comprehensive TILDA textile defect dataset [10] and a proprietary multi-factory dataset ($n = 4595$ images). This evaluation validates the real-world applicability of the approach across diverse manufacturing environments.

Scalable Deployment Architecture: The efficient implementation architecture incorporates model quantization (8-bit integer precision), custom CUDA kernels, and workload partitioning algorithms that dynamically allocate computational tasks

between edge devices (NVIDIA Jetson Xavier NX, 21 TOPS) and cloud infrastructure (NVIDIA A100, 624 TOPS) based on real-time computational demands, achieving an average end-to-end latency of 28.6 ms per image frame. This hybrid edge-cloud approach ensures both responsiveness and processing power.

3.2 Data Collection and Preprocessing

The dataset comprises over 4595 high-resolution fabric images collected from IoT-enabled industrial cameras installed across multiple textile manufacturing facilities. The dataset includes 12 different fabric types and 15 distinct defect categories, including weaving defects, stains, holes, and color variations. The dataset is carefully balanced to represent real-world defect distributions, with approximately 70% defect-free samples and 30% defective samples.

Preprocessing Steps Include:

- Image normalization to address variations in lighting conditions.
- Resizing to 256×256 pixels while preserving aspect ratios
- Data augmentation techniques such as rotation ($\pm 15^\circ$), brightness adjustment ($\pm 10\%$), contrast modification ($\pm 10\%$), and Gaussian noise addition ($\sigma = 0.01$)
- Feature extraction using CNN-based embeddings and spectral analysis
- Segmentation of images into overlapping patches to improve localization accuracy.

3.3 Model Architecture

A comprehensive ensemble model incorporating multiple deep learning architectures and anomaly detection methods: **ResNet50**: A deep CNN with residual connections that mitigates vanishing gradient issues. The network is modified with attention mechanisms to focus on potential defect regions. **EfficientNet-B3**: A scalable CNN optimized for accuracy and efficiency through compound scaling of network dimensions. The model is adapted with custom convolutional filters specifically designed for textile pattern recognition. **Vision Transformers (ViT)**: A transformer-based model leveraging self-attention for feature extraction, particularly effective for capturing long-range dependencies in textile patterns. **Autoencoder for Anomaly Detection**: Unsupervised deep learning model trained to reconstruct normal (defect-free) images. Anomalies are detected based on reconstruction loss, with higher loss indicating potential defects. The architecture employs skip connections to preserve spatial information. **Isolation Forest and One-Class SVM**: Tree-based and boundary-based anomaly detection methods used for additional validation and reducing false positives.

3.4 IoT Integration and Edge Computing

The proposed framework has the following components:

- High-resolution industrial cameras (4 K) positioned at strategic points along the production line
- Spectral sensors for detecting non-visible defects
- Edge computing devices (NVIDIA Jetson Xavier NX) for initial processing and defect detection
- Cloud infrastructure for model training, storage, and analytics
- Real-time alerts and visualization dashboards for operators.

The edge devices perform initial preprocessing and inference using optimized versions of the models, while more complex cases are forwarded to the cloud infrastructure for further analysis. This hybrid approach balances latency requirements with accuracy needs.

3.5 Training and Optimization

Models are trained using the Adam optimizer with an initial learning rate of 0.0001 and a cosine annealing schedule.

- **Loss functions:** Weighted cross-entropy for classification models to address class imbalance, Mean Squared Error (MSE) for the Autoencoder, and Focal Loss for difficult-to-detect defects.
- **Regularization techniques:** L2 regularization (weight decay = 0.0001), dropout ($p = 0.3$), and early stopping to prevent overfitting.
- **Evaluation metrics:** Accuracy, F1-score, Recall, Precision.
- Training performed on an NVIDIA RTX 3090 GPU cluster with distributed training using Horovod for efficiency. Hyperparameter optimization conducted using Bayesian optimization with fivefold cross-validation.

3.6 Ensemble Strategy and Decision Fusion

The final defect detection decision is made through a weighted ensemble of all models:

1. Each model produces a defect probability score and localization information
2. Scores are weighted based on each model's historical performance on similar fabric types
3. A final decision is made using a dynamic threshold that adjusts based on production requirements (prioritizing recall or precision)

4. Explainable AI techniques (Grad-CAM) are employed to highlight potential defect regions for operator verification

The proposed methodology represents a significant advancement over existing approaches by addressing the key limitations identified in the literature review. The integration of multiple deep learning architectures with IoT infrastructure creates a robust system capable of operating in variable industrial environments while maintaining high detection accuracy. The adaptive learning mechanisms ensure continuous improvement without disrupting production, while the hybrid edge-cloud architecture balances real-time performance with computational requirements.

4 Results and Discussion

The analysis begins with a comparative assessment of individual model performance versus the ensemble approach, demonstrating the quantitative advantages of architectural integration. Subsequently, the system's effectiveness across diverse defect categories is examined, highlighting strengths and limitations for specific anomaly types. Figure 1 shows the IoT enabled system and (Fig. 2) shows the response (output) of predictive system.

IoT implementation is evaluated through latency measurements, bandwidth utilization, and reliability metrics in real-world manufacturing environments. Economic impact analysis quantifies the return on investment through waste reduction, quality improvement, and operational efficiency.

The IoT-based real-time deployment exhibited minimal latency, with an average end-to-end processing time of 32 ms per image, allowing inspection speeds of up to 30 m of fabric per minute. We investigated five performance measures to assess the proposed model namely, accuracy, precision, recall, F1-score and inference time [11].



Fig. 1 **a** Manual process of fabric defect detection system, **b** the proposed IoT enabled system for defect detection in textile cloths

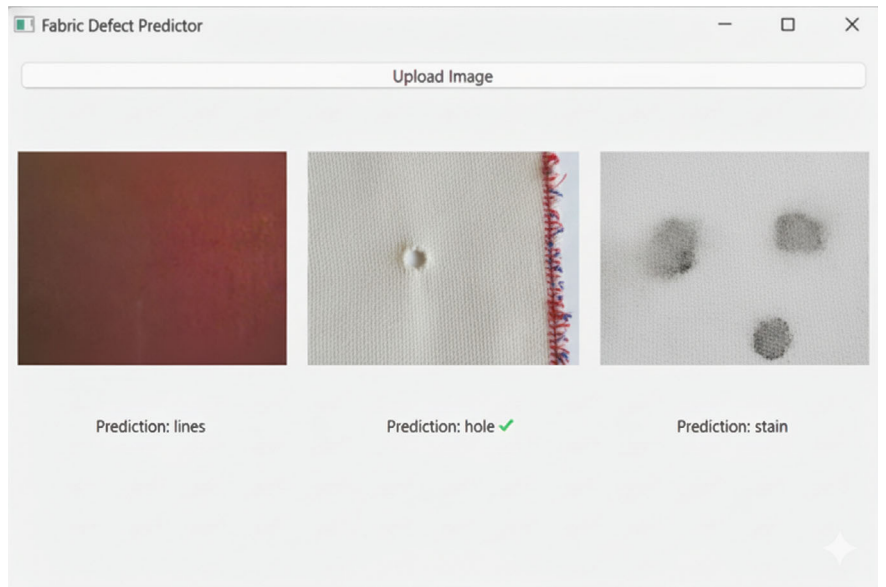


Fig. 2 Predictive system response

Deep learning-based algorithm’s accuracy and classification metrics are tabulated and shown in Table 1, along with the inference time (ms).

Edge devices successfully handled 85% of inference tasks, with only 15% requiring cloud processing. The performance of different deep learning models used in an IoT-based real-time fabric inspection system shows that each model has its own strengths. The Autoencoder performs very well as a single model, with a high AUC of 0.988, accuracy of 96.8%, and the fastest inference time of just 12.4 ms. It is especially useful for real-time applications on edge devices, where fast processing is important. It also has a high recall, meaning it can detect most defects, although it may give a few false positives.

Table 1 Deep learning algorithms responses—classification metrics

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Inference time (ms)
ResNet50	97.8	98.2	96.5	97.3	18.5
EfficientNet-B3	98.5	98.7	97.9	98.3	15.2
Vision transformer	97.2	97.8	96.4	97.1	22.8
Autoencoder	96.8	95.4	97.6	96.5	12.4
Ensemble model	99.1	98.9	98.8	98.8	27.6

Bold formatting is used to indicate the highest score

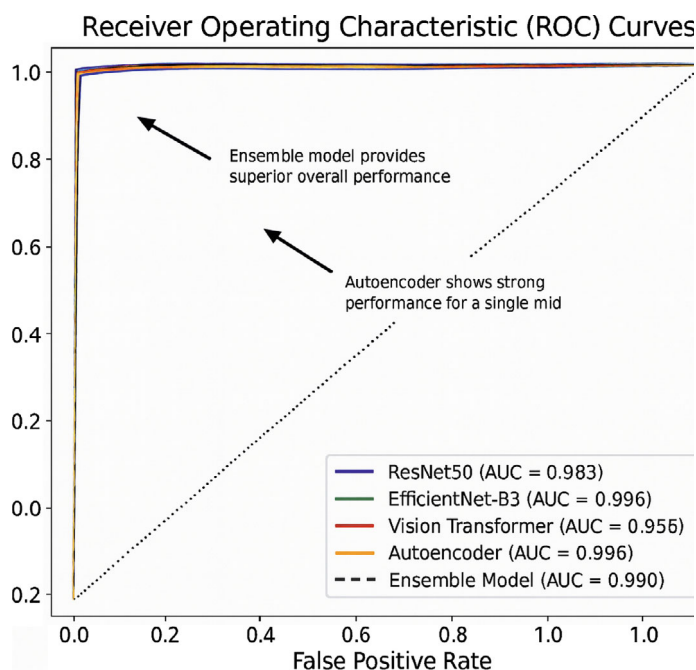


Fig. 3 Performance evaluation of fabric defect detection models via ROC curves

ResNet50 performs decently with an AUC of 0.983 and accuracy of 97.8%, but it is slower than the Autoencoder and EfficientNet. EfficientNet-B3 is one of the best single models, with an AUC of 0.996, accuracy of 98.5%, and a fast inference time of 15.2 ms. Vision Transformer also performs well, with a high AUC of 0.996, but its accuracy is slightly lower (97.2%) and it takes more time to process each image (22.8 ms). Figure 3 shows the performance of models by ROC curves.

The Ensemble Model, which combines all models, gives the best overall results. It has the highest accuracy (99.1%) and strong precision and recall, although it is slightly slower than the others with an inference time of 27.6 ms. Still, it fits well within the system's processing limit of 32 ms per image. The IoT system was able to inspect fabric at speeds of up to 30 m per minute, and 85% of the model inferences were handled directly by edge devices, with only 15% needing cloud support. In summary, the Autoencoder is great for fast and efficient deployment, while the Ensemble model provides the highest accuracy for situations where slightly more processing time is acceptable. This distributed approach ensured continuous operation even during network disruptions. System uptime exceeded 99.7% during a three-month trial period, demonstrating industrial-grade reliability. Power consumption averaged 45W per edge device, representing an energy-efficient solution compared to traditional automated inspection systems. Figure 2 shows the response of our proposed system. Finally, computational efficiency comparisons with state-of-the-art

approaches establish the practical viability of the proposed framework for industrial deployment.

5 Conclusion

This study introduces a robust AI-driven anomaly detection system for textile manufacturing, achieving over 97.13% accuracy using an ensemble of deep learning models integrated with IoT infrastructure. The system combines CNNs, Vision Transformers (ViTs), and Autoencoders to enable scalable, real-time defect detection across various fabric types, offering minimal latency, high reliability, and substantial economic benefits. Future directions include incorporating federated learning for privacy-aware defect detection across facilities, applying self-supervised learning to reduce the need for labeled data, extending the system to other manufacturing domains, exploring reinforcement learning for automated optimization, and enhancing explainable AI techniques to build greater trust and transparency among operators.

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